

Saudi Sanitary Code-Plumbing

SBC 701 - CR

Code Requirements

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Saudi Plumbing Code

SBC 701

Key List of the Saudi Codes: Designations and brief titles

Title	Code Req. ¹	Code & Com. ²	Arabic Prov. ³
The General Building Code	SBC 201-CR	SBC 201-CC	SBC 201-AR
Structural – Loading and Forces	SBC 301-CR	SBC 301-CC	SBC 301-AR
Structural – Construction	SBC 302- CR		SBC 302-AR
Structural – Soil and Foundations	SBC 303- CR	SBC 303-CC	SBC 303-AR
Structural – Concrete Structures	SBC 304- CR	SBC 304-CC	SBC 304-AR
Structural – Masonry Structures	SBC 305- CR	SBC 305-CC	SBC 305-AR
Structural – Steel Structures			
Electrical Code	SBC 401- CR		SBC 401-AR
Mechanical Code	SBC 501-CR	SBC 501-CC	SBC 501-AR
Energy Conservation-Nonresidential	SBC 601- CR	SBC 601- CC	SBC 601- AR
Energy Conservation-Residential	SBC 602- CR	SBC 602- CC	SBC 602- AR
Plumbing Code	SBC 701- CR	SBC 701-CC	SBC 701-AR
Private sewage Code	SBC 702- CR		SBC 702-AR
Fire Code	SBC 801- CR	SBC 801-CC	SBC 801-AR
Existing Buildings Code	SBC 901- CR	SBC 901-CC	SBC 901-AR
Green Construction Code	SBC 1001- CR	SBC 1001-CC	SBC 1001-AR
Residential Building Code*	SBC 1101- CR	SBC 1101-CC	SBC 1101-AR
Fuel Gas Code*	SBC 1201- CR	SBC 1201-CC	SBC 1201-AR
1. CR: Code Requirements without Commentary 2. CC: Code Requirements with Commentary 3. AR: Arabic Code Provisions * Under Development			

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PREFACE

The Saudi Plumbing Code (SBC 701) is a Code that regulates the design and installation of plumbing systems including the plumbing fixtures in the buildings. SBC 701 addresses general plumbing regulations, fixture requirements, water heater installations and systems for water distribution, sanitary drainage, special wastes, venting, storm drainage and medical gases. The purpose of the SBC 701 is to establish the minimum acceptable level of safety to protect life and property from the potential dangers associated with supplying potable water to plumbing fixtures and outlets and the conveyance of bacteria-laden wastewater from fixtures. In summary, the SBC 701 sets forth the minimum requirements for providing safe water to a building as well as a safe manner in which liquid-borne wastes are carried away from a building.

2015 International Plumbing Code (IPC 2015), published by the International Code Council (ICC), is the base code in the development of this Code. Saudi Building Code National Committee (SBCNC) has made an agreement with the ICC to use their materials and modify them as per the local construction needs and regulatory requirements of Saudi Arabia. The ICC is not responsible or liable in any way to SBCNC or to any other party or entity for any modifications or changes that SBCNC makes to such documents.

The contents of the current edition of the Code is substantially updated from its 2007 edition. The writing process of SBC 701-18 followed the methodology approved by the Saudi Building Code National Committee. Many changes and modifications were made in its base code (IPC 2015) to meet the local weather, materials, construction and regulatory requirements.

The committees responsible for SBC 701 Code have taken all precautions to avoid ambiguities, omissions, and errors in the document. Despite these efforts, the users of SBC 701 may find information or requirements that may be subject to more than one interpretation or may be incomplete. The SBCNC alone possesses the authority and responsibility for updating, modifying and interpreting the Code.

It is a common assumption that engineering knowledge is a prerequisite in understanding code provisions and requirements; thus, the code is oriented towards individuals who possess the background knowledge to evaluate the significance and limitations of its content and recommendations. They shall be able to determine the applicability of all regulatory limitations before applying the Code and must comply with all applicable laws and regulations.

The requirements related to administration and enforcement of this Code are advisory only. SBCNC and governmental organizations, in charge of enforcing this Code, possess the authority to modify these administrative requirements.



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CHAPTER 1 SCOPE AND ADMINISTRATION

PART 1—SCOPE AND APPLICATION

SECTION 101 GENERAL

101.1 Title.

These regulations shall be known as the Saudi Plumbing Code (SBC 701) hereinafter referred to as “this code.”

101.2 Scope.

The provisions of this code shall apply to the erection, installation, alteration, repairs, relocation, replacement, addition to, use or maintenance of plumbing systems within this jurisdiction. This code shall regulate non-flammable medical gas, inhalation anesthetic, vacuum piping, nonmedical oxygen systems and sanitary and condensate vacuum collection systems. The installation of fuel gas distribution piping and equipment, fuel-gas-fired water heaters and water heater venting systems shall be regulated by the Saudi *Fuel Gas Code* (SBC 1201). Provisions in the appendices shall not apply unless specifically adopted.

Exception: Detached one and two-family dwellings and multiple single-family dwellings (townhouses) not more than three stories high with separate means of egress and their accessory structures shall comply with the *SBC 1101*.

101.3 Intent.

The purpose of this code is to establish minimum standards to provide a reasonable level of safety, health, property protection and public welfare by regulating and controlling the design, construction, installation, quality of materials, location, operation and maintenance or use of plumbing equipment and systems

101.4 Severability.

If any section, subsection, sentence, clause or phrase of this code is for any reason held to be unconstitutional, such decision shall not affect the validity of the remaining portions of this code.

SECTION 102 APPLICABILITY

102.1 General.

Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall govern. Where, in any specific case, different sections of this code specify different

materials, methods of construction or other requirements, the most restrictive shall govern.

102.2 Existing installations.

Plumbing systems lawfully in existence at the time of the adoption of this code shall be permitted to have their use and maintenance continued if the use, maintenance or repair is in accordance with the original design and hazard to life, health or property is not created by such plumbing system.

102.3 Maintenance.

Plumbing systems, materials and appurtenances, both existing and new, and parts thereof, shall be maintained in proper operating condition in accordance with the original design in a safe and sanitary condition. Devices or safeguards required by this code shall be maintained in compliance with the edition of the code under which they were installed.

The owner or the owner’s authorized agent shall be responsible for maintenance of plumbing systems. To determine compliance with this provision, the code official shall have the authority to require any plumbing system to be re-inspected.

102.4 Additions, alterations or repairs.

Additions, alterations, renovations or repairs to any plumbing system shall conform to that required for a new plumbing system without requiring the existing plumbing system to comply with all the requirements of this code. Additions, alterations or repairs shall not cause an existing system to become unsafe, insanitary or overloaded.

Minor additions, alterations, renovations and repairs to existing plumbing systems shall meet the provisions for new construction, unless such work is done in the same manner and arrangement as was in the existing system, is not hazardous and is *approved*.

102.5 Change in occupancy.

It shall be unlawful to make any change in the *occupancy* of any structure that will subject the structure to any special provision of this code applicable to the new *occupancy* without approval of the code official. The code official shall certify that such structure meets the intent of the provisions of law governing building construction for the proposed new *occupancy* and that such change of *occupancy* does not result in any hazard to the

public health, safety or welfare.

102.6 Historic buildings.

The provisions of this code relating to the construction, alteration, repair, enlargement, restoration, relocation or moving of buildings or structures shall not be mandatory for existing buildings or structures identified and classified by the state or local jurisdiction as historic buildings where such buildings or structures are judged by the code official to be safe and in the public interest of health, safety and welfare regarding any proposed construction, alteration, repair, enlargement, restoration, relocation or moving of buildings.

102.7 Moved buildings.

Except as determined by Section 102.2, plumbing systems that are a part of buildings or structures moved into or within the jurisdiction shall comply with the provisions of this code for new installations.

102.8 Referenced codes and standards.

The codes and standards referenced in this code shall be those that are listed in Chapter 15 and such codes and standards shall be considered as part of the requirements of this code to the prescribed extent of each such reference and as further regulated in Sections 102.8.1 and 102.8.2.

102.8.1 Conflicts.

Where conflicts occur between provisions of this code and the referenced standards, the provisions of this code shall apply.

102.8.2 Provisions in referenced codes and standards.

Where the extent of the reference to a referenced code or standard includes subject matter that is within the scope of this code, the provisions of this code, as applicable, shall take precedence over the provisions in the referenced code or standard.

102.9 Requirements not covered by code.

Any requirements necessary for the strength, stability or proper operation of an existing or proposed plumbing system, or for the public safety, health and general welfare, not specifically covered by this code shall be determined by the code official.

102.10 Other laws.

The provisions of this code shall not be deemed to nullify any provisions of local, state or federal law.

102.11 Application of references.

Reference to chapter section numbers, or to provisions not specifically identified by number, shall be construed to refer to such chapter, section or provision of this code.

PART 2—ADMINISTRATION AND ENFORCEMENT

SECTION 103 DEPARTMENT OF PLUMBING INSPECTION

103.1 General.

The department of plumbing inspection is hereby created and the executive official in charge thereof shall be known as the code official.

103.2 Appointment.

The code official shall be appointed by the chief appointing authority of the jurisdiction.

103.3 Deputies.

In accordance with the prescribed procedures of this jurisdiction and with the concurrence of the appointing authority, the code official shall have the authority to appoint a deputy code official, other related technical officers, inspectors and other employees. Such employees shall have powers as delegated by the code official.

103.4 Liability.

The code official, member of the board of appeals or employee charged with the enforcement of this code, while acting for the jurisdiction in good faith and without malice in the discharge of the duties required by this code or other pertinent law or ordinance, shall not thereby be rendered civilly or criminally liable personally, and is hereby relieved from all personal liability for any damage accruing to persons or property as a result of any act or by reason of an act or omission in the discharge of official duties.

103.4.1 Legal defense.

Any suit or criminal complaint instituted against any officer or employee because of an act performed by that officer or employee in the lawful discharge of duties and under the provisions of this code shall be defended by the legal representative of the jurisdiction until the final termination of the proceedings. The code official or any subordinate shall not be liable for costs in any action, suit or proceeding that is instituted in pursuance of the provisions of this code.

SECTION 104 DUTIES AND POWERS OF THE CODE OFFICIAL

104.1 General.

The code official is hereby authorized and directed to enforce the provisions of this code. The code official shall have the authority to render interpretations of this code and to adopt policies and procedures in order to clarify the application of its provisions. Such interpretations, policies and procedures shall be in compliance with the intent and purpose of this code. Such policies and procedures shall not have the effect of waiving requirements specifically provided for in this code.

104.2 Applications and permits.

The code official shall receive applications, review construction documents and issue permits for the installation and alteration of plumbing systems, inspect the premises for which such permits have been issued, and enforce compliance with the provisions of this code.

104.3 Inspections.

The code official shall make all the required inspections, or shall accept reports of inspection by *approved* agencies or individuals. Reports of such inspections shall be in writing and be certified by a responsible officer of such *approved agency* or by the responsible individual. The code official is authorized to engage such expert opinion as deemed necessary to report on unusual technical issues that arise, subject to the approval of the appointing authority.

104.4 Right of entry.

Where it is necessary to make an inspection to enforce the provisions of this code, or where the code official has reasonable cause to believe that there exists in any building or upon any premises any conditions or violations of this code that make the building or premises unsafe, insanitary, dangerous or hazardous, the code official shall have the authority to enter the building or premises at all reasonable times to inspect or to perform the duties imposed upon the code official by this code. If such building or premises is occupied, the code official shall present credentials to the occupant and request entry. If such building or premises is unoccupied, the code official shall first make a reasonable effort to locate the owner, the owner's authorized agent or other person having charge or control of the building or premises and request entry. If entry is refused, the code official shall have recourse to every remedy provided by law to secure entry.

Where the code official shall have first obtained a proper inspection warrant or other remedy provided by law to secure entry, no owner, owner's authorized agent, occupant or person having charge, care or control of any building or premises shall fail or neglect, after proper request is made as herein provided, to promptly permit entry therein by the code official for the purpose of inspection and examination pursuant to this code.

104.5 Identification.

The code official shall carry proper identification when inspecting structures or premises in the performance of duties under this code.

104.6 Notices and orders.

The code official shall issue all necessary notices or orders to ensure compliance with this code.

104.7 Department records.

The code official shall keep official records of applications received, permits and certificates issued, fees collected, reports of inspections, and notices and orders issued. Such records shall be retained in the official records for the period required for the retention of public records.

SECTION 105 APPROVAL

105.1 Modifications.

Where there are practical difficulties involved in carrying out the provisions of this code, the code official shall have the authority to grant modifications for individual cases, upon application of the owner or owner's authorized agent, provided the code official shall first find that special individual reason makes the strict letter of this code impractical and the modification conforms to the intent and purpose of this code and that such modification does not lessen health, life and fire safety requirements. The details of action granting modifications shall be recorded and entered in the files of the plumbing inspection department.

105.2 Alternative materials, methods and equipment.

The provisions of this code are not intended to prevent the installation of any material or to prohibit any method of construction not specifically prescribed by this code, provided that any such alternative has been *approved*. An alternative material or method of construction shall be *approved* where the code official finds that the proposed alternative material, method or equipment complies with the intent of the provisions of this

code and is not less than the equivalent of that prescribed in this code. Where the alternative material, design or method of construction is not *approved*, the code official shall respond in writing, stating the reasons why the alternative was not *approved*.

105.2.1 Research reports.

Supporting data, where necessary to assist in the approval of materials or assemblies not specifically provided for in this code, shall consist of valid research reports from *approved* sources.

105.3 Required testing.

Where there is insufficient evidence of compliance with the provisions of this code, or evidence that a material or method does not conform to the requirements of this code, or in order to substantiate claims for alternate materials or methods, the code official shall have the authority to require tests as evidence of compliance to be made at no expense to the jurisdiction.

105.3.1 Test methods.

Test methods shall be as specified in this code or by other recognized test standards. In the absence of recognized and accepted test methods, the code official shall approve the testing procedures.

105.3.2 Testing agency.

Tests shall be performed by an *approved* agency.

105.3.3 Test reports.

Reports of tests shall be retained by the code official for the period required for retention of public records.

105.4 Approved materials and equipment.

Materials, equipment and devices approved by the code official shall be constructed and installed in accordance with such approval.

105.4.1 Material and equipment reuse.

Materials, equipment and devices shall not be reused unless such elements have been reconditioned, tested, placed in good and proper working condition and *approved*.

SECTION 106 PERMITS

106.1 When Required.

Any owner, owner's authorized agent or contractor who desires to construct, enlarge, alter, repair, move, demolish or change the *occupancy* of a building or structure, or to erect, install, enlarge, alter, repair, remove, convert or replace any

plumbing system, the installation of which is regulated by this code, or to cause any such work to be performed, shall first make application to the code official and obtain the required permit for the work.

106.1.1 Annual permit.

Instead of an individual construction permit for each alteration to an already *approved* system or equipment or appliance installation, the code official is authorized to issue an annual permit upon application therefor to any person, firm or corporation regularly employing one or more qualified tradespersons in the building, structure or on the premises owned or operated by the applicant for the permit.

106.1.2 Annual permit records.

The person to whom an annual permit is issued shall keep a detailed record of alterations made under such annual permit. The code official shall have access to such records at all times or such records shall be filed with the code official as designated.

106.2 Exempt work.

The following work shall be exempt from the requirement for a permit:

- 1 The stopping of leaks in drains, water, soil, waste or vent pipe provided, however, that if any concealed trap, drainpipe, water, soil, waste or vent pipe becomes defective and it becomes necessary to remove and replace the same with new material, such work shall be considered as new work and a permit shall be obtained and inspection made as provided in this code.
- 2 The clearing of stoppages or the repairing of leaks in pipes, valves or fixtures, and the removal and reinstallation of water closets, provided such repairs do not involve or require the replacement or rearrangement of valves, pipes or fixtures.

Exemption from the permit requirements of this code shall not be deemed to grant authorization for any work to be done in violation of the provisions of this code or any other laws or ordinances of this jurisdiction.

106.3 Application for permit.

Each application for a permit, with the required fee, shall be filed with the code official on a form furnished for that purpose and shall contain a general description of the proposed work and its

location. The application shall be signed by the owner or owner's authorized agent. The permit application shall indicate the proposed occupancy of all parts of the building and of that portion of the site or lot, if any, not covered by the building or structure and shall contain such other information required by the code official.

106.3.1 Construction documents.

Construction documents, engineering calculations, diagrams and other such data shall be submitted in two or more sets with each application for a permit. The code official shall require construction documents, computations and specifications to be prepared and designed by a registered design professional where required by state law. Construction documents shall be drawn to scale and shall be of sufficient clarity to indicate the location, nature and extent of the work proposed and show in detail that the work conforms to the provisions of this code. Construction documents for buildings more than two stories in height shall indicate where penetrations will be made for pipes, fittings and components and shall indicate the materials and methods for maintaining required structural safety, fire-resistance rating and fire blocking.

Exception: The code official shall have the authority to waive the submission of construction documents, calculations or other data if the nature of the work applied for is such that reviewing of construction documents is not necessary to determine compliance with this code.

106.3.2 Preliminary inspection.

Before a permit is issued, the code official shall be authorized to inspect and evaluate the systems, equipment, buildings, devices, premises and spaces or areas to be used.

106.3.3 Time limitation of application.

An application for a permit for any proposed work shall be deemed to have been abandoned 180 days after the date of filing, unless such application has been pursued in good faith or a permit has been issued; except that the code official shall have the authority to grant one or more extensions of time for additional periods not exceeding 180 days each. The extension shall be requested in writing and justifiable cause demonstrated.

106.4 By whom application is made.

Application for a permit shall be made by the person or agent to install all or part of any plumbing system. The applicant shall meet all qualifications

established by statute, or by rules promulgated by this code, or by ordinance or by resolution. The full name and address of the applicant shall be stated in the application.

106.5 Permit issuance.

The application, construction documents and other data filed by an applicant for permit shall be reviewed by the code official. If the code official finds that the proposed work conforms to the requirements of this code and all laws and ordinances applicable thereto, and that the fees specified in Section 106.6 have been paid, a permit shall be issued to the applicant.

106.5.1 Approved construction documents.

When the code official issues the permit where construction documents are required, the construction documents shall be endorsed in writing and stamped "APPROVED." Such *approved* construction documents shall not be changed, modified or altered without authorization from the code official. Work shall be done in accordance with the *approved* construction documents.

The code official shall have the authority to issue a permit for the construction of a part of a plumbing system before the entire construction documents for the whole system have been submitted or *approved*, provided adequate information and detailed statements have been filed complying with all pertinent requirements of this code. The holders of such permit shall proceed at their own risk without assurance that the permit for the entire plumbing system will be granted.

106.5.2 Validity.

The issuance of a permit or approval of construction documents shall not be construed to be a permit for, or an approval of, any violation of any of the provisions of this code or any other ordinance of the jurisdiction. A permit presuming to give authority to violate or cancel the provisions of this code shall not be valid.

The issuance of a permit based upon construction documents and other data shall not prevent the code official from thereafter requiring the correction of errors in said construction documents and other data or from preventing building operations being carried on there under where in violation of this code or of other ordinances of this jurisdiction.

106.5.3 Expiration.

Every permit issued by the code official under the provisions of this code shall expire by limitation

and become null and void if the work authorized by such permit is not commenced within 180 days from the date of such permit, or if the work authorized by such permit is suspended or abandoned at any time after the work is commenced for a period of 180 days. Before such work can be recommenced, a new permit shall be first obtained and the fee therefore shall be one-half the amount required for a new permit for such work, provided changes have not been made and will not be made in the original construction documents for such work, and provided further that such suspension or abandonment has not exceeded 1 year.

106.5.4 Extensions.

Any permittee holding an unexpired permit shall have the right to apply for an extension of the time within which the permittee will commence work under that permit when work is unable to be commenced within the time required by this section for good and satisfactory reasons. The code official shall extend the time for action by the permittee for a period not exceeding 180 days if there is reasonable cause. A permit shall not be extended more than once. The fee for an extension shall be one-half the amount required for a new permit for such work.

106.5.5 Suspension or revocation of permit.

The code official shall have the authority to suspend or revoke a permit issued under the provisions of this code wherever the permit is issued in error or on the basis of incorrect, inaccurate or incomplete information, or in violation of any ordinance or regulation or any of the provisions of this code.

106.5.6 Retention of construction documents.

One set of *approved* construction documents shall be retained by the code official for a period of not less than 180 days from date of completion of the permitted work, or as required by state or local laws.

One set of approved construction documents shall be returned to the applicant, and said set shall be kept on the site of the building or work at all times during which the work authorized thereby is in progress.

106.5.7 Previous approvals.

This code shall not require changes in the construction documents, construction or designated occupancy of a structure for which a lawful permit has been heretofore issued or otherwise lawfully authorized, and the construction of which has been pursued in good faith within 180 days after the

effective date of this code and has not been abandoned.

106.5.8 Posting of permit.

The permit or a copy shall be kept on the site of the work until the completion of the project.

106.6 Fees.

A permit shall not be issued until the fees prescribed in Section 106.6.2 have been paid, and an amendment to a permit shall not be released until the additional fee, if any, due to an increase of the plumbing systems, has been paid.

106.6.1 Work commencing before permit issuance.

Any person who commences any work on a plumbing system before obtaining the necessary permits shall be subject to 100 percent of the usual permit fee in addition to the required permit fees.

106.6.2 Fee schedule.

The fees for all plumbing work shall be as indicated in the following:

106.6.3 Fee refunds.

The code official shall authorize the refunding of fees as follows:

- 1 The full amount of any fee paid hereunder that was erroneously paid or collected.
- 2 Not more than [SPECIFY PERCENTAGE] percent of the permit fee paid where work has been done under a permit issued in accordance with this code.
- 3 Not more than [SPECIFY PERCENTAGE] percent of the plan review fee paid where an application for a permit for which a plan review fee has been paid is withdrawn or canceled before any plan review effort has been expended.

The code official shall not authorize the refunding of any fee paid except upon written application filed by the original permittee not later than 180 days after the date of fee payment.

SECTION 107 INSPECTIONS AND TESTING

107.1 General.

The code official is authorized to conduct such inspections as are deemed necessary to determine compliance with the provisions of this code. Construction or work for which a permit is required

shall be subject to inspection by the code official, and such construction or work shall remain accessible and exposed for inspection purposes until *approved*. Approval as a result of an inspection shall not be construed to be an approval of a violation of the provisions of this code or of other ordinances of the jurisdiction. Inspections presuming to give authority to violate or cancel the provisions of this code or of other ordinances of the jurisdiction shall not be valid. It shall be the duty of the permit applicant to cause the work to remain accessible and exposed for inspection purposes. Neither the code official nor the jurisdiction shall be liable for expense entailed in the removal or replacement of any material required to allow inspection.

107.2 Required inspections and testing.

The code official, upon notification from the permit holder or the permit holder's agent, shall make the following inspections and such other inspections as necessary, and shall either release that portion of the construction or shall notify the permit holder or an agent of any violations that must be corrected. The holder of the permit shall be responsible for the scheduling of such inspections.

- 1 Underground inspection shall be made after trenches or ditches are excavated and bedded, piping installed, and before any backfill is put in place.
- 2 Rough-in inspection shall be made after the roof, framing, fire blocking, fire stopping, draft stopping and bracing is in place and all sanitary, storm and water distribution piping is roughed-in, and prior to the installation of wall or ceiling membranes.
- 3 Final inspection shall be made after the building is complete, all plumbing fixtures are in place and properly connected, and the structure is ready for occupancy.

107.2.1 Other inspections.

In addition to the inspections specified above, the code official shall be authorized to make or require other inspections of any construction work to ascertain compliance with the provisions of this code and other laws that are enforced.

107.2.2 Inspection requests.

It shall be the duty of the holder of the permit or their duly authorized agent to notify the code official when work is ready for inspection. It shall be the duty of the permit holder to provide *access* to

and means for inspections of such work that are required by this code.

107.2.3 Approval required.

Work shall not be done beyond the point indicated in each successive inspection without first obtaining the approval of the code official. The code official, upon notification, shall make the requested inspections and shall either indicate the portion of the construction that is satisfactory as completed, or notify the permit holder or his or her agent wherein the same fails to comply with this code. Any portions that do not comply shall be corrected and such portion shall not be covered or concealed until authorized by the code official.

107.2.4 Approved agencies.

The code official is authorized to accept reports of *approved* inspection agencies, provided that such agencies satisfy the requirements as to qualifications and reliability.

107.2.5 Evaluation and follow-up inspection services.

Prior to the approval of a closed, prefabricated plumbing system and the issuance of a plumbing permit, the code official shall require the submittal of an evaluation report on each prefabricated plumbing system indicating the complete details of the plumbing system, including a description of the system and its components, the basis upon which the plumbing system is being evaluated, test results and similar information, and other data as necessary for the code official to determine conformance to this code.

107.2.5.1 Evaluation service.

The code official shall designate the evaluation service of an *approved agency* as the evaluation agency, and review such agency's evaluation report for adequacy and conformance to this code.

107.2.5.2 Follow-up inspection.

Except where *ready access* is provided to all plumbing systems, service equipment and accessories for complete inspection at the site without disassembly or dismantling, the code official shall conduct the frequency of in-plant inspections necessary to ensure conformance to the *approved* evaluation report or shall designate an independent, *approved* inspection agency to conduct such inspections. The inspection agency shall furnish the code official with the follow-up inspection manual and a report of inspections upon request, and the plumbing system shall have an

identifying label permanently affixed to the system indicating that factory inspections have been performed.

107.2.5.3 Test and inspection records.

Required test and inspection records shall be available to the code official at all times during the fabrication of the plumbing system and the erection of the building, or such records as the code official designates shall be filed.

107.3 Special inspections.

Special inspections of *alternative engineered design* plumbing systems shall be conducted in accordance with Sections 107.3.1 and 107.3.2.

107.3.1 Periodic inspection.

The registered design professional or designated inspector shall periodically inspect and observe the *alternative engineered design* to determine that the installation is in accordance with the *approved* construction documents. Discrepancies shall be brought to the immediate attention of the plumbing contractor for correction. Records shall be kept of all inspections.

107.3.2 Written report.

The registered design professional shall submit a final report in writing to the code official upon completion of the installation, certifying that the alternative engineered design conforms to the approved construction documents. A notice of approval for the plumbing system shall not be issued until a written certification has been submitted.

107.4 Testing.

Plumbing work and systems shall be tested as required in Section 312 and in accordance with Sections 107.4.1 through 107.4.3. Tests shall be made by the permit holder and observed by the code official.

107.4.1 New, altered, extended or repaired systems.

New plumbing systems and parts of existing systems that have been altered, extended or repaired shall be tested as prescribed herein to disclose leaks and defects, except that testing is not required in the following cases:

- 1 In any case that does not include addition to, replacement, alteration or relocation of any water supply, drainage or vent piping.

- 2 In any case where plumbing equipment is set up temporarily for exhibition purposes.

107.4.2 Equipment, material and labor for tests.

Equipment, material and labor required for testing a plumbing system or part thereof shall be furnished by the permit holder.

107.4.3 Re-inspection and testing.

Where any work or installation does not pass any initial test or inspection, the necessary corrections shall be made to comply with this code. The work or installation shall then be resubmitted to the code official for inspection and testing.

107.5 Approval.

After the prescribed tests and inspections indicate that the work complies in all respects with this code, a notice of approval shall be issued by the code official.

107.5.1 Revocation.

The code official is authorized to, in writing, suspend or revoke a notice of approval issued under the provisions of this code wherever the notice is issued in error, or on the basis of incorrect information supplied, or where it is determined that the building or structure, premise or portion thereof is in violation of any ordinance or regulation or any of the provisions of this code.

107.6 Temporary connection.

The code official shall have the authority to authorize the temporary connection of the building or system to the utility source for the purpose of testing plumbing systems or for use under a temporary certificate of occupancy.

107.7 Connection of service utilities.

A person shall not make connections from a utility, source of energy, fuel, power, water system or sewer system to any building or system that is regulated by this code for which a permit is required until authorized by the code official.

SECTION 108 VIOLATIONS

108.1 Unlawful acts.

It shall be unlawful for any person, firm or corporation to erect, construct, alter, repair, remove, demolish or utilize any plumbing system, or cause same to be done, in conflict with or in violation of any of the provisions of this code.

108.2 Notice of violation.

The code official shall serve a notice of violation or order to the person responsible for the erection, installation, alteration, extension, repair, removal or demolition of plumbing work in violation of the provisions of this code, or in violation of a detail statement or the *approved* construction documents thereunder, or in violation of a permit or certificate issued under the provisions of this code. Such order shall direct the discontinuance of the illegal action or condition and the abatement of the violation.

108.3 Prosecution of violation.

If the notice of violation is not complied with promptly, the code official shall request the legal counsel of the jurisdiction to institute the appropriate proceeding at law or in equity to restrain, correct or abate such violation, or to require the removal or termination of the unlawful occupancy of the structure in violation of the provisions of this code or of the order or direction made pursuant thereto.

108.4 Violation penalties.

Any person who violates a provision of this code or fails to comply with any of the requirements thereof or who erects, constructs, alters or repairs a building or structure in violation of the approved construction documents or directive of the building official, or of a permit or certificate issued under the provisions of this code, shall be subject to penalties as prescribed by law. Each day that a violation continues after due notice has been served shall be deemed a separate offense. The jurisdiction must establish penalties for a variety of violations that may occur. Violations specifically addressed by the code include failure to comply with the code, approved plans or directives of the building official

108.5 Stop work orders.

Upon notice from the code official, work on any plumbing system that is being performed contrary to the provisions of this code or in a dangerous or unsafe manner shall immediately cease. Such notice shall be in writing and shall be given to the owner of the property, or to the owner's authorized agent, or to the person performing the work. The notice shall state the conditions under which work is authorized to resume. Where an emergency exists, the code official shall not be required to give a written notice prior to stopping the work. Any person who shall continue any work in or about the structure after having been served with a stop work order, except such work as that person is directed to perform to remove a violation or unsafe condition, shall be liable to a fine.

108.6 Abatement of violation.

The imposition of the penalties herein prescribed shall not preclude the legal officer of the jurisdiction from instituting appropriate action to prevent unlawful construction or to restrain, correct or abate a violation, or to prevent illegal occupancy of a building, structure or premises, or to stop an illegal act, conduct, business or utilization of the plumbing on or about any premises.

108.7 Unsafe plumbing.

Any plumbing regulated by this code that is unsafe or that constitutes a fire or health hazard, insanitary condition, or is otherwise dangerous to human life is hereby declared unsafe. Any use of plumbing regulated by this code constituting a hazard to safety, health or public welfare by reason of inadequate maintenance, dilapidation, obsolescence, fire hazard, disaster, damage or abandonment is hereby declared an unsafe use. Any such unsafe equipment is hereby declared to be a public nuisance and shall be abated by repair, rehabilitation, demolition or removal.

108.7.1 Authority to condemn equipment.

Whenever the code official determines that any plumbing, or portion thereof, regulated by this code has become hazardous to life, health or property or has become insanitary, the code official shall order in writing that such plumbing either be removed or restored to a safe or sanitary condition. A time limit for compliance with such order shall be specified in the written notice. A person shall not use or maintain defective plumbing after receiving such notice.

Where such plumbing is to be disconnected, written notice as prescribed in Section 108.2 shall be given. In cases of immediate danger to life or property, such disconnection shall be made immediately without such notice.

108.7.2 Authority to disconnect service utilities.

The code official shall have the authority to authorize disconnection of utility service to the building, structure or system regulated by the technical codes in case of an emergency, where necessary, to eliminate an immediate danger to life or property. Where possible, the owner or the owner's authorized agent and occupant of the building, structure or service system shall be notified of the decision to disconnect utility service prior to taking such action. If not notified prior to disconnecting, the owner, the owner's authorized

agent or occupant of the building, structure or service systems shall be notified in writing, as soon as practical thereafter.

108.7.3 Connection after order to disconnect.

A person shall not make connections from any energy, fuel, power supply or water distribution system or supply energy, fuel or water to any equipment regulated by this code that has been disconnected or ordered to be disconnected by the code official or the use of which has been ordered to be discontinued by the code official until the code official authorizes the reconnection and use of such equipment.

SECTION 109 MEANS OF APPEAL

109.1 Application for appeal.

Any person shall have the right to appeal a decision of the code official to the board of appeals. An application for appeal shall be based on a claim that the true intent of this code or the rules legally adopted thereunder have been incorrectly interpreted, the provisions of this code do not fully apply, or an equally good or better form of construction is proposed. The application shall be filed on a form obtained from the code official within 20 days after the notice was served.

109.2 Membership of board.

The board of appeals shall consist of five members appointed by the chief appointing authority as follows: one for 5 years, one for 4 years, one for 3 years, one for 2 years and one for 1 year. Thereafter, each new member shall serve for 5 years or until a successor has been appointed.

109.2.1 Qualifications.

The board of appeals shall consist of five individuals, one from each of the following professions or disciplines:

- 1 Registered design professional who is a registered architect; or a builder or superintendent of building construction with not less than 10 years' experience, 5 years of which shall have been in responsible charge of work.
- 2 Registered design professional with structural engineering or architectural experience.
- 3 Registered design professional with mechanical and plumbing engineering experience; or a mechanical and plumbing contractor with not less than 10 years'

experience, 5 years of which shall have been in responsible charge of work.

- 4 Registered design professional with electrical engineering experience; or an electrical contractor with not less than 10 years' experience, 5 years of which shall have been in responsible charge of work.
- 5 Registered design professional with fire protection engineering experience; or a fire protection contractor with not less than 10 years' experience, 5 years of which shall have been in responsible charge of work.

109.2.2 Alternate members.

The chief appointing authority shall appoint two alternate members who shall be called by the board chairman to hear appeals during the absence or disqualification of a member. Alternate members shall possess the qualifications required for board membership, and shall be appointed for 5 years or until a successor has been appointed.

109.2.3 Chairman.

The board shall annually select one of its members to serve as chairman.

109.2.4 Disqualification of member.

A member shall not hear an appeal in which that member has any personal, professional or financial interest.

109.2.5 Secretary.

The chief administrative officer shall designate a qualified clerk to serve as secretary to the board. The secretary shall file a detailed record of all proceedings in the office of the chief administrative officer.

109.2.6 Compensation of members.

Compensation of members shall be determined by law.

109.3 Notice of meeting.

The board shall meet upon notice from the chairman, within 10 days of the filing of an appeal or at stated periodic meetings.

109.4 Open hearing.

Hearings before the board shall be open to the public. The appellant, the appellant's representative, the code official and any person whose interests are affected shall be given an opportunity to be heard.

109.4.1 Procedure.

The board shall adopt and make available to the public through the secretary procedures under which a hearing will be conducted. The procedures shall not require compliance with strict rules of evidence, but shall mandate that only relevant information be received.

109.5 Postponed hearing.

When five members are not present to hear an appeal, either the appellant or the appellant's representative shall have the right to request a postponement of the hearing.

109.6 Board decision.

The board shall modify or reverse the decision of the code official by a concurring vote of three members.

109.6.1 Resolution.

The decision of the board shall be by resolution. Certified copies shall be furnished to the appellant and to the code official.

109.6.2 Administration.

The code official shall take immediate action in accordance with the decision of the board.

109.7 Court review.

Any person, whether or not a previous party of the appeal, shall have the right to apply to the appropriate court for a writ of certiorari to correct errors of law. Application for review shall be made in the manner and time required by law following the filing of the decision in the office of the chief administrative officer.

SECTION 110 TEMPORARY EQUIPMENT, SYSTEMS AND USES**110.1 General.**

The code official is authorized to issue a permit for temporary equipment, systems and uses. Such permits shall be limited as to time of service, but shall not be permitted for more than 180 days. The code official is authorized to grant extensions for demonstrated cause.

110.2 Conformance.

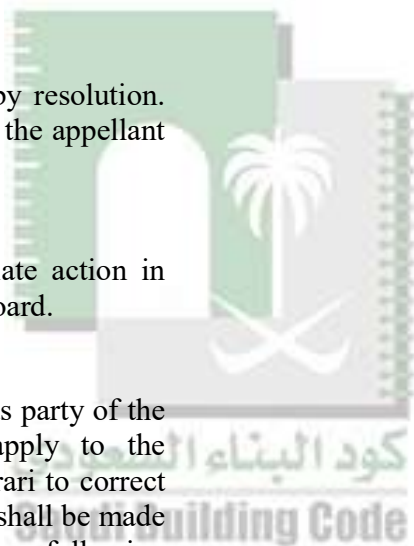
Temporary equipment, systems and uses shall conform to the structural strength, fire safety, means of egress, accessibility, light, ventilation and sanitary requirements of this code as necessary to ensure the public health, safety and general welfare.

110.3 Temporary utilities.

The code official is authorized to give permission to temporarily supply utilities before an installation has been fully completed and the final certificate of completion has been issued. The part covered by the temporary certificate shall comply with the requirements specified for temporary lighting, heat or power in the code.

110.4 Termination of approval.

The code official is authorized to terminate such permit for temporary equipment, systems or uses and to order the temporary equipment, systems or uses to be discontinued.



CHAPTER 2 DEFINITIONS

SECTION 201 GENERAL

201.1 Scope.

Unless otherwise expressly stated, the following words and terms shall, for the purposes of this code, have the meanings shown in this chapter.

201.2 Interchangeability.

Words stated in the present tense include the future; words stated in the masculine gender include the feminine; the singular number includes the plural and the plural the singular.

201.3 Terms defined in other codes.

Where terms are not defined in this code and are defined in code, such terms shall have the meanings ascribed to them as in those codes.

201.4 Terms not defined.

Where terms are not defined through the methods authorized by this section, such terms shall have ordinarily accepted meanings such as the context implies.

SECTION 202 GENERAL DEFINITIONS

ACCEPTED ENGINEERING PRACTICE.

That which conforms to accepted principles, tests or standards of nationally recognized technical or scientific authorities.

ACCESS (TO).

That which enables a fixture, appliance or equipment to be reached by ready access or by a means that first requires the removal or movement of a panel, door or similar obstruction (see "Ready access").

ACCESS COVER.

A removable plate, usually secured by bolts or screws, to permit access to a pipe or pipe fitting for the purposes of inspection, repair or cleaning.

ADAPTER FITTING.

An approved connecting device that suitably and properly joins or adjusts pipes and fittings that do not otherwise fit together.

AIR ADMITTANCE VALVE.

One-way valve designed to allow air to enter the plumbing drainage system when negative pressures develop in the piping system. The device shall close by gravity and seal the vent terminal at zero differential pressure (no flow conditions) and under positive internal pressures. The purpose of an air admittance valve is to provide a method of allowing air to enter the plumbing drainage system without the use of a vent extended to open air and to prevent sewer gases from escaping into a building.

AIR BREAK (Drainage System).

A piping arrangement in which a drain from a fixture, appliance or device discharges indirectly into another fixture, receptacle or interceptor at a point below the flood level rim and above the trap seal.

AIR GAP (Drainage System).

The unobstructed vertical distance through the free atmosphere between the outlet of the waste pipe and the flood level rim of the receptacle into which the waste pipe is discharging.

AIR GAP (Water Distribution System).

The unobstructed vertical distance through the free atmosphere between the lowest opening from any pipe or faucet supplying water to a tank, plumbing fixture or other device and the flood level rim of the receptacle.

ALTERNATE ON-SITE NON-POTABLE WATER.

Non-potable water from other than public utilities, on-site surface sources and subsurface natural freshwater sources. Examples of such water are gray water, on-site reclaimed water, collected rainwater, captured condensate and rejected water from reverse osmosis systems.

ALTERNATIVE ENGINEERED DESIGN.

A plumbing system that performs in accordance with the intent of Chapters 3 through 12 and provides an equivalent level of performance for the protection of public health, safety and

welfare. The system design is not specifically regulated by Chapters 3 through 12.

ANCHORS. See “Supports.”

ANTISIPHON. A term applied to valves or mechanical devices that eliminate siphonage.

APPROVED. Acceptable to the code official or other authority having jurisdiction.

APPROVED AGENCY.

An established and recognized agency that is regularly engaged in conducting tests or furnishing inspection services, where such agency has been approved by the code official.

AREA DRAIN. A receptacle designed to collect surface or storm water from an open area.

ASPIRATOR.

A fitting or device supplied with water or other fluid under positive pressure that passes through an integral orifice or constriction, causing a vacuum. Aspirators are also referred to as suction apparatus, and are similar in operation to an ejector.

BACKFLOW.

Pressure created by any means in the water distribution system, which by being in excess of the pressure in the water supply mains causes a potential backflow condition.

Backpressure, low-head.

A pressure less than or equal to 30 kPa or the pressure exerted by a 3000 mm column of water.

Backsiphonage.

The backflow of potentially contaminated water into the potable water system as a result of the pressure in the potable water system falling below atmospheric pressure of the plumbing fixtures, pools, tanks or vats connected to the potable water distribution piping.

Drainage. A reversal of flow in the drainage system.

Water supply system.

The flow of water or other liquids, mixtures or substances into the distribution pipes of a potable water supply from any source except the intended source.

BACKFLOW CONNECTION.

Any arrangement whereby backflow is possible.

BACKFLOW PREVENTER.

A backflow prevention assembly, a backflow prevention device or other means or method to prevent backflow into the potable water supply.

BACKWATER VALVE.

A device or valve installed in the building drain or sewer pipe where a sewer is subject to backflow, and that prevents drainage or waste from backing up into a lower level or fixtures and causing a flooding condition.

BASE FLOOD ELEVATION.

A reference point, determined in accordance with the building code, based on the depth or peak elevation of flooding, including wave height, which has a 1 percent (100-year flood) or greater chance of occurring in any given year.

BATHROOM GROUP.

A group of fixtures consisting of a water closet, lavatory, bathtub or shower, including or excluding a bidet, an emergency floor drain or both. Such fixtures are located together on the same floor level.

BEDPAN STEAMER OR BOILER.

A fixture utilized for scalding bedpans or urinals by direct application of steam or boiling water.

BEDPAN WASHER AND STERILIZER.

A fixture designed to wash bedpans and to flush the contents into the sanitary drainage system. Included are fixtures of this type that provide for disinfecting utensils by scalding with steam or hot water.

BEDPAN WASHER HOSE.

A device supplied with hot and cold water and located adjacent to a water closet or clinical sink to be utilized for cleansing bedpans.

BRANCH. Any part of the piping system except a riser, main or stack.

BRANCH INTERVAL.

A vertical measurement of distance, 2400 mm or more in developed length, between the

connections of horizontal branches to a drainage stack. Measurements are taken down the stack from the highest horizontal branch connection.

BRANCH VENT. A vent connecting one or more individual vents with a vent stack or stack vent.

BUILDING. Any structure occupied or intended for supporting or sheltering any occupancy.

BUILDING DRAIN.

That part of the lowest piping of a drainage system that receives the discharge from soil, waste and other drainage pipes inside and that extends 750 mm in developed length of pipe beyond the exterior walls of the building and conveys the drainage to the building sewer.

Combined. A building drain that conveys both sewage and storm water or other drainage.

Sanitary. A building drain that conveys sewage only.

Storm. A building drain that conveys storm water or other drainage, but not sewage.

BUILDING SEWER.

That part of the drainage system that extends from the end of the building drain and conveys the discharge to a public sewer, private sewer, individual sewage disposal system or other point of disposal.

Combined. A building sewer that conveys both sewage and storm water or other drainage.

Sanitary. A building sewer that conveys sewage only.

Storm. A building sewer that conveys storm water or other drainage, but not sewage.

BUILDING SUBDRAIN.

That portion of a drainage system that does not drain by gravity into the building sewer.

BUILDING TRAP.

A device, fitting or assembly of fittings installed in the building drain to prevent circulation of air between the drainage system of the building and the building sewer.

CIRCUIT VENT.

A vent that connects to a horizontal drainage

branch and vents two traps to a maximum of eight traps or trapped fixtures connected into a battery.

CIRCULATING HOT WATER SYSTEM.

A specifically designed water distribution system where one or more pumps are operated in the service hot water piping to circulate heated water from the water-heating equipment to fixture supply and back to the water-heating equipment.

CISTERN.

A small covered tank for storing water for a home or farm. Generally, this tank stores rainwater to be utilized for purposes other than in the potable water supply, and such tank is placed underground in most cases.

CLEANOUT.

An access opening in the drainage system utilized for the removal of obstructions. Types of cleanouts include a removable plug or cap, and a removable fixture or fixture trap.

CODE.

These regulations, subsequent amendments thereto, or any emergency rule or regulation that the administrative authority having jurisdiction has lawfully adopted.

CODE OFFICIAL.

The officer or other designated authority charged with the administration and enforcement of this code, or a duly authorized representative.

COLLECTION PIPE.

Unpressurized pipe used within the collection system that drains on-site non-potable water or rainwater to a storage tank by gravity.

COMBINATION FIXTURE.

A fixture combining one sink and laundry tray or a two- or three-compartment sink or laundry tray in one unit.

COMBINATION WASTE AND VENT SYSTEM.

A specially designed system of waste piping embodying the horizontal wet venting of one or more sinks, lavatories, drinking fountains or floor drains by means of a common waste and vent pipe adequately sized to provide free movement of air

above the flow line of the drain.

COMBINED BUILDING DRAIN. See “Building drain, combined.”

COMBINED BUILDING SEWER. See “Building sewer, combined.”

COMMON VENT.

A vent connecting at the junction of two fixture drains or to a fixture branch and serving as a vent for both fixtures.

CONCEALED FOULING SURFACE.

Any surface of a plumbing fixture that is not readily visible and is not scoured or cleansed with each fixture operation.

CONDUCTOR.

A pipe inside the building that conveys storm water from the roof to a storm or combined building drain.

CONSTRUCTION DOCUMENT.

All of the written, graphic and pictorial documents prepared or assembled for describing the design, location and physical characteristics of the elements of the project necessary for obtaining a building permit. The construction drawings shall be drawn to an appropriate scale.

CONTAMINATION.

An impairment of the quality of the potable water that creates an actual hazard to the public health through poisoning or the spread of disease by sewage, industrial fluids or waste.

CRITICAL LEVEL (C-L).

An elevation (height) reference point that determines the minimum height at which a backflow preventer or vacuum breaker is installed above the flood level rim of the fixture or receptor served by the device. The critical level is the elevation level below which there is a potential for backflow to occur. If the critical level marking is not indicated on the device, the bottom of the device shall constitute the critical level.

CROSS CONNECTION.

Any physical connection or arrangement between two otherwise separate piping systems, one of

which contains potable water and the other either water of unknown or questionable safety or steam, gas or chemical, whereby there exists the possibility for flow from one system to the other, with the direction of flow depending on the pressure differential between the two systems (see “Backflow”).

DEMAND RECIRCULATION WATER SYSTEM.

A water distribution system where one or more pumps prime the service hot water piping with heated water upon a demand for hot water.

DEPTH OF TRAP SEAL.

The depth of liquid that would have to be removed from a full trap before air could pass through the trap.

DESIGN FLOOD ELEVATION.

The elevation of the “design flood,” including wave height, relative to the datum specified on the community’s legally designated flood hazard map. In areas designated as Zone AO, the design flood elevation shall be the elevation of the highest existing grade of the building’s perimeter plus the depth number (in meter) specified on the flood hazard map. In areas designated as Zone AO where a depth number is not specified on the map, the depth number shall be taken as being equal to 600 mm.

DEVELOPED LENGTH.

The length of a pipeline measured along the centerline of the pipe and fittings.

DISCHARGE PIPE. A pipe that conveys the discharge from plumbing fixtures or appliances.

DRAIN. Any pipe that carries waste water or water-borne wastes in a building drainage system.

DRAINAGE FITTING.

The type of fitting or fittings utilized in the drainage system. Drainage fittings are similar to cast-iron fittings, except that instead of having a bell and spigot, drainage fittings are recessed and tapped to eliminate ridges on the inside of the installed pipe.

DRAINAGE FIXTURE UNIT.

Drainage (dfu). A measure of the probable discharge into the drainage system by various types of plumbing fixtures. The drainage fixture-unit value for a particular fixture depends on its volume rate of drainage discharge, on the time duration of a single drainage operation and on the average time between successive operations.

DRAINAGE SYSTEM

Piping within a public or private premise that conveys sewage, rainwater or other liquid waste to a point of disposal. A drainage system does not include the mains of a public sewer system or a private or public sewage treatment or disposal plant.

Building gravity. A drainage system that drains by gravity into the building sewer.

Sanitary. A drainage system that carries sewage and excludes storm, surface and ground water.

Storm. A drainage system that carries rainwater, surface water, subsurface water and similar liquid waste.

DRINKING FOUNTAIN.

A plumbing fixture that is connected to the potable water distribution system and the drainage system. The fixture allows the user to obtain a drink directly from a stream of flowing water without the use of any accessories.

EFFECTIVE OPENING.

The minimum cross-sectional area at the point of water supply discharge, measured or expressed in terms of the diameter of a circle or, if the opening is not circular, the diameter of a circle of equivalent cross sectional area. For faucets and similar fittings, the effective opening shall be measured at the smallest orifice in the fitting body or in the supply piping to the fitting.

EMERGENCY FLOOR DRAIN.

A floor drain that does not receive the discharge of any drain or indirect waste pipe, and that protects against damage from accidental spills, fixture overflows and leakage.

ESSENTIALLY NONTOXIC TRANSFER FLUID.

Fluids having a Gosselin rating of 1, including propylene glycol; mineral oil;

polydimethylsiloxane; hydrochlorofluorocarbon, chlorofluorocarbon and carbon refrigerants; and FDA approved boiler water additives for steam boilers.

ESSENTIALLY TOXIC TRANSFER FLUID.

Soil, waste or gray water and fluids having a Gosselin rating of 2 or more, including ethylene glycol, hydrocarbon oils, ammonia refrigerants and hydrazine.

EXISTING INSTALLATION.

Any plumbing system regulated by this code that was legally installed prior to the effective date of this code, or for which a permit to install has been issued.

FAUCET.

A valve end of a water pipe through which water is drawn from or held within the pipe.

FILL VALVE.

A water supply valve, opened or closed by means of a float or similar device, utilized to supply water to a tank. An anti-siphon fill valve contains an anti-siphon device in the form of an approved air gap or vacuum breaker that is an integral part of the fill valve unit and that is positioned on the discharge side of the water supply control valve.

FIXTURE. See "Plumbing fixture."

FIXTURE BRANCH. A drain serving two or more fixtures that discharges to another drain or to a stack.

FIXTURE DRAIN.

The drain from the trap of a fixture to a junction with any other drain pipe.

FIXTURE FITTING.

Supply fitting.

A fitting that controls the volume, direction of flow or both, of water and is either attached to or accessible from a fixture, or is used with an open or atmospheric discharge.

Waste fitting.

A combination of components that conveys the sanitary waste from the outlet of a fixture to the connection to the sanitary drainage system.

FIXTURE SUPPLY.

The water supply pipe connecting a fixture to a branch water supply pipe or directly to a main water supply pipe.

FLOOD HAZARD AREA.

The greater of the following two areas:

1. The area within a flood plain subject to a 1-percent or greater chance of flooding in any given year.
- 6 The area designated as a *flood hazard area* on a community's flood hazard map or as otherwise legally designated.

FLOOD LEVEL RIM. The edge of the receptacle from which water overflows.

FLOW CONTROL (Vented).

A device installed upstream from the interceptor having an orifice that controls the rate of flow through the interceptor and an air intake (vent) downstream from the orifice that allows air to be drawn into the flow stream.

FLOW PRESSURE.

The pressure in the water supply pipe near the faucet or water outlet while the faucet or water outlet is wide open and flowing.

FLUSH TANK.

A tank designed with a fill valve and flush valve to flush the contents of the bowl or usable portion of the fixture.

FLUSHOMETER TANK.

A device integrated within an air accumulator vessel that is designed to discharge a predetermined quantity of water to fixtures for flushing purposes.

FLUSHOMETER VALVE.

A valve attached to a pressurized water supply pipe and so designed that when activated it opens the line for direct flow into the fixture at a rate and quantity to operate the fixture properly, and then gradually closes to reseal fixture traps and avoid water hammer.

GRAY WATER.

Waste discharged from lavatories, bathtubs,

showers, clothes washers and laundry trays.

GREASE INTERCEPTOR.

Fats, oils and greases (FOG) disposal system.

A plumbing appurtenance that reduces nonpetroleum fats, oils and greases in effluent by separation or mass and volume reduction.

Gravity.

Plumbing appurtenances of not less than 1900 L capacity that are installed in the sanitary drainage system to intercept free-floating fats, oils and grease from waste water discharge. Separation is accomplished by gravity during a retention time of not less than 30 minutes.

Hydro-mechanical.

Plumbing appurtenances that are installed in the sanitary drainage system to intercept free floating fats, oils and grease from waste water discharge. Continuous separation is accomplished by air entrainment, buoyancy and interior baffling.

GREASE-LADEN WASTE.

Effluent discharge that is produced from food processing, food preparation or other sources where grease, fats and oils enter automatic dishwasher prerinse stations, sinks or other appurtenances.

GREASE REMOVAL DEVICE, AUTOMATIC (GRD).

A plumbing appurtenance that is installed in the sanitary drainage system to intercept free-floating fats, oils and grease from waste water discharge. Such a device operates on a time- or event-controlled basis and has the ability to remove free-floating fats, oils and grease automatically without intervention from the user except for maintenance.

GRIDDED WATER DISTRIBUTION SYSTEM.

A water distribution system where every water distribution pipe is interconnected so as to provide two or more paths to each fixture supply pipe.

HANGERS. See "Supports."

HORIZONTAL BRANCH DRAIN.

A drainage branch pipe extending laterally from a soil or waste stack or building drain, with or without vertical sections or branches, that receives the discharge from two or more fixture drains or branches and conducts the discharge to the soil or waste stack or to the building drain.

HORIZONTAL PIPE.

Any pipe or fitting that makes an angle of less than 45 degrees with a horizontal plane.

HOT WATER. Water at a temperature greater than or equal to 43°C.

HOUSE TRAP. See “Building trap.”

INDIRECT WASTE PIPE.

A waste pipe that does not connect directly with the drainage system, but that discharges into the drainage system through an air break or air gap into a trap, fixture, receptor or interceptor.

INDIVIDUAL SEWAGE DISPOSAL SYSTEM.

A system for disposal of domestic sewage by means of a septic tank, cesspool or mechanical treatment, designed for utilization apart from a public sewer to serve a single establishment or building.

INDIVIDUAL VENT.

A pipe installed to vent a fixture trap and that connects with the vent system above the fixture served or terminates in the open air.

INDIVIDUAL WATER SUPPLY.

A water supply that serves one or more families, and that is not an approved public water supply.

INTERCEPTOR.

A device designed and installed to separate and retain for removal, by automatic or manual means, deleterious, hazardous or undesirable matter from normal wastes, while permitting normal sewage or wastes to discharge into the drainage system by gravity.

JOINT.

Expansion. A loop,

return bend or return offset that provides for the expansion and contraction in a piping system and is utilized in tall buildings or where there is a rapid change of temperature, as in power plants,

steam rooms and similar occupancies.

Flexible.

Any joint between two pipes that permits one pipe to be deflected or moved without movement or deflection of the other pipe.

Mechanical. See “Mechanical joint.”

Slip.

A type of joint made by means of a washer or a special type of packing compound in which one pipe is slipped into the end of an adjacent pipe.

LEAD-FREE SOLDER AND FLUX. Containing not more than 0.2-percent lead.

LEADER.

An exterior drainage pipe for conveying storm water from roof or gutter drains to an approved means of disposal.

LOCAL VENT STACK.

A vertical pipe to which connections are made from the fixture side of traps and through which vapor or foul air is removed from the fixture or device utilized on bedpan washers.

MACERATING TOILET SYSTEM.

An assembly consisting of a water closet and sump with a macerating pump that is designed to collect, grind and pump wastes from the water closet and up to two other fixtures connected to the sump.

MAIN. The principal pipe artery to which branches are connected.

MANIFOLD. See “Plumbing appurtenance.”

MECHANICAL JOINT.

A connection between pipes, fittings, or pipes and fittings that is not screwed, caulked, threaded, soldered, solvent cemented, brazed, welded or heat fused. A joint in which compression is applied along the centerline of the pieces being joined. In some applications, the joint is part of a coupling, fitting or adapter.

MEDICAL GAS SYSTEM.

The complete system to convey medical gases for direct patient application from central supply systems (bulk tanks, manifolds and medical air

compressors), with pressure and operating controls, alarm warning systems, related components and piping networks extending to station outlet valves at patient use points.

MEDICAL VACUUM SYSTEM.

A system consisting of central-vacuum-producing equipment with pressure and operating controls, shutoff valves, alarm-warning systems, gauges and a network of piping extending to and terminating with suitable station inlets at locations where patient suction may be required.

METER. A measuring device used to collect data and indicate water usage.

NON-POTABLE WATER. Water not safe for drinking, personal or culinary utilization.

NUISANCE.

Public nuisance as known in common law or in equity jurisprudence; whatever is dangerous to human life or detrimental to health; whatever structure or premises is not sufficiently ventilated, sewerage, drained, cleaned or lighted, with respect to its intended occupancy; and whatever renders the air, or human food, drink or water supply unwholesome.

OCCUPANCY. The purpose for which a building or portion thereof is utilized or occupied.

OFFSET. A combination of approved bends that makes two changes in direction bringing one section of the pipe out of line but into a line parallel with the other section.

ON-SITE NON-POTABLE WATER REUSE SYSTEM.

A water system for the collection, treatment, storage, distribution and reuse of non-potable water generated on site, including but not limited to a gray water system. This definition does not include a rainwater harvesting system.

OPEN AIR. Outside the structure.

PLUMBING.

The practice, materials and fixtures utilized in the installation, maintenance, extension and alteration of all piping, fixtures, plumbing appliances and plumbing appurtenances, within or adjacent to any structure, in connection with sanitary drainage or storm drainage facilities;

venting systems; and public or private water supply systems.

PLUMBING APPLIANCE.

Water or drain-connected devices intended to perform a special function. These devices have their operation or control dependent on one or more energized components, such as motors, controls or heating elements. Such devices are manually adjusted or controlled by the owner or operator, or are operated automatically through one or more of the following actions: a time cycle, a temperature range, a pressure range, a measured volume or weight.

PLUMBING APPURTENANCE.

A manufactured device, prefabricated assembly or on-the-job assembly of component parts that is an adjunct to the basic piping system and plumbing fixtures. An appurtenance demands no additional water supply and does not add any discharge load to a fixture or to the drainage system.

PLUMBING FIXTURE.

A receptacle or device that is connected to a water supply system or discharges to a drainage system or both. Such receptacles or devices require a supply of water; or discharge liquid waste or liquid-borne solid waste; or require a supply of water and discharge waste to a drainage system.

PLUMBING SYSTEM.

A system that includes the water distribution pipes; plumbing fixtures and traps; water-treating or water-using equipment; soil, waste and vent pipes; and building drains; in addition to their respective connections, devices and appurtenances within a structure or premises; and the water service, building sewer and building storm sewer serving such structure or premises.

POLLUTION.

An impairment of the quality of the potable water to a degree that does not create a hazard to public health but that does adversely and unreasonably affect the aesthetic qualities of such potable water for domestic use.

POTABLE WATER.

Water free from impurities present in amounts

sufficient to cause disease or harmful physiological effects and conforming to the bacteriological and chemical quality requirements of the Public Health Service Drinking Water Standards or the regulations of the public health authority having jurisdiction.

PRIVATE.

In the classification of plumbing fixtures, “private” applies to fixtures in residences and apartments, and to fixtures in nonpublic toilet rooms of hotels and motels and similar installations in buildings where the plumbing fixtures are intended for utilization by a family or an individual.

PUBLIC OR PUBLIC UTILIZATION.

In the classification of plumbing fixtures, “public” applies to fixtures in general toilet rooms of schools, gymnasiums, hotels, airports, bus and railroad stations, public buildings, public comfort stations, office buildings, stadiums, stores, restaurants and other installations where a number of fixtures are installed so that their utilization is similarly unrestricted.

PUBLIC WATER MAIN.

A water supply pipe for public utilization controlled by public authority.

QUICK-CLOSING VALVE.

A valve or faucet that closes automatically when released manually or that is controlled by a mechanical means for fast-action closing.

RAINWATER. Water from natural precipitation.

READY ACCESS.

That which enables a fixture, appliance or equipment to be directly reached without requiring the removal or movement of any panel, door or similar obstruction and without the use of a portable ladder, step stool or similar device.

RECLAIMED WATER.

Non-potable water that has been derived from the treatment of waste water by a facility or system licensed or permitted to produce water meeting the jurisdiction’s water requirements for its intended uses. Also known as “recycled water.”

REDUCED PRESSURE PRINCIPLE

BACKFLOW PREVENTION ASSEMBLY.

A backflow prevention device consisting of two independently acting check valves, internally force-loaded to a normally closed position and separated by an intermediate chamber (or zone) in which there is an automatic relief means of venting to the atmosphere, internally loaded to a normally open position between two tightly closing shutoff valves and with a means for testing for tightness of the checks and opening of the relief means.

REGISTERED DESIGN PROFESSIONAL.

An individual who is registered or licensed to practice professional architecture or engineering as defined by the statutory requirements of the professional registration laws of the state or jurisdiction in which the project is to be constructed.

RELIEF VALVE.

Pressure relief valve.

A pressure-actuated valve held closed by a spring or other means and designed to relieve pressure automatically at the pressure at which such valve is set.

Temperature and pressure relief (T&P) valve.

A combination relief valve designed to function as both a temperature relief and a pressure relief valve.

Temperature relief valve.

A temperature-actuated valve designed to discharge automatically at the temperature at which such valve is set.

RELIEF VENT.

A vent whose primary function is to provide circulation of air between drainage and vent systems.

RIM. An unobstructed open edge of a fixture.

RISER. See “Water pipe, riser.”

ROOF DRAIN.

A drain installed to receive water collecting on the surface of a roof and to discharge such water into a leader or a conductor.

ROUGH-IN.

Parts of the plumbing system that are installed prior to the installation of fixtures. This includes drainage, water supply, vent piping and the necessary fixture supports and any fixtures that are built into the structure.

SELF-CLOSING FAUCET.

A faucet containing a valve that automatically closes upon deactivation of the opening means.

SEPARATOR. See “Interceptor.”

SEWAGE.

Any liquid waste containing animal or vegetable matter in suspension or solution, including liquids containing chemicals in solution.

SEWAGE EJECTOR.

A device for lifting sewage by entraining the sewage in a high-velocity jet of steam, air or water.

SEWER.

Building sewer. See “Building sewer.”

Public sewer.

That part of the drainage system of pipes, installed and maintained by a city, township, county, public utility company or other public entity, and located on public property, in the street or in an approved dedicated easement of public or community use.

Sanitary sewer. A sewer that carries sewage and excludes storm, surface and ground water.

Storm sewer. A sewer that conveys rainwater, surface water, subsurface water and similar liquid wastes.

SLOPE. The fall (pitch) of a line of pipe in reference to a horizontal plane. In drainage, the slope is expressed as the fall in units vertical per units horizontal (percent) for a length of pipe.

SOIL PIPE. A pipe that conveys sewage containing fecal matter to the building drain or building sewer.

SPILLPROOF VACUUM BREAKER.

An assembly consisting of one check valve force-loaded closed and an air-inlet vent valve force-loaded open to atmosphere, positioned downstream of the check valve, and located

between and including two tightly closing shutoff valves and a test cock.

STACK.

A general term for any vertical line of soil, waste, vent or inside conductor piping that extends through at least one story with or without offsets.

STACK VENT.

The extension of a soil or waste stack above the highest horizontal drain connected to the stack.

STACK VENTING. A method of venting a fixture or fixtures through the soil or waste stack.

STERILIZER.

Boiling type.

A boiling-type sterilizer is a fixture of a non-pressure type utilized for boiling instruments, utensils or other equipment for disinfection. These devices are portable or are connected to the plumbing system.

Instrument. A device for the sterilization of various instruments.

Pressure (autoclave). A pressure vessel fixture designed to utilize steam under pressure for sterilizing.

Pressure instrument washer sterilizer.

A pressure vessel fixture designed to both wash and sterilize instruments during the operating cycle of the fixture.

Utensil. A device for the sterilization of utensils as utilized in health care services.

Water. A device for sterilizing water and storing water.

STERILIZER VENT.

A separate pipe or stack, indirectly connected to the building drainage system at the lower terminal, that receives the vapors from non-pressure sterilizers, or the exhaust vapors from pressure sterilizers, and conducts the vapors directly to the open air. Also called vapor, steam, atmospheric or exhaust vent.

STORM DRAIN. See “Drainage system, storm.”

STORM WATER. Natural precipitation, including snowmelt, that has contacted a surface

at or below grade.

STRUCTURE. That which is built or constructed or a portion thereof.

SUBSOIL DRAIN.

A drain that collects subsurface water or seepage water and conveys such water to a place of disposal.

SUMP.

A tank or pit that receives sewage or liquid waste, located below the normal grade of the gravity system and that must be emptied by mechanical means.

SUMP PUMP.

An automatic water pump powered by an electric motor for the removal of drainage, except raw sewage, from a sump, pit or low point.

SUMP VENT.

A vent from pneumatic sewage ejectors, or similar equipment, that terminates separately to the open air.

SUPPORTS. Devices for supporting and securing pipe, fixtures and equipment.

SWIMMING POOL.

Any structure, basin, chamber or tank containing an artificial body of water for swimming, diving or recreational bathing having a depth of 600 mm or more at any point.

TEMPERED WATER. Water having a temperature range between 29°C and 43°C.

THIRD-PARTY CERTIFICATION AGENCY.

An approved agency operating a product or material certification system that incorporates initial product testing, assessment and surveillance of a manufacturer's quality control system.

THIRD-PARTY CERTIFIED.

Certification obtained by the manufacturer indicating that the function and performance characteristics of a product or material have been determined by testing and ongoing surveillance by an approved third-party certification agency. Assertion of certification is in the form of identification in accordance with the

requirements of the third-party certification agency.

TOILET FACILITY. A room or space that contains not less than one water closet and one lavatory.

TRAP.

A fitting or device that provides a liquid seal to prevent the emission of sewer gases without materially affecting the flow of sewage or waste water through the trap.

TRAP SEAL.

The vertical distance between the weir and the top of the dip of the trap.

UNSTABLE GROUND.

Earth that does not provide a uniform bearing for the barrel of the sewer pipe between the joints at the bottom of the pipe trench.

VACUUM. Any pressure less than that exerted by the atmosphere.

VACUUM BREAKER.

A type of backflow preventer installed on openings subject to normal atmospheric pressure that prevents backflow by admitting atmospheric pressure through ports to the discharge side of the device.

VENT PIPE. See "Vent system."

VENT STACK.

A vertical vent pipe installed primarily for the purpose of providing circulation of air to and from any part of the drainage system.

VENT SYSTEM.

A pipe or pipes installed to provide a flow of air to or from a drainage system, or to provide a circulation of air within such system to protect trap seals from siphonage and backpressure.

VERTICAL PIPE. Any pipe or fitting that makes an angle of 45 degrees or more with the horizontal.

WALL-HUNG WATER CLOSET.

A wall-mounted water closet installed in such a way that the fixture does not touch the floor.

WASTE.

The discharge from any fixture, appliance, area or appurtenance that does not contain fecal matter.

WASTE PIPE.

A pipe that conveys only waste.

WASTE RECEPTOR.

A floor sink, standpipe, hub drain or floor drain that receives the discharge of one or more indirect waste pipes.

WATER COOLER.

A drinking fountain that incorporates a means of reducing the temperature of the water supplied to it from the potable water distribution system.

WATER DISPENSER.

A plumbing fixture that is manually controlled by the user for the purpose of dispensing potable drinking water into a receptacle such as a cup, glass or bottle. Such fixture is connected to the potable water distribution system of the premises. This definition also includes a free-standing apparatus for the same purpose that is not connected to the potable water distribution system and that is supplied with potable water from a container, bottle or reservoir.

WATER-HAMMER ARRESTOR.

A device utilized to absorb the pressure surge (water hammer) that occurs when water flow is suddenly stopped in a water supply system.

WATER HEATER.

Any heating appliance or equipment that heats potable water and supplies such water to the potable hot water distribution system.

WATER MAIN.

A water supply pipe or system of pipes, installed and maintained by a city, township, county, public utility company or other public entity, on public property, in the street or in an approved dedicated easement of public or community use.

WATER OUTLET.

A discharge opening through which water is supplied to a fixture, into the atmosphere (except into an open tank that is part of the water supply system), to a boiler or heating system, or to any devices or equipment requiring water to operate

but which are not part of the plumbing system.

WATER PIPE.

Riser. A water supply pipe that extends one full story or more to convey water to branches or to a group of fixtures.

Water distribution pipe. A pipe within the structure or on the premises that conveys water from the water service pipe, or from the meter when the meter is at the structure, to the points of utilization.

Water service pipe. The pipe from the water main or other source of potable water supply, or from the meter when the meter is at the public right of way, to the water distribution system of the building served.

WATER SUPPLY SYSTEM.

The water service pipe, water distribution pipes, and the necessary connecting pipes, fittings, control valves and all appurtenances in or adjacent to the structure or premises.

WELL.

Bored.

A well-constructed by boring a hole in the ground with an auger and installing a casing.

Drilled.

A well-constructed by making a hole in the ground with a drilling machine of any type and installing a casing and screen.

Driven.

A well-constructed by driving a pipe in the ground. The drive pipe is usually fitted with a well point and screen.

Dug.

A well-constructed by excavating a large-diameter shaft and installing a casing.

WHIRLPOOL BATHTUB.

A plumbing appliance consisting of a bathtub fixture that is equipped and fitted with a circulating piping system designed to accept, circulate and discharge bathtub water upon each use.

YOKE VENT.

A pipe connecting upward from a soil or waste stack to a vent stack for the purpose of preventing pressure changes in the stacks.



CHAPTER 3 GENERAL REGULATIONS

SECTION 301 GENERAL

301.1 Scope.

The provisions of this chapter shall govern the general regulations regarding the installation of plumbing not specific to other chapters.

301.2 System installation.

Plumbing shall be installed with due regard to preservation of the strength of structural members and prevention of damage to walls and other surfaces through fixture usage.

301.3 Connections to drainage system.

Plumbing fixtures, drains, appurtenances and appliances used to receive, or discharge liquid waste or sewage shall be directly connected to the sanitary drainage system of the building or premises, in accordance with the requirements of this code (SBC 701). This section shall not be construed to prevent indirect waste systems required by Chapter 8.

Exception: Bathtubs, showers, lavatories, clothes washers and laundry trays shall not be required to discharge to the sanitary drainage system where such fixtures discharge to an *approved* system in accordance with Chapters 13 and 14.

301.4 Connections to water supply.

Every plumbing fixture, device or appliance requiring or using water for its proper operation shall be directly or indirectly connected to the water supply system in accordance with the provisions of this code.

301.5 Pipe, tube and fitting sizes.

Unless otherwise indicated, the pipe, tube and fitting sizes specified in this code are expressed in nominal or standard sizes as designated in the referenced material standards.

301.6 Prohibited locations.

Plumbing systems shall not be located in an elevator shaft or in an elevator equipment room.

Exception: Floor drains, sumps and sump pumps shall be permitted at the base of the shaft,

provided that they are indirectly connected to the plumbing system and comply with Section 1003.4.

301.7 Conflicts.

In instances where conflicts occur between this code and the manufacturer's installation instructions, the more restrictive provisions shall apply.

SECTION 302 EXCLUSION OF MATERIALS DETRIMENTAL TO THE SEWER SYSTEM

302.1 Detrimental or dangerous materials.

Ashes, cinders or rags; flammable, poisonous or explosive liquids or gases; oil, grease or any other insoluble material capable of obstructing, damaging or overloading the building drainage or sewer system, or capable of interfering with the normal operation of the sewage treatment processes, shall not be deposited, by any means, into such systems.

302.2 Industrial wastes.

Waste products from manufacturing or industrial operations shall not be introduced into the public sewer until it has been determined by the code official or other authority having jurisdiction that the introduction thereof will not damage the public sewer system or interfere with the functioning of the sewage treatment plant.

SECTION 303 MATERIALS

303.1 Identification.

Each length of pipe and each pipe fitting, trap, fixture, material and device utilized in a plumbing system shall bear the identification of the manufacturer and any markings required by the applicable referenced standards.

303.2 Installation of materials.

All materials used shall be installed in strict accordance with the standards under which the materials are accepted and approved. In the absence of such installation procedures, the manufacturer's instructions shall be followed.

Where the requirements of referenced standards or manufacturer's installation instructions do not conform to minimum provisions of this code, the provisions of this code shall apply.

303.3 Plastic pipe, fittings and components.

All plastic pipe, fittings and components shall be third-party certified as conforming to NSF 14.

303.4 Third-party certification.

All plumbing products and materials shall be listed by a third-party certification agency as complying with the referenced standards. Products and materials shall be identified in accordance with Section 303.1.

SECTION 304 RODENTPROOFING

304.1 General.

Plumbing systems shall be designed and installed in accordance with Sections 304.2 through 304.4 to prevent rodents from entering structures.

304.2 Strainer plates.

All strainer plates on drain inlets shall be designed and installed so that all openings are not greater than 13 mm in least dimension.

304.3 Meter boxes.

Meter boxes shall be constructed in such a manner that rodents are prevented from entering a structure by way of the water service pipes connecting the meter box and the structure.

304.4 Openings for pipes.

In or on structures where openings have been made in walls, floors or ceilings for the passage of pipes, the annular space between the pipe and the sides of the opening shall be sealed with caulking materials or closed with gasketing systems compatible with the piping materials and locations.

SECTION 305 PROTECTION OF PIPES AND PLUMBING SYSTEM COMPONENTS

305.1 Corrosion.

Pipes passing through concrete or cinder walls and floors or other corrosive material shall be protected against external corrosion by a protective sheathing or wrapping or other means

that will withstand any reaction from the lime and acid of concrete, cinder or other corrosive material. Sheathing or wrapping shall allow for movement including expansion and contraction of piping. The wall thickness of the material shall be not less than 0.65 mm.

305.2 Stress and strain.

Piping in a plumbing system shall be installed so as to prevent strains and stresses that exceed the structural strength of the pipe. Where necessary, provisions shall be made to protect piping from damage resulting from expansion, contraction and structural settlement.

305.3 Pipes through foundation walls.

Any pipe that passes through a foundation wall shall be provided with a relieving arch, or a pipe sleeve pipe shall be built into the foundation wall. The sleeve shall be two pipe sizes greater than the pipe passing through the wall.

305.4 Freezing.

Water, soil and waste pipes shall not be installed outside of a building, in attics or crawl spaces, concealed in outside walls, or in any other place subjected to freezing temperatures unless adequate provision is made to protect such pipes from freezing by insulation or heat or both. Exterior water supply system piping shall be installed not less than 150 mm below the frost line and not less than 300 mm below grade.

305.4.1 Sewer depth.

Building sewers that connect to private sewage disposal systems shall be installed not less than 500 mm below finished grade at the point of septic tank connection. Building sewers shall be installed not less than 300 mm below grade.

305.5 Waterproofing of openings.

Joints at the roof and around vent pipes shall be made water tight by the use of lead, copper, galvanized steel, aluminum, plastic or other approved flashings or flashing material. Exterior wall openings shall be made water tight.

305.6 Protection against physical damage.

In concealed locations where piping, other than cast-iron or galvanized steel, is installed through holes or notches in studs, joists, rafters or similar

members less than 40 mm from the nearest edge of the member, the pipe shall be protected by steel shield plates. Such shield plates shall have a thickness of not less than 1.463 mm (No. 16 gage). Such plates shall cover the area of the pipe where the member is notched or bored, and shall extend not less than 50 mm above sole plates and below top plates.

305.7 Protection of components of plumbing system.

Components of a plumbing system installed along alleyways, driveways, parking garages or other locations exposed to damage shall be recessed into the wall or otherwise protected in an approved manner.

SECTION 306 TRENCHING, EXCAVATION AND BACKFILL

306.1 Support of piping.

Buried piping shall be supported throughout its entire length.

306.2 Trenching and bedding.

Where trenches are excavated such that the bottom of the trench forms the bed for the pipe, solid and continuous load-bearing support shall be provided between joints. Bell holes, hub holes and coupling holes shall be provided at points where the pipe is joined. Such pipe shall not be supported on blocks to grade. In instances where the materials manufacturer's installation instructions are more restrictive than those prescribed by the code, the material shall be installed in accordance with the more restrictive requirement.

306.2.1 Over excavation.

Where trenches are excavated below the installation level of the pipe such that the bottom of the trench does not form the bed for the pipe, the trench shall be backfilled to the installation level of the bottom of the pipe with sand or fine gravel placed in layers not greater than 150 mm in depth and such backfill shall be compacted after each placement.

306.2.2 Rock removal.

Where rock is encountered in trenching, the rock shall be removed to not less than 75 mm below

the installation level of the bottom of the pipe, and the trench shall be backfilled to the installation level of the bottom of the pipe with sand tamped in place so as to provide uniform load-bearing support for the pipe between joints. The pipe, including the joints, shall not rest on rock at any point.

306.2.3 Soft load-bearing materials.

If soft materials of poor load-bearing quality are found at the bottom of the trench, stabilization shall be achieved by over excavating not less than two pipe diameters and backfilling to the installation level of the bottom of the pipe with fine gravel, crushed stone or a concrete foundation. The concrete foundation shall be bedded with sand tamped into place so as to provide uniform load-bearing support for the pipe between joints.

306.3 Backfilling.

Backfill shall be free from discarded construction material and debris. Loose earth free from rocks, broken concrete and frozen chunks shall be placed in the trench in 150 mm layers and tamped in place until the crown of the pipe is covered by 300 mm of tamped earth. The backfill under and beside the pipe shall be compacted for pipe support. Backfill shall be brought up evenly on both sides of the pipe so that the pipe remains aligned. In instances where the manufacturer's instructions for materials are more restrictive than those prescribed by the code, the material shall be installed in accordance with the more restrictive requirement.

306.4 Tunneling.

Where pipe is to be installed by tunneling, jacking or a combination of both, the pipe shall be protected from damage during installation and from subsequent uneven loading. Where earth tunnels are used, adequate supporting structures shall be provided to prevent future settling or caving.

SECTION 307 STRUCTURAL SAFETY

307.1 General.

In the process of installing or repairing any part of a plumbing and drainage installation, the finished floors, walls, ceilings, tile work or any

other part of the building or premises that must be changed or replaced shall be left in a safe structural condition in accordance with the requirements of the SBC 201, and the SBC 301 through SBC 305.

307.2 Cutting, notching or bored holes.

A framing member shall not be cut, notched or bored in excess of limitations specified in the SBC 201, and the SBC 301 through SBC 305.

307.3 Penetrations of floor/ceiling assemblies and fire-resistance-rated assemblies.

Penetrations of floor/ceiling assemblies and assemblies required to have a fire-resistance rating shall be protected in accordance with SBC 201, the SBC 301 through SBC 305, and the SBC 801.

307.4 Alterations to trusses.

Truss members and components shall not be cut, drilled, notched, spliced or otherwise altered in any way without written concurrence and approval of a registered design professional. Alterations resulting in the addition of loads to any member (e.g., HVAC equipment, water heater) shall not be permitted without verification that the truss is capable of supporting such additional loading.

307.5 Protection of footings.

Trenching installed parallel to footings and walls shall not extend into the bearing plane of a footing or wall. The upper boundary of the bearing plane is a line that extends downward, at an angle of 45 degrees from horizontal, from the outside bottom

edge of the footing or wall.

307.6 Piping materials exposed within plenums.

Piping materials exposed within plenums shall comply with the provisions of SBC 501.

SECTION 308 PIPING SUPPORT

308.1 General.

Plumbing piping shall be supported in accordance with this section.

308.2 Piping seismic supports.

Where earthquake loads are applicable in accordance with SBC 301, plumbing piping supports shall be designed and installed for the seismic forces in accordance with the SBC 301.

308.3 Materials.

Hangers, anchors and supports shall support the piping and the contents of the piping. Hangers and strapping material shall be of approved material that will not promote galvanic action.

308.4 Structural attachment.

Hangers and anchors shall be attached to the building construction in an approved manner.

308.5 Interval of support.

Pipe shall be supported in accordance with Table 308.5.

Exception: The interval of support for piping systems designed to provide for expansion/contraction shall conform to the engineered design in accordance with Section 316.1.

TABLE 308.5 HANGER SPACING

PIPING MATERIAL	MAXIMUM HORIZONTAL SPACING (m)	MAXIMUM VERTICAL SPACING (m)
Acrylonitrile butadiene styrene (ABS) pipe	1.2	3.0b
Aluminum tubing	3.0	4.5
Brass pipe	3.0	3.0
Cast-iron pipe	1.5a	4.5
Chlorinated polyvinyl chloride (CPVC) pipe and tubing, 25 mm and smaller	3.6	3.0b
Chlorinated polyvinyl chloride (CPVC) pipe and tubing, 32 mm and larger	1.2	3.0b
Copper or copper-alloy pipe	3.6	3.0
Copper or copper-alloy tubing, 32 mm diameter and smaller	1.8	3.0
Copper or copper-alloy tubing, 38 mm diameter and larger	3.0	3.0
Cross-linked polyethylene (PEX) pipe	0.8	3.0b
Cross-linked polyethylene/ aluminum/cross-linked polyethylene (PEX-AL- PEX) pipe	0.8	1.2
Lead pipe	Continuous	1.2
Polyethylene/aluminum/ polyethylene (PE-AL-PE) pipe	0.8	1.2
Polyethylene of raised temperature (PE-RT) pipe	0.8	3.0b
Polypropylene (PP) pipe or tubing 25 mm and smaller	0.8	3.0b
Polypropylene (PP) pipe or tubing, 32 mm and larger	1.2	3.0b
Polyvinyl chloride (PVC) pipe	1.2	3.0b
Stainless steel drainage systems	3.0	3.0b
Steel pipe	3.6	4.5

a. The maximum horizontal spacing of cast-iron pipe hangers shall be increased to 3.0 m where 3.0 m lengths of pipe are installed.

b. For sizes 50 mm millimeters and smaller, a guide shall be installed midway between required vertical supports. Such guides shall prevent pipe movement in a direction perpendicular to the axis of the pipe

308.6 Sway bracing.

Rigid support sway bracing shall be provided at changes in direction greater than 45 degrees for pipe sizes 100 mm and larger.

308.7 Anchorage.

Anchorage shall be provided to restrain drainage piping from axial movement.

308.7.1 Location.

For pipe sizes greater than 100 mm, restraints shall be provided for drain pipes at all changes in direction and at all changes in diameter greater

than two pipe sizes. Braces, blocks, rodding and other suitable methods as specified by the coupling manufacturer shall be utilized.

308.8 Expansion joint fittings.

Expansion joint fittings shall be used only where necessary to provide for expansion and contraction of the pipes. Expansion joint fittings shall be of the typical material suitable for use with the type of piping in which such fittings are installed.

308.9 Parallel water distribution systems.

Piping bundles for manifold systems shall be supported in accordance with Table 308.5. Support at changes in direction shall be in accordance with the manufacturer's instructions. Where hot water piping is bundled with cold or hot water piping, each hot water pipe shall be insulated.

SECTION 309 FLOOD HAZARD RESISTANCE

309.1 General.

Plumbing systems and equipment in structures erected in *flood hazard areas* shall be constructed in accordance with the requirements of this section and the SBC 201 and the SBC 301.

309.2 Flood hazard.

For structures located in flood hazard areas, the following systems and equipment shall be located and installed as required by 1612 of SBC 201.

- 1 Water service pipes.
- 2 Pump seals in individual water supply systems where the pump is located below the design flood elevation.
- 3 Covers on potable water wells shall be sealed, except where the top of the casing well or pipe sleeve is elevated to not less than 300 mm above the design flood elevation.
- 4 Sanitary drainage piping.
- 5 Storm drainage piping.
- 6 Manhole covers shall be sealed, except where elevated to or above the design flood elevation.
- 7 Other plumbing fixtures, faucets, fixture fittings, piping systems and equipment.
- 8 Water heaters.
- 9 Vents and vent systems.

Exception: The systems listed in this section are permitted to be located below the elevation required by Section 1612 of SBC 201 for utilities and attendant equipment, provided that the systems are designed and installed to prevent water from entering or accumulating within their components and the systems are constructed to

resist hydrostatic and hydrodynamic loads and stresses, including the effects of buoyancy, during the occurrence of flooding up to such elevation.

309.3 Coastal high-hazard areas and coastal A zones.

Structures located in coastal high-hazard areas and coastal A zones shall meet the requirements of Section 309.2. The plumbing systems, pipes and fixtures shall not be mounted on or penetrate through walls intended to break away under flood loads.

SECTION 310 WASHROOM AND TOILET ROOM REQUIREMENTS

310.1 Light and ventilation.

Washrooms and toilet rooms shall be illuminated and ventilated in accordance with the SBC 201 and the SBC 501.

310.2 Location of fixtures and compartments.

The location of plumbing fixtures and the requirements for compartments and partitions shall be in accordance with Section 405.3.

310.3 Interior finish.

Interior finish surfaces of toilet rooms shall comply with the with the SBC 201.

SECTION 311 TOILET FACILITIES FOR WORKERS

311.1 General.

Toilet facilities shall be provided for construction workers and such facilities shall be maintained in a sanitary condition. Construction worker toilet facilities of the nonsewer type shall conform to ANSI Z4.3.

SECTION 312 TESTS AND INSPECTIONS

312.1 Required tests.

The permit holder shall make the applicable tests prescribed in Sections 312.2 through 312.10 to determine compliance with the provisions of this code. The permit holder shall give reasonable advance notice to the code official when the plumbing work is ready for tests. The equipment,

material, power and labor necessary for the inspection and test shall be furnished by the permit holder and he or she shall be responsible for determining that the work will withstand the test pressure prescribed in the following tests. All plumbing system piping shall be tested with either water or, for piping systems other than plastic, by air. After the plumbing fixtures have been set and their traps filled with water, the entire drainage system shall be submitted to final tests. The code official shall require the removal of any cleanouts if necessary to ascertain whether the pressure has reached all parts of the system.

312.1.1 Test gauges.

Gauges used for testing shall be as follows:

- 1 Tests requiring a pressure of 69 kPa or less shall utilize a testing gauge having increments of 0.69 kPa or less.
- 2 Tests requiring a pressure of greater than 69 kPa but less than or equal to 690 kPa shall utilize a testing gauge having increments of 6.9 kPa or less.
- 3 Tests requiring a pressure of greater than 690 kPa shall utilize a testing gauge having increments of 14 kPa or less.

312.2 Drainage and vent water test.

A water test shall be applied to the drainage system either in its entirety or in sections. If applied to the entire system, all openings in the piping shall be tightly closed, except the highest opening, and the system shall be filled with water to the point of overflow. If the system is tested in sections, each opening shall be tightly plugged except the highest openings of the section under test, and each section shall be filled with water, but no section shall be tested with less than a 3000 mm head of water. In testing successive sections, at least the upper 3000 mm of the next preceding section shall be tested so that no joint or pipe in the building, except the uppermost 3000 mm of the system, shall have been submitted to a test of less than a 3000 mm head of water. This pressure shall be held for not less than 15 minutes. The system shall then be tight at all points.

312.3 Drainage and vent air test.

Plastic piping shall not be tested using air. An air

test shall be made by forcing air into the system until there is a uniform gauge pressure of 34.5 kPa or sufficient to balance a 250 mm column of mercury. This pressure shall be held for a test period of not less than 15 minutes. Any adjustments to the test pressure required because of changes in ambient temperatures or the seating of gaskets shall be made prior to the beginning of the test period.

312.4 Drainage and vent final test.

The final test of the completed drainage and vent systems shall be visual and in sufficient detail to determine compliance with the provisions this code. Where a smoke test is utilized, it shall be made by filling all traps with water and then introducing into the entire system a pungent, thick smoke produced by one or more smoke machines. When the smoke appears at stack openings on the roof, the stack openings shall be closed and a pressure equivalent to 248.8 Pa shall be held for a test period of not less than 15 minutes.

312.5 Water supply system test.

Upon completion of a section of or the entire water supply system, the system, or portion completed, shall be tested and proved tight under a water pressure not less than the working pressure of the system; or, for piping systems other than plastic, by an air test of not less than 344 kPa. This pressure shall be held for not less than 15 minutes. The water utilized for tests shall be obtained from a potable source of supply. The required tests shall be performed in accordance with this section and Section 107.

312.6 Gravity sewer test.

Gravity sewer tests shall consist of plugging the end of the building sewer at the point of connection with the public sewer, filling the building sewer with water, testing with not less than a 3000 mm head of water and maintaining such pressure for 15 minutes.

312.7 Forced sewer test.

Forced sewer tests shall consist of plugging the end of the building sewer at the point of connection with the public sewer and applying a pressure of 35 kPa greater than the pump rating, and maintaining such pressure for 15 minutes.

312.8 Storm drainage system test.

Storm drain systems within a building shall be tested by water or air in accordance with Section 312.2 or 312.3.

312.9 Shower liner test.

Where shower floors and receptors are made water tight by the application of materials required by Section 417.5.2, the completed liner installation shall be tested. The pipe from the shower drain shall be plugged water tight for the test. The floor and receptor area shall be filled with potable water to a depth of not less than 50 mm measured at the threshold. Where a threshold of at least 50 mm high does not exist, a temporary threshold shall be constructed to retain the test water in the lined floor or receptor area to a level not less than 50 mm deep measured at the threshold. The water shall be retained for a test period of not less than 15 minutes, and there shall not be evidence of leakage.

312.10 Inspection and testing of backflow prevention assemblies.

Inspection and testing shall comply with Sections 312.10.1 and 312.10.2.

312.10.1 Inspections.

Annual inspections shall be made of all backflow prevention assemblies and air gaps to determine whether they are operable.

312.10.2 Testing.

Reduced pressure principle, double check, pressure vacuum breaker, reduced pressure detector fire protection, double check detector fire protection, and spill-resistant vacuum breaker backflow preventer assemblies and hose connection backflow preventers shall be tested at the time of installation, immediately after repairs or relocation and at least annually. The testing procedure shall be performed in accordance with one of the following standards: ASSE 5013, ASSE 5015, ASSE 5020, ASSE 5047, ASSE 5048, ASSE 5052, ASSE 5056, CSA B64.10 or CSA B64.10.1.

SECTION 313 EQUIPMENT EFFICIENCIES**313.1 General.**

Equipment efficiencies shall be in accordance with the SBC 601 and SBC 602.

SECTION 314 CONDENSATE DISPOSAL**314.1 Fuel-burning appliances.**

Liquid combustion byproducts of condensing appliances shall be collected and discharged to an approved plumbing fixture or disposal area in accordance with the manufacturer's instructions. Condensate piping shall be of approved corrosion-resistant material and shall not be smaller than the drain connection on the appliance. Such piping shall maintain a horizontal slope in the direction of discharge of not less than one-eighth unit vertical in 12 units horizontal (1-percent slope).

314.2 Evaporators and cooling coils.

Condensate drain systems shall be provided for equipment and appliances containing evaporators or cooling coils. Condensate drain systems shall be designed, constructed and installed in accordance with Sections 314.2.1 through 314.2.5.

314.2.1 Condensate disposal.

Condensate from all cooling coils and evaporators shall be conveyed from the drain pan outlet to an approved place of disposal. Such piping shall maintain a horizontal slope in the direction of discharge of not less than one-eighth unit vertical in 12 units horizontal (1-percent slope). Condensate shall not discharge into a street, alley or other areas so as to cause a nuisance.

314.2.2 Drain pipe materials and sizes.

Components of the condensate disposal system shall be cast iron, galvanized steel, copper, cross-linked polyethylene, polyethylene, ABS, CPVC or PVC or polypropylene pipe or tubing. All components shall be selected for the pressure and temperature rating of the installation. Joints and connections shall be made in accordance with the applicable provisions of Chapter 7 relative to the material type. Condensate waste and drain line size shall be not less than 20 mm internal diameter and shall not decrease in size from the drain pan connection to the place of condensate

disposal. Where the drain pipes from more than one unit are manifolded together for condensate drainage, the pipe or tubing shall be sized in accordance with Table 314.2.2.

TABLE 314.2.2 CONDENSATE DRAIN SIZING

EQUIPMENT CAPACITY	MINIMUM CONDENSATE PIPE DIAMETER (mm)
Up to 20 tons of refrigeration	20 mm
Over 20 tons to 40 tons of refrigeration	25 mm
Over 40 tons to 90 tons of refrigeration	32 mm
Over 90 tons to 125 tons of refrigeration	38 mm
Over 125 tons to 250 tons of refrigeration	50 mm

314.2.3 Auxiliary and secondary drain systems.

In addition to the requirements of Section 314.2.1, where damage to any building components could occur as a result of overflow from the equipment primary condensate removal system, one of the following auxiliary protection methods shall be provided for each cooling coil or fuel fired appliance that produces condensate:

- 1 An auxiliary drain pan with a separate drain shall be provided under the coils on which condensation will occur. The auxiliary pan drain shall discharge to a conspicuous point of disposal to alert occupants in the event of a stoppage of the primary drain. The pan shall have a depth of not less than 38 mm, shall be not less than 75 mm larger than the unit or the coil dimensions in width and length and shall be constructed of corrosion-resistant material. Galvanized sheet metal pans shall have a thickness of not less than 0.6010 mm (No. 24 gage) galvanized sheet metal. Nonmetallic pans shall have a thickness of not less than 1.6 mm.

- 2 A separate overflow drain line shall be connected to the drain pan provided with the equipment. Such overflow drain shall discharge to a conspicuous point of disposal to alert occupants in the event of a stoppage of the primary drain. The overflow drain line shall connect to the drain pan at a higher level than the primary drain connection.
- 3 An auxiliary drain pan without a separate drain line shall be provided under the coils on which condensate will occur. Such pan shall be equipped with a water-level detection device conforming to UL 508 that will shut off the equipment served prior to overflow of the pan. The auxiliary drain pan shall be constructed in accordance with Item 1 of this section.
- 4 A water-level detection device conforming to UL 508 shall be provided that will shut off the equipment served in the event that the primary drain is blocked. The device shall be installed in the primary drain line, the overflow drain line or in the equipment-supplied drain pan, located at a point higher than the primary drain line connection and below the overflow rim of such pan.

Exception: Fuel-fired appliances that automatically shut down operation in the event of a stoppage in the condensate drainage system.

314.2.3.1 Water-level monitoring devices.

On down-flow units and all other coils that do not have a secondary drain or provisions to install a secondary or auxiliary drain pan, a water-level monitoring device shall be installed inside the primary drain pan. This device shall shut off the equipment served in the event that the primary drain becomes restricted. Devices installed in the drain line shall not be permitted.

314.2.3.2 Appliance, equipment and insulation in pans.

Where appliances, equipment or insulation are subject to water damage when auxiliary drain pans fill such portions of the appliances, equipment and insulation shall be installed above the *flood level rim* of the pan. Supports located

inside of the pan to support the appliance or equipment shall be water resistant and *approved*.

314.2.4 Traps.

Condensate drains shall be trapped as required by the equipment or appliance manufacturer.

314.2.4.1 Ductless mini-split system traps.

Ductless mini-split equipment that produces condensation shall be provided with an in-line check valve located in the drain line or a trap.

314.2.5 Cleanouts.

Condensate drain lines shall be configured to permit the clearing of blockages and performance of maintenance without requiring the drain line to be cut.

SECTION 315 PENETRATIONS

315.1 Sealing of annular spaces.

The annular space between the outside of a pipe and the inside of a pipe sleeve or between the outside of a pipe and an opening in a building envelope wall, floor, or ceiling assembly penetrated by a pipe shall be sealed in an *approved* manner with caulking material, foam sealant or closed with a gasketing system. The caulking material, foam sealant or gasketing system shall be designed for the conditions at the penetration location and shall be compatible with the pipe, sleeve and building materials in contact with the sealing materials. Annular spaces created by pipes penetrating fire-resistance-rated assemblies or membranes of such assemblies shall be sealed or closed in accordance with Section 714 of the SBC 201.

SECTION 316 ALTERNATIVE ENGINEERED DESIGN

316.1 Alternative engineered design.

The design, documentation, inspection, testing and approval of an alternative engineered design plumbing system shall comply with Sections 316.1.1 through 316.1.6.

316.1.1 Design criteria.

An alternative engineered design shall conform to

the intent of the provisions of this code and shall provide an equivalent level of quality, strength, effectiveness, fire resistance, durability and safety. Material, equipment or components shall be designed and installed in accordance with the manufacturer's instructions.

316.1.2 Submittal.

The registered design professional shall indicate on the permit application that the plumbing system is an alternative engineered design. The permit and permanent permit records shall indicate that an alternative engineered design was part of the approved installation.

316.1.3 Technical data.

The registered design professional shall submit sufficient technical data to substantiate the proposed alternative engineered design and to prove that the performance meets the intent of this code.

316.1.4 Construction documents.

The registered design professional shall submit to the code official two complete sets of signed and sealed construction documents for the alternative engineering design. The construction documents shall include floor plans and a riser diagram of the work. Where appropriate, the construction documents shall indicate the direction of flow, all pipe sizes, grade of horizontal piping, loading and location of fixtures and appliances.

316.1.5 Design approval.

Where the code official determines that the alternative engineered design conforms to the intent of this code, the plumbing system shall be approved. If the alternative engineered design is not approved, the code official shall notify the registered design professional in writing, stating the reasons thereof.

316.1.6 Inspection and testing.

The alternative engineered design shall be tested and inspected in accordance with the requirements of Sections 107 and 312.

CHAPTER 4 FIXTURES, FAUCETS AND FIXTURE FITTINGS

SECTION 401 GENERAL

401.1 Scope.

This chapter shall govern the materials, design and installation of plumbing fixtures, faucets and fixture fittings in accordance with the type of occupancy, and shall provide for the minimum number of fixtures for various types of occupancies.

401.2 Prohibited fixtures and connections.

Water closets having a concealed trap seal or an unventilated space or having walls that are not thoroughly washed at each discharge in accordance with SASO 1258; SASO 1473; ASME A112.19.2/CSA B45.1 shall be prohibited. Any water closet that permits siphonage of the contents of the bowl back into the tank shall be prohibited. Trough urinals shall be prohibited.

401.3 Water conservation.

The maximum water flow rates and flush volume for plumbing fixtures and fixture fittings shall comply with Section 604.4.

SECTION 402 FIXTURE MATERIALS

402.1 Quality of fixtures.

Plumbing fixtures shall be constructed of approved materials, with smooth, impervious surfaces, free from defects and concealed fouling surfaces, and shall conform to standards cited in this code. All porcelain enameled surfaces on plumbing fixtures shall be acid resistant.

402.2 Materials for specialty fixtures.

Materials for specialty fixtures not otherwise covered in this code shall be of stainless steel, soapstone, chemical stoneware or plastic, or shall be lined with lead, copper-base alloy, nickel-copper alloy, corrosion-resistant steel or other material especially suited to the application for which the fixture is intended.

402.3 Sheet copper.

Sheet copper for general applications shall

conform to ASTM B 152 and shall not weigh less than 3.7 kg per m².

402.4 Sheet lead.

Sheet lead for pans shall not weigh less than 19.5 kg per m² and shall be coated with an asphalt paint or other approved coating.

SECTION 403 MINIMUM PLUMBING FACILITIES

403.1 Minimum number of fixtures.

Plumbing fixtures shall be provided in the minimum number as shown in Table 403.1, based on the actual use of the building or space. Uses not shown in Table 403.1 shall be considered individually by the code official. The number of occupants shall be determined by the SBC 201. Occupancy classification shall be determined in accordance with the SBC 201.

403.1.1 Fixture calculations.

To determine the occupant load of each sex, the total occupant load shall be divided in half. To determine the required number of fixtures, the fixture ratio or ratios for each fixture type shall be applied to the occupant load of each sex in accordance with Table 403.1. Fractional numbers resulting from applying the fixture ratios of Table 403.1 shall be rounded up to the next whole number. For calculations involving multiple occupancies, such fractional numbers for each occupancy shall first be summed and then rounded up to the next whole number.

Exception: The total occupant load shall not be required to be divided in half where *approved* statistical data indicates a distribution of the sexes of other than 50 percent of each sex.

403.1.2 Family or assisted-use toilet and bath fixtures.

Fixtures located within family or assisted-use toilet and bathing rooms required by Section 1109.2.1 of the SBC 201 are permitted to be included in the number of required fixtures for either the male or female occupants in assembly and mercantile occupancies.

Table 403.1 Minimum Number of Required Plumbing Fixtures^a
(See Sections 403.1.1 And 403.2)

No.	Classification	Occupancy	Description	Water closets (urinals: see section 419.2)		Lavatories		Bathtubs/ showers	Drinking fountain (see section 410)	Other
				Male	Female	Male	Female			
1	Assembly	A-1 ^d	Theaters and other buildings for the performing arts and motion pictures	1 per 125	1 per 65	1 per 200	—	—	1 per 500	1 service sink
		A-2 ^d	Dance halls and buildings for similar purposes	1 per 40	1 per 40	1 per 75	—	—	1 per 500	1 service sink
			Restaurants, banquet halls and food courts	1 per 75	1 per 75	1 per 200	—	—	1 per 500	1 service sink
		A-3 ^d	Auditoriums without permanent seating, art galleries, exhibition halls, museums, lecture halls, libraries, arcades and gymnasiums	1 per 125	1 per 65	1 per 200	—	—	1 per 500	1 service sink
			Passenger terminals and transportation facilities	1 per 500	1 per 500	1 per 750	—	—	1 per 1,000	1 service sink
			Mosques	1 per 150	1 per 75	1 per 200	—	—	1 per 1,000	1 service sink

(continued)



TABLE 403.1 —continued
Minimum Number of Required Plumbing Fixtures^a
(See Sections 403.1.1 And 403.2)

No.	Classification	Occupancy	Description	Water closets (urinals: see section 419.2)		Lavatories		Bathtubs/showers	Drinking fountain (see section 410)	Other
				Male	Female	Male	Female			
1 (cont.)	Assembly	A-4	Coliseums, arenas, skating rinks, pools and tennis courts for indoor sporting events and activities	1 per 75 for the first 1,500 and 1 per 120 for the remainder exceeding 1,500	1 per 40 for the first 1,520 and 1 per 60 for the remainder exceeding 1,520	1 per 200	1 per 150	—	1 per 1,000	1 service sink
		A-5	Stadiums, amusement parks, bleachers and grandstands for outdoor sporting events and activities	1 per 75 for the first 1,500 and 1 per 120 for the remainder exceeding 1,500	1 per 40 for the first 1,520 and 1 per 60 for the remainder exceeding 1,520	1 per 200	1 per 150	—	1 per 1,000	1 service sink
2	Business	B	Buildings for the transaction of business, professional services, other services involving merchandise, office buildings, banks, light industrial and similar uses	1 per 25 for the first 50 and 1 per 50 for the remainder exceeding 50		1 per 40 for the first 80 and 1 per 80 for the remainder exceeding 80		—	1 per 100	1 service sink ^e
3	Educational	E	Educational facilities	1 per 50		1 per 50		—	1 per 100	1 service sink
4	Factory and industrial	F-1 and F-2	Structures in which occupants are engaged in work fabricating, assembly or processing of products or materials	1 per 100		1 per 100		(see Section 411)	1 per 400	1 service sink
5	Institutional	I-1	Residential care	1 per 10		1 per 10		1 per 8	1 per 100	1 service sink
		I-2	Hospitals, ambulatory nursing home care recipient	1 per room ^c		1 per room ^c		1 per 15	1 per 100	1 service sink per floor
			Employees, other than residential care ^b	1 per 25		1 per 35		—	1 per 100	—
			Visitors, other than residential care	1 per 75		1 per 100		—	1 per 500	—
		I-3	Prisons ^b	1 per cell		1 per cell		1 per 15	1 per 100	1 service sink
			Reformatories, detention centers, and correctional centers ^b	1 per 15		1 per 15		1 per 15	1 per 100	1 service sink
			Employees ^b	1 per 25		1 per 35		—	1 per 100	—
		I-4	Adult day care and child day care	1 per 15		1 per 15		1	1 per 100	1 service sink

(continued)

TABLE 403.1 —continued
Minimum Number of Required Plumbing Fixtures^a
(See Sections 403.1.1 and 403.2)

No.	Classification	Occupancy	Description	Water closets (urinals: see section 419.2)		Lavatories		Bathtubs/showers	Drinking fountain (see section 410)	Other
				Male	Female	Male	Female			
6	Mercantile	M	Retail stores, service stations, shops, salesrooms, markets and shopping centers	1 per 500		1 per 750		—	1 per 1,000	1 service sink ^e
7	Residential	R-1	Hotels, motels, boarding houses (transient)	1 per sleeping unit		1 per sleeping unit		1 per sleeping unit	—	1 service sink
		R-2	Dormitories, and boarding houses (not transient)	1 per 10		1 per 10		1 per 8	1 per 100	1 service sink
		R-2	Apartment house	1 per dwelling unit		1 per dwelling unit		1 per dwelling unit	—	1 kitchen sink per dwelling unit; 1 auto-matic clothes washer connection per 20 dwelling units
		R-3	Congregate living facilities with 16 or fewer persons	1 per 10		1 per 10		1 per 8	1 per 100	1 service sink
		R-3	One- and two-family dwellings and lodging houses with five or fewer guestrooms	1 per dwelling unit		1 per dwelling unit		1 per dwelling unit	—	1 kitchen sink per dwelling unit; 1 automatic clothes washer connection per dwelling unit
		R-4	Congregate living facilities with 16 or fewer persons	1 per 10		1 per 10		1 per 8	1 per 100	1 service sink
8	Storage	S-1 S-2	Structures for the storage of goods, warehouses, storehouse and freight depots. Low and Moderate Hazard.	1 per 100		1 per 100		See Section 411	1 per 1,000	1 service sink

- The fixtures shown are based on one fixture being the minimum required for the number of persons indicated or any fraction of the number of persons indicated. The number of occupants shall be determined by the by the SBC 201.
- Toilet facilities for employees shall be separate from facilities for inmates or care recipients.
- A single-occupant toilet room with one water closet and one lavatory serving not more than two adjacent patient sleeping units shall be permitted provided that each patient sleeping unit has direct access to the toilet room and provision for privacy for the toilet room user is provided.
- The occupant load for seasonal outdoor seating and entertainment areas shall be included when determining the minimum number of facilities required.
- For business and mercantile occupancies with an occupant load of 15 or fewer, service sinks shall not be required.

403.2 Separate facilities.

Where plumbing fixtures are required, separate facilities shall be provided for each sex.

Exceptions:

- 1 Separate facilities shall not be required for dwelling units and sleeping units.

403.2.1 Family or assisted-use toilet facilities serving as separate facilities.

Where a building or tenant space requires a separate toilet facility for each sex and each toilet facility is required to have only one water closet, two family or assisted-use toilet facilities shall be permitted to serve as the required separate facilities. Family or assisted use toilet facilities shall not be required to be identified for exclusive use by either sex as required by Section 403.4.

403.3 Required public toilet facilities.

Customers, patrons and visitors shall be provided with public toilet facilities in structures and tenant spaces intended for public utilization. The number of plumbing fixtures located within the required toilet facilities shall be provided in accordance with Section 403 for all users. Employees shall be provided with toilet facilities in all occupancies. Employee toilet facilities shall be either separate or combined employee and public toilet facilities.

Exception: Public toilet facilities shall not be required in:

- 1 Open or enclosed parking garages where there are no parking attendants.
- 2 Structures and tenant spaces intended for quick transactions, including takeout, pickup and drop-off, having a public access area less than or equal to 28 m².

403.3.1 Access.

The route to the public toilet facilities required by Section 403.3 shall not pass through kitchens, storage rooms or closets. Access to the required facilities shall be from within the building or from the exterior of the building. Routes shall comply with the accessibility requirements of the SBC 201. The public shall have access to the required toilet facilities at all times that the building is occupied.

403.3.2 Prohibited toilet room location.

Toilet rooms shall not open directly into a room used for the preparation of food for service to the public.

403.3.3 Location of toilet facilities in occupancies other than malls.

In occupancies other than covered and open mall buildings, the required public and employee toilet facilities shall be located not more than one story above or below the space required to be provided with toilet facilities, and the path of travel to such facilities shall not exceed a distance of 152 m.

Exception: The location and maximum distances of travel to required employee facilities in factory and industrial occupancies are permitted to exceed that required by this section, provided that the location and maximum distance of travel are *approved*.

403.3.4 Location of toilet facilities in malls.

In covered and open mall buildings, the required public and employee toilet facilities shall be located not more than one story above or below the space required to be provided with toilet facilities, and the path of travel to such facilities shall not exceed a distance of 91 m. In mall buildings, the required facilities shall be based on total square footage within a covered mall building or within the perimeter line of an open mall building, and facilities shall be installed in each individual store or in a central toilet area located in accordance with this section. The maximum distance of travel to central toilet facilities in mall buildings shall be measured from the main entrance of any store or tenant space. In mall buildings, where employees' toilet facilities are not provided in the individual store, the maximum distance of travel shall be measured from the employees' work area of the store or tenant space.

403.3.5 Pay facilities.

Where pay facilities are installed, such facilities shall be in excess of the required minimum facilities. Required facilities shall be free of charge.

403.3.6 Door locking.

Where a toilet room is provided for the use of multiple occupants, the egress door for the room shall not be lockable from the inside of

the room. This section does not apply to family or assisted-use toilet rooms.

403.4 Signage.

Required public facilities shall be provided with signs that designate the sex, as required by Section 403.2. Signs shall be readily visible and located near the entrance to each toilet facility. Signs for accessible toilet facilities shall comply with Section 1111 of the SBC 201.

403.4.1 Directional signage.

Directional signage indicating the route to the required public toilet facilities shall be posted in a lobby, corridor, aisle or similar space, such that the sign can be readily seen from the main entrance to the building or tenant space.

403.5 Drinking fountain location.

Drinking fountains shall not be required to be located in individual tenant spaces provided that public drinking fountains are located within a distance of travel of 152 m of the most remote location in the tenant space and not more than one story above or below the tenant space. Where the tenant space is in a covered or open mall, such distance shall not exceed 91 m. Drinking fountains shall be located on an accessible route.

SECTION 404 ACCESSIBLE PLUMBING FACILITIES

404.1 Where required.

Accessible plumbing facilities and fixtures shall be provided in accordance with the SBC 201.

404.2 Accessible fixture requirements.

Accessible plumbing fixtures shall be installed with the clearances, heights, spacings and arrangements in accordance with ICC A117.1.

404.3 Exposed pipes and surfaces.

Water supply and drain pipes under accessible lavatories and sinks shall be covered or otherwise configured to protect against contact. Pipe coverings shall comply with ASME A112.18.9.

SECTION 405 INSTALLATION OF FIXTURES

405.1 Water supply protection.

The supply lines and fittings for every plumbing fixture shall be installed so as to

prevent backflow.

405.2 Access for cleaning.

Plumbing fixtures shall be installed so as to afford easy access for cleaning both the fixture and the area around the fixture.

405.3 Setting.

Fixtures shall be set level and in proper alignment with reference to adjacent walls.

405.3.1 Water closets, urinals, lavatories and bidets.

A water closet, urinal, lavatory or bidet shall not be set closer than 375 mm from its center to any side wall, partition, vanity or other obstruction, or closer than 750 mm center to center between adjacent fixtures. There shall be not less than a 525 mm clearance in front of the water closet, urinal, lavatory or bidet to any wall, fixture or door. Water closet compartments shall be not less than 750 mm in width and not less than 1500 mm in depth for floor mounted water closets and not less than 750 mm in width and 1400 mm in depth for wall-hung water closets.

405.3.2 Public lavatories.

In employee and public toilet rooms, the required lavatory shall be located in the same room as the required water closet.

405.3.3 Location of fixtures and piping.

Piping, fixtures or equipment shall not be located in such a manner as to interfere with the normal operation of windows, doors or other means of egress openings.

405.3.4 Water closet compartment.

Each water closet utilized by the public or employees shall occupy a separate compartment with walls or partitions and a door enclosing the fixtures to ensure privacy.

Exceptions:

- 1 Water closet compartments shall not be required in a single-occupant toilet room with a lockable door.
- 2 Toilet rooms located in child day care facilities and containing two or more water closets shall be permitted to have one water closet without an enclosing compartment.

- 3 This provision is not applicable to toilet areas located within Group I-3 housing areas.

405.3.5 Urinal partitions.

Each urinal utilized by the public or employees shall occupy a separate area with walls or partitions to provide privacy. The walls or partitions shall begin at a height not greater than 300 mm from and extend not less than 1500 mm above the finished floor surface. The walls or partitions shall extend from the wall surface at each side of the urinal not less than 500 mm or to a point not less than 150 mm beyond the outermost front lip of the urinal measured from the finished backwall surface, whichever is greater.

Exceptions:

- 1 Urinal partitions shall not be required in a single-occupant or family/assisted-use toilet room with a lockable door.
- 2 Toilet rooms located in child day care facilities and containing two or more urinals shall be permitted to have one urinal without partitions.

405.4 Floor and wall drainage connections.

Connections between the drain and floor outlet plumbing fixtures shall be made with a floor flange or a waste connector and sealing gasket. The waste connector and sealing gasket joint shall comply with the joint tightness test of ASME A112.4.3 and shall be installed in accordance with the manufacturer's instructions. The flange shall be attached to the drain and anchored to the structure. Connections between the drain and wall-hung water closets shall be made with an approved extension nipple or horn adaptor. The water closet shall be bolted to the hanger with corrosion-resistant bolts or screws. Joints shall be sealed with an approved elastomeric gasket, flange-to-fixture connection complying with ASME A112.4.3 or an approved setting compound.

405.4.1 Floor flanges.

Floor flanges for water closets or similar fixtures shall be not less than 3.2 mm thick for brass, 6.4 mm thick for plastic and 6.4 mm thick and not less than a 50 mm caulking depth for cast iron or galvanized malleable iron.

Floor flanges of hard lead shall weigh not less than 0.7 kg and shall be composed of lead alloy with not less than 7.75-percent antimony by weight. Closet screws and bolts shall be of brass. Flanges shall be secured to the building structure with corrosion-resistant screws or bolts.

405.4.2 Securing floor outlet fixtures.

Floor outlet fixtures shall be secured to the floor or floor flanges by screws or bolts of corrosion-resistant material.

405.4.3 Securing wall-hung water closet bowls.

Wall-hung water closet bowls shall be supported by a concealed metal carrier that is attached to the building structural members so that strain is not transmitted to the closet connector or any other part of the plumbing system. The carrier shall conform to ASME A112.6.1M or ASME A112.6.2.

405.5 Water-tight joints.

Joints formed where fixtures come in contact with walls or floors shall be sealed.

405.6 Plumbing in mental health centers.

In mental health centers, pipes or traps shall not be exposed, and fixtures shall be bolted through walls.

405.7 Design of overflows.

Where any fixture is provided with an overflow, the waste shall be designed and installed so that standing water in the fixture will not rise in the overflow when the stopper is closed, and no water will remain in the overflow when the fixture is empty.

405.7.1 Connection of overflows.

The overflow from any fixture shall discharge into the drainage system on the inlet or fixture side of the trap.

Exception: The overflow from a flush tank serving a water closet or urinal shall discharge into the fixture served.

405.8 Slip joint connections.

Slip joints shall be made with an approved elastomeric gasket and shall only be installed on the trap outlet, trap inlet and within the trap seal. Fixtures with concealed slip-joint connections shall be provided with an access panel or utility space not less than 300 mm in

its smallest dimension or other approved arrangement so as to provide access to the slip joint connections for inspection and repair.

405.9 Design and installation of plumbing fixtures.

SECTION 406 AUTOMATIC CLOTHES WASHERS

406.1 Water connection.

The water supply to an automatic clothes washer shall be protected against backflow by an air gap that is integral with the machine or a backflow preventer shall be installed in accordance with Section 608. Air gaps shall comply with ASME A112.1.2 or A112.1.3.

406.2 Waste connection.

The waste from an automatic clothes washer shall discharge through an air break into a standpipe in accordance with Section 802.3.3 or into a laundry sink. The trap and fixture drain for an automatic clothes washer standpipe shall be not less than 50 mm in diameter. The fixture drain for the standpipe serving an automatic clothes washer shall connect to a 75 mm or larger diameter fixture branch or stack. Automatic clothes washers that discharge by gravity shall be permitted to drain to a waste receptor or an approved trench drain.

SECTION 407 BATHTUBS

407.1 Approval.

Bathtubs shall conform to ASME A112.19.1/CSA B45.2, ASME A112.19.2/CSA B45.1, ASME A112.19.3/CSA B45.4 or CSA B45.5/IAPMO Z124 or approved equivalent standards.

407.2 Bathtub waste outlets and overflows.

Bathtubs shall be equipped with a waste outlet and an overflow outlet. The outlets shall be connected to waste tubing or piping not less than 38 mm in diameter. The waste outlet shall be equipped with a water-tight stopper.

407.3 Glazing.

Windows and doors within a bathtub enclosure shall conform to the safety glazing requirements of the SBC 201.

407.4 Bathtub enclosure.

Integral fixture fitting mounting surfaces on manufactured plumbing fixtures or plumbing fixtures constructed on site shall meet the design requirements of ASME A112.19.2/CSA B45.1 or ASME A112.19.3/CSA B45.4.

Doors in a bathtub enclosure shall conform to ASME A112.19.15.

SECTION 408 BIDETS

408.1 Approval.

Bidets shall conform to ASME A112.19.2/CSA B45.1.

408.2 Water connection.

The water supply to a bidet shall be protected against backflow by an air gap or backflow preventer in accordance with Section 608.13.1, 608.13.2, 608.13.3, 608.13.5, 608.13.6 or 608.13.8.

408.3 Bidet water temperature.

The discharge water temperature from a bidet fitting shall be limited to a maximum temperature of 43°C by a water temperature limiting device conforming to ASSE 1070 or CSA B125.3.

SECTION 409 DISHWASHING MACHINES

409.1 Approval.

Commercial dishwashing machines shall conform to ASSE 1004 and NSF 3.

409.2 Water connection.

The water supply to a dishwashing machine shall be protected against backflow by an air gap that is integral with the machine or a backflow preventer shall be installed in accordance with Section 608. Air gaps shall comply with ASME A112.1.2 or A112.1.3.

409.3 Waste connection.

The waste connection of a dishwashing machine shall comply with Section 802.1.6 or 802.1.7, as applicable.

SECTION 410 DRINKING FOUNTAINS

410.1 Approval.

Drinking fountains shall conform to ASME A112.19.1/CSA B45.2 or ASME

A112.19.2/CSA B45.1 and water coolers shall conform to AHRI 1010. Drinking fountains and water coolers shall conform to NSF 61, Section 9. Electrically operated, refrigerated drinking water coolers shall be listed and labeled in accordance with UL 399.

410.2 Small occupancies.

Drinking fountains shall not be required for an occupant load of 15 or fewer.

410.3 Provide high and low drinking fountains.

Where drinking fountains are required, not fewer than two drinking fountains shall be provided. One drinking fountain shall comply with the requirements for people who use a wheelchair and one drinking fountain shall comply with the requirements for standing persons.

Exception:

- 1 A single drinking fountain with two separate spouts that complies with the requirements for people who use a wheelchair and standing persons shall be permitted to be substituted for two separate drinking fountains.
- 2 Where drinking fountains are primarily for children's use, the drinking fountains for people using wheel chairs shall be permitted to comply with the children's provisions in ICC A117.1 and drinking fountains for standing children shall be permitted to provide the spout at 750 mm minimum above floor.

410.4 Substitution.

Where restaurants provide drinking water in a container free of charge, drinking fountains shall not be required in those restaurants. In other occupancies where drinking fountains are required, water dispensers shall be permitted to be substituted for not more than 50 percent of the required number of drinking fountains.

410.5 Prohibited location.

Drinking fountains, water coolers and water dispensers shall not be installed in public restrooms.

SECTION 411 EMERGENCY SHOWERS AND EYEWASH STATIONS

411.1 Approval.

Emergency showers and eyewash stations shall conform to ISEA Z358.1.

411.2 Waste connection.

Waste connections shall not be required for emergency showers and eyewash stations.

SECTION 412 FLOOR AND TRENCH DRAINS

412.1 Approval.

Floor drains shall conform to ASME A112.3.1, ASME A112.6.3 or CSA B79. Trench drains shall comply with ASME A112.6.3.

412.2 Floor drains.

Floor drains shall have removable strainers. The floor drain shall be constructed so that the drain is capable of being cleaned. Access shall be provided to the drain inlet. Ready access shall be provided to floor drains.

Exception: Floor drains serving refrigerated display cases shall be provided with *access*.

412.3 Size of floor drains.

Floor drains shall have a drain outlet not less than 50 mm in diameter.

412.4 Public laundries and central washing facilities.

In public coin-operated laundries and in the central washing facilities of multiple-family dwellings, the rooms containing automatic clothes washers shall be provided with floor drains located to readily drain the entire floor area. Such drains shall have an outlet of not less than 75 mm in diameter.

SECTION 413 FOOD WASTE DISPOSER UNITS

413.1 Approval.

Domestic food waste disposers shall conform to ASSE 1008 and shall be listed and labeled in accordance with UL 430 or ASSE 1009. Food waste disposers shall not increase the *drainage fixture unit* load on the sanitary drainage system.

413.2 Domestic food waste disposer waste outlets.

Domestic food waste disposers shall be connected to a drain of not less than 40 mm in diameter.

413.3 Commercial food waste disposer waste outlets.

Commercial food waste disposers shall be connected to a drain not less than 40 mm in diameter. Commercial food waste disposers shall be connected and trapped separately from any other fixtures or sink compartments.

413.4 Water supply required.

Food waste disposers shall be provided with a supply of cold water. The water supply shall be protected against backflow by an air gap or backflow preventer in accordance with Section 608.

SECTION 414 GARBAGE CAN WASHERS**414.1 Water connection.**

The water supply to a garbage can washer shall be protected against backflow by an air gap or a backflow preventer in accordance with Section 608.13.1, 608.13.2, 608.13.3, 608.13.5, 608.13.6 or 608.13.8.

414.2 Waste connection.

Garbage can washers shall be trapped separately. The receptacle receiving the waste from the washer shall have a removable basket or strainer to prevent the discharge of large particles into the drainage system.

SECTION 415 LAUNDRY TRAYS**415.1 Approval.**

Laundry trays shall conform to ASME A112.19.1/CSA B45.2, ASME A112.19.2/CSA B45.1, ASME A112.19.3/CSA B45.4 or CSA B45.5/IAPMO Z124.

415.2 Waste outlet.

Each compartment of a laundry tray shall be provided with a waste outlet not less than 40 mm in diameter and a strainer or crossbar to restrict the clear opening of the waste outlet.

SECTION 416 LAVATORIES (WASH BASINS)**416.1 Approval**

Lavatories shall conform to ASME A112.19.1/CSA B45.2, ASME A112.19.2/CSA B45.1, ASME A112.19.3/CSA B45.4 or CSA B45.5/IAPMO Z124. Group wash-up equipment shall conform to the requirements of Section 402. Every 500 mm of rim space shall be considered as one lavatory (wash basin).

416.2 Cultured marble lavatories.

Cultured marble vanity tops with an integral lavatory shall conform to CSA B45.5/ IAPMO Z124.

416.3 Lavatory waste outlets.

Lavatories shall have waste outlets not less than 32 mm in diameter. A strainer, pop-up stopper, crossbar or other device shall be provided to restrict the clear opening of the waste outlet.

416.4 Moveable lavatory systems.

Moveable lavatory systems shall comply with ASME A112.19.12.

416.5 Tempered water for public hand-washing facilities.

Tempered water shall be delivered from lavatories and group wash fixtures located in public toilet facilities provided for customers, patrons and visitors. Tempered water shall be delivered through an approved water-temperature limiting device that conforms to ASSE 1070 or CSA B125.3.

SECTION 417 SHOWERS**417.1 Approval.**

Prefabricated showers and shower compartments shall conform to ASME A112.19.2/CSA B45.1 or CSA B45.5/IAPMO Z124. Shower valves for individual showers shall conform to the requirements of Section 424.3.

417.2 Water supply risers.

Water supply risers from the shower valve to the shower head outlet, whether exposed or concealed, shall be attached to the structure. The attachment to the structure shall be made by the use of support devices designed for use with the specific piping material or by fittings

anchored with screws.

417.3 Shower waste outlet.

Waste outlets serving showers shall be not less than 40 mm in diameter and, for other than waste outlets in bathtubs, shall have removable strainers not less than 75 mm in diameter with strainer openings not less than 6.4 mm in least dimension. Where each shower space is not provided with an individual waste outlet, the waste outlet shall be located and the floor pitched so that waste from one shower does not flow over the floor area serving another shower. Waste outlets shall be fastened to the waste pipe in an approved manner.

417.4 Shower compartments.

Shower compartments shall be not less than 0.60 m² in interior cross sectional area. Shower compartments shall be not less than 0.75 m in least dimension as measured from the finished interior dimension of the compartment, exclusive of fixture valves, showerheads, soap dishes and safety grab bars or rails. Except as required in Section 404, the minimum required area and dimension shall be measured from the finished interior dimension at a height equal to the top of the threshold and at a point tangent to its centerline and shall be continued to a height not less than 1.8 m above the shower drain outlet.

Exception: Shower compartments having not less than 650 mm in minimum dimension measured from the finished interior dimension of the compartment, provided that the shower compartment has not less than 0.84 m² of cross-sectional area.

417.4.1 Floor and wall area.

Bathtub floors, shower floors, wall areas above built-in tubs that have installed shower heads and walls in shower compartments shall be constructed of smooth, corrosion-resistant and nonabsorbent waterproof materials. Wall materials shall extend to a height of not less than 1850 mm above the room floor level, and not less than 1.8 m above the drain of the tub or shower. Such walls shall form a water-tight joint with each other and with either the tub or shower floor.

417.4.2 Access.

The shower compartment access and egress opening shall have a clear and unobstructed

finished width of not less than 600 mm. Shower compartments required to be designed in conformance to accessibility provisions shall comply with Section 404.1.

417.5 Shower floors or receptors.

Floor surfaces shall be constructed of impervious, noncorrosive, nonabsorbent and waterproof materials.

417.5.1 Support.

Floors or receptors under shower compartments shall be laid on, and supported by, a smooth and structurally sound base.

417.5.2 Shower lining.

Floors under shower compartments, except where prefabricated receptors have been provided, shall be lined and made water tight utilizing material complying with Sections 417.5.2.1 through 417.5.2.6. Such liners shall turn up on all sides not less than 50 mm above the finished threshold level. Liners shall be recessed and fastened to an *approved* backing so as not to occupy the space required for wall covering, and shall not be nailed or perforated at any point less than 25 mm above the finished threshold. Liners shall be pitched one-fourth unit vertical in 12 units horizontal (2-percent slope) and shall be sloped toward the fixture drains and be securely fastened to the waste outlet at the seepage entrance, making a water-tight joint between the liner and the outlet. The completed liner shall be tested in accordance with Section 312.9.

Exceptions:

1. Floor surfaces under shower heads provided for rinsing laid directly on the ground are not required to comply with this section.
2. Where a sheet-applied, load-bearing, bonded, waterproof membrane is installed as the shower lining, the membrane shall not be required to be recessed.

417.5.2.1 PVC sheets.

Plasticized polyvinyl chloride (PVC) sheets shall meet the requirements of ASTM D 4551. Sheets shall be joined by solvent welding in accordance with the manufacturer's installation instructions.

417.5.2.2 Chlorinated polyethylene (CPE) sheets.

Non plasticized chlorinated polyethylene sheet shall meet the requirements of ASTM D 4068. The liner shall be joined in accordance with the manufacturer's installation instructions.

417.5.2.3 Sheet lead.

Sheet lead shall weigh not less than 19.5 kg per m² and shall be coated with an asphalt paint or other *approved* coating. The lead sheet shall be insulated from conducting substances other than the connecting drain by 7.0 kg asphalt felt or an equivalent. Sheet lead shall be joined by burning.

417.5.2.4 Sheet copper.

Sheet copper shall conform to ASTM B 152 and shall weigh not less than 3.5 kg per m². The copper sheet shall be insulated from conducting substances other than the connecting drain by 7.0 kg asphalt felt or an equivalent. Sheet copper shall be joined by brazing or soldering.

417.5.2.5 Sheet-applied, load-bearing, bonded, waterproof membranes.

Sheet-applied, load-bearing, bonded, waterproof membranes shall meet requirements of ANSI A118.10 and shall be applied in accordance with the manufacturer's installation instructions.

417.5.2.6 Liquid-type, trowel-applied, load-bearing, bonded waterproof materials.

Liquid-type, trowel applied, load-bearing, bonded waterproof materials shall meet the requirements of ANSI A118.10 and shall be applied in accordance with the manufacturer's instructions.

417.6 Glazing.

Windows and doors within a shower enclosure shall conform to the safety glazing requirements of the SBC 201. If a glass or plastic window or door is installed in a tub/shower or shower compartment, the SBC 201 (Section 2406) requires the glazing (the glass or plastic) to be of safety glazing materials. This is required so that if a bather slips or falls and breaks the glazing, he or she will not be exposed to sharp, broken glazing material. Safety glazing is made to break into small rounded pieces that have minimal capacity to severely cut human skin

SECTION 418 SINKS

418.1 Approval.

Sinks shall conform to ASME A112.19.1/ CSA B45.2, ASME A112.19.2/CSA B45.1, ASME A112.19.3/CSA B45.4 or CSA B45.5/IAPMO Z124.

418.2 Sink waste outlets.

Sinks shall be provided with waste outlets having a diameter not less than 40 mm. A strainer or crossbar shall be provided to restrict the clear opening of the waste outlet.

418.3 Moveable sink systems.

Moveable sink systems shall comply with ASME A112.19.12.

Movable sinks allow the user to adjust the elevation of the sink for easier use.

SECTION 419 URINALS

419.1 Approval.

Urinals shall conform to ASME A112.19.2/ CSA B45.1, ASME A112.19.19 or CSA B45.5/IAPMO Z124. Urinals shall conform to the water consumption requirements of Section 604.4. Water-supplied urinals shall conform to the hydraulic performance requirements of ASME A112.19.2/CSA B45.1 or CSA B45.5/IAPMO Z124.

419.2 Substitution for water closets.

In each bathroom or toilet room, urinals shall not be substituted for more than 67 percent of the required water closets in assembly and educational occupancies. Urinals shall not be substituted for more than 50 percent of the required water closets in all other occupancies.

419.3 Surrounding material.

Wall and floor space to a point 0.60 m in front of a urinal lip and 1.2 m above the floor and at least 0.6 m to each side of the urinal shall be waterproofed with a smooth, readily cleanable, nonabsorbent material.

SECTION 420 WATER CLOSETS

420.1 Approval.

Water closets shall conform to the water consumption requirements of Section 604.4 and shall conform to ASME A112.19.2/CSA B45.1, ASME A112.19.3/ CSA B45.4 or CSA B45.5/IAPMO Z124. Water closets shall

conform to the hydraulic performance requirements of ASME A112.19.2/CSA B45.1. Water closet tanks shall conform to ASME A112.19.2/CSA B45.1, ASME A112.19.3/CSA B45.4 or CSA B45.5/IAPMO Z124. Electro-hydraulic water closets shall comply with ASME A112.19.2/CSA B45.1. Water closets equipped with a dual flushing device shall comply with ASME A112.19.14.

420.2 Water closets for public or employee toilet facilities.

Water closet bowls for public or employee toilet facilities shall be of the elongated type.

420.3 Water closet seats.

Water closets shall be equipped with seats of smooth, nonabsorbent material. All seats of water closets provided for public or employee toilet facilities shall be of the hinged open-front type. Integral water closet seats shall be of the same material as the fixture. Water closet seats shall be sized for the water closet bowl type.

420.4 Water closet connections.

A 100 mm by 75 mm closet bend shall be acceptable. Where a 75 mm bend is utilized on water closets, a 100 mm by 75 mm flange shall be installed to receive the fixture horn.

SECTION 421 WHIRLPOOL BATHTUBS

421.1 Approval.

Whirlpool bathtubs shall comply with ASME A112.19.7/CSA B45.10 and shall be listed and labeled in accordance with UL 1795.

421.2 Installation.

Whirlpool bathtubs shall be installed and tested in accordance with the manufacturer's instructions. The pump shall be located above the weir of the fixture trap.

421.3 Drain.

The pump drain and circulation piping shall be sloped to drain the water in the volute and the circulation piping when the whirlpool bathtub is empty.

421.4 Suction fittings.

Suction fittings for whirlpool bathtubs shall comply with ASME A112.19.7/CSA B45.10.

421.5 Access to pump.

Access shall be provided to circulation pumps in accordance with the fixture or pump manufacturer's installation instructions. Where the manufacturer's instructions do not specify the location and minimum size of field-fabricated access openings, an opening not less than 300 mm by 300 mm shall be installed to provide access to the circulation pump. Where pumps are located more than 600 mm from the access opening, an opening not less than 450 mm by 450 mm shall be installed. A door or panel shall be permitted to close the opening. In all cases, the access opening shall be unobstructed and of the size necessary to permit the removal and replacement of the circulation pump.

421.6 Whirlpool enclosure.

Doors within a whirlpool enclosure shall conform to ASME A112.19.15.

SECTION 422 HEALTH CARE FIXTURES AND EQUIPMENT

422.1 Scope.

This section shall govern those aspects of health care plumbing systems that differ from plumbing systems in other structures. Health care plumbing systems shall conform to the requirements of this section in addition to the other requirements of this code. The provisions of this section shall apply to the special devices and equipment installed and maintained in the following *occupancies*: nursing homes, homes for the aged, orphanages, infirmaries, first aid stations, psychiatric facilities, clinics, professional offices of dentists and doctors, mortuaries, educational facilities, surgery, dentistry, research and testing laboratories, establishments manufacturing pharmaceutical drugs and medicines and other structures with similar apparatus and equipment classified as plumbing.

422.2 Approval.

All special plumbing fixtures, equipment, devices and apparatus shall be of an approved type.

422.3 Protection.

All devices, appurtenances, appliances and apparatus intended to serve some special function, such as sterilization, distillation, processing, cooling, or storage of ice or foods, and that connect to either the water supply or

drainage system, shall be provided with protection against backflow, flooding, fouling, contamination of the water supply system and stoppage of the drain.

422.4 Materials.

Fixtures designed for therapy, special cleansing or disposal of waste materials, combinations of such purposes, or any other special purpose, shall be of smooth, impervious, corrosion-resistant materials and, where subjected to temperatures in excess of 82°C, shall be capable of withstanding, without damage, higher temperatures.

422.5 Access.

Access shall be provided to concealed piping in connection with special fixtures where such piping contains steam traps, valves, relief valves, check valves, vacuum breakers or other similar items that require periodic inspection, servicing, maintenance or repair. Access shall be provided to concealed piping that requires periodic inspection, maintenance or repair.

422.6 Clinical sink.

A clinical sink shall have an integral trap in which the upper portion of a visible trap seal provides a water surface. The fixture shall be designed so as to permit complete removal of the contents by siphonic or blowout action and to reseal the trap. A flushing rim shall provide water to cleanse the interior surface. The fixture shall have the flushing and cleansing characteristics of a water closet.

422.7 Prohibited usage of clinical sinks and service sinks.

A clinical sink serving a soiled utility room shall not be considered as a substitute for, or be utilized as, a service sink. A service sink shall not be utilized for the disposal of urine, fecal matter or other human waste.

422.8 Ice prohibited in soiled utility room.

Machines for manufacturing ice, or any device for the handling or storage of ice, shall not be located in a soiled utility room.

422.9 Sterilizer equipment requirements.

The approval and installation of all sterilizers shall conform to the requirements of the SBC 501.

422.9.1 Sterilizer piping.

Access for the purposes of inspection and maintenance shall be provided to all sterilizer piping and devices necessary for the operation of sterilizers.

422.9.2 Steam supply.

Steam supplies to sterilizers, including those connected by pipes from overhead mains or branches, shall be drained to prevent any moisture from reaching the sterilizer. The condensate drainage from the steam supply shall be discharged by gravity.

422.9.3 Steam condensate return.

Steam condensate returns from sterilizers shall be a gravity return system.

422.9.4 Condensers.

Pressure sterilizers shall be equipped with a means of condensing and cooling the exhaust steam vapors. Non-pressure sterilizers shall be equipped with a device that will automatically control the vapor, confining the vapors within the vessel.

422.10 Special elevations.

Control valves, vacuum outlets and devices protruding from a wall of an operating, emergency, recovery, examining or delivery room, or in a corridor or other location where patients are transported on a wheeled stretcher, shall be located at an elevation that prevents bumping the patient or stretcher against the device.

SECTION 423 SPECIALTY PLUMBING FIXTURES

423.1 Water connections.

Baptisteries, ornamental and lily pools, aquariums, ornamental fountain basins, swimming pools, and similar constructions, where provided with water supplies, shall be protected against backflow in accordance with Section 608.

423.2 Approval.

Specialties requiring water and waste connections shall be submitted for approval.

423.3 Footbaths, pedicure baths and head shampoo sinks.

The water supplied to specialty plumbing fixtures, such as pedicure chairs having an integral foot bathtub, footbaths, and head

shampoo sinks, shall be limited to a maximum temperature of 49°C by a water temperature limiting device that conforms to ASSE 1070 or CSA B125.3.

SECTION 424 FAUCETS AND OTHER FIXTURE FITTINGS

424.1 Approval.

Faucets and fixture fittings shall conform to ASME A112.18.1/CSA B125.1. Faucets and fixture fittings that supply drinking water for human ingestion shall conform to the requirements of NSF 61, Section 9. Flexible water connectors exposed to continuous pressure shall conform to the requirements of Section 605.6.

424.1.1 Faucets and supply fittings.

Faucets and supply fittings shall conform to the water consumption requirements of Section 604.4.

424.1.2 Waste fittings.

Waste fittings shall conform to ASME A112.18.2/CSA B125.2, ASTM F 409 or to one of the standards listed in Tables 702.1 and 702.4 for above-ground drainage and vent pipe and fittings.

424.2 Hand showers.

Hand-held showers shall conform to ASME A112.18.1/CSA B125.1. Hand-held showers shall provide backflow protection in accordance with ASME A112.18.1/CSA B125.1 or shall be protected against backflow by a device complying with ASME A112.18.3.

424.3 Individual shower valves.

Individual shower and tub-shower combination valves shall be balanced-pressure, thermostatic or combination balanced-pressure/thermostatic valves that conform to the requirements of ASSE 1016/ ASME A112.1016/CSA B125.16 or ASME A112.18.1/CSA B125.1 and shall be installed at the point of use. Shower and tub-shower combination valves required by this section shall be equipped with a means to limit the maximum setting of the valve to 50°C, which shall be field adjusted in accordance with the manufacturer's instructions. In-line thermostatic valves shall not be utilized for compliance with this section.

424.4 Reserved for numbering sequence

424.5 Bathtub and whirlpool bathtub valves.

The hot water supplied to bathtubs and whirlpool bathtubs shall be limited to a maximum temperature of 50°C by a water-temperature limiting device that conforms to ASSE 1070 or CSA B125.3, except where such protection is otherwise provided by a combination tub/shower valve in accordance with Section 424.3.

424.6 Hose-connected outlets.

Faucets and fixture fittings with hose-connected outlets shall conform to ASME A112.18.3 or ASME A112.18.1/CSA B125.1.

424.7 Temperature-actuated, flow reduction valves for individual fixture fittings.

Temperature-actuated, flow reduction devices, where installed for individual fixture fittings, shall conform to ASSE 1062. Such valves shall not be used alone as a substitute for the balanced-pressure, thermostatic or combination shower valves required in Section 424.3.

424.8 Transfer valves.

Deck-mounted bath/shower transfer valves containing an integral atmospheric vacuum breaker shall conform to the requirements of ASME A112.18.1/CSA B125.1.

424.9 Water closet personal hygiene devices.

Personal hygiene devices integral to water closets or water closet seats shall conform to the requirements of ASME A112.4.2.

SECTION 425 FLUSHING DEVICES FOR WATER CLOSETS AND URINALS

425.1 Flushing devices required.

Each water closet, urinal, clinical sink and any plumbing fixture that depends on trap siphonage to discharge the fixture contents to the drainage system shall be provided with a flushometer valve, flushometer tank or a flush tank designed and installed to supply water in quantity and rate of flow to flush the contents of the fixture, cleanse the fixture and refill the fixture trap.

425.1.1 Separate for each fixture.

A flushing device shall not serve more than one fixture.

425.2 Flushometer valves and tanks.

Flushometer valves and tanks shall comply with ASSE 1037 or CSA B125.3. Vacuum breakers on flushometer valves shall conform to the performance requirements of ASSE 1001 or CSA B64.1.1. Access shall be provided to vacuum breakers. Flushometer valves shall be of the water conservation type and shall not be used where the water pressure is lower than the minimum required for normal operation. When operated, the valve shall automatically complete the cycle of operation, opening fully and closing positively under the water supply pressure. Each flushometer valve shall be provided with a means for regulating the flow through the valve. The trap seal to the fixture shall be automatically refilled after each flushing cycle.

425.3 Flush tanks.

Flush tanks equipped for manual flushing shall be controlled by a device designed to refill the tank after each discharge and to shut off completely the water flow to the tank when the tank is filled to operational capacity. The trap seal to the fixture shall be automatically refilled after each flushing. The water supply to flush tanks equipped for automatic flushing shall be controlled with a timing device or sensor control devices.

425.3.1 Fill valves.

Flush tanks shall be equipped with an anti-siphon fill valve conforming to ASSE 1002 or CSA B125.3. The fill valve backflow preventer shall be located not less than 25 mm above the full opening of the overflow pipe.

425.3.2 Overflows in flush tanks.

Flush tanks shall be provided with overflows discharging to the water closet or urinal connected thereto and shall be sized to prevent flooding the tank at the maximum rate at which the tanks are supplied with water according to the manufacturer's design conditions. The opening of the overflow pipe shall be located above the flood level rim of the water closet or urinal or above a secondary overflow in the flush tank.

425.3.3 Sheet copper.

Sheet copper utilized for flush tank linings shall conform to ASTM B 152 and shall not weigh less than 3.05 kg per m².

425.3.4 Access required.

All parts in a flush tank shall be accessible for repair and replacement.

425.4 Flush pipes and fittings.

Flush pipes and fittings shall be of nonferrous material and shall conform to ASME A112.19.5/CSA B45.15.

SECTION 426 MANUAL FOOD AND BEVERAGE DISPENSING EQUIPMENT

426.1 Approval.

Manual food and beverage dispensing equipment shall conform to the requirements of NSF 18.

SECTION 427 FLOOR SINKS

427.1 Approval.

Sanitary floor sinks shall conform to the requirements of ASME A112.6.7.

CHAPTER 5 WATER HEATERS

SECTION 501 GENERAL

501.1 Scope.

The provisions of this chapter shall govern the materials, design and installation of water heaters and the related safety devices and appurtenances.

501.2 Water heater as space heater.

Where a combination potable water heating and space heating system requires water for space heating at temperatures greater than 60°C, a master thermostatic mixing valve complying with ASSE 1017 shall be provided to limit the water supplied to the potable hot water distribution system to a temperature of 60°C or less. The potability of the water shall be maintained throughout the system.

501.3 Drain valves.

Drain valves for emptying shall be installed at the bottom of each tank-type water heater and hot water storage tank. The drain valve inlet shall be not less than 19 mm nominal iron pipe size and the outlet shall be provided with male garden hose threads.

501.4 Location.

Water heaters and storage tanks shall be located and connected so as to provide access for observation, maintenance, servicing and replacement.

501.5 Water heater labeling.

All water heaters shall be third-party certified.

501.6 Water temperature control in piping from tankless heaters.

The temperature of water from tankless water heaters shall be not greater than 60°C where intended for domestic uses. This provision shall not supersede the requirement for protective shower valves in accordance with Section 424.3.

501.7 Pressure marking of storage tanks.

Storage tanks and water heaters installed for domestic hot water shall have the maximum

allowable working pressure clearly and indelibly stamped in the metal or marked on a plate welded thereto or otherwise permanently attached. Such markings shall be in an accessible position outside of the tank so as to make inspection or re-inspection readily possible.

501.8 Temperature controls.

Hot water supply systems shall be equipped with automatic temperature controls capable of adjustments from the lowest to the highest acceptable temperature settings for the intended temperature operating range.

SECTION 502 INSTALLATION

502.1 General.

Water heaters shall be installed in accordance with the manufacturer's instructions. Oil-fired water heaters shall conform to the requirements of this code and the SBC 501. Electric water heaters shall conform to the requirements of this code and provisions of SBC 401. Gas-fired water heaters shall conform to the requirements of the Saudi Fuel Gas Code SBC 1201.

502.1.1 Elevation and protection.

Elevation of water heater ignition sources and mechanical damage protection requirements for water heaters shall be in accordance with the SBC 501 and the SBC 1201.

502.2 Rooms used as a plenum.

Water heaters using solid, liquid or gas fuel shall not be installed in a room containing air-handling machinery where such room is used as a plenum.

502.3 Water heaters installed in attics.

Attics containing a water heater shall be provided with an opening and unobstructed passageway large enough to allow removal of the water heater. The passageway shall be not less than 750 mm in height and 550 mm in width and not more than 6000 mm in length when measured along the centerline of the passageway from the opening to the water heater. The passageway shall have continuous

solid flooring not less than 600 mm in width. A level service space not less than 750 mm in length and 750 mm in width shall be present at the front or service side of the water heater. The clear access opening dimensions shall be not less than 500 mm by 750 mm where such dimensions are large enough to allow removal of the water heater.

502.4 Seismic supports.

Where earthquake loads are applicable in accordance with the SBC 201, water heater supports shall be designed and installed for the seismic forces in accordance with the SBC 201 and SBC 301.

502.5 Clearances for maintenance and replacement.

Appliances shall be provided with access for inspection, service, repair and replacement without disabling the function of a fire-resistance-rated assembly or removing permanent construction, other appliances or any other piping or ducts not connected to the appliance being inspected, serviced, repaired or replaced. A level working space not less than 750 mm in length and 750 mm in width shall be provided in front of the control side to service an appliance.

SECTION 503 CONNECTIONS

503.1 Cold water line valve.

The cold water branch line from the main water supply line to each hot water storage tank or water heater shall be provided with a valve, located near the equipment and serving only the hot water storage tank or water heater. The valve shall not interfere or cause a disruption of the cold water supply to the remainder of the cold water system. The valve shall be provided with access on the same floor level as the water heater served.

503.2 Water circulation.

The method of connecting a circulating water heater to the tank shall provide proper circulation of water through the water heater. The pipe or tubes required for the installation of appliances that will draw from the water heater or storage tank shall comply with the provisions of this code for material and installation.

SECTION 504 SAFETY DEVICES

504.1 Antisiphon devices.

An *approved* means, such as a cold water “dip” tube with a hole at the top or a vacuum relief valve installed in the cold water supply line above the top of the heater or tank, shall be provided to prevent siphoning of any storage water heater or tank.

504.2 Vacuum relief valve.

Bottom fed water heaters and bottom fed tanks connected to water heaters shall have a vacuum relief valve installed. The vacuum relief valve shall comply with ANSI Z21.22.

504.3 Shutdown.

A means for disconnecting an electric hot water supply system from its energy supply shall be provided in accordance with NFPA 70. A separate valve shall be provided to shut off the energy fuel supply to all other types of hot water supply systems.

504.4 Relief valve.

Storage water heaters operating above atmospheric pressure shall be provided with an approved, self-closing (levered) pressure relief valve and temperature relief valve or combination thereof. The relief valve shall conform to ANSI Z21.22. The relief valve shall not be used as a means of controlling thermal expansion.

504.4.1 Installation.

Such valves shall be installed in the shell of the water heater tank. Temperature relief valves shall be so located in the tank as to be actuated by the water in the top 150 mm of the tank served. For installations with separate storage tanks, the approved, self-closing (levered) pressure relief valve and temperature relief valve or combination thereof conforming to ANSI Z21.22 valves shall be installed on both the storage water heater and storage tank. There shall not be a check valve or shutoff valve between a relief valve and the heater or tank served.

504.5 Relief valve approval.

Temperature and pressure relief valves, or combinations thereof, and energy cutoff devices shall bear the label of an approved agency and shall have a temperature setting of not more than 99°C and a pressure setting not exceeding the tank or water heater manufacturer’s rated working pressure or 1035 kPa, whichever is less. The relieving capacity

of each pressure relief valve and each temperature relief valve shall equal or exceed the heat input to the water heater or storage tank.

504.6 Requirements for discharge piping.

The discharge piping serving a pressure relief valve, temperature relief valve or combination thereof shall:

- 1 Not be directly connected to the drainage system.
- 2 Discharge through an air gap located in the same room as the water heater.
- 3 Not be smaller than the diameter of the outlet of the valve served and shall discharge full size to the air gap.
- 4 Serve a single relief device and shall not connect to piping serving any other relief device or equipment.
- 5 Discharge to the floor, to the pan serving the water heater or storage tank, to a waste receptor or to the outdoors.
- 6 Discharge in a manner that does not cause personal injury or structural damage.
- 7 Discharge to a termination point that is readily observable by the building occupants.
- 8 Not be trapped.
- 9 Be installed so as to flow by gravity.
- 10 Terminate not more than 150 mm above and not less than two times the discharge pipe diameter above the floor or flood level rim of the waste receptor.
- 11 Not have a threaded connection at the end of such piping.

- 12 Not have valves or tee fittings.

- 13 Be constructed of those materials listed in Section 605.4 or materials tested, rated and *approved* for such use in accordance with ASME A112.4.1.

504.7 Required pan.

Where a storage tank-type water heater or a hot water storage tank is installed in a location where water leakage from the tank will cause damage, the tank shall be installed in a galvanized steel pan having a material thickness of not less than 0.60 mm (No. 24 gage), or other pans approved for such use.

504.7.1 Pan size and drain.

The pan shall be not less than 38 mm in depth and shall be of sufficient size and shape to receive all dripping or condensate from the tank or water heater. The pan shall be drained by an indirect waste pipe having a diameter of not less than 19 mm. Piping for safety pan drains shall be of those materials listed in Table 605.4.

504.7.2 Pan drain termination.

The pan drain shall extend full size and terminate over a suitably located indirect waste receptor or floor drain or extend to the exterior of the building and terminate not less than 150 mm and not more than 600 mm above the adjacent ground surface. Where a pan drain was not previously installed, a pan drain shall not be required for a replacement water heater installation.

SECTION 505 INSULATION

505.1 Unfired vessel insulation.

Unfired hot water storage tanks shall be insulated to R-2.2 ($\text{m}^2 \cdot \text{K/W}$).

CHAPTER 6 WATER SUPPLY AND DISTRIBUTION

SECTION 601 GENERAL

601.1 Scope.

This chapter shall govern the materials, design and installation of water supply systems, both hot and cold, for utilization in connection with human occupancy and habitation and shall govern the installation of individual water supply systems.

601.2 Solar energy utilization.

Solar energy systems used for heating potable water or using an independent medium for heating potable water shall comply with the applicable requirements of this code. The use of solar energy shall not compromise the requirements for cross connection or protection of the potable water supply system required by this code.

601.3 Existing piping used for grounding.

Existing metallic water service piping used for electrical grounding shall not be replaced with nonmetallic pipe or tubing until other approved means of grounding is provided.

601.4 Tests.

The potable water distribution system shall be tested in accordance with Section 312.5.

601.5 Rehabilitation of piping systems.

Where pressure piping systems are rehabilitated using an epoxy lining system, such lining system shall comply with ASTM F 2831.

SECTION 602 WATER REQUIRED

602.1 General.

Structures equipped with plumbing fixtures and utilized for human occupancy or habitation shall be provided with a potable supply of water in the amounts and at the pressures specified in this chapter

602.2 Potable water required.

Only potable water shall be supplied to plumbing fixtures that provide water for drinking, bathing or culinary purposes, or for the processing of food, medical or pharmaceutical products. Unless otherwise provided in this code, potable water shall be supplied to all plumbing fixtures.

602.3 Individual water supply.

Where a potable public water supply is not

available, individual sources of potable water supply shall be utilized.

602.3.1 Sources.

Dependent on geological and soil conditions and the amount of rainfall, individual water supplies are of the following types: drilled well, driven well, dug well, bored well, spring, stream or cistern. Surface bodies of water and land cisterns shall not be sources of individual water supply unless properly treated by approved means to prevent contamination.

602.3.2 Minimum quantity.

The combined capacity of the source and storage in an individual water supply system shall supply the fixtures with water at rates and pressures as required by this chapter.

602.3.3 Water quality.

Water from an individual water supply shall be approved as potable by the authority having jurisdiction prior to connection to the plumbing system.

602.3.4 Disinfection of system.

After construction, the individual water supply system shall be purged of deleterious matter and disinfected in accordance with Section 610.

602.3.5 Pumps.

Pumps shall be rated for the transport of potable water. Pumps in an individual water supply system shall be constructed and installed so as to prevent contamination from entering a potable water supply through the pump units. Pumps shall be sealed to the well casing or covered with a water-tight seal. Pumps shall be designed to maintain a prime and installed such that ready access is provided to the pump parts of the entire assembly for repairs.

602.3.5.1 Pump enclosure.

The pump room or enclosure around a well pump shall be drained and protected from freezing by heating or other approved means. Where pumps are installed in basements, such pumps shall be mounted on a block or shelf not less than 500 mm above the basement floor. Well pits shall be prohibited.

SECTION 603 WATER SERVICE

603.1 Size of water service pipe.

The water service pipe shall be sized to supply water to the structure in the quantities and at the pressures required in this code. The water service pipe shall be not less than 20 mm in diameter.

603.2 Separation of water service and building sewer.

Where water service piping is located in the same trench with the building sewer, such sewer shall be constructed of materials listed in Table 702.2. Where the building sewer piping is not constructed of materials listed in Table 702.2, the water service pipe and the building sewer shall be horizontally separated by not less than 1500 mm of undisturbed or compacted earth. The required separation distance shall not apply where a water service pipe crosses a sewer pipe, provided the water service is sleeved to a point not less than 1500 mm horizontally from the sewer pipe centerline on both sides of such crossing. The sleeve shall be of pipe materials listed in Table 605.3, 702.2 or 702.3. The required separation distance shall not apply where the bottom of the water service pipe, located within 1500 mm of the sewer, is not less than 300 mm above the highest point of the top of the building sewer.

603.2.1 Water service near sources of pollution.

Potable water service pipes shall not be located in, under or above cesspools, septic tanks, septic tank drainage fields or seepage pits (see Section 605.1 for soil and ground water conditions).

SECTION 604 DESIGN OF BUILDING WATER DISTRIBUTION SYSTEM

604.1 General.

The design of the water distribution system shall conform to accepted engineering practice. Methods utilized to determine pipe sizes shall be approved.

604.2 System interconnection.

At the points of interconnection between the hot and cold water supply piping systems and the individual

fixtures, appliances or devices, provisions shall be made to prevent flow between such piping systems.

604.3 Water distribution system design criteria.

The water distribution system shall be designed, and pipe sizes shall be selected such that under conditions of peak demand, the capacities at the fixture supply pipe outlets shall be not less than shown in Table 604.3. The minimum flow rate and flow pressure provided to fixtures and appliances not listed in Table 604.3 shall be in accordance with the manufacturer's installation instructions.

604.4 Maximum flow and water consumption.

The maximum water consumption flow rates and quantities for all plumbing fixtures and fixture fittings shall be in accordance with Table 604.4.

Exceptions:

- 1 Vegetable sprays.
- 2 Clinical sinks having a water consumption not greater than 17 L per flushing cycle.
- 3 Service sinks.
- 4 Emergency showers.

604.5 Size of fixture supply.

The minimum size of a fixture supply pipe shall be as shown in Table 604.5. The fixture supply pipe shall terminate not more than 800 mm from the point of connection to the fixture. A reduced size flexible water connector installed between the supply pipe and the fixture shall be of an approved type. The supply pipe shall extend to the floor or wall adjacent to the fixture. The minimum size of individual distribution lines utilized in gridded or parallel water distribution systems shall be as shown in Table 604.5.

TABLE 604.3 WATER DISTRIBUTION SYSTEM DESIGN CRITERIA REQUIRED
CAPACITY AT FIXTURE SUPPLY PIPE OUTLETS

FIXTURE SUPPLY OUTLET SERVING	FLOW RATE ^a (L/min)	FLOW PRESSURE (kPa)
Bathtub, balanced-pressure, thermostatic or combination balanced- pressure/thermostatic mixing valve	15	138
Bidet, thermostatic mixing valve	7.5	138
Combination fixture	15	55
Dishwasher, residential	10	55
Drinking fountain	3.0	55
Laundry tray	15	55
Lavatory, private	3.0	55
Lavatory, private, mixing valve	3.0	55
Lavatory, public	1.5	55
Shower	9.5	55
Shower, balanced-pressure, thermostatic or combination balanced- pressure/thermostatic mixing valve	9.5 ^b	138
Sillcock, hose bibb	20	55
Sink, residential	6.5	55
Sink, service	12	55
Urinal, valve	45	170
Water closet, blow out, flushometer valve	95	310
Water closet, flushometer tank	6.0	138
Water closet, siphonic, flushometer valve	95	240
Water closet, tank, close coupled	12	138
Water closet, tank, one piece	22	138

- a. For additional requirements for flow rates and quantities, see Section 604.4.
- b. Where the shower mixing valve manufacturer indicates a lower flow rating for the mixing valve, the lower value shall be applied.

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Table 604.4 Maximum Flow Rates and Consumption for Plumbing Fixtures and Fixture Fittings

PLUMBING FIXTURE OR FIXTURE FITTING	MAXIMUM FLOW RATE OR QUANTITY ^b		
Lavatory (wash-basin), private	6 L/min at 415 kPa		
Lavatory (wash-basin), public (metering)	1 L per metering cycle		
Lavatory (wash-basin), public (other than metering)	2 L/min at 415 kPa		
Shower head ^a	9.5 L/min at 550 kPa		
Sink faucet	6 L/min at 415 kPa		
Urinal	1 L per flushing cycle		
Water closet	Single flush	S trap	3.0+0.5 L per flushing cycle
		P trap	4.0+0.5 L per flushing cycle
	Dual flush	S trap	3.0+0.5 L per flushing cycle as calculated by $(f+(5x))/6$
		P trap	4.0+0.5 L per flushing cycle as calculated by $(f+(5x))/6$

a. A hand-held shower spray is a shower head.

b. Consumption tolerances shall be determined from referenced standards.

Table 604.5 Minimum Sizes of Fixture Water Supply Pipes

FIXTURE	MINIMUM PIPE SIZE (mm)
Bathtubs ^a (1.5 m × 0.8 m) and smaller	13
Bathtubs ^a (larger than 1.5 m × 0.8 m)	13
Bidet	10
Combination sink and tray	13
Dishwasher, domestic ^a	13
Drinking fountain	10
Hose bibbs	13
Kitchen sink ^a	13
Laundry, 1, 2 or 3 compartments ^a	13
Lavatory	10
Shower, single head ^a	13
Sinks, flushing rim	20
Sinks, service	13
Urinal, flush tank	13
Urinal, flushometer valve	20
Wall hydrant	13
Water closet, flush tank	10
Water closet, flushometer tank	10
Water closet, flushometer valve	25
Water closet, one piece ^a	13

a. Where the developed length of the distribution line is 15,000 mm or less, and the available pressure at the meter is 240 kPa or greater, the minimum size of an individual distribution line supplied from a manifold and installed as part of a parallel water distribution system shall be one nominal tube size smaller than the sizes indicated.

604.6 Variable street pressures.

Where street water main pressures fluctuate, the building water distribution system shall be designed for the minimum pressure available.

604.7 Inadequate water pressure.

Wherever water pressure from the street main or other source of supply is insufficient to provide flow pressures at fixture outlets as required under Table 604.3, a water pressure booster system conforming to Section 606.5 shall be installed on the building water supply system.

604.8 Water pressure-reducing valve or regulator.

Where water pressure within a building exceeds 550 kPa static, an approved water pressure-reducing valve conforming to ASSE 1003 or CSA B356 with strainer shall be installed to reduce the pressure in the building water distribution piping to not greater than 550 kPa static.

Exception: Service lines to sill cocks and outside hydrants, and main supply risers where pressure from the mains is reduced to 550 kPa or less at individual fixtures.

604.8.1 Valve design.

The pressure-reducing valve shall be designed to remain open to permit uninterrupted water flow in case of valve failure.

604.8.2 Repair and removal.

Water pressure-reducing valves, regulators and strainers shall be so constructed and installed as to permit repair or removal of parts without breaking a pipeline or removing the valve and strainer from the pipeline.

604.9 Water hammer.

The flow velocity of the water distribution system shall be controlled to reduce the possibility of water hammer. A water-hammer arrestor shall be installed where quick-closing valves are utilized. Water-hammer arrestors shall be installed in accordance with the manufacturer's instructions. Water-hammer arrestors shall conform to ASSE 1010.

604.10 Gridded and parallel water distribution system manifolds.

Hot water and cold water manifolds installed with gridded or parallel connected individual distribution lines to each fixture or fixture fitting shall be designed in accordance with Sections 604.10.1 through 604.10.3.

604.10.1 Manifold sizing.

Hot water and cold water manifolds shall be sized in accordance with Table 604.10.1. The total liters per minute is the demand of all outlets supplied.

TABLE 604.10.1 Manifold Sizing

NOMINAL SIZE INTERNAL DIAMETER (mm)	MAXIMUM DEMAND (L/min)	
	Velocity at 1.2 m/sec	Velocity at 2.4 m/sec
15	7.5	19
20	23	42
25	38	76
32	57	117
40	83	167

604.10.2 Valves.

Individual fixture shutoff valves installed at the manifold shall be identified as to the fixture being supplied.

604.10.3 Access.

Access shall be provided to manifolds with

integral factory- or field-installed valves.

604.11 Individual pressure balancing in-line valves for individual fixture fittings.

Where individual pressure balancing in-line valves for individual fixture fittings are installed, such valves shall comply with ASSE 1066. Such valves shall be installed in an

accessible location and shall not be utilized alone as a substitute for the balanced pressure, thermostatic or combination shower valves required in Section 424.3.

SECTION 605 MATERIALS, JOINTS AND CONNECTIONS

605.1 Soil and ground water.

The installation of a water service or water distribution pipe shall be prohibited in soil and ground water contaminated with solvents, fuels, organic compounds or other detrimental materials causing permeation, corrosion, degradation or structural failure of the piping material. Where detrimental conditions are suspected, a chemical analysis of the soil and ground water conditions shall be required to ascertain the acceptability of the water service or water distribution piping material for the specific installation. Where detrimental conditions exist, approved alternative materials or routing shall be required.

605.2 Lead content of water supply pipe and fittings.

Pipe and pipe fittings, including valves and faucets, utilized in the water supply system shall have not more than 8-percent lead content.

605.2.1 Lead content of drinking water pipe and fittings.

Pipe, pipe fittings, joints, valves, faucets and fixture fittings utilized to supply water for drinking or cooking purposes shall comply with NSF 372 and shall have a weighted average lead content of 0.25 percent or less.

605.3 Water service pipe.

Water service pipe shall conform to NSF 61 and shall conform to one of the standards listed in Table 605.3. Water service pipe or tubing, installed underground and outside of the structure, shall have a working pressure rating of not less than 1100 kPa at 25°C. Where the water pressure exceeds 1100 kPa, piping material shall have a working pressure rating not less than the highest available pressure. Water service piping materials not third-party certified for water distribution shall terminate at or before the full open valve located at the entrance to the structure. Ductile iron water service piping shall be cement mortar lined in accordance with AWWA C104.

605.3.1 Dual check-valve-type backflow preventer.

Dual check-valve backflow preventers installed on the water supply system shall comply with ASSE 1024 or CSA B64.6.

605.4 Water distribution pipe.

Water distribution pipe shall conform to NSF 61 and shall conform to one of the standards listed in Table 605.4. Hot water distribution pipe and tubing shall have a pressure rating of not less than 700 kPa at 80°C.

605.5 Fittings.

Pipe fittings shall be approved for installation with the piping material installed and shall comply with the applicable standards listed in Table 605.5. Pipe fittings utilized in water supply systems shall also comply with NSF 61. Ductile and gray iron pipe and pipe fittings utilized in water service piping systems shall be cement mortar lined in accordance with AWWA C104.

TABLE 605.3 Water Service Pipe

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe	ASTM D 1527; ASTM D 2282
Brass pipe	ASTM B 43
Chlorinated polyvinyl chloride (CPVC) plastic pipe	ASTM D 2846; ASTM F 441; ASTM F 442; CSA B137.6
Chlorinated polyvinyl chloride/aluminum/chlorinated polyvinyl chloride (CPVC/AL/CPVC)	ASTM F 2855
Copper or copper-alloy pipe	ASTM B 42; ASTM B 302
Copper or copper-alloy tubing (Type K, WK, L, WL, M or WM)	ASTM B 75; ASTM B 88; ASTM B 251; ASTM B 447
Cross-linked polyethylene (PEX) plastic pipe and tubing	ASTM F 876; ASTM F 877; AWWA C904; CSA B137.5
Cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL- PEX) pipe	ASTM F 1281; ASTM F 2262; CSA B137.10
Cross-linked polyethylene/aluminum/high-density polyethylene (PEX-AL-HDPE)	ASTM F 1986
Ductile iron water pipe	AWWA C151/A21.51; AWWA C115/A21.15
Galvanized steel pipe	ASTM A 53
Polyethylene (PE) plastic pipe	ASTM D 2239; ASTM D 3035; AWWA C901; CSA B137.11
Polyethylene (PE) plastic tubing	ASTM D 2737; AWWA C901; CSA B137.1
Polyethylene/aluminum/polyethylene (PE-AL-PE) pipe	ASTM F 1282; CSA B137.9
Polyethylene of raised temperature (PE-RT) plastic tubing	ASTM F 2769
Polypropylene (PP) plastic pipe or tubing	ASTM F 2389; CSA B137.11
Polyvinyl chloride (PVC) plastic pipe	ASTM D 1785; ASTM D 2241; ASTM D 2672; CSA B137.3
Stainless steel pipe (Type 304/304L)	ASTM A 312; ASTM A 778
Stainless steel pipe (Type 316/316L)	ASTM A 312; ASTM A 778

TABLE 605.4 WATER DISTRIBUTION PIPE

MATERIAL	STANDARD
Brass pipe	ASTM B 43
Chlorinated polyvinyl chloride (CPVC) plastic pipe and tubing	ASTM D 2846; ASTM F 441; ASTM F 442; CSA B137.6
Chlorinated polyvinyl chloride/aluminum/chlorinated polyvinyl chloride (CPVC/AL/CPVC)	ASTM F 2855
Copper or copper-alloy pipe	ASTM B 42; ASTM B 302
Copper or copper-alloy tubing (Type K, WK, L, WL, M or WM)	ASTM B 75; ASTM B 88; ASTM B 251; ASTM B 447
Cross-linked polyethylene (PEX) plastic tubing	ASTM F 876; ASTM F 877; CSA B137.5
Cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX) pipe	ASTM F 1281; ASTM F 2262; CSA B137.10
Cross-linked polyethylene/aluminum/high-density polyethylene (PEX-AL-HDPE)	ASTM F 1986; CAN/CSA B137.10M
Ductile iron pipe	AWWA C151/A21.51; AWWA C115/A21.15
Galvanized steel pipe	ASTM A 53
Polyethylene/aluminum/polyethylene (PE-AL-PE) composite pipe	ASTM F 1282; CAN/CSA-B137.9
Polyethylene of raised temperature (PE-RT) plastic tubing	ASTM F 2769
Polypropylene (PP) plastic pipe or tubing	ASTM F 2389; CSA B137.11
Stainless steel pipe (Type 304/304L)	ASTM A 312; ASTM A 778
Stainless steel pipe (Type 316/316L)	ASTM A 312; ASTM A 778

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Table 605.5 Pipe Fittings

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic	ASTM D 2468
Cast iron	ASME B16.4
Chlorinated polyvinyl chloride (CPVC) plastic	ASSE 1061; ASTM D 2846; ASTM F 437; ASTM F 438; ASTM F 439; CSA B137.6
Copper or copper alloy	ASME B16.15; ASME B16.18; ASME B16.22; ASME B16.26; ASME B16.51; ASSE 1061; ASTM F 1476; ASTM F 1548
Cross-linked polyethylene/aluminum/high-density polyethylene (PEX-AL-HDPE)	ASTM F 1986
Fittings for cross-linked polyethylene (PEX) plastic tubing	ASSE 1061, ASTM F 877; ASTM F 1807; ASTM F 1960; ASTM F 2080; ASTM F 2098, ASTM F 2159; ASTM F 2434; ASTM F 2735; CSA B137.5
Fittings for polyethylene of raised temperature (PE-RT) plastic tubing	ASTM F 1807; ASTM F 2098; ASTM F 2159; ASTM F 2735; ASTM F 2769
Gray iron and ductile iron	ASTM F 1476; ASTM F 1548; AWWA C110/A21.10; AWWA C153/A21.53;
Insert fittings for polyethylene/aluminum/polyethylene (PE-AL-PE) and cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX)	ASTM F 1974; ASTM F 1281; ASTM F 1282; CSA B137.9; CSA B137.10M
Malleable iron	ASME B16.3
Metal (brass) insert fittings for polyethylene/aluminum/polyethylene (PE-AL-PE) and cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX)	ASTM F 1974
Polyethylene (PE) plastic pipe	ASTM D 2609; ASTM D 2683; ASTM D 3261; ASTM F 1055; CSA B137.1
Polypropylene (PP) plastic pipe or tubing	ASTM F 2389; CSA B137.11
Polyvinyl chloride (PVC) plastic	ASTM D 2464; ASTM D 2466; ASTM D 2467; CSA B137.2; CSA B137.3
Stainless steel (Type 304/304L)	ASTM A 312; ASTM A 778; ASTM F 1476; ASTM F 1548
Stainless steel (Type 316/316L)	ASTM A 312; ASTM A 778; ASTM F 1476; ASTM F 1548
Steel	ASME B16.9; ASME B16.11; ASME B16.28; ASTM F 1476; ASTM F 1548

605.5.1 Mechanically formed tee fittings.

Mechanically extracted outlets shall have a height not less than three times the thickness of the branch tube wall.

605.5.1.1 Full flow assurance.

Branch tubes shall not restrict the flow in the run tube. A dimple serving as a depth stop shall be formed in the branch tube to ensure that penetration into the collar is of the correct depth. For inspection purposes, a second dimple shall be placed 6.4 mm above the first dimple. Dimples shall be aligned with the tube

run.

605.5.1.2 Brazed joints.

Mechanically formed tee fittings shall be brazed in accordance with Section 605.14.1.

605.6 Flexible water connectors.

Flexible water connectors exposed to continuous pressure shall conform to ASME A112.18.6/CSA B125.6. Access shall be provided to all flexible water connectors.

605.7 Valves.

Valves shall be compatible with the type of piping material installed in the system. Valves shall conform to one of the standards listed in Table 605.7 or shall be approved. Valves intended to supply drinking water shall meet the requirements of NSF 61.

605.8 Manufactured pipe nipples.

Manufactured pipe nipples shall conform to one of the standards listed in Table 605.8.

605.9 Prohibited joints and connections.

The following types of joints and connections shall be prohibited:

1. Cement or concrete joints.
- 5 Joints made with fittings not *approved* for the specific installation.
- 6 Solvent-cement joints between different types of plastic pipe.
- 7 Saddle-type fittings.

605.10 ABS plastic.

Joints between ABS plastic pipe and fittings shall comply with Sections 605.10.1 through 605.10.3.

605.10.1 Mechanical joints.

Mechanical joints on water pipes shall be made with an elastomeric seal conforming to ASTM D 3139. Mechanical joints shall only be installed in underground systems, unless otherwise approved. Joints shall be installed only in accordance with the manufacturer's instructions.

605.10.2 Solvent cementing.

Joint surfaces shall be clean and free from moisture. Solvent cement that conforms to ASTM D 2235 shall be applied to all joint

surfaces. The joint shall be made while the cement is wet. Joints shall be made in accordance with ASTM D 2235. Solvent-cement joints shall be permitted above or below ground.

605.10.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Schedule 80 or heavier pipe shall be permitted to be threaded with dies specifically designed for plastic pipe. Approved thread lubricant or tape shall be applied on the male threads only.

605.11 Brass.

Joints between brass pipe and fittings shall comply with Sections 605.11.1 through 605.11.4.

605.11.1 Brazed joints.

All joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

605.11.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.11.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

605.11.4 Welded joints.

All joint surfaces shall be cleaned. The joint shall be welded with an approved filler metal.

605.12 Gray iron and ductile iron joints.

Joints for gray and ductile iron pipe and fittings shall comply with AWWA C111/A21.11 and shall be installed in accordance with the manufacturer's instructions.

605.13 Copper pipe.

Joints between copper or copper-alloy pipe and fittings shall comply with Sections 605.13.1 through 605.13.5.

605.13.1 Brazed joints.

All joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

TABLE 605.7

VALVES

MATERIAL	STANDARD
Chlorinated polyvinyl chloride (CPVC) plastic	ASME A112.4.14; ASME A112.18.1/CSA B125.1; ASTM F 1970; CSA B125.3
Copper or copper alloy	ASME A112.4.14; ASME A112.18.1/CSA B125.1; ASME B16.34; CSA B125.3; MSS SP-67; MSS SP-80; MSS SP-110
Cross-linked polyethylene (PEX) plastic	ASME A112.4.14; ASME A112.18.1/CSA B125.1; CSA B125.3; NSF 359
Gray iron and ductile iron	AWWA C500; AWWA C504; AWWA C507; MSS SP-67; MSS SP-70; MSS SP-71; MSS SP-72; MSS SP-78
Polypropylene (PP) plastic	ASME A112.4.14; ASTM F 2389
Polyvinyl chloride (PVC) plastic	ASME A112.4.14; ASTM F 1970

TABLE 605.8 MANUFACTURED PIPE NIPPLES

MATERIAL	STANDARD
Brass-, copper-, chromium-plated	ASTM B 687
Steel	ASTM A 733

605.13.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.13.3 Solder joints.

Solder joints shall be made in accordance with ASTM B 828. Cut tube ends shall be reamed to the full inside diameter of the tube end. Joint surfaces shall be cleaned. A flux conforming to ASTM B 813 shall be applied. The joint shall be soldered with a solder conforming to ASTM B 32. The joining of water supply piping shall be made with lead-free solder and fluxes. "Lead free" shall mean a chemical composition equal to or less than 0.2-percent lead.

605.13.4 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

605.13.5 Welded joints.

Joint surfaces shall be cleaned. The joint shall be welded with an approved filler metal.

605.14 Copper tubing.

Joints between copper or copper alloy tubing and fittings shall comply with Sections 605.14.1 through 605.14.5.

605.14.1 Brazed joints.

Joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

605.14.2 Flared joints.

Flared joints for water pipe shall be made by a tool designed for that operation.

605.14.3 Grooved and shouldered mechanical joints.

Grooved and shouldered mechanical joints shall comply with ASTM F 1476, shall be made with an approved elastomeric seal and shall be installed in accordance with the manufacturer's instructions. Such joints shall be exposed or concealed.

605.14.4 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.14.5 Press-connect joints.

Press-connect joints shall conform to one of the standards listed in Table 605.5, and shall be installed in accordance with the manufacturer's instructions. Cut tube ends shall be reamed to the full inside diameter of the tube end. Joint surfaces shall be cleaned. The tube shall be

fully inserted into the press connect fitting. Press-connect joints shall be pressed with a tool certified by the manufacturer.

605.14.6 Solder joints.

Solder joints shall be made in accordance with the methods of ASTM B 828. All cut tube ends shall be reamed to the full inside diameter of the tube end. All joint surfaces shall be cleaned. A flux conforming to ASTM B 813 shall be applied. The joint shall be soldered with a solder conforming to ASTM B 32. The joining of water supply piping shall be made with lead-free solders and fluxes. "Lead free" shall mean a chemical composition equal to or less than 0.2-percent lead.

605.15 CPVC plastic.

Joints between CPVC plastic pipe and fittings shall comply with Sections 605.15.1 through 605.15.3.

605.15.1 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.15.2 Solvent cementing.

Joint surfaces shall be clean and free from moisture. Joints shall be made in accordance with the pipe manufacturer's installation instructions. Where such instructions require that a primer be used, the primer shall be applied to the joint surfaces and a solvent cement orange in color and conforming to ASTM F 493 shall be applied to the joint surfaces. Where such instructions allow for a one-step solvent cement, yellow in color and conforming to ASTM F 493, to be used, the joint surfaces shall not require application of a primer before the solvent cement is applied. The joint shall be made while the cement is wet and in accordance with ASTM D 2846 or ASTM F 493. Solvent cemented joints shall be permitted above or below ground.

605.15.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Schedule 80 or heavier pipe shall be permitted to be threaded with dies specifically designed for plastic pipe, but the pressure rating of the pipe shall be reduced by 50 percent. Thread by socket molded fittings shall be permitted. Approved thread lubricant or tape shall be applied on the male threads only.

605.16 Chlorinated polyvinyl chloride/aluminum/chlorinated polyvinyl chloride (CPVC/AL/CPVC) pipe and tubing.

Joints between CPVC/AL/CPVC plastic pipe or CPVC fittings shall comply with Sections 605.16.1 and 605.16.2.

605.16.1 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.16.2 Solvent cementing.

Joint surfaces shall be clean and free from moisture, and an approved primer shall be applied. Solvent cement, orange in color and conforming to ASTM F 493, shall be applied to joint surfaces. The joint shall be made while the cement is wet, and in accordance with ASTM D 2846 or ASTM F 493. Solvent cement joints shall be permitted above or below ground.

Exception: A primer is not required where all of the following conditions apply:

1. The solvent cement used is third-party certified as conforming to ASTM F 493.
- 8 The solvent cement used is yellow in color.
- 9 The solvent cement is used only for joining 13 mm through diameter (50 mm) CPVC/AL/CPVC pipe and CPVC fittings.
- 10 The CPVC fittings are manufactured in accordance with ASTM D 2846.

605.17 PEX plastic.

Joints between cross-linked polyethylene plastic tubing and fittings shall comply with Sections 605.17.1 and 605.17.2.

605.17.1 Flared joints.

Flared pipe ends shall be made by a tool designed for that operation.

605.17.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions. Fittings for cross-linked polyethylene (PEX) plastic tubing shall comply with the applicable standards listed in Table 605.5 and shall be installed in accordance with the manufacturer's instructions. PEX tubing

shall be factory marked with the appropriate standards for the fittings that the PEX manufacturer specifies for use with the tubing.

605.18 Steel.

Joints between galvanized steel pipe and fittings shall comply with Sections 605.18.1 through 605.18.3.

605.18.1 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

605.18.2 Mechanical joints.

Joints shall be made with an approved elastomeric seal. Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.18.3 Grooved and shouldered mechanical joints.

Grooved and shouldered mechanical joints shall comply with ASTM F 1476, shall be made with an approved elastomeric seal and shall be installed in accordance with the manufacturer's instructions. Such joints shall be exposed or concealed.

605.19 PE plastic.

Joints between polyethylene plastic pipe or tubing and fittings shall comply with Sections 605.19.1 through 605.19.4.

605.19.1 Flared joints.

Flared joints shall be permitted where so indicated by the pipe manufacturer. Flared joints shall be made by a tool designed for that operation.

605.19.2 Heat-fusion joints.

Joint surfaces shall be clean and free from moisture. All joint surfaces shall be heated to melt temperature and joined. The joint shall be undisturbed until cool. Joints shall be made in accordance with ASTM D 2657.

605.19.3 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.19.4 Installation.

Polyethylene pipe shall be cut square, with a cutter designed for plastic pipe. Except where

joined by heat fusion, pipe ends shall be chamfered to remove sharp edges. Kinked pipe shall not be installed. The minimum pipe bending radius shall be not less than 30 pipe diameters, or the minimum coil radius, whichever is greater. Piping shall not be bent beyond straightening of the curvature of the coil. Bends shall be prohibited within 10 pipe diameters of any fitting or valve. Stiffener inserts installed with compression-type couplings and fittings shall not extend beyond the clamp or nut of the coupling or fitting.

605.20 Polypropylene (PP) plastic.

Joints between PP plastic pipe and fittings shall comply with Section 605.20.1 or 605.20.2.

605.20.1 Heat-fusion joints.

Heat-fusion joints for polypropylene pipe and tubing joints shall be installed with socket-type heat-fused polypropylene fittings, butt-fusion polypropylene fittings or electro fusion polypropylene fittings. Joint surfaces shall be clean and free from moisture. The joint shall be undisturbed until cool. Joints shall be made in accordance with ASTM F 2389.

605.20.2 Mechanical and compression sleeve joints.

Mechanical and compression sleeve joints shall be installed in accordance with the manufacturer's instructions.

605.21 Polyethylene/aluminum/polyethylene (PE-AL-PE) and cross-linked polyethylene/ aluminum/cross-linked polyethylene (PEX-AL-PEX).

Joints between PE-AL-PE or PEX-AL-PEX pipe and fittings shall comply with Section 605.21.1.

605.21.1 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions. Fittings for PE-AL-PE and PEX-AL-PEX as described in ASTM F 1974, ASTM F 1281, ASTM F 1282, CSA B137.9 and CSA B137.10 shall be installed in accordance with the manufacturer's instructions.

605.22 PVC plastic.

Joints between PVC plastic pipe and fittings shall comply with Sections 605.22.1 through 605.22.3.

605.22.1 Mechanical joints.

Mechanical joints on water pipe shall be made with an elastomeric seal conforming to ASTM D 3139. Mechanical joints shall not be installed in above-ground systems unless otherwise approved. Joints shall be installed in accordance with the manufacturer's instructions.

605.22.2 Grooved and shouldered mechanical joints.

Grooved and shouldered mechanical joints shall comply with ASTM F 1476, shall be made with an approved elastomeric seal and shall be installed in accordance with the manufacturer's instructions. Such joints shall be exposed or concealed.

605.22.3 Solvent cementing.

Joint surfaces shall be clean and free from moisture. A purple primer that conforms to ASTM F 656 shall be applied. Solvent cement not purple in color and conforming to ASTM D 2564 or CSA B137.3 shall be applied to all joint surfaces. The joint shall be made while the cement is wet and shall be in accordance with ASTM D 2855. Solvent-cement joints shall be permitted above or below ground.

605.22.4 Threaded joints.

Threads shall conform to ASME B1.20.1. Schedule 80 or heavier pipe shall be permitted to be threaded with dies specifically designed for plastic pipe, but the pressure rating of the pipe shall be reduced by 50 percent. Thread by socket molded fittings shall be permitted. Approved thread lubricant or tape shall be applied on the male threads only.

605.23 Stainless steel.

Joints between stainless steel pipe and fittings shall comply with Sections 605.23.1 and 605.23.3.

605.23.1 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

605.23.2 Welded joints.

All joint surfaces shall be cleaned. The joint shall be welded autogenously or with an approved filler metal as referenced in ASTM A 312.

605.23.3 Grooved and shouldered mechanical joints.

Grooved and shouldered mechanical joints shall comply with ASTM F 1476, shall be made with an approved elastomeric seal and shall be installed in accordance with the manufacturer's instructions. Such joints shall be exposed or concealed.

605.24 Joints between different materials.

Joints between different piping materials shall be made with a mechanical joint of the compression or mechanical-sealing type, or as permitted in Sections 605.24.1, 605.24.2 and 605.24.3. Connectors or adapters shall have an elastomeric seal conforming to ASTM F 477. Joints shall be installed in accordance with the manufacturer's instructions.

605.24.1 Copper or copper-alloy tubing to galvanized steel pipe.

Joints between copper or copper-alloy tubing and galvanized steel pipe shall be made with a brass fitting or dielectric fitting or a dielectric union conforming to ASSE 1079. The copper tubing shall be soldered to the fitting in an approved manner, and the fitting shall be screwed to the threaded pipe.

605.24.2 Plastic pipe or tubing to other piping material.

Joints between different types of plastic pipe or between plastic pipe and other piping material shall be made with approved adapters or transition fittings.

605.24.3 Stainless steel.

Joints between stainless steel and different piping materials shall be made with a mechanical joint of the compression or mechanical sealing type or a dielectric fitting or a dielectric union conforming to ASSE 1079.

605.25 PE-RT plastic.

Joints between polyethylene of raised temperature plastic tubing and fittings shall be in accordance with Section 605.25.1.

605.25.1 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions. Fittings for polyethylene of raised temperature plastic tubing shall comply with the applicable standards listed in Table 605.5 and shall be installed in accordance with the

manufacturer's instructions. Polyethylene of raised temperature plastic tubing shall be factory marked with the applicable standards for the fittings that the manufacturer of the tubing specifies for use with the tubing.

SECTION 606 INSTALLATION OF THE BUILDING WATER DISTRIBUTION SYSTEM

606.1 Location of full-open valves.

Full-open valves shall be installed in the following locations:

- 1 On the building water service pipe from the public water supply near the curb.
- 2 On the water distribution supply pipe at the entrance into the structure.
- 3 On the discharge side of every water meter.
- 4 On the base of every water riser pipe in occupancies other than multiple-family residential occupancies that are two stories or less in height and in one and two family residential occupancies.
- 5 On the top of every water down-feed pipe in occupancies other than one- and two-family residential occupancies.
- 6 On the entrance to every water supply pipe to a dwelling unit, except where supplying a single fixture equipped with individual stops.
- 7 On the water supply pipe to a gravity or pressurized water tank.
- 8 On the water supply pipe to every water heater.

606.2 Location of shutoff valves.

Shutoff valves shall be installed in the following locations:

On the fixture supply to each plumbing fixture other than bathtubs and showers in one- and two-family residential occupancies, and other than in individual sleeping units that are provided with unit shutoff valves in hotels, motels, boarding houses and similar occupancies.

On the water supply pipe to each sill cock.

On the water supply pipe to each appliance or mechanical equipment.

606.3 Access to valves.

Access shall be provided to all full open valves and shutoff valves.

606.4 Valve identification.

Service and hose bibb valves shall be identified. All other valves installed in locations that are not adjacent to the fixture or appliance shall be identified, indicating the fixture or appliance served.

606.5 Water pressure booster systems.

Water pressure booster systems shall be provided as required by Sections 606.5.1 through 606.5.10.

606.5.1 Water pressure booster systems required.

Where the water pressure in the public water main or individual water supply system is insufficient to supply the minimum pressures and quantities specified in this code, the supply shall be supplemented by an elevated water tank, a hydropneumatic pressure booster system or a water pressure booster pump installed in accordance with Section 606.5.5.

606.5.2 Support.

All water supply tanks shall be supported in accordance with the SBC 201.

606.5.3 Covers.

All water supply tanks shall be covered to keep out unauthorized persons, dirt and vermin. The covers of gravity tanks shall be vented with a return bend vent pipe with an area not less than the area of the down-feed riser pipe, and the vent shall be screened with a corrosion resistant screen of not less than 630 by 787 mesh per m.

606.5.4 Overflows for water supply tanks.

A gravity or suction water supply tank shall be provided with an overflow with a diameter not less than that shown in Table 606.5.4. The overflow outlet shall discharge at a point not less than 150 mm above the roof or roof drain; floor or floor drain; or over an open water-supplied fixture. The overflow outlet shall be covered with a corrosion-resistant screen of not less than 630 by 787 mesh per m and by 6.4 mm hardware cloth or shall terminate in a horizontal angle seat check valve. Drainage from overflow pipes shall be directed so as not to freeze on roof walks.

606.5.5 Low-pressure cutoff required on booster pumps.

A low-pressure cutoff shall be installed on all booster pumps in a water pressure booster system to prevent creation of a vacuum or negative pressure on the suction side of the pump when a positive pressure of 70 kPa or less occurs on the suction side of the pump.

606.5.6 Potable water inlet control and location.

Potable water inlets to gravity tanks shall be controlled by a fill valve or other automatic

supply valve installed so as to prevent the tank from overflowing. The inlet shall be terminated so as to provide an air gap not less than 100 mm above the overflow.

606.5.7 Tank drain pipes.

A valved pipe shall be provided at the lowest point of each tank to permit emptying of the tank. The tank drain pipe shall discharge as required for overflow pipes and shall not be smaller in size than specified in Table 605.5.7.

Table 606.5.4 Sizes for Overflow Pipes for Water Supply Tanks

MAXIMUM CAPACITY OF WATER SUPPLY LINE TO TANK (L/min)	DIAMETER OF OVERFLOW PIPE (mm)
0 – 200	50
200 – 600	65
600 – 750	75
750 – 1,500	100
1,500 – 2,650	125
2,650 – 3,800	150
Over 3,800	200

TABLE 606.5.7 SIZE OF DRAIN PIPES FOR WATER TANKS

TANK CAPACITY (Liters)	DRAIN PIPE (mm)
Up to 2,850	25
2,851 to 5,700	40
5,701 to 11,400	50
11,401 to 20,000	65
20,001 to 30,000	75
Over 30,000	100

606.5.8 Prohibited location of potable supply tanks.

Potable water gravity tanks or manholes of potable water pressure tanks shall not be located directly under any soil or waste piping or any source of contamination.

606.5.9 Pressure tanks, vacuum relief.

All water pressure tanks shall be provided with a vacuum relief valve at the top of the tank that will operate up to a maximum water pressure of 1400 kPa and up to a maximum temperature of 90°C. The size of such vacuum relief valve shall be not less than 15 mm.

Exception: This section shall not apply to pressurized captive air diaphragm/bladder tanks.

606.5.10 Pressure relief for tanks.

Every pressure tank in a hydropneumatic pressure booster system shall be protected with a pressure relief valve. The pressure relief valve shall be set at a maximum pressure equal to the rating of the tank. The relief valve shall be installed on the supply pipe to the tank or on the tank. The relief valve shall discharge by gravity to a safe place of disposal.

606.6 Water supply system test.

Upon completion of a section of or the entire water supply system, the system, or portion

completed, shall be tested in accordance with Section 312.

606.7 Labeling of water distribution pipes in bundles.

Where water distribution piping is bundled at installation, each pipe in the bundle shall be identified using stenciling or commercially available pipe labels. The identification shall indicate the pipe contents and the direction of flow in the pipe. The interval of the identification markings on the pipe shall not exceed 7500 mm. There shall be not less than one identification label on each pipe in each room, space or story.

SECTION 607 HOT WATER SUPPLY SYSTEM

607.1 Where required.

In residential occupancies, hot water shall be supplied to plumbing fixtures and equipment utilized for bathing, washing, culinary purposes, cleansing, laundry or building maintenance. In nonresidential occupancies, hot water shall be supplied for culinary purposes, cleansing, laundry or building maintenance purposes. In nonresidential occupancies, hot water or tempered water shall be supplied for bathing and washing purposes.

607.1.1 Temperature limiting means.

A thermostat control for a water heater shall not serve as the temperature limiting means for the purposes of complying with the requirements of this code for maximum allowable hot or tempered water delivery temperature at fixtures.

607.1.2 Tempered water temperature control.

Tempered water shall be supplied through a water temperature limiting device that conforms to ASSE 1070 and shall limit the tempered water to a maximum of 43°C. This provision shall not supersede the requirement for protective shower valves in accordance with Section 424.3.

607.2 Hot or tempered water supply to fixtures.

The developed length of hot or tempered water piping, from the source of hot water to the fixtures that require hot or tempered water, shall not exceed 15,000 mm. Recirculating

system piping and heat-traced piping shall be considered to be sources of hot or tempered water.

607.2.1 Circulation systems and heat trace systems for maintaining heated water temperature in distribution systems.

For Group R2, R3 and R4 occupancies that are three stories or less in height above grade plane, the installation of heated water circulation and temperature maintenance systems shall be in accordance with Section R403.5.1 of the International Energy Conservation Code (or its Equivalent in SBC 601/SBC 602). For other than Group R2, R3 and R4 occupancies that are three stories or less in height above grade plane, the installation of heated water circulation and heat trace systems shall be in accordance with Section C404.6 of the International Energy Conservation Code (or its Equivalent in SBC 601/SBC 602).

607.2.1.1 Pump controls for hot water storage systems.

The controls on pumps that circulate water between a water heater and a storage tank for heated water shall limit operation of the pump from heating cycle startup to not greater than 5 minutes after the end of the cycle.

607.2.1.2 Demand recirculation controls for distribution systems.

A water distribution system having one or more recirculation pumps that pump water from a heated water supply pipe back to the heated water source through a cold water supply pipe shall be a demand recirculation water system. Pumps shall have controls that comply with both of the following:

- 1 The control shall start the pump upon receiving a signal from the action of a user of a fixture or appliance, sensing the presence of a user of a fixture, or sensing the flow of hot or tempered water to a fixture fitting or appliance.
- 2 The control shall limit the temperature of the water entering the cold water piping to 40°C.

607.2.2 Piping for recirculation systems having master thermostatic valves.

Where a thermostatic mixing valve is used in a

system with a hot water recirculating pump, the hot water or tempered water return line shall be routed to the cold water inlet pipe of the water heater and the cold water inlet pipe or the hot water return connection of the thermostatic mixing valve.

607.3 Thermal expansion control.

Where a storage water heater is supplied with cold water that passes through a check valve, pressure reducing valve or backflow preventer, a thermal expansion tank shall be connected to the water heater cold water supply pipe at a point that is downstream of all check valves, pressure reducing valves and backflow preventers. Thermal expansion tanks shall be sized in accordance with the tank manufacturer's instructions and shall be sized such that the pressure in the water distribution system shall not exceed that required by Section 604.8.

607.4 Flow of hot water to fixtures.

Fixture fittings, faucets and diverters shall be installed and adjusted so that the flow of hot water from the fittings corresponds to the left-hand side of the fixture fitting.

Exception: Shower and tub/shower mixing valves conforming to ASSE 1016/ASME A112.1016/CSA B125.16 or ASME A112.18.1/CSA B125.1, where the flow of hot water corresponds to the markings on the device.

607.5 Insulation of piping.

For other than Group R2, R3 and R4 occupancies that are three stories or less in height above grade plane, piping to the inlet of a water heater and piping conveying water heated by a water heater shall be insulated in accordance with Section C404.4 of the International Energy Conservation Code (or its equivalent in SBC 601/SBC 602). For Group R2, R3 and R4 occupancies that are three stories or less in height above grade plane, piping to the inlet of a water heater and piping conveying water heated by a water heater shall be insulated in accordance with Section R403.5.3 of the International Energy Conservation Code (or its Equivalent in SBC 601/SBC 602).

SECTION 608 PROTECTION OF POTABLE WATER SUPPLY

608.1 General.

A potable water supply system shall be designed, installed and maintained in such a manner so as to prevent contamination from non-potable liquids, solids or gases being introduced into the potable water supply through cross connections or any other piping connections to the system. Backflow preventer applications shall conform to Table 608.1, except as specifically stated in Sections 608.2 through 608.16.10.

608.2 Plumbing fixtures.

The supply lines and fittings for plumbing fixtures shall be installed so as to prevent backflow. Plumbing fixture fittings shall provide backflow protection in accordance with ASME A112.18.1/CSA B125.1.

608.3 Devices, appurtenances, appliances and apparatus.

Devices, appurtenances, appliances and apparatus intended to serve some special function, such as sterilization, distillation, processing, cooling, or storage of ice or foods, and that connect to the water supply system, shall be provided with protection against backflow and contamination of the water supply system. Water pumps, filters, softeners, tanks and other appliances and devices that handle or treat potable water shall be protected against contamination.

608.3.1 Special equipment, water supply protection.

The water supply for hospital fixtures shall be protected against backflow with a reduced pressure principle backflow prevention assembly, an atmospheric or spill-resistant vacuum breaker assembly, or an air gap. Vacuum breakers for bedpan washer hoses shall not be located less than 1500 mm above the floor. Vacuum breakers for hose connections in health care or laboratory areas shall not be less than 1800 mm above the floor.

608.4 Water service piping.

Water service piping shall be protected in accordance with Sections 603.2 and 603.2.1.

608.5 Chemicals and other substances.

Chemicals and other substances that produce either toxic conditions, taste, odor or

discoloration in a potable water system shall not be introduced into, or utilized in, such systems.



TABLE 608.1 APPLICATION OF BACKFLOW PREVENTERS

DEVICE	DEGREE OF HAZARD ^a	APPLICATION ^b	APPLICABLE STANDARDS
Backflow prevention assemblies:			
Double check backflow prevention assembly and double check fire protection backflow prevention assembly	Low hazard	Backpressure or backsiphonage Sizes 10 mm - 400 mm	ASSE 1015, AWWA C510, CSA B64.5, CSA B64.5.1
Double check detector fire protection backflow prevention assemblies	Low hazard	Backpressure or backsiphonage Sizes 50 mm - 400 mm	ASSE 1048
Pressure vacuum breaker assembly	High or low hazard	Backsiphonage only Sizes 13 – 25 mm	ASSE 1020, CSA B64.1.2
Reduced pressure principle backflow prevention assembly and reduced pressure principle fire protection backflow assembly	High or low hazard	Backpressure or backsiphonage Sizes 10 mm - 400 mm	ASSE 1013, AWWA C511, CSA B64.4, CSA B64.4.1
Reduced pressure detector fire protection backflow prevention assemblies	High or low hazard	Backsiphonage or backpressure (Fire sprinkler systems)	ASSE 1047
Spill-resistant vacuum breaker assembly	High or low hazard	Backsiphonage only Sizes 6 mm - 50 mm	ASSE 1056
Backflow preventer plumbing devices:			
Antisiphon-type fill valves for gravity water closet flush tanks	High hazard	Backsiphonage only	ASSE 1002, CSA B125.3
Backflow preventer for carbonated beverage machines	Low hazard	Backpressure or backsiphonage Sizes 6 mm - 10 mm	ASSE 1022
Backflow preventer with intermediate atmospheric vents	Low hazard	Backpressure or backsiphonage Sizes 6 mm - 20 mm	ASSE 1012, CSA B64.3
Dual-check-valve-type backflow preventer	Low hazard	Backpressure or backsiphonage Sizes 6 mm – 25 mm	ASSE 1024, CSA B64.6
Hose connection backflow preventer	High or low hazard	Low head backpressure, rated working pressure, backpressure or backsiphonage Sizes 13 mm - 25 mm	ASME A112.21.3, ASSE 1052, CSA B64.2.1.1
Hose connection vacuum breaker	High or low hazard	Low head backpressure or backsiphonage Sizes 13 mm, 20 mm, 25 mm	ASME A112.21.3, ASSE 1011, CSA B64.2, CSA B64.2.1
Laboratory faucet backflow preventer	High or low hazard	Low head backpressure and backsiphonage	ASSE 1035, CSA B64.7
Pipe-applied atmospheric-type vacuum breaker	High or low hazard	Backsiphonage only Sizes 6 mm - 100 mm	ASSE 1001, CSA B64.1.1
Vacuum breaker wall hydrants, frost-resistant, automatic-draining-type	High or low hazard	Low head backpressure or backsiphonage Sizes 20 mm - 25 mm	ASME A112.21.3, ASSE 1019, CSA B64.2.2
Other means or methods:			
Air gap	High or low hazard	Backsiphonage or backpressure	ASME A112.1.2
Air gap fittings for use with plumbing fixtures, appliances and appurtenances	High or low hazard	Backsiphonage or backpressure	ASME A112.1.3
Barometric loop	High or low hazard	Backsiphonage only	(See Section 608.13.4)

a. Low hazard—See Pollution (Section 202), High hazard—See Contamination (Section 202).

b. See Backpressure, low head (Section 202).

608.6 Water service piping.

Water service piping shall be protected in accordance with Sections 603.2 and 603.2.1.

608.7 Chemicals and other substances.

Chemicals and other substances that produce either toxic conditions, taste, odor or discoloration in a potable water system shall not be introduced into, or utilized in, such systems.

608.8 Cross connection control.

Cross connections shall be prohibited, except where approved backflow prevention assemblies, backflow prevention devices or other means or methods are installed to protect the potable water supply.

608.8.1 Private water supplies.

Cross connections between a private water supply and a potable public supply shall be prohibited.

608.9 Valves and outlets prohibited below grade.

Potable water outlets and combination stop-and-waste valves shall not be installed underground or below grade. Freeze proof yard hydrants that drain the riser into the ground are considered to be stop-and-waste valves.

Exception: Freeze proof yard hydrants that drain the riser into the ground shall be permitted to be installed, provided that the potable water supply to such hydrants is protected upstream of the hydrants in accordance with Section 608 and the hydrants are permanently identified as non-potable outlets by approved signage that reads as follows: "Caution, Non-potable Water. Do Not Drink."

608.10 Identification of non-potable water systems.

Where non-potable water systems are installed, the piping conveying the non-potable water shall be identified either by color marking, metal tags or tape in accordance with Sections 608.8.1 through 608.8.2.3.

608.10.1 Signage required.

Non-potable water outlets, such as hose connections, open ended pipes and faucets, shall be identified with signage that reads as follows: "Non-potable water is utilized for [application name]. CAUTION: NON-POTABLE WATER – DO NOT DRINK." The words shall be legibly and indelibly printed on a tag or sign constructed of corrosion-resistant waterproof material or shall be indelibly printed on the fixture. The letters of the words shall be not less than 13 mm in height and in colors in contrast to the background on which they are applied. In addition to the required wordage, the pictograph shown in Figure 608.8.1 shall appear on the required signage.

608.10.2 Distribution pipe labeling and marking.

Non-potable distribution piping shall be purple in color and shall be embossed, or integrally stamped or marked, with the words: "CAUTION: NON-POTABLE WATER – DO NOT DRINK" or the piping shall be installed with a purple identification tape or wrap. Pipe identification shall include the contents of the piping system and an arrow indicating the direction of flow. Hazardous piping systems shall also contain information addressing the nature of the hazard. Pipe identification shall be repeated at intervals not exceeding 7500 mm and at each point where the piping passes through a wall, floor or roof. Lettering shall be readily observable within the room or space where the piping is located.

608.10.2.1 Color.

The color of the pipe identification shall be discernable and consistent throughout the building. The color purple shall be used to identify reclaimed, rain and gray water distribution systems.

608.10.2.2 Lettering size.

The size of the background color field and lettering shall comply with Table 608.8.2.2.

TABLE 608.8.2.2 Size of Pipe Identification

PIPE DIAMETER (mm)	LENGTH BACKGROUND COLOR FIELD (mm)	SIZE OF LETTERS (mm)
20 to 32	200	13
40 to 50	200	20
65 to 150	300	30
200 to 250	600	64
over 250	800	89

608.10.2.3 Identification tape.

Where used, identification tape shall be at least 75 mm wide and have white or black lettering on a purple field stating "CAUTION: NON-POTABLE WATER – DO NOT DRINK." Identification tape shall be installed on top of non-potable rainwater distribution pipes, fastened at least every 3000 mm to each pipe length and run continuously the entire length of the pipe.

608.11 Reutilization prohibited.

Water utilized for the cooling of equipment or other processes shall not be returned to the potable water system. Such water shall be discharged into a drainage system through an air gap or shall be utilized for non-potable purposes.

608.12 Reuse of piping.

Piping that has been utilized for any purpose other than conveying potable water shall not be utilized for conveying potable water.

608.13 Painting of water tanks.

The interior surface of a potable water tank shall not be lined, painted or repaired with any material that changes the taste, odor, color or potability of the water supply when the tank is placed in, or returned to, service.

608.14 Pumps and other appliances.

Water pumps, filters, softeners, tanks and other devices that handle or treat potable water shall be protected against contamination.

608.15 Backflow protection.

Means of protection against backflow shall be provided in accordance with Sections 608.13.1 through 608.13.10.

608.15.1 Air gap.

The minimum required air gap shall be measured vertically from the lowest end of a potable water outlet to the flood level rim of the fixture or receptacle into which such potable water outlet discharges. Air gaps shall comply with ASME A112.1.2 and air gap fittings shall comply with ASME A112.1.3.

608.15.2 Reduced pressure principle backflow prevention assemblies.

Reduced pressure principle backflow prevention assemblies shall conform to ASSE 1013, AWWA C511, CSA B64.4 or CSA B64.4.1. Reduced pressure detector assembly backflow preventers shall conform to ASSE 1047. These devices shall be permitted to be installed where subject to continuous pressure conditions. The relief opening shall discharge by air gap and shall be prevented from being submerged.

608.15.3 Backflow preventer with intermediate atmospheric vent.

Backflow preventers with intermediate atmospheric vents shall conform to ASSE 1012 or CSA B64.3. These devices shall be permitted to be installed where subject to continuous pressure conditions. The relief opening shall discharge by air gap and shall be prevented from being submerged.

608.15.4 Barometric loop.

Barometric loops shall precede the point of connection and shall extend vertically to a height of 10,500 mm. A barometric loop shall only be utilized as an atmospheric-type or pressure-type vacuum breaker.

608.15.5 Pressure vacuum breaker assemblies.

Pressure vacuum breaker assemblies shall conform to ASSE 1020 or CSA B64.1.2. Spill-

resistant vacuum breaker assemblies shall comply with ASSE 1056. These assemblies are designed for installation under continuous pressure conditions where the critical level is installed at the required height. Pressure vacuum breaker assemblies shall not be installed in locations where spillage could cause damage to the structure.

608.15.6 Atmospheric-type vacuum breakers.

Pipe applied atmospheric-type vacuum breakers shall conform to ASSE 1001 or CSA B64.1.1. Hose-connection vacuum breakers shall conform to ASME A112.21.3, ASSE 1011, ASSE 1019, ASSE 1035, ASSE 1052, CSA B64.2, CSA B64.2.1, CSA B64.2.1.1, CSA B64.2.2 or CSA B64.7. These devices shall operate under normal atmospheric pressure when the critical level is installed at the required height.

608.15.7 Double check backflow prevention assemblies.

Double check backflow prevention assemblies shall conform to ASSE 1015, CSA B64.5, CSA B64.5.1 or AWWA C510. Double check detector fire protection backflow prevention assemblies shall conform to ASSE 1048. These assemblies shall be capable of operating under continuous pressure conditions.

608.15.8 Spill-resistant pressure vacuum breaker assemblies.

Spill-resistant pressure vacuum breaker assemblies shall conform to ASSE 1056 or CSA B64.1.3. These assemblies are designed for installation under continuous-pressure conditions where the critical level is installed at the required height.

608.15.9 Chemical dispenser backflow devices.

Backflow devices for chemical dispensers shall

comply with ASSE 1055 or shall be equipped with an air gap fitting.

608.15.10 Dual check backflow preventer.

Dual check backflow preventers shall conform to ASSE 1024 or CSA B64.6.

608.16 Location of backflow preventers.

Access shall be provided to backflow preventers as specified by the manufacturer's instructions.

608.16.1 Outdoor enclosures for backflow prevention devices.

Outdoor enclosures for backflow prevention devices shall comply with ASSE 1060.

608.16.2 Protection of backflow preventers.

Backflow preventers shall not be located in areas subject to freezing except where they can be removed by means of unions or are protected from freezing by heat, insulation or both.

608.16.2.1 Relief port piping.

The termination of the piping from the relief port or air gap fitting of a backflow preventer shall discharge to an approved indirect waste receptor or to the outdoors where it will not cause damage or create a nuisance.

608.17 Protection of potable water outlets.

All potable water openings and outlets shall be protected against backflow in accordance with Section 608.15.1, 608.15.2, 608.15.3, 608.15.4, 608.15.4.1 or 608.15.4.2.

608.17.1 Protection by air gap.

Openings and outlets shall be protected by an air gap between the opening and the fixture flood level rim as specified in Table 608.15.1. Openings and outlets equipped for hose connection shall be protected by means other than an air gap.

TABLE 608.15.1 Minimum Required Air Gaps

FIXTURE	MINIMUM AIR GAP	
	Away from a wall ^a (mm)	Close to a wall (mm)
Lavatories and other fixtures with effective openings not greater than 12 mm in diameter	25	38
Sinks, laundry trays, gooseneck back faucets and other fixtures with effective openings not greater than 20 mm in diameter	38	65
Over-rim bath fillers and other fixtures with effective openings not greater than 25 mm in diameter	50	75
Drinking water fountains, single orifice not greater than 12 mm in diameter or multiple orifices with a total area of 100 mm ² (area of circle 12 mm in diameter)	25	38
Effective openings greater than 25 mm	Two times the diameter of the effective opening	Three times the diameter of the effective opening

- a. Applicable where walls or obstructions are spaced from the nearest inside-edge of the spout opening a distance greater than three times the diameter of the effective opening for a single wall, or a distance greater than four times the diameter of the effective opening for two intersecting walls.

608.17.2 Protection by reduced pressure principle backflow prevention assembly.

Openings and outlets shall be protected by a reduced pressure principle backflow prevention assembly or a reduced pressure principle fire protection backflow prevention assembly on potable water supplies.

608.17.3 Protection by a backflow preventer with intermediate atmospheric vent.

Openings and outlets shall be protected by a backflow preventer with an intermediate atmospheric vent.

608.17.4 Protection by a vacuum breaker.

Openings and outlets shall be protected by atmospheric-type or pressure type vacuum breakers. The critical level of the vacuum breaker shall be set not less than 150 mm above the flood level rim of the fixture or device. Fill valves shall be set in accordance with Section 425.3.1. Vacuum breakers shall not be installed under exhaust hoods or similar locations that will contain toxic fumes or vapors. Pipe applied vacuum breakers shall be installed not less than 150 mm above the flood level rim of the fixture,

receptor or device served.

608.17.4.1 Deck-mounted and integral vacuum breakers.

Approved deck-mounted or equipment mounted vacuum breakers and faucets with integral atmospheric vacuum breakers or spill-resistant vacuum breaker assemblies shall be installed in accordance with the manufacturer's instructions and the requirements for labeling with the critical level not less than 25 mm above the flood level rim.

608.17.4.2 Hose connections.

Sillcocks, hose bibbs, wall hydrants and other openings with a hose connection shall be protected by an atmospheric-type or pressure-type vacuum breaker or a permanently attached hose connection vacuum breaker.

Exceptions:

- 1 This section shall not apply to water heater and boiler drain valves that are provided with hose connection threads and that are intended only for tank or vessel draining.
- 2 This section shall not apply to water supply valves intended for connection

of clothes washing machines where backflow prevention is otherwise provided or is integral with the machine.

608.18 Connections to the potable water system.

Connections to the potable water system shall conform to Sections 608.16.1 through 608.16.10.

608.18.1 Beverage dispensers.

The water supply connection to beverage dispensers shall be protected against backflow by a backflow preventer conforming to ASSE 1022 or by an air gap. The portion of the backflow preventer device downstream from the second check valve and the piping downstream there from shall not be affected by carbon dioxide gas.

608.18.2 Connections to boilers.

The potable supply to the boiler shall be equipped with a backflow preventer with an intermediate atmospheric vent complying with ASSE 1012 or CSA B64.3. Where conditioning chemicals are introduced into the system, the potable water connection shall be protected by an air gap or a reduced pressure principle backflow preventer, complying with ASSE 1013, CSA B64.4 or AWWA C511.

608.18.3 Heat exchangers.

Heat exchangers utilizing an essentially toxic transfer fluid shall be separated from the potable water by double-wall construction. An air gap open to the atmosphere shall be provided between the two walls. Heat exchangers utilizing an essentially nontoxic transfer fluid shall be permitted to be of single-wall construction.

608.18.4 Connections to automatic fire sprinkler systems and standpipe systems.

The potable water supply to automatic fire sprinkler and standpipe systems shall be protected against backflow by a double check backflow prevention assembly, a double check fire protection backflow prevention assembly or a reduced pressure principle fire protection backflow prevention assembly.

Exceptions:

- 1 Where systems are installed as a portion of the water distribution system in accordance with the requirements of this code and are not provided with a fire department connection, isolation of the water supply system shall not be required.
- 2 Isolation of the water distribution system is not required for deluge, preaction or dry pipe systems.

608.18.4.1 Additives or non-potable source.

Where systems under continuous pressure contain chemical additives or antifreeze, or where systems are connected to a non-potable secondary water supply, the potable water supply shall be protected against backflow by a reduced pressure principle backflow prevention assembly or a reduced pressure principle fire protection backflow prevention assembly. Where chemical additives or antifreeze are added to only a portion of an automatic fire sprinkler or standpipe system, the reduced pressure principle backflow prevention assembly or the reduced pressure principle fire protection backflow prevention assembly shall be permitted to be located so as to isolate that portion of the system. Where systems are not under continuous pressure, the potable water supply shall be protected against backflow by an air gap or an atmospheric vacuum breaker conforming to ASSE 1001 or CSA B64.1.1.

608.18.5 Connections to lawn irrigation systems.

The potable water supply to lawn irrigation systems shall be protected against backflow by an atmospheric vacuum breaker, a pressure vacuum breaker assembly or a reduced pressure principle backflow prevention assembly. Valves shall not be installed downstream from an atmospheric vacuum breaker. Where chemicals are introduced into the system, the potable water supply shall be protected against backflow by a reduced pressure principle backflow prevention assembly.

608.18.6 Connections subject to backpressure.

Where a potable water connection is made to a non-potable line, fixture, tank, vat, pump or other equipment subject to high hazard backpressure, the potable water connection shall be protected by a reduced pressure

principle backflow prevention assembly.

608.18.7 Chemical dispensers.

Where chemical dispensers connect to the potable water distribution system, the water supply system shall be protected against backflow in accordance with Section 608.13.1, 608.13.2, 608.13.5, 608.13.6, 608.13.8 or 608.13.9.

608.18.8 Portable cleaning equipment.

Where the portable cleaning equipment connects to the water distribution system, the water supply system shall be protected against backflow in accordance with Section 608.13.1, 608.13.2, 608.13.3, 608.13.7 or 608.13.8.

608.18.9 Dental pump equipment.

Where dental pumping equipment connects to the water distribution system, the water supply system shall be protected against backflow in accordance with Section 608.13.1, 608.13.2, 608.13.5, 608.13.6 or 608.13.8.

608.18.10 Coffee machines and noncarbonated beverage dispensers.

The water supply connection to coffee

machines and noncarbonated beverage dispensers shall be protected against backflow by a backflow preventer conforming to ASSE 1022 or by an air gap.

608.19 Protection of individual water supplies.

An individual water supply shall be located and constructed so as to be safeguarded against contamination in accordance with Sections 608.17.1 through 608.17.8.

608.19.1 Well locations.

A potable ground water source or pump suction line shall not be located closer to potential sources of contamination than the distances shown in Table 608.17.1. In the event the underlying rock structure is limestone or fragmented shale, the local or state health department shall be consulted on well site location. The distances in Table 608.17.1 constitute minimum separation and shall be increased in areas of creviced rock or limestone, or where the direction of movement of the ground water is from sources of contamination toward the well.

TABLE 608.17.1 Distance from Contamination to Private Water Supplies and Pump Suction Lines

SOURCE OF CONTAMINATION	DISTANCE (m)
Barnyard	30
Farm silo	8
Pasture	30
Pump house floor drain of cast iron draining to ground surface	0.6
Seepage pits	15
Septic tank	8
Sewer	3
Subsurface disposal fields	15
Subsurface pits	15

608.19.2 Elevation.

Well sites shall be positively drained and shall be at higher elevations than potential sources of contamination.

608.19.3 Depth.

Private potable well supplies shall not be developed from a water table less than 3000 mm below the ground surface.

608.19.4 Water-tight casings.

Each well shall be provided with a water-tight casing extending to not less than 3000 mm below the ground surface. Casings shall extend not less than 150 mm above the well platform. Casings shall be large enough to permit installation of a separate drop pipe. Casings shall be sealed at the bottom in an impermeable stratum or extend several meters into the water-bearing stratum.

608.19.5 Drilled or driven well casings.

Drilled or driven well casings shall be of steel or other approved material. Where drilled wells extend into a rock formation, the well casing shall extend to and set firmly in the formation. The annular space between the earth and the outside of the casing shall be filled with cement grout to a depth of not less than 3000 mm below the ground surface. In an instance of casing to rock installation, the grout shall extend to the rock surface.

608.19.6 Dug or bored well casings.

Dug or bored well casings shall be of water-tight concrete, tile or galvanized or corrugated metal pipe extending to not less than 3000 mm below the ground surface. Where the water table is more than 3000 mm below the ground surface, the water-tight casing shall extend below the table surface. Well casings for dug wells or bored wells constructed with sections of concrete, tile or galvanized or corrugated metal pipe shall be surrounded by 150 mm of grout poured into the hole between the outside of the casing and the ground and extending not less than 3000 mm below the ground surface.

608.19.7 Cover.

Potable water wells shall be equipped with an overlapping water-tight cover at the top of the well casing or pipe sleeve such that contaminated water or other substances are prevented from entering the well through the annular opening at the top of the well casing, wall or pipe sleeve. Covers shall extend downward not less than 50 mm over the outside of the well casing or wall. A dug well cover shall be provided with a pipe sleeve permitting the withdrawal of the pump suction pipe, cylinder or jet body without disturbing the cover. Where pump sections or discharge pipes enter or leave a well through the side of the casing, the circle of contact shall be water tight.

608.19.8 Drainage.

Potable water wells and springs shall be constructed such that surface drainage will be diverted away from the well or spring.

SECTION 609 HEALTH CARE PLUMBING

609.1 Scope.

This section shall govern those aspects of health care plumbing systems that differ from

plumbing systems in other structures. Health care plumbing systems shall conform to the requirements of this section in addition to the other requirements of this code. The provisions of this section shall apply to the special devices and equipment installed and maintained in the following occupancies: nursing homes, homes for the aged, orphanages, infirmaries, first aid stations, psychiatric facilities, clinics, professional offices of dentists and doctors, mortuaries, educational facilities, surgery, dentistry, research and testing laboratories, establishments manufacturing pharmaceutical drugs and medicines and other structures with similar apparatus and equipment classified as plumbing.

609.2 Water service.

Hospitals shall have two water service pipes installed in such a manner so as to minimize the potential for an interruption of the supply of water in the event of a water main or water service pipe failure.

609.3 Hot water.

Hot water shall be provided to supply all of the hospital fixture, kitchen and laundry requirements. Special fixtures and equipment shall have hot water supplied at a temperature specified by the manufacturer. The hot water system shall be installed in accordance with Section 607.

609.4 Vacuum breaker installation.

Vacuum breakers shall be installed not less than 150 mm above the flood level rim of the fixture or device in accordance with Section 608. The flood level rim of hose connections shall be the maximum height at which any hose is utilized.

609.5 Prohibited water closet and clinical sink supply.

Jet-or water-supplied orifices, except those supplied by the flush connections, shall not be located in or connected with a water closet bowl or clinical sink. This section shall not prohibit an approved bidet installation.

609.6 Clinical, hydrotherapeutic and radiological equipment.

Clinical, hydrotherapeutic, radiological or any equipment that is supplied with water or that discharges to the waste system shall conform to the requirements of this section and Section 608.

609.7 Condensate drain trap seal.

A water supply shall be provided for cleaning, flushing and resealing the condensate trap, and the trap shall discharge through an air gap in accordance with Section 608.

609.8 Valve leakage diverter.

Each water sterilizer filled with water through directly connected piping shall be equipped with an approved leakage diverter or bleed line on the water supply control valve to indicate and conduct any leakage of unsterile water away from the sterile zone.

SECTION 610 DISINFECTION OF POTABLE WATER SYSTEM

610.1 General.

New potable water systems shall be purged of deleterious matter and disinfected prior to utilization. The method to be followed shall be that prescribed by the health authority or water purveyor having jurisdiction or, in the absence of a prescribed method, the procedure described in either AWWA C651 or AWWA C652, or as described in this section. This requirement shall apply to “on-site” or “inplant” fabrication of a system or to a modular portion of a system.

- 1 The pipe system shall be flushed with clean, potable water until dirty water does not appear at the points of outlet.
- 2 The system or part thereof shall be filled with a water/chlorine solution containing not less than 50 parts per million (50 mg/L) of chlorine, and the system or part thereof shall be valved off and allowed to stand for 24 hours; or the system or part thereof shall be filled with a water/chlorine solution containing not less than 200 parts per million (200 mg/L) of chlorine and allowed to stand for 3 hours.
- 3 Following the required standing time, the system shall be flushed with clean potable water until the chlorine is purged from the system.

- 4 The procedure shall be repeated where shown by a bacteriological examination that contamination remains present in the system.

SECTION 611 DRINKING WATER TREATMENT UNITS

611.1 Design.

Drinking water treatment units shall meet the requirements of NSF 42, NSF 44, NSF 53, NSF 62 or CSA B483.1.

611.2 Reverse osmosis systems.

The discharge from a reverse osmosis drinking water treatment unit shall enter the drainage system through an air gap or an air gap device that meets the requirements of NSF 58 or CSA B483.1.

611.3 Connection tubing.

The tubing to and from drinking water treatment units shall be of a size and material as recommended by the manufacturer. The tubing shall comply with NSF 14, NSF 42, NSF 44, NSF 53, NSF 58 or NSF 61.

SECTION 612 SOLAR SYSTEMS

612.1 Solar systems.

The construction, installation, alterations and repair of systems, equipment and appliances intended to utilize solar energy for space heating or cooling, domestic hot water heating, swimming pool heating or process heating shall be in accordance with the SBC 501.

SECTION 613 TEMPERATURE CONTROL DEVICES AND VALVES

613.1 Temperature-actuated mixing valves.

Temperature actuated mixing valves, which are installed to reduce water temperatures to defined limits, shall comply with ASSE 1017. Such valves shall be installed at the hot water source.

Figures for Chapter 6



Figure 608.8.1
“PICTOGRAPH—DO NOT DRINK”



CHAPTER 7 SANITARY DRAINAGE

SECTION 701 GENERAL

701.1 Scope.

The provisions of this chapter shall govern the materials, design, construction and installation of sanitary drainage systems.

701.2 Sewer required.

Buildings in which plumbing fixtures are installed and premises having drainage piping shall be connected to a public sewer, where available, or an approved private sewage disposal system in accordance with SBC 702.

701.3 Separate sewer connection.

A building having plumbing fixtures installed and intended for human habitation, occupancy or use on premises abutting on a street, alley or easement in which there is a public sewer shall have a separate connection with the sewer. Where located on the same lot, multiple buildings shall not be prohibited from connecting to a common building sewer that connects to the public sewer.

701.4 Sewage treatment.

Sewage or other waste from a plumbing system that is deleterious to surface or subsurface waters shall not be discharged into the ground or into any waterway unless it has first been rendered innocuous through subjection to an approved form of treatment.

701.5 Damage to drainage system or public sewer.

Waste detrimental to the public sewer system or to the functioning of the sewage-treatment plant shall be treated and disposed of in accordance with Section 1003 as directed by the code official.

701.6 Tests.

The sanitary drainage system shall be tested in accordance with Section 312.

701.7 Engineered systems.

Engineered sanitary drainage systems shall conform to the provisions of Sections 316 and 714.

701.8 Drainage piping in food service areas.

Exposed soil or waste piping shall not be installed above any working, storage or eating surfaces in food service establishments.

SECTION 702 MATERIALS

702.1 Above-ground sanitary drainage and vent pipe.

Above-ground soil, waste and vent pipe shall conform to one of the standards listed in Table 702.1.

702.2 Underground building sanitary drainage and vent pipe.

Underground building sanitary drainage and vent pipe shall conform to one of the standards listed in Table 702.2.

702.3 Building sewer pipe.

Building sewer pipe shall conform to one of the standards listed in Table 702.3.

702.4 Fittings.

Pipe fittings shall be approved for installation with the piping material installed and shall comply with the applicable standards listed in Table 702.4.

702.5 Temperature rating.

Where the waste water temperature will be greater than 60°C, the sanitary drainage piping material shall be rated for the highest temperature of the waste water.

702.6 Chemical waste system.

A chemical waste system shall be completely separated from the sanitary drainage system. The chemical waste shall be treated in accordance with Section 803.2 before discharging to the sanitary drainage system. Separate drainage systems for chemical wastes and vent pipes shall be of an approved material that is resistant to corrosion and degradation for the concentrations of chemicals involved.

702.7 Lead bends and traps.

The wall thickness of lead bends and traps shall be not less than 3.2 mm.

TABLE 702.1 Above-Ground Drainage and Vent Pipe

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200) and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2661; ASTM F 628; ASTM F 1488; CSA B181.1
Brass pipe	ASTM B 43
Cast-iron pipe	ASTM A 74; ASTM A 888; CISPI 301
Copper or copper-alloy pipe	ASTM B 42; ASTM B 302
Copper or copper-alloy tubing (Type K, L, M or DWV)	ASTM B 75; ASTM B 88; ASTM B 251; ASTM B 306
Galvanized steel pipe	ASTM A 53
Glass pipe	ASTM C 1053
Polyolefin pipe	ASTM F 1412; CSA B181.3
Polyvinyl chloride (PVC) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200), and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2665; ASTM F 891; ASTM F 1488; CSA B181.2
Polyvinyl chloride (PVC) plastic pipe with a 88-mm O.D. and a solid, cellular core or composite wall	ASTM D 2949, ASTM F 1488
Polyvinylidene fluoride (PVDF) plastic pipe	ASTM F 1673; CSA B181.3
Stainless steel drainage systems, Types 304 and 316L	ASME A112.3.1

TABLE 702.2 Underground Building Drainage and Vent Pipe

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200) and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2661; ASTM F 628; ASTM F 1488; CSA B181.1
Cast-iron pipe	ASTM A 74; ASTM A 888; CISPI 301
Copper or copper-alloy tubing (Type K, L, M or DWV)	ASTM B 75; ASTM B 88; ASTM B 251; ASTM B 306
Polyolefin pipe	ASTM F 1412; CSA B181.3
Polyvinyl chloride (PVC) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200) and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2665; ASTM F 891; ASTM F 1488; CSA B181.2
Polyvinyl chloride (PVC) plastic pipe with a 88-mm O.D. and a solid, cellular core or composite wall	ASTM D 2949, ASTM F 1488
Polyvinylidene fluoride (PVDF) plastic pipe	ASTM F 1673; CSA B181.3
Stainless steel drainage systems, Type 316L	ASME A 112.3.1

TABLE 702.3 Building Sewer Pipe

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200) and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2661; ASTM F 628; ASTM F 1488; CSA B181.1
Acrylonitrile butadiene styrene (ABS) plastic pipe in sewer and drain diameters, including SDR 42 (PS 20), PS 35, SDR 35 (PS 45), PS 50, PS 100, PS 140, SDR 23.5 (PS 150) and PS 200; with a solid, cellular core or composite wall	ASTM F 1488; ASTM D 2751
Cast-iron pipe	ASTM A 74; ASTM A 888; CISPI 301
Concrete pipe	ASTM C 14; ASTM C 76; CSA A257.1M; CSA A257.2M
Copper or copper-alloy tubing (Type K or L)	ASTM B 75; ASTM B 88; ASTM B 251
Polyethylene (PE) plastic pipe (SDR-PR)	ASTM F 714
Polyvinyl chloride (PVC) plastic pipe in IPS diameters, including Schedule 40, DR 22 (PS 200) and DR 24 (PS 140); with a solid, cellular core or composite wall	ASTM D 2665; ASTM F 891; ASTM F 1488
Polyvinyl chloride (PVC) plastic pipe in sewer and drain diameters, including PS 25, SDR 41 (PS 28), PS 35, SDR 35 (PS 46), PS 50, PS 100, SDR 26 (PS 115), PS 140 and PS 200; with a solid, cellular core or composite wall	ASTM F 891; ASTM F 1488; ASTM D 3034; CSA B182.2; CSA B182.4
Polyvinyl chloride (PVC) plastic pipe with a 88-mm O.D. and a solid, cellular core or composite wall	ASTM D 2949, ASTM F 1488
Polyvinylidene fluoride (PVDF) plastic pipe	ASTM F 1673; CSA B181.3
Stainless steel drainage systems, Types 304 and 316L	ASME A112.3.1
Vitrified clay pipe	ASTM C 4; ASTM C 700

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TABLE 702.4 Pipe Fittings

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe in IPS diameters	ASTM D 2661; ASTM F 628; CSA B181.1
Acrylonitrile butadienemstyrene (ABS) plastic pipe in sewer and drain diameters	ASTM D 2751
Cast iron	ASME B 16.4; ASME B 16.12; ASTM A 74; ASTM A 888; CISPI 301
Copper or copper alloy	ASME B 16.15; ASME B 16.18; ASME B 16.22; ASME B 16.23; ASME B 16.26; ASME B 16.29
Glass	ASTM C 1053
Gray iron and ductile iron	AWWA C 110/A21.10
Malleable iron	ASME B 16.3
Polyolefin	ASTM F 1412; CSA B181.3
Polyvinyl chloride (PVC) plastic in IPS diameters	ASTM D 2665; ASTM F 1866
Polyvinyl chloride (PVC) plastic pipe in sewer and drain diameters	ASTM D 3034
Polyvinyl chloride (PVC) plastic pipe with a 88-mm O.D.	ASTM D 2949
Polyvinylidene fluoride (PVDF) plastic pipe	ASTM F 1673; CSA B181.3
Stainless steel drainage systems, Types 304 and 316L	ASME A 112.3.1
Steel	ASME B 16.9; ASME B 16.11; ASME B 16.28
Vitrified clay	ASTM C 700

SECTION 703 BUILDING SEWER

703.1 Building sewer pipe near the water service.

The proximity of a sewer to a water service shall comply with Section 603.2.

703.2 Drainage pipe in filled ground.

Where a building sewer or building drain is installed on filled or unstable ground, the drainage pipe shall conform to one of the standards for ABS plastic pipe, cast-iron pipe, copper or copper alloy tubing, or PVC plastic pipe listed in Table 702.3.

703.3 Sanitary and storm sewers.

Where separate systems of sanitary drainage and storm drainage are installed in the same property, the sanitary and storm building sewers or drains shall be permitted to be laid side by side in one trench.

703.4 Existing building sewers and drains.

Existing building sewers and drains shall connect with new building sewer and drainage

systems only where found by examination and test to conform to the new system in quality of material. The code official shall notify the owner to make the changes necessary to conform to this code.

703.5 Cleanouts on building sewers.

Cleanouts on building sewers shall be located as set forth in Section 708.

Improperly located cleanouts may create difficulties in the cleaning of stoppages in the drainage system. Cleanout locations and spacing requirements are stated in Section 708.

703.6 Combined sanitary and storm public sewer.

Where the public sewer is a combined system for both sanitary and storm water, the sanitary sewer shall be connected independently to the public sewer.

SECTION 704 DRAINAGE PIPING INSTALLATION

704.1 Slope of horizontal drainage piping.

Horizontal drainage piping shall be installed in uniform alignment at uniform slopes. The slope of a horizontal drainage pipe shall be not less than that indicated in Table 704.1.

TABLE 704.1 Slope of Horizontal Drainage Pipe

SIZE (mm)	MINIMUM SLOPE (%)
65 or less	2
80 to 150	1
200 or larger	0.5

704.2 Change in size.

The size of the drainage piping shall not be reduced in size in the direction of the flow. A 100 mm by 75 mm water closet connection shall not be considered as a reduction in size.

704.3 Connections to offsets and bases of stacks.

Horizontal branches shall connect to the bases of stacks at a point located not less than 10 times the diameter of the drainage stack downstream from the stack. Horizontal branches shall connect to horizontal stack offsets at a point located not less than 10 times the diameter of the drainage stack downstream from the upper stack.

704.4 Future fixtures.

Drainage piping for future fixtures shall terminate with an approved cap or plug.

SECTION 705 JOINTS

705.1 General.

This section contains provisions applicable to joints specific to sanitary drainage piping.

705.2 ABS plastic.

Joints between ABS plastic pipe or fittings shall comply with Sections 705.2.1 through 705.2.3.

705.2.1 Mechanical joints.

Mechanical joints on drainage pipes shall be made with an elastomeric seal conforming to ASTM C 1173, ASTM D 3212 or CSA B602. Mechanical joints shall be installed only in underground systems unless otherwise approved. Joints shall be installed in accordance with the manufacturer's instructions.

705.2.2 Solvent cementing.

Joint surfaces shall be clean and free from moisture. Solvent cement that conforms to ASTM D 2235 or CSA B181.1 shall be applied to all joint surfaces. The joint shall be made while the cement is wet. Joints shall be made in accordance with ASTM D 2235, ASTM D 2661, ASTM F 628 or CSA B181.1. Solvent cement joints shall be permitted above or below ground.

705.2.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Schedule 80 or heavier pipe shall be permitted to be threaded with dies specifically designed for plastic pipe. Approved thread lubricant or tape shall be applied on the male threads only.

705.3 Brass.

Joints between brass pipe or fittings shall comply with Sections 705.3.1 through 705.3.4.

705.3.1 Brazed joints.

All joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

705.3.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

705.3.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

705.3.4 Welded joints.

All joint surfaces shall be cleaned. The joint shall be welded with an approved filler metal.

705.4 Cast iron.

Joints between cast-iron pipe or fittings shall comply with Sections 705.4.1 through 705.4.3.

705.4.1 Caulked joints.

Joints for hub and spigot pipe shall be firmly packed with oakum or hemp. Molten lead shall be poured in one operation to a depth of not less than 25 mm. The lead shall not recede more than 3.2 mm below the rim of the hub and shall be caulked tight. Paint, varnish or other coatings shall not be permitted on the jointing material until after the joint has been tested and

approved. Lead shall be run in one pouring and shall be caulked tight. Acid-resistant rope and acid proof cement shall be permitted.

705.4.2 Compression gasket joints.

Compression gaskets for hub and spigot pipe and fittings shall conform to ASTM C 564 and shall be tested to ASTM C 1563. Gaskets shall be compressed when the pipe is fully inserted.

705.4.3 Mechanical joint coupling.

Mechanical joint couplings for hubless pipe and fittings shall consist of an elastomeric sealing sleeve and a metallic shield that comply with CISPI 310, ASTM C 1277 or ASTM C 1540. The elastomeric sealing sleeve shall conform to ASTM C 564 or CSA B602 and shall be provided with a center stop. Mechanical joint couplings shall be installed in accordance with the manufacturer's instructions.

705.5 Concrete joints.

Joints between concrete pipe and fittings shall be made with an elastomeric seal conforming to ASTM C 443, ASTM C 1173, CSA A257.3M or CSA B602.

705.6 Copper pipe.

Joints between copper or copper-alloy pipe or fittings shall comply with Sections 705.6.1 through 705.6.5.

705.6.1 Brazed joints.

All joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

705.6.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

705.6.3 Solder joints.

Solder joints shall be made in accordance with the methods of ASTM B 828. Cut tube ends shall be reamed to the full inside diameter of the tube end. All joint surfaces shall be cleaned. A flux conforming to ASTM B 813 shall be applied. The joint shall be soldered with a solder conforming to ASTM B 32.

705.6.4 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the

male threads only.

705.6.5 Welded joints.

All joint surfaces shall be cleaned. The joint shall be welded with an approved filler metal.

705.7 Copper tubing.

Joints between copper or copper-alloy tubing or fittings shall comply with Sections 705.7.1 through 705.7.3.

705.7.1 Brazed joints.

All joint surfaces shall be cleaned. An approved flux shall be applied where required. The joint shall be brazed with a filler metal conforming to AWS A5.8.

705.7.2 Mechanical joints.

Mechanical joints shall be installed in accordance with the manufacturer's instructions.

705.7.3 Solder joints.

Solder joints shall be made in accordance with the methods of ASTM B 828. Cut tube ends shall be reamed to the full inside diameter of the tube end. All joint surfaces shall be cleaned. A flux conforming to ASTM B 813 shall be applied. The joint shall be soldered with a solder conforming to ASTM B 32.

705.8 Borosilicate glass joints.

Glass-to-glass connections shall be made with a bolted compression-type, 300 series stainless steel coupling with contoured acid-resistant elastomeric compression ring and a fluorocarbon polymer inner seal ring; or with caulked joints in accordance with Section 705.8.1.

705.8.1 Caulked joints.

Lead-caulked joints for hub and spigot soil pipe shall be firmly packed with oakum or hemp and filled with molten lead not less than 25 mm in depth and not to recede more than 3.2 mm below the rim of the hub. Paint, varnish or other coatings shall not be permitted on the jointing material until after the joint has been tested and approved. Lead shall be run in one pouring and shall be caulked tight. Acid-resistant rope and acid proof cement shall be permitted.

705.9 Steel.

Joints between galvanized steel pipe or fittings shall comply with Sections 705.9.1 and 705.9.2.

705.9.1 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

705.9.2 Mechanical joints.

Joints shall be made with an approved elastomeric seal. Mechanical joints shall be installed in accordance with the manufacturer's instructions.

705.9.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only.

705.9.4 Mechanical joints.

Joints shall be made with an approved elastomeric seal. Mechanical joints shall be installed in accordance with the manufacturer's instructions.

705.10 Lead.

Joints between lead pipe or fittings shall comply with Sections 705.10.1 and 705.10.2.

705.10.1 Burned.

Burned joints shall be uniformly fused together into one continuous piece. The thickness of the joint shall be at least as thick as the lead being joined. The filler metal shall be of the same material as the pipe.

705.10.2 Wiped.

Joints shall be fully wiped, with an exposed surface on each side of the joint not less than 19 mm. The joint shall be not less than 9.5 mm thick at the thickest point.

705.11 PVC plastic.

Joints between PVC plastic pipe or fittings shall comply with Sections 705.11.1 through 705.11.3.

705.11.1 Mechanical joints.

Mechanical joints on drainage pipe shall be made with an elastomeric seal conforming to ASTM C 1173, ASTM D 3212 or CSA B602. Mechanical joints shall not be installed in above-ground systems, unless otherwise approved. Joints shall be installed in accordance with the manufacturer's instructions.

705.11.2 Solvent cementing.

Joint surfaces shall be clean and free from moisture. A purple primer that conforms to ASTM F 656 shall be applied. Solvent cement not purple in color and conforming to ASTM D 2564, CSA B137.3, CSA B181.2 or CSA B182.1 shall be applied to all joint surfaces. The joint shall be made while the cement is wet and shall be in accordance with ASTM D 2855. Solvent cement joints shall be permitted above or below ground.

Exception:

A primer is not required where both of the following conditions apply:

1. The solvent cement used is third-party certified as conforming to ASTM D 2564.
2. The solvent cement is used only for joining PVC drain, waste and vent pipe and fittings in non-pressure applications in sizes up to and including 100 mm in diameter.

705.11.3 Threaded joints.

Threads shall conform to ASME B1.20.1. Schedule 80 or heavier pipe shall be permitted to be threaded with dies specifically designed for plastic pipe. Approved thread lubricant or tape shall be applied on the male threads only.

705.12 Vitrified clay.

Joints between vitrified clay pipe or fittings shall be made with an elastomeric seal conforming to ASTM C 425, ASTM C 1173 or CSA B602.

705.13 Polyethylene plastic pipe.

Joints between polyethylene plastic pipe and fittings shall be underground and shall comply with Section 705.13.1 or 705.13.2.

705.13.1 Heat-fusion joints.

Joint surfaces shall be clean and free from moisture. All joint surfaces shall be cut, heated to melting temperature and joined using tools specifically designed for the operation. Joints shall be undisturbed until cool. Joints shall be made in accordance with ASTM D 2657 and the manufacturer's instructions.

705.13.2 Mechanical joints.

Mechanical joints in drainage piping shall be made with an elastomeric seal conforming to ASTM C 1173, ASTM D 3212 or CSA B602. Mechanical joints shall be installed in

accordance with the manufacturer's instructions.

705.14 Polyolefin plastic.

Joints between polyolefin plastic pipe and fittings shall comply with Sections 705.14.1 and 705.14.2.

705.14.1 Heat-fusion joints.

Heat-fusion joints for polyolefin pipe and tubing joints shall be installed with socket type heat-fused polyolefin fittings or electrofusion polyolefin fittings. Joint surfaces shall be clean and free from moisture. The joint shall be undisturbed until cool. Joints shall be made in accordance with ASTM F 1412 or CSA B181.3.

705.14.2 Mechanical and compression sleeve joints.

Mechanical and compression sleeve joints shall be installed in accordance with the manufacturer's instructions.

705.15 Polyvinylidene fluoride plastic.

Joints between polyvinylidene plastic pipe and fittings shall comply with Sections 705.15.1 and 705.15.2.

705.15.1 Heat-fusion joints.

Heat-fusion joints for polyvinylidene fluoride pipe and tubing joints shall be installed with socket-type heat-fused polyvinylidene fluoride fittings or electrofusion polyvinylidene fittings and couplings. Joint surfaces shall be clean and free from moisture. The joint shall be undisturbed until cool. Joints shall be made in accordance with ASTM F 1673.

705.15.2 Mechanical and compression sleeve joints.

Mechanical and compression sleeve joints shall be installed in accordance with the manufacturer's instructions.

705.16 Joints between different materials.

Joints between different piping materials shall be made with a mechanical joint of the compression or mechanical-sealing type conforming to ASTM C 1173, ASTM C 1460 or ASTM C 1461. Connectors and adapters shall be approved for the application and such joints shall have an elastomeric seal conforming to ASTM C 425, ASTM C 443, ASTM C 564, ASTM C 1440, ASTM F 477,

CSA A257.3M or CSA B602, or as required in Sections 705.16.1 through 705.16.7. Joints between glass pipe and other types of materials shall be made with adapters having a TFE seal. Joints shall be installed in accordance with the manufacturer's instructions.

705.16.1 Copper or copper-alloy tubing to cast-iron hub pipe.

Joints between copper or copper-alloy tubing and cast-iron hub pipe shall be made with a brass ferrule or compression joint. The copper or copper-alloy tubing shall be soldered to the ferrule in an approved manner, and the ferrule shall be joined to the cast-iron hub by a caulked joint or a mechanical compression joint.

705.16.2 Copper or copper-alloy tubing to galvanized steel pipe.

Joints between copper or copper-alloy tubing and galvanized steel pipe shall be made with a brass converter fitting or dielectric fitting. The copper tubing shall be soldered to the fitting in an approved manner, and the fitting shall be screwed to the threaded pipe.

705.16.3 Cast-iron pipe to galvanized steel or brass pipe.

Joints between cast-iron and galvanized steel or brass pipe shall be made by either caulked or threaded joints or with an approved adapter fitting.

705.16.4 Plastic pipe or tubing to other piping material.

Joints between different types of plastic pipe or between plastic pipe and other piping material shall be made with an approved adapter fitting. Joints between plastic pipe and cast-iron hub pipe shall be made by a caulked joint or a mechanical compression joint.

705.16.5 Lead pipe to other piping material.

Joints between lead pipe and other piping material shall be made by a wiped joint to a caulking ferrule, soldering nipple or bushing or shall be made with an approved adapter fitting.

705.16.6 Borosilicate glass to other materials.

Joints between glass pipe and other types of materials shall be made with adapters having a TFE seal and shall be installed in accordance with the manufacturer's instructions.

705.16.7 Stainless steel drainage systems to other materials.

Joints between stainless steel drainage systems and other piping materials shall be made with approved mechanical couplings.

705.17 Drainage slip joints.

Slip joints shall comply with Section 405.8.

705.18 Caulking ferrules.

Ferrules shall be of red brass and shall be in accordance with Table 705.18.

TABLE 705.18 Caulking Ferrule Specifications

PIPE SIZES (mm)	INSIDE DIAMETER (mm)	LENGTH (mm)	MINIMUM WEIGHT EACH (gram)
50	55	115	450
80	85	115	800
100	115	115	1100

705.19 Soldering bushings.

Soldering bushings shall be of red brass and shall be in accordance with Table 705.19.

TABLE 705.19 Soldering Bushing Specifications

PIPE SIZES (mm)	MINIMUM WEIGHT EACH (gram)
32	170
40	225
50	400
65	625
80	900
100	1600

705.20 Stainless steel drainage systems.

O-ring joints for stainless steel drainage systems shall be made with an approved elastomeric seal.

SECTION 706 CONNECTIONS BETWEEN DRAINAGE PIPING AND FITTINGS

706.1 Connections and changes in direction.

All connections and changes in direction of the sanitary drainage system shall be made with approved drainage fittings. Connections between drainage piping and fixtures shall conform to Section 405.

706.2 Obstructions.

The fittings shall not have ledges, shoulders or reductions capable of retarding or obstructing flow in the piping. Threaded drainage pipe fittings shall be of the recessed drainage type. This section shall not be applicable to tubular waste fittings used to convey vertical flow upstream of the trap seal liquid level of a fixture trap.

706.3 Installation of fittings.

Fittings shall be installed to guide sewage and waste in the direction of flow. Change in direction shall be made by fittings installed in accordance with Table 706.3. Change in direction by combination fittings, side inlets or increasers shall be installed in accordance with Table 706.3 based on the pattern of flow created by the fitting. Double sanitary tee patterns shall not receive the discharge of back-to-back water closets and fixtures or appliances with pumping action discharge.

Exception: Back-to-back water closet connections to double sanitary tees shall be permitted where the horizontal *developed length* between the outlet of the water closet and the connection to the double sanitary tee pattern is 450 mm or greater.

706.4 Heelor side-inlet quarter bends.

Heel-inlet quarter bends shall be an acceptable means of connection, except where the quarter bend serves a water closet. A low-heel inlet shall not be used as a wet-vented connection. Side-inlet quarter bends shall be an acceptable means of connection for drainage, wet venting and *stack* venting arrangements.

TABLE 706.3 Fittings for Change in Direction

TYPE OF FITTING PATTERN	CHANGE IN DIRECTION		
	Horizontal to vertical	Vertical to horizontal	Horizontal to horizontal
Sixteenth bend	X	X	X
Eighth bend	X	X	X
Sixth bend	X	X	X
Quarter bend	X	X ^a	X ^a
Short sweep	X	X ^{a,b}	X ^a
Long sweep	X	X	X
Sanitary tee	X ^c	—	—
Wye	X	X	X
Combination wye and eighth bend	X	X	X

- The fittings shall only be permitted for a 50 mm or smaller fixture drain.
- 80 mm or larger.
- For a limitation on double sanitary tees, see Section 705.3.

SECTION 707 PROHIBITED JOINTS AND CONNECTIONS

707.1 Prohibited joints.

The following types of joints and connections shall be prohibited:

- Cement or concrete joints.
- Mastic or hot-pour bituminous joints.
- Joints made with fittings not approved for the specific installation.
- Joints between different diameter pipes made with elastomeric rolling O-rings.
- Solvent-cement joints between different types of plastic pipe.
- Saddle-type fittings.

SECTION 708 CLEANOUTS

708.1 Cleanouts required.

Cleanouts shall be provided for drainage piping in accordance with Sections 708.1.1 through 708.1.11.

708.1.1 Horizontal drains and building drains.

Horizontal drainage pipes in buildings shall have cleanouts located at intervals of not more than 30 m. Building drains shall have cleanouts located at intervals of not more than 30 m except where manholes are used instead of cleanouts, the manholes shall be located at intervals of not more than 122 m. The interval

length shall be measured from the cleanout or manhole opening, along the developed length of the piping to the next drainage fitting providing access for cleaning, the end of the horizontal drain or the end of the building drain.

Exception: Horizontal *fixture drain* piping serving a non-removable trap shall not be required to have a clean-out for the section of piping between the trap and the vent connection for such trap.

708.1.2 Building sewers.

Building sewers smaller than 200 mm shall have cleanouts located at intervals of not more than 30 m. Building sewers 200 mm and larger shall have a manhole located not more than 60 m from the junction of the building drain and building sewer and at intervals of not more than 120 m. The interval length shall be measured from the cleanout or manhole opening, along the developed length of the piping to the next drainage fitting providing access for cleaning, a manhole or the end of the building sewer.

708.1.3 Building drain and building sewer junction.

The junction of the building drain and the building sewer shall be served by a cleanout that is located at the junction or within 3000 mm of the developed length of piping upstream of the junction. For the requirements of this section, the removal of the water closet shall not be required to provide cleanout access.

708.1.4 Changes of direction.

Where a horizontal drainage pipe, a building drain or a building sewer has a change of horizontal direction greater than 45 degrees, a cleanout shall be installed at the change of direction. Where more than one change of horizontal direction greater than 45 degrees occurs within 12 m of developed length of piping, the cleanout installed for the first change of direction shall serve as the cleanout for all changes in direction within that 12 m of developed length of piping.

708.1.5 Cleanout size.

Cleanouts shall be the same size as the piping served by the cleanout, except that cleanouts for piping larger than 100 mm need not be larger than 100 mm.

Exceptions:

1. A removable P-trap with slip or ground joint connections can serve as a cleanout for drain piping that is one size larger than the P-trap size.
2. Cleanouts located on stacks can be one size smaller than the stack size.
3. The size of cleanouts for cast-iron piping can be in accordance with the referenced standards for cast-iron fittings as indicated in Table 702.4.

708.1.6 Cleanout plugs.

Cleanout plugs shall be of brass, plastic or other approved materials. Cleanout plugs for borosilicate glass piping systems shall be of borosilicate glass. Brass cleanout plugs shall conform to ASTM A 74 and shall be limited for use only on metallic piping systems. Plastic cleanout plugs shall conform to the referenced standards for plastic pipe fittings, as indicated in Table 702.4. Cleanout plugs shall have a raised square head, a countersunk square head or a countersunk slot head. Where a cleanout plug will have a trim cover screw installed into the plug, the plug shall be manufactured with a blind end threaded hole for such purpose.

708.1.7 Manholes.

Manholes and manhole covers shall be of an approved type. Manholes located inside of a building shall have gas-tight covers that require tools for removal.

708.1.8 Installation arrangement.

The installation arrangement of a cleanout shall enable cleaning of drainage piping only in the direction of drainage flow.

Exceptions:

1. Test tees serving as cleanouts.
2. A two-way cleanout installation that is *approved* for meeting the requirements of Section 708.1.3.

708.1.9 Required clearance.

Cleanouts for 150 mm and smaller piping shall be provided with a clearance of not less than 500 mm from, and perpendicular to, the face of the opening to any obstruction. Cleanouts for 200 mm and larger piping shall be provided with a clearance of not less than 900 mm from, and perpendicular to, the face of the opening to any obstruction.

708.1.10 Cleanout access.

Required cleanouts shall not be installed in concealed locations. For the purposes of this section, concealed locations include, but are not limited to, the inside of plenums, within walls, within floor/ceiling assemblies, below grade and in crawl spaces where the height from the crawl space floor to the nearest obstruction along the path from the crawl space opening to the cleanout location is less than 600 mm. Cleanouts with openings at a finished wall shall have the face of the opening located within 38 mm of the finished wall surface. Cleanouts located below grade shall be extended to grade level so that the top of the cleanout plug is at or above grade. A cleanout installed in a floor or walkway that will not have a trim cover installed shall have a countersunk plug installed so the top surface of the plug is flush with the finished surface of the floor or walkway.

708.1.10.1 Cleanout plug trim covers.

Trim covers and access doors for cleanout plugs shall be designed for such purposes and shall be *approved*. Trim cover fasteners that thread into cleanout plugs shall be corrosion resistant. Cleanout plugs shall not be covered with mortar, plaster or any other permanent material.

708.1.10.2 Floor cleanout assemblies.

Where it is necessary to protect a cleanout plug from the loads of vehicular traffic, cleanout assemblies in accordance with ASME A12.36.2M shall be installed.

708.1.11 Prohibited use.

The use of a threaded cleanout opening to add a fixture or to extend piping shall be prohibited except where another cleanout of equal size is installed with the required access and clearance.

SECTION 709 FIXTURE UNITS

709.1 Values for fixtures.

Drainage fixture unit values as given in Table 709.1 designate the relative load weight of different kinds of fixtures that shall be employed in estimating the total load carried by a soil or waste pipe, and shall be used in connection with Tables 710.1(1) and 710.1(2) of sizes for soil, waste and vent pipes for which the permissible load is given in terms of fixture units.

709.2 Fixtures not listed in Table 709.1.

Fixtures not listed in Table 709.1 shall have a drainage fixture unit load based on the outlet size of the fixture in accordance with Table 709.2. The minimum trap size for unlisted fixtures shall be the size of the drainage outlet but not less than 32 mm.

709.3 Values for continuous and semicontinuous flow.

Drainage fixture unit values for continuous and semicontinuous flow into a drainage system shall be computed on the basis that 0.06 L/s of flow is equivalent to two fixture units.

709.4 Values for indirect waste receptor.

The drainage fixture unit load of an indirect waste receptor receiving the discharge of indirectly connected fixtures shall be the sum of the drainage fixture unit values of the fixtures that discharge to the receptor, but not less than the drainage fixture unit value given for the indirect waste receptor in Table 709.1 or 709.2.

709.4.1 Clear-water waste receptors.

Where waste receptors such as floor drains, floor sinks and hub drains receive only clear-water waste from display cases, refrigerated display cases, ice bins, coolers and freezers, such receptors shall have a *drainage fixture unit* value of one half.

SECTION 710 DRAINAGE SYSTEM SIZING

710.1 Maximum fixture unit load.

The maximum number of drainage fixture units connected to a given size of building sewer, building drain or horizontal branch of the building drain shall be determined using Table 710.1(1). The maximum number of drainage fixture units connected to a given size of horizontal branch or vertical soil or waste stack shall be determined using Table 710.1(2).

710.1.1 Horizontal stack offsets.

Horizontal stack offsets shall be sized as required for building drains in accordance with Table 710.1(1), except as required by Section 7011.3.

710.1.2 Vertical stack offsets.

Vertical stack offsets shall be sized as required for straight stacks in accordance with Table 710.1(2), except where required to be sized as a building drain in accordance with Section 711.1.1.

710.2 Future fixtures.

Where provision is made for the future installation of fixtures, those provided for shall be considered in determining the required sizes of drain pipes.

TABLE 709.1 Drainage Fixture Units for Fixtures And Groups

FIXTURE TYPE	DRAINAGE FIXTURE UNIT VALUE AS LOAD FACTORS	MINIMUM SIZE OF TRAP (mm)
Automatic clothes washers, commercial ^{a,g}	3	50
Automatic clothes washers, residential ^g	2	50
Bathroom group as defined in Section 202 (6 Lpf water closet) ^f	5	—
Bathroom group as defined in Section 202 (water closet flushing greater than 6 Lpf) ^f	6	—
Bathtub (with or without overhead shower or whirlpool attachments)	2	40
Bidet	1	40
Combination sink and tray	2	40
Dental lavatory	1	40
Dental unit or cuspidor	1	40
Dishwashing machine ^e , domestic	2	40
Drinking fountain	1/2	40
Emergency floor drain	0	50
Floor drains ^h	2 ^h	50
Floor sinks	Note h	50
Kitchen sink, domestic	2	40
Kitchen sink, domestic with food waste disposer and/or dishwasher	2	40
Laundry tray (1 or 2 compartments)	2	40
Lavatory	1	40
Shower (based on the total flow rate through showerheads and body sprays)		
Flow rate:		
22 Lpm or less	2	40
Greater than 22 Lpm to 46 Lpm	3	50
Greater than 46 Lpm to 98 Lpm	5	76
Greater than 98 Lpm to 210 Lpm	6	102
Service sink	2	40
Sink	2	40
Urinal	4	Note d
Urinal, 3.8 L per flush or less	2 ^e	Note d
Urinal, nonwater supplied	1/2	Note d
Wash sink (circular or multiple) each set of faucets	2	40
Water closet, flushometer tank, public or private	4 ^e	Note d
Water closet, private (6 Lpf)	3 ^e	Note d
Water closet, private (flushing greater than 6 Lpf)	4 ^e	Note d
Water closet, public (6 Lpf)	4 ^e	Note d
Water closet, public (flushing greater than 6 Lpf)	6 ^e	Note d

- For traps larger than 80 mm, use Table 709.2.
- A showerhead over a bathtub or whirlpool bathtub attachment does not increase the *drainage fixture unit* value.
- See Sections 709.2 through 709.4.1 for methods of computing unit value of fixtures not listed in this table or for rating of devices with intermittent flows.
- Trap size shall be consistent with the fixture outlet size.
- For the purpose of computing loads on *building drains* and *sewers*, water closets and urinals shall not be rated at a lower *drainage fixture unit* unless the lower values are confirmed by testing.
- For fixtures added to a *bathroom group*, add the dfu value of those additional fixtures to the *bathroom group* fixture count.
- See Section 406.2 for sizing requirements for *fixture drain*, branch drain and drainage *stack* for an automatic clothes washer standpipe.
- See Sections 709.4 and 709.4.1.

TABLE 709.2 Drainage Fixture Units for Fixture Drains or Traps

FIXTURE DRAIN OR TRAP SIZE (mm)	DRAINAGE FIXTURE UNIT VALUE
40	2
50	3
65	4
80	5
100	6

TABLE 710.1(1) Building Drains and Sewers

DIAMETER OF PIPE (mm)	Maximum Number Of Drainage Fixture Units Connected To Any Portion Of The Building Drain Or The Building Sewer, Including Branches Of The Building Drain ^a			
	Slope mm per meter			
	5 mm	10 mm	20 mm	40 mm
40	—	—	3	3
50	—	—	21	26
65	—	—	24	31
75	—	36	42	50
100	—	180	216	250
125	—	390	480	575
150	—	700	840	1,000
200	1,400	1,600	1,920	2,300
250	2,500	2,900	3,500	4,200
300	3,900	4,600	5,600	6,700
375	7000	8300	10000	12,000

- a. The minimum size of any *building drain* serving a water closet shall be 75 mm.

TABLE 710.1(2) Horizontal Fixture Branches and Stacks^a

DIAMETER OF PIPE (mm)	Maximum number of drainage fixture units (dfu)			
	Total for horizontal branch	Stacks ^b		
		Total discharge into one branch interval	Total for stack of three branch Intervals or less	Total for stack greater than three branch intervals
40	3	2	4	8
50	6	6	10	24
65	12	9	20	42
80	20	20	48	72
100	160	90	240	500
125	360	200	540	1,100
150	620	350	960	1,900
200	1,400	600	2,200	3,600
250	2,500	1,000	3,800	5,600
300	3,900	1,500	6,000	8,400
375	7,000	Note c	Note c	Note c

- a. Does not include branches of the building drain. Refer to Table 710.1(1).
- b. Stacks shall be sized based on the total accumulated connected load at each story or branch interval. As the total accumulated connected load decreases, stacks are permitted to be reduced in size. Stack diameters shall not be reduced to less than one-half of the diameter of the largest stack size required.
- c. Sizing load based on design criteria

SECTION 711 OFFSETS IN DRAINAGE PIPING IN BUILDINGS OF FIVE STORIES OR MORE

711.1 Horizontal branch connections above or below vertical stack offsets.

If a horizontal branch connects to the stack within 600 mm above or below a vertical stack offset, and the offset is located more than four branch intervals below the top of the stack, the offset shall be vented in accordance with Section 907.

711.1.1 Omission of vents for vertical stack offsets.

Vents for vertical offsets required by Section 711.1 shall not be required where the stack and its offset are sized as a building drain [see Table 710.1(1)].

711.2 Horizontal stack offsets.

A stack with a horizontal offset located more than four branch intervals below the top of the stack shall be vented in accordance with Section 907 and sized as follows:

- 1 The portion of the stack above the offset shall be sized as for a vertical stack based on the total number of drainage fixture units above the offset.
- 2 The offset shall be sized in accordance with Section 710.1.1.
- 3 The portion of the stack below the offset shall be sized as for the offset or based on the total number of drainage fixture units on the entire stack, whichever is larger [see Table 710.1(2), Column 5].

711.2.1 Omission of vents for horizontal stack offsets.

Vents for horizontal stack offsets required by Section 711.2 shall not be required where the stack and its offset are one pipe size larger than required for a building drain [see Table 710.1(1)] and the entire stack and offset are not less in cross-sectional area than that required for a straight stack plus the area of an offset vent as provided for in Section 907.

711.3 Offsets below lowest branch.

Where a vertical offset occurs in a soil or waste stack below the lowest horizontal branch, a

change in diameter of the stack because of the offset shall not be required. If a horizontal offset occurs in a soil or waste stack below the lowest horizontal branch, the required diameter of the offset and the stack below it shall be determined as for a building drain in accordance with Table 710.1(1).

SECTION 712 SUMPS AND EJECTORS

712.1 Building sub-drains.

Building sub-drains that cannot be discharged to the sewer by gravity flow shall be discharged into a tightly covered and vented sump from which the liquid shall be lifted and discharged into the building gravity drainage system by automatic pumping equipment or other approved method. In other than existing structures, the sump shall not receive drainage from any piping within the building capable of being discharged by gravity to the building sewer.

712.2 Valves required.

A check valve and a full open valve located on the discharge side of the check valve shall be installed in the pump or ejector discharge piping between the pump or ejector and the gravity drainage system. Access shall be provided to such valves. Such valves shall be located above the sump cover required by Section 712.1 or, where the discharge pipe from the ejector is below grade, the valves shall be accessibly located outside the sump below grade in an access pit with a removable access cover.

712.3 Sump design.

The sump pump, pit and discharge piping shall conform to the requirements of Sections 712.3.1 through 712.3.5.

712.3.1 Sump pump.

The sump pump capacity and head shall be appropriate to anticipated use requirements.

712.3.2 Sump pit.

The sump pit shall be not less than 500 mm in diameter and not less than 600 mm in depth, unless otherwise approved. The pit shall be accessible and located such that all drainage flows into the pit by gravity. The sump pit shall be constructed of tile, concrete, steel, plastic or other approved materials. The pit bottom shall

be solid and provide permanent support for the pump. The sump pit shall be fitted with a gastight removable cover that is installed flush with grade or floor level, or above grade or floor level. The cover shall be adequate to support anticipated loads in the area of use. The sump pit shall be vented in accordance with Chapter 9.

712.3.3 Discharge pipe and fittings.

Discharge pipe and fittings serving sump pumps and ejectors shall be constructed of materials in accordance with Sections 712.3.3.1 and 712.3.3.2 and shall be approved.

712.3.3.1 Materials.

Pipe and fitting materials shall be constructed of brass, copper, CPVC, ductile iron, PE, or PVC.

712.3.3.2 Ratings.

Pipe and fittings shall be rated for the maximum system operating pressure and temperature. Pipe fitting materials shall be compatible with the pipe material. Where pipe and fittings are buried in the earth, they shall be suitable for burial.

712.3.4 Maximum effluent level.

The effluent level control shall be adjusted and maintained to at all times prevent the effluent in the sump from rising to within 50 mm of the invert of the gravity drain inlet into the sump.

712.3.5 Pump connection to the drainage system.

Pumps connected to the drainage system shall connect to a building sewer, building drain, soil stack, waste stack or horizontal branch drain. Where the discharge line connects into horizontal drainage piping, the connection shall be made through a wye fitting into the top of the drainage piping and such wye fitting shall be located not less than 10 pipe diameters from the base of any soil stack, waste stack or fixture drain.

712.4 Sewage pumps and sewage ejectors.

A sewage pump or sewage ejector shall automatically discharge the contents of the sump to the building drainage system.

712.4.1 Macerating toilet systems.

Macerating toilet systems shall comply with

ASME A112.3.4/CSA B45.9 and shall be installed in accordance with the manufacturer's instructions.

712.4.2 Capacity.

A sewage pump or sewage ejector shall have the capacity and head for the application requirements. Pumps or ejectors that receive the discharge of water closets shall be capable of handling spherical solids with a diameter of up to and including 50 mm. Other pumps or ejectors shall be capable of handling spherical solids with a diameter of up to and including 25 mm. The capacity of a pump or ejector based on the diameter of the discharge pipe shall be not less than that indicated in Table 712.4.2.

Exceptions:

1. Grinder pumps or grinder ejectors that receive the discharge of water closets shall have a discharge opening of not less than 32 mm.
2. Macerating toilet assemblies that serve single water closets shall have a discharge opening of not less than 20 mm.

TABLE 712.4.2 Minimum Capacity of Sewage Pump or Sewage Ejector

DIAMETER OF THE DISCHARGE PIPE (mm)	CAPACITY OF PUMP OR EJECTOR (L/min)
50	80
65	115
80	175

SECTION 713 HEALTH CARE PLUMBING

713.1 Scope.

This section shall govern those aspects of health care plumbing systems that differ from plumbing systems in other structures. Health care plumbing systems shall conform to this section in addition to the other requirements of this code. The provisions of this section shall apply to the special devices and equipment installed and maintained in the following occupancies: nursing homes; homes for the aged; orphanages; infirmaries; first aid stations; psychiatric facilities; clinics; professional offices of dentists and doctors; mortuaries; educational facilities; surgery, dentistry, research and testing laboratories;

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vent stack.

713.9.3 Trap seal maintenance.

A water supply pipe not less than 6.4 mm in diameter shall be taken from the flush supply of each bedpan washer on the discharge or fixture side of the vacuum breaker, shall be trapped to form not less than a 75 mm water seal and shall be connected to the local vent stack on each floor. The water supply shall be installed so as to provide a supply of water to the local vent stack for cleansing and drain trap seal maintenance each time a bedpan washer is flushed.

713.10 Sterilizer vents and stacks.

Multiple installations of pressure and non-pressure sterilizers shall have the vent connections to the sterilizer vent stack made by means of inverted wye fittings. Access shall be provided to vent connections for the purpose of inspection and maintenance.

713.10.1 Drainage.

The connection between sterilizer vent or exhaust openings and the sterilizer vent stack shall be designed and installed to drain to the funnel or basket type waste fitting. In multiple installations, the sterilizer vent stack shall be drained separately to the lowest sterilizer funnel or basket-type waste fitting or receptor.

713.11 Sterilizer vent stack sizes.

Sterilizer vent stack sizes shall comply with Sections 713.11.1 through 713.11.4.

713.11.1 Bedpan steamers.

The minimum size of a sterilizer vent serving a bedpan steamer shall be 38 mm in diameter. Multiple installations shall be sized in accordance with Table 713.11.1.

713.11.2 Boiling-type sterilizers.

The size of a sterilizer vent stack shall be not less than 50 mm in diameter where serving a utensil sterilizer and not less than 38 mm in diameter where serving an instrument sterilizer. Combinations of boiling-type sterilizer vent connections shall be sized in accordance with Table 713.11.1.

713.11.3 Pressure sterilizers.

Pressure sterilizer vent stacks shall be 64 mm minimum. Those serving combinations of pressure sterilizer exhaust connections shall be sized in accordance with Table 713.11.3.

713.11.4 Pressure instrument washer sterilizer sizes.

The diameter of a sterilizer vent stack serving an instrument washer sterilizer shall be not less than 50 mm. Not more than two sterilizers shall be installed on a 50 mm stack, and not more than four sterilizers shall be installed on a 75 mm stack.

TABLE 713.11.1

Stack Sizes For Bedpan Steamers And Boiling-Type Sterilizers
(Number of Connections of Various Sizes Permitted to Various-Sized Sterilizer Vent Stacks)

STACK SIZE (mm)	CONNECTION SIZE (mm)	
	40	50
40 ^a	1	0
50 ^a	2	1
50 ^b	1	1
80 ^a	4	2
80 ^b	2	2
100 ^a	8	4
100 ^b	4	4

- Total of each size.
- Combination of sizes.

TABLE 713.11.3
Stack Sizes For Pressure Sterilizers
(Number of Connections of Various Sizes Permitted To Various-Sized Vent Stacks)

STACK SIZE (mm)	CONNECTION SIZE (mm)			
	20	25	32	40
40 ^a	3 or	2 or	1	—
40 ^b	2 and	1	—	—
50 ^a	6 or	3 or	2 or	1
50 ^b	3 and	2	—	—
50 ^b	2 and	1 and	1	—
50 ^b	1 and	1 and	—	1
80 ^a	15 or	7 or	5 or	3
80 ^b	1 and	1 and 5 and	2 and	2 1

a. Total of each size.

b. Combination of sizes.

SECTION 714 COMPUTERIZED DRAINAGE DESIGN

714.1 Design of drainage system.

The sizing, design and layout of the drainage system shall be permitted to be designed by approved computer design methods.

714.2 Load on drainage system.

The load shall be computed from the simultaneous or sequential discharge conditions from fixtures, appurtenances and appliances or the peak usage design condition.

714.2.1 Fixture discharge profiles.

The discharge profiles for flow rates versus time from fixtures and appliances shall be in accordance with the manufacturer's specifications.

714.3 Selections of drainage pipe sizes.

Pipe shall be sized to prevent full-bore flow.

714.3.1 Selecting pipe wall roughness.

Pipe size calculations shall be conducted with the pipe wall roughness factor (ks), in accordance with the manufacturer's specifications and as modified for aging roughness factors with deposits and corrosion.

714.3.2 Slope of horizontal drainage piping.

Horizontal drainage piping shall be designed and installed at slopes in accordance with Table

704.1.

SECTION 715 BACKWATER VALVES

715.1 Sewage backflow.

Where plumbing fixtures are installed on a floor with a finished floor elevation below the elevation of the manhole cover of the next upstream manhole in the public sewer, such fixtures shall be protected by a backwater valve installed in the building drain, or horizontal branch serving such fixtures. Plumbing fixtures installed on a floor with a finished floor elevation above the elevation of the manhole cover of the next upstream manhole in the public sewer shall not discharge through a backwater valve.

Exception: In existing buildings, fixtures above the elevation of the manhole cover of the next upstream manhole in the public sewer shall not be prohibited from discharging through a backwater valve.

715.2 Material.

Bearing parts of backwater valves shall be of corrosion-resistant material. Backwater valves shall comply with ASME A112.14.1, CSA B181.1 or CSA B181.2.

715.3 Seal.

Backwater valves shall be so constructed as to provide a mechanical seal against backflow.

715.4 Diameter.

Backwater valves, when fully opened, shall have a capacity not less than that of the pipes in which they are installed.

715.5 Location.

Backwater valves shall be installed so that access is provided to the working parts for service and repair.

SECTION 716 VACUUM DRAINAGE SYSTEMS

716.1 Scope.

Vacuum drainage systems shall be in accordance with Sections 716.2 through 716.4.

716.2 System design.

Vacuum drainage systems shall be designed in accordance with the vacuum drainage system manufacturer's instructions. The system layout, including piping layout, tank assemblies, vacuum pump assembly and other components necessary for proper function of the system shall be in accordance with the manufacturer's instructions. Plans, specifications and other data for such systems shall be submitted to the code official for review and approval prior to installation.

716.2.1 Fixtures.

Gravity-type fixtures installed in vacuum drainage systems shall comply with Chapter 4.

716.2.2 Drainage fixture units.

Drainage fixture units for gravity drainage systems that discharge into, or receive discharge from, vacuum drainage systems shall be based on the values in this chapter.

716.2.3 Water supply fixture units.

Water supply fixture units shall be based on the values in Chapter 6 of this code, except that the water supply fixture unit for a vacuum-type water closet shall be 1.

716.2.4 Traps and cleanouts.

Gravity drainage fixtures shall be provided with traps and cleanouts in accordance with this chapter and Chapter 10.

716.2.5 Materials.

Vacuum drainage pipe, fitting and valve materials shall be in accordance with the vacuum drainage system manufacturer's instructions and the requirements of this

chapter.

716.3 Testing and demonstrations.

After completion of the entire system installation, the system shall be subjected to a vacuum test of 475 mm of mercury and shall be operated to function as required by the code official and the manufacturer of the vacuum drainage system. Recorded proof of all tests shall be submitted to the code official.

716.4 Written instructions.

Written instructions for the operation, maintenance, safety and emergency procedures shall be provided to the building owner. The code official shall verify that the building owner is in receipt of such instructions.

SECTION 717 REPLACEMENT OF UNDERGROUND SEWERS BY PIPE-BURSTING METHODS

717.1 General.

This section shall govern the replacement of existing building sewer piping by pipe-bursting methods.

717.2 Applicability.

The replacement of building sewer piping by pipe-bursting methods shall be limited to gravity drainage piping of sizes 150 mm and smaller. The replacement piping shall be of the same nominal size as the existing piping.

717.3 Pre-installation inspection.

The existing piping sections to be replaced shall be inspected internally by a recorded video camera survey. The survey shall include notations of the position of cleanouts and the depth of connections to the existing piping.

717.4 Pipe.

The replacement piping shall be manufactured with a standard dimension ratio (SDR) of 17 and in compliance with ASTM F 714.

717.5 Pipe fittings.

Pipe fittings to be connected to the replacement piping shall be of extra-high molecular weight PE3408 material and shall be manufactured with an SDR of 17 and in compliance with ASTM D 2683

717.6 Cleanouts.

Where the existing building sewer did not have

cleanouts meeting the requirements of this code, cleanout fittings shall be installed as required by this code.

717.7 Post-installation inspection.

The completed replacement piping section shall be inspected internally by a recorded video camera survey. The video survey shall be

reviewed and approved by the code official prior to pressure testing of the replacement piping system.

717.8 Pressure testing.

The replacement piping system as well as the connections to the replacement piping shall be tested in accordance with Section 312.



CHAPTER 8 INDIRECT/SPECIAL WASTE

SECTION 801 GENERAL

801.1 Scope.

This chapter shall govern matters concerning indirect waste piping and special wastes. This chapter shall further control matters concerning food-handling establishments, sterilizers, clear-water waste, swimming pools, methods of providing air breaks or air gaps, and neutralizing devices for corrosive wastes.

801.2 Protection.

Devices, appurtenances, appliances and apparatus intended to serve some special function, such as sterilization, distillation, processing, cooling, or storage of ice or foods, and that discharge to the drainage system, shall be provided with protection against backflow, flooding, fouling, contamination and stoppage of the drain.

SECTION 802 INDIRECT WASTES

802.1 Where required.

Food-handling equipment, in other than dwelling units, clear-water waste, dishwashing machines and utensils, pots, pans and dishwashing sinks shall discharge through an indirect waste pipe as specified in Sections 802.1.1 through 802.1.8. Health-care related fixtures, devices and equipment shall discharge to the drainage system through an indirect waste pipe by means of an air gap in accordance with this chapter and Section 713.3. Fixtures not required by this section to be indirectly connected shall be directly connected to the plumbing system in accordance with Chapter 7.

802.1.1 Food handling.

Equipment and fixtures utilized for the storage, preparation and handling of food shall discharge through an indirect waste pipe by means of an air gap. Each well of a multiple-compartment sink shall discharge independently to a waste receptor.

802.1.2 Floor drains in food storage areas.

Floor drains located within walk-in refrigerators or freezers in food service and food establishments shall be indirectly connected to the sanitary drainage system by

means of an air gap. Where a floor drain is located within an area subject to freezing, the waste line serving the floor drain shall not be trapped and shall indirectly discharge into a waste receptor located outside of the area subject to freezing.

Exception: Where protected against backflow by a backwater valve, such floor drains shall be indirectly connected to the sanitary drainage system by means of an *air break* or an *air gap*.

802.1.3 Potable clear-water waste.

Where devices and equipment, such as sterilizers and relief valves, discharge potable water to the building drainage system, the discharge shall be through an indirect waste pipe by means of an air gap.

802.1.4 Swimming pools.

Where waste water from swimming pools, backwash from filters and water from pool deck drains discharge to the building drainage system, the discharge shall be through an indirect waste pipe by means of an air gap.

802.1.5 Non-potable clear-water waste.

Where devices and equipment such as process tanks, filters, drips and boilers discharge non-potable water to the building drainage system, the discharge shall be through an indirect waste pipe by means of an air break or an air gap.

802.1.6 Domestic dishwashing machines.

Domestic dishwashing machines shall discharge indirectly through an air gap or air break into a waste receptor in accordance with Section 802.2, or discharge into a wye branch fitting on the tailpiece of the kitchen sink or the dishwasher connection of a food waste disposer. The waste line of a domestic dishwashing machine discharging into a kitchen sink tailpiece or food waste disposer shall connect to a deck-mounted air gap or the waste line shall rise and be securely fastened to the underside of the sink rim or counter.

802.1.7 Commercial dishwashing machines.

The discharge from a commercial dishwashing machine shall be through an air gap or air break into a waste receptor in accordance with Section 802.2.

802.1.8 Food utensils, dishes, pots and pans sinks.

Sinks, in other than dwelling units, used for the washing, rinsing or sanitizing of utensils, dishes, pots, pans or service ware used in the preparation, serving or eating of food shall discharge indirectly through an air gap or an air break to the drainage system.

802.2 Installation.

Indirect waste piping shall discharge through an air gap or air break into a waste receptor. Waste receptors shall be trapped and vented and shall connect to the building drainage system. Indirect waste piping that exceeds 750 mm in developed length measured horizontally, or 1350 mm in total developed length, shall be trapped.

Exception: Where a waste receptor receives only clear water waste and does not directly connect to a sanitary drainage system, the receptor shall not require a trap.

802.2.1 Air gap.

The air gap between the indirect waste pipe and the flood level rim of the waste receptor shall be not less than twice the effective opening of the indirect waste pipe.

802.2.2 Air break.

An air break shall be provided between the indirect waste pipe and the trap seal of the waste receptor.

802.3 Waste receptors.

For other than hub drains that receive only clear-water waste and standpipes, a removable strainer or basket shall cover the outlet of waste receptors. Waste receptors shall not be installed in concealed spaces. Waste receptors shall not be installed in plenums, crawl spaces, attics, interstitial spaces above ceilings and below floors. Ready access shall be provided to waste receptors.

802.3.1 Size of receptors.

A waste receptor shall be sized for the maximum discharge of all indirect waste pipes served by the receptor. Receptors shall be installed to prevent splashing or flooding.

802.3.2 Hub drains.

A hub drain shall be in the form of a hub or a pipe extending not less than 25 mm above a water-impervious floor.

802.3.3 Standpipes.

Standpipes shall be individually trapped. Standpipes shall extend not less than 450 mm but not greater than 1050 mm above the trap weir. Access shall be provided to standpipes and drains for rodding.

SECTION 803 SPECIAL WASTES

803.1 Neutralizing device required for corrosive wastes.

Corrosive liquids, spent acids or other harmful chemicals that destroy or injure a drain, sewer, soil or waste pipe, or create noxious or toxic fumes or interfere with sewage treatment processes shall not be discharged into the plumbing system without being thoroughly diluted, neutralized or treated by passing through an approved dilution or neutralizing device. Such devices shall be automatically provided with a sufficient supply of diluting water or neutralizing medium so as to make the contents non injurious before discharge into the drainage system. The nature of the corrosive or harmful waste and the method of its treatment or dilution shall be approved prior to installation.

803.2 System design.

A chemical drainage and vent system shall be designed and installed in accordance with this code. Chemical drainage and vent systems shall be completely separated from the sanitary systems. Chemical waste shall not discharge to a sanitary drainage system until such waste has been treated in accordance with Section 803.1.

SECTION 804 MATERIALS, JOINTS AND CONNECTIONS

804.1 General.

The materials and methods utilized for the construction and installation of indirect waste pipes and systems shall comply with the applicable provisions of Chapter 7.

CHAPTER 9 VENTS

SECTION 901 GENERAL

901.1 Scope.

The provisions of this chapter shall govern the materials, design, construction and installation of vent systems.

901.2 Trap seal protection.

The plumbing system shall be provided with a system of vent piping that will permit the admission or emission of air so that the seal of any fixture trap shall not be subjected to a pressure differential of more than 250 Pa.

901.2.1 Venting required.

Traps and trapped fixtures shall be vented in accordance with one of the venting methods specified in this chapter.

901.3 Chemical waste vent systems.

The vent system for a chemical waste system shall be independent of the sanitary vent system and shall terminate separately through the roof to the outdoors or to an air admittance valve that complies with ASSE 1049. Air admittance valves for chemical waste systems shall be constructed of materials approved in accordance with Section 702.5 and shall be tested for chemical resistance in accordance with ASTM F 1412.

901.4 Use limitations.

The plumbing vent system shall not be utilized for purposes other than the venting of the plumbing system.

901.5 Tests.

The vent system shall be tested in accordance with Section 312.

901.6 Engineered systems.

Engineered venting systems shall conform to the provisions of Section 919.

SECTION 902 MATERIALS

902.1 Vents.

The materials and methods utilized for the construction and installation of venting systems shall comply with the applicable provisions of Section 702.

902.2 Sheet copper.

Sheet copper for vent pipe flashings shall conform to ASTM B 152 and shall weigh not less than 2.5 kg per m².

902.3 Sheet lead.

Sheet lead for vent pipe flashings shall weigh not less than 15 kg per m² for field-constructed flashings and not less than 12 kg per m² for prefabricated flashings.

SECTION 903 VENT TERMINALS

903.1 Roof extension.

Open vent pipes that extend through a roof shall be terminated not less than 300 mm above the roof. Where a roof is to be used for assembly or as a promenade, observation deck or similar purposes, open vent pipes shall terminate not less than 2 m above the roof.

903.2 Frost closure.

Where the 97.5-percent value for outside design temperature is -18°C or less, vent extensions through a roof or wall shall be not less than 75 mm in diameter. Any increase in the size of the vent shall be made not less than 300 mm inside the thermal envelope of the building.

903.3 Flashings.

The juncture of each vent pipe with the roof line shall be made water tight by an approved flashing.

903.4 Prohibited use.

A vent terminal shall not be used for any purpose other than a vent terminal.

903.5 Location of vent terminal.

An open vent terminal from a drainage system shall not be located directly beneath any door, openable window, or other air intake opening of the building or of an adjacent building, and any such vent terminal shall not be within 3000 mm horizontally of such an opening unless it is 1000 mm or more above the top of such opening.

903.6 Extension through the wall.

Vent terminals extending through the wall shall terminate at a point not less than 3 m from a lot line and not less than 3 m above average ground level. Vent terminals shall not terminate under the overhang of a structure with soffit vents.

Side wall vent terminals shall be protected to prevent birds or rodents from entering or blocking the vent opening.

903.7 Extension outside a structure.

In climates where the 97.5-percent value for outside design temperature is less than -18°C , vent pipes installed on the exterior of the structure shall be protected against freezing by insulation, heat or both.

SECTION 904 OUTDOOR VENT EXTENSIONS

904.1 Required vent extension.

The vent system serving each building drain shall have not less than one vent pipe that extends to the outdoors.

904.1.1 Installation.

The required vent shall be a dry vent that connects to the building drain or an extension of a drain that connects to the building drain. Such vent shall not be an island fixture vent as allowed by Section 916.

904.1.2 Size.

The required vent shall be sized in accordance with Section 906.2 based on the required size of the building drain.

904.2 Vent stack required.

A vent stack shall be required for every drainage stack that has five branch intervals or more.

Exception: Drainage stacks installed in accordance with Section 913.

904.3 Vent termination.

Vent stacks or stack vents shall terminate outdoors to the open air or to a stack-type air admittance valve in accordance with Section 918.

904.4 Vent connection at base.

Vent stacks shall connect to the base of the drainage stack. The vent stack shall connect at or below the lowest horizontal branch. Where the vent stack connects to the building drain, the connection shall be located downstream of the drainage stack and within a distance of 10 times the diameter of the drainage stack.

904.5 Vent headers.

Stack vents and vent stacks connected into a

common vent header at the top of the stacks and extending to the open air at one point shall be sized in accordance with the requirements of Section 906.1. The number of fixture units shall be the sum of all fixture units on all stacks connected thereto, and the developed length shall be the longest vent length from the intersection at the base of the most distant stack to the vent terminal in the open air, as a direct extension of one stack.

SECTION 905 VENT CONNECTIONS AND GRADES

905.1 Connection.

Individual, branch and circuit vents shall connect to a vent stack, stack vent, air admittance valve or extend to the open air.

905.2 Grade.

Vent and branch vent pipes shall be so graded and connected as to drain back to the drainage pipe by gravity.

905.3 Vent connection to drainage system.

Every dry vent connecting to a horizontal drain shall connect above the centerline of the horizontal drain pipe.

905.4 Vertical rise of vent.

Every dry vent shall rise vertically to a point not less than 150 mm above the flood level rim of the highest trap or trapped fixture being vented.

Exception: Vents for interceptors located outdoors.

905.5 Height above fixtures.

A connection between a vent pipe and a vent stack or stack vent shall be made at not less than 150 mm above the flood level rim of the highest fixture served by the vent. Horizontal vent pipes forming branch vents, relief vents or loop vents shall be located not less than 150 mm above the flood level rim of the highest fixture served.

905.6 Vent for future fixtures.

Where the drainage piping has been roughed-in for future fixtures, a rough-in connection for a vent shall be installed. The vent size shall be not less than one-half the diameter of the rough-in drain to be served. The vent rough-in shall connect to the vent system, or shall be vented by other means as provided for in this chapter.

The connection shall be identified to indicate that it is a vent.

SECTION 906 VENT PIPE SIZING

906.1 Size of stack vents and vent stacks.

The minimum required diameter of stack vents and vent stacks shall be determined from the developed length and the total of drainage fixture units connected thereto in accordance with Table 906.1, but in no case shall the diameter be less than one-half the diameter of the drain served or less than 32 mm.

906.2 Vents other than stack vents or vent stacks.

The diameter of individual vents, branch vents, circuit vents and relief vents shall be not less than one-half the required diameter of the drain served. The required size of the drain shall be determined in accordance with Table 710.1(2). Vent pipes shall not be less than 32 mm in diameter. Vents exceeding 12 m in developed length shall be increased by one nominal pipe size for the entire developed length of the vent pipe. Relief vents for soil and waste stacks in buildings having more than 10 branch intervals shall be sized in accordance with Section 908.2.

906.3 Developed length.

The developed length of individual, branch, circuit and relief vents shall be measured from the farthest point of vent connection to the drainage system to the point of connection to the vent stack, stack vent or termination outside of the building.

906.4 Multiple branch vents.

Where multiple branch vents are connected to a common branch vent, the common branch vent shall be sized in accordance with this section based on the size of the common horizontal drainage branch that is or would be required to serve the total drainage fixture unit load being vented.

906.5 Sump vents.

Sump vent sizes shall be determined in

accordance with Sections 906.5.1 and 906.5.2.

906.5.1 Sewage pumps and sewage ejectors other than pneumatic.

Drainage piping below sewer level shall be vented in the same manner as that of a gravity system. Building sump vent sizes for sumps with sewage pumps or sewage ejectors, other than pneumatic, shall be determined in accordance with Table 906.5.1.

906.5.2 Pneumatic sewage ejectors.

The air pressure relief pipe from a pneumatic sewage ejector shall be connected to an independent vent stack terminating as required for vent extensions through the roof. The relief pipe shall be sized to relieve air pressure inside the ejector to atmospheric pressure, but shall be not less than 32 mm in size.

SECTION 907 VENTS FOR STACK OFFSETS

907.1 Vent for horizontal offset of drainage stack.

Horizontal offsets of drainage stacks shall be vented where five or more branch intervals are located above the offset. The offset shall be vented by venting the upper section of the drainage stack and the lower section of the drainage stack.

907.2 Upper section.

The upper section of the drainage stack shall be vented as a separate stack with a vent stack connection installed in accordance with Section 904.4. The offset shall be considered the base of the stack.

907.3 Lower section.

The lower section of the drainage stack shall be vented by a yoke vent connecting between the offset and the next lower horizontal branch. The yoke vent connection shall be permitted to be a vertical extension of the drainage stack. The size of the yoke vent and connection shall be a minimum of the size required for the vent stack of the drainage stack.

TABLE 906.1 Size And Developed Length of Stack Vents and Vent Stacks

DIAMETER OF SOIL OR WASTE STACK ^b (mm)	TOTAL FIXTURE UNITS BEING VENTED (dfu)	MAXIMUM DEVELOPED LENGTH OF VENT (m) ^a										
		DIAMETER OF VENT (mm)										
		32	40	50	65	75	100	125	150	200	250	300
40	8	15	45	—	—	—	—	—	—	—	—	—
40	10	9	30	—	—	—	—	—	—	—	—	—
50	12	—	23	60	—	—	—	—	—	—	—	—
50	10	9	15	45	—	—	—	—	—	—	—	—
65	42	8	9	30	90	—	—	—	—	—	—	—
75	10	—	13	45	110	317	—	—	—	—	—	—
75	21	—	10	36	82	247	—	—	—	—	—	—
75	53	—	8	29	70	207	—	—	—	—	—	—
75	102	—	8	—	64	189	—	—	—	—	—	—
100	43	—	—	26	26	76	299	—	—	—	—	—
100	140	—	—	11.8	110	60	229	—	—	—	—	—
100	320	—	—	7	17	52	195	—	—	—	—	—
100	540	—	—	7	15	45	177	—	—	—	—	—
125	190	—	—	7	9	25	98	302	—	—	—	—
125	490	—	—	—	7	19	76	232	—	—	—	—
125	940	—	—	—	5	16.2	64	204	—	—	—	—
125	1,400	—	—	—	5	14.9	58	180	—	—	—	—
150	500	—	—	—	—	10.1	40	122	305	—	—	—
150	1,100	—	—	—	—	8	30	94	238	—	—	—
150	2,000	—	—	—	—	7	26	79	201	—	—	—
150	2,900	—	—	—	—	6	24	73	183	—	—	—
200	1,800	—	—	—	—	—	10	29	73	287	—	—
200	3,400	—	—	—	—	—	7	22	58	219	—	—
200	5,600	—	—	—	—	—	6	19	49	186	—	—
200	7,600	—	—	—	—	—	5	17.1	171	171	—	—
250	4,000	—	—	—	—	—	—	10	94	94	293	—
250	7,200	—	—	—	—	—	—	7.3	18.3	73	226	—
250	11,000	—	—	—	—	—	—	6	15	60	192	—
250	15,000	—	—	—	—	—	—	5	14	55	174	—
300	7,300	—	—	—	—	—	—	—	10	37	116	287
300	13,000	—	—	—	—	—	—	—	7	29	90	219
300	20,000	—	—	—	—	—	—	—	6	24	76	186
300	26,000	—	—	—	—	—	—	—	5	22	70	152
375	15,000	—	—	—	—	—	—	—	—	12	39	93
375	25,000	—	—	—	—	—	—	—	—	9.3	28.8	72
375	38,000	—	—	—	—	—	—	—	—	7.8	24.3	60
375	50,000	—	—	—	—	—	—	—	—	7.2	22.2	54

- a. The *developed length* shall be measured from the vent connection to the open air.
- b. Shown sizes are nominal sizes based on internal diameters. If the pipe nominal size is based on the external diameter, find the internal diameter (from pipe's manufacturer) and use a pipe size with an internal diameter equal to or greater than the size shown in the table.

TABLE 906.5.1 Size And Length of Sump Vents

DISCHARGE CAPACITY OF PUMP (L/min)	MAXIMUM DEVELOPED LENGTH OF VENT (m) ^a					
	Diameter of vent (mm)					
	32	40	50	65	75	100
38	No limit ^b	No limit	No limit	No limit	No limit	No limit
76	82	No limit	No limit	No limit	No limit	No limit
150	22	49	No limit	No limit	No limit	No limit
250	10	23	82	No limit	No limit	No limit
300	5	13	45	116	No limit	No limit
375	3 ^c	8	30	76	No limit	No limit
550	Not permitted	3 ^c	14	34	113	No limit
750	Not permitted	Not permitted	6	18	64	No limit
950	Not permitted	Not permitted	3	11	41	No limit
1150	Not permitted	Not permitted	3 ^c	7	27	116
1500	Not permitted	Not permitted	Not permitted	3 ^c	14	64
1900	Not permitted	Not permitted	Not permitted	Not permitted	8	40

- Developed length plus an appropriate allowance for entrance losses and friction due to fittings, changes in direction and diameter. Suggested allowances shall be obtained from NBS Monograph 31 or other approved sources. An allowance of 50 percent of the developed length shall be assumed if a more precise value is not available.
- Actual values greater than 150 m.
- Less than 3 m.

SECTION 908 RELIEF VENTS— STACKS OF MORE THAN 10 BRANCH INTERVALS

908.1 Where required.

Soil and waste stacks in buildings having more than 10 branch intervals shall be provided with a relief vent at each tenth interval installed, beginning with the top floor.

908.2 Size and connection.

The size of the relief vent shall be equal to the size of the vent stack to which it connects. The lower end of each relief vent shall connect to the soil or waste stack through a wye below the horizontal branch serving the floor, and the upper end shall connect to the vent stack through a wye not less than 900 mm above the floor.

SECTION 909 FIXTURE VENTS

909.1 Distance of trap from vent.

Each fixture trap shall have a protecting vent located so that the slope and the developed length in the fixture drain from the trap weir to

the vent fitting are within the requirements set forth in Table 909.1.

Exception: The developed length of the fixture drain from the trap weir to the vent fitting for self-siphoning fixtures, such as water closets, shall not be limited.

909.2 Venting of fixture drains.

The total fall in a fixture drain due to pipe slope shall not exceed the diameter of the fixture drain, nor shall the vent connection to a fixture drain, except for water closets, be below the weir of the trap.

909.3 Crown vent.

A vent shall not be installed within two pipe diameters of the trap weir.

TABLE 909.1 Maximum Distance of Fixture Trap from Vent

SIZE OF TRAP (mm)	SIZE OF FIXTURE DRAIN (mm)	SLOPE (%)	DISTANCE FROM TRAP (m)
32	40	2	1.5
40	40	2	1.8
45	50	2	2.1
50	50	2	2.4
75	75	1	3.6
100	100	1	4.8

SECTION 910 INDIVIDUAL VENT

910.1 Individual vent permitted.

Each trap and trapped fixture is permitted to be provided with an individual vent. The individual vent shall connect to the fixture drain of the trap or trapped fixture being vented.

SECTION 911 COMMON VENT

911.1 Individual vent as common vent.

An individual vent is permitted to vent two traps or trapped fixtures as a common vent. The traps or trapped fixtures being common vented shall be located on the same floor level.

911.2 Connection at the same level.

Where the fixture drains being common vented connect at the same level, the vent connection shall be at the interconnection of the fixture drains or downstream of the interconnection.

911.3 Connection at different levels.

Where the fixture drains connect at different levels, the vent shall connect as a vertical extension of the vertical drain. The vertical drain pipe connecting the two fixture drains shall be considered the vent for the lower fixture drain, and shall be sized in accordance with Table 911.3. The upper fixture shall not be a water closet.

TABLE 911.3 COMMON VENT SIZES

PIPE SIZE (mm)	MAXIMUM DISCHARGE FROM UPPER FIXTURE DRAIN (dfu)
40	1
50	4
65 to 80	6

SECTION 912 WET VENTING

912.1 Horizontal wet vent permitted.

Any combination of fixtures within two bathroom groups located on the same floor level is permitted to be vented by a horizontal wet vent. The wet vent shall be considered the vent for the fixtures and shall extend from the connection of the dry vent along the direction of the flow in the drain pipe to the most downstream fixture drain connection to the horizontal branch drain. Each wet-vented fixture drain shall connect independently to the horizontal wet vent. Only the fixtures within the bathroom groups shall connect to the wet-vented horizontal branch drain. Any additional fixtures shall discharge downstream of the horizontal wet vent.

912.1.1 Vertical wet vent permitted.

Any combination of fixtures within two bathroom groups located on the same floor level is permitted to be vented by a vertical wet vent. The vertical wet vent shall be considered the vent for the fixtures and shall extend from the connection of the dry vent down to the lowest fixture drain connection. Each wet-vented fixture shall connect independently to the vertical wet vent. Water closet drains shall connect at the same elevation. Other fixture drains shall connect above or at the same elevation as the water closet fixture drains. The dry-vent connection to the vertical wet vent shall be an individual or common vent serving one or two fixtures.

912.2 Dry vent connection.

The required dry-vent connection for wet-vented systems shall comply with Sections 912.2.1 and 912.2.2.

912.2.1 Horizontal wet vent.

The dry-vent connection for a horizontal wet-vent system shall be an individual vent or a common vent for any bathroom group fixture, except an emergency floor drain. Where the

dry-vent connects to a water closet fixture drain, the drain shall connect horizontally to the horizontal wet-vent system. Not more than one wet-vented fixture drain shall discharge upstream of the dry-vented fixture drain connection.

912.2.2 Vertical wet vent.

The dry-vent connection for a vertical wet-vent system shall be an individual vent or common vent for the most upstream fixture drain.

912.3 Size.

The dry vent serving the wet vent shall be sized based on the largest required diameter of pipe within the wet-vent system served by the dry vent. The wet vent shall be of a size not less than that specified in Table 912.3, based on the fixture unit discharge to the wet vent.

TABLE 912.3 Wet Vent Size

WET VENT PIPE SIZE (mm)	DRAINAGE FIXTURE UNIT LOAD (dfu)
40	1
50	4
65	6
80	12

SECTION 913 WASTE STACK VENT

913.1 Waste stack vent permitted.

A waste stack shall be considered a vent for all of the fixtures discharging to the stack where installed in accordance with the requirements of this section.

913.2 Stack installation.

The waste stack shall be vertical, and both horizontal and vertical offsets shall be prohibited between the lowest fixture drain connection and the highest fixture drain connection. Fixture drains shall connect separately to the waste stack. The stack shall not receive the discharge of water closets or urinals.

913.3 Stack vent.

A stack vent shall be provided for the waste stack. The size of the stack vent shall be not less than the size of the waste stack. Offsets shall be permitted in the stack vent, shall be located not less than 150 mm above the flood level of the highest fixture and shall be in accordance with

Section 905.2. The stack vent shall be permitted to connect with other stack vents and vent stacks in accordance with Section 904.5.

913.4 Waste stack size.

The waste stack shall be sized based on the total discharge to the stack and the discharge within a branch interval in accordance with Table 913.4. The waste stack shall be the same size throughout its length.

TABLE 913.4 Waste Stack Vent Size

STACK SIZE (mm)	MAXIMUM NUMBER OF DRAINAGE FIXTURE UNITS (dfu)	
	Total discharge into one branch interval	Total discharge for stack
40	1	2
50	2	4
65	No limit	8
80	No limit	24
100	No limit	50
125	No limit	75
150	No limit	100

SECTION 914 CIRCUIT VENTING

914.1 Circuit vent permitted.

A maximum of eight fixtures connected to a horizontal branch drain shall be permitted to be circuit vented. Each fixture drain shall connect horizontally to the horizontal branch being circuit vented. The horizontal branch drain shall be classified as a vent from the most downstream fixture drain connection to the most upstream fixture drain connection to the horizontal branch.

914.1.1 Multiple circuit-vented branches.

Circuit-vented horizontal branch drains are permitted to be connected together. Each group of a maximum of eight fixtures shall be considered a separate circuit vent and shall conform to the requirements of this section.

914.2 Vent connection.

The circuit vent connection shall be located between the two most upstream fixture drains. The vent shall connect to the horizontal branch and shall be installed in accordance with Section 905. The circuit vent pipe shall not receive the discharge of any soil or waste.

914.3 Slope and size of horizontal branch.

The slope of the vent section of the horizontal branch drain shall be not greater than one unit vertical in 12 units horizontal (8.3-percent slope). The entire length of the vent section of the horizontal branch drain shall be sized for the total drainage discharge to the branch.

914.3.1 Size of multiple circuit vent.

Each separate circuit-vented horizontal branch that is interconnected shall be sized independently in accordance with Section 914.3. The downstream circuit-vented horizontal branch shall be sized for the total discharge into the branch, including the upstream branches and the fixtures within the branch.

914.4 Relief vent.

A relief vent shall be provided for circuit-vented horizontal branches receiving the discharge of four or more water closets and connecting to a drainage stack that receives the discharge of soil or waste from upper horizontal branches.

914.4.1 Connection and installation.

The relief vent shall connect to the horizontal branch drain between the stack and the most downstream fixture drain of the circuit vent. The relief vent shall be installed in accordance with Section 905.

914.4.2 Fixture drain or branch.

The relief vent is permitted to be a fixture drain or fixture branch for fixtures located within the same branch interval as the circuit-vented horizontal branch. The maximum discharge to

a relief vent shall be four fixture units.

914.5 Additional fixtures.

Fixtures, other than the circuit vented fixtures, are permitted to discharge to the horizontal branch drain. Such fixtures shall be located on the same floor as the circuit-vented fixtures and shall be either individually or common vented.

SECTION 915 COMBINATION WASTE AND VENT SYSTEM**915.1 Type of fixtures.**

A combination waste and vent system shall not serve fixtures other than floor drains, sinks, lavatories and drinking fountains. Combination waste and vent systems shall not receive the discharge from a food waste disposer or clinical sink.

915.2 Installation.

The only vertical pipe of a combination waste and vent system shall be the connection between the fixture drain and the horizontal combination waste and vent pipe. The vertical distance shall not exceed 2.5 m.

915.2.1 Slope.

The slope of a horizontal combination waste and vent pipe shall not exceed one-half unit vertical in 12 units horizontal (4-percent slope) and shall not be less than that indicated in Table 704.1.

915.2.2 Size and length.

The size of a combination waste and vent pipe shall be not less than that indicated in Table 915.2.2. The horizontal length of a combination waste and vent system shall be unlimited.

TABLE 915.2.2 Size of Combination Waste and Vent Pipe

DIAMETER PIPE (mm)	MAXIMUM NUMBER OF DRAINAGE FIXTURE UNITS (dfu)	
	Connecting to a horizontal branch or stack	Connecting to a building drain or building subdrain
50	3	4
65	6	26
80	12	31
100	20	50
125	160	250
150	360	575

915.2.3 Connection.

The combination waste and vent system shall be provided with a dry vent connected at any point within the system or the system shall connect to a horizontal drain that serves vented fixtures located on the same floor. Combination waste and vent systems connecting to building drains receiving only the discharge from one or more stacks shall be provided with a dry vent. The vent connection to the combination waste and vent pipe shall extend vertically to a point not less than 150 mm above the flood level rim of the highest fixture being vented before offsetting horizontally.

915.2.4 Vent size.

The vent shall be sized for the total drainage fixture unit load in accordance with Section 906.2.

915.2.5 Fixture branch or drain.

The fixture branch or fixture drain shall connect to the combination waste and vent within a distance specified in Table 909.1. The combination waste and vent pipe shall be considered the vent for the fixture.

SECTION 916 ISLAND FIXTURE VENTING

916.1 Limitation.

Island fixture venting shall not be permitted for fixtures other than sinks and lavatories. Residential kitchen sinks with a dishwasher waste connection, a food waste disposer, or both, in combination with the kitchen sink waste, shall be permitted to be vented in accordance with this section.

916.2 Vent connection.

The island fixture vent shall connect to the fixture drain as required for an individual or common vent. The vent shall rise vertically to

above the drainage outlet of the fixture being vented before offsetting horizontally or vertically downward. The vent or branch vent for multiple island fixture vents shall extend to a point not less than 150 mm above the highest island fixture being vented before connecting to the outside vent terminal.

916.3 Vent installation below the fixture flood level rim.

The vent located below the flood level rim of the fixture being vented shall be installed as required for drainage piping in accordance with Chapter 7, except for sizing. The vent shall be sized in accordance with Section 906.2. The lowest point of the island fixture vent shall connect full size to the drainage system. The connection shall be to a vertical drain pipe or to the top half of a horizontal drain pipe. Cleanouts shall be provided in the island fixture vent to permit rodding of all vent piping located below the flood level rim of the fixtures. Rodding in both directions shall be permitted through a cleanout.

SECTION 917 SINGLE STACK VENT SYSTEM

917.1 Where permitted.

A drainage stack shall serve as a single stack vent system where sized and installed in accordance with Sections 917.2 through 917.9. The drainage stack and branch piping shall be the vents for the drainage system. The drainage stack shall have a stack vent.

917.2 Stack size.

Drainage stacks shall be sized in accordance with Table 917.2. Stacks shall be uniformly sized based on the total connected drainage fixture unit load. The stack vent shall be the same size as the drainage stack. A 75 mm stack shall serve not more than two water closets.

TABLE 917.2 Single Stack Size

STACK SIZE (mm)	MAXIMUM CONNECTED DRAINAGE FIXTURE UNITS		
	Stacks less than 25 meters in height	Stacks 25 meters to less than 50 meters in height	Stacks 50 meters and greater in height
75	24	NP	NP
100	225	24	NP
125	480	225	24
150	1,015	480	225
200	2,320	1,015	480
250	4,500	2,320	1,015
300	8,100	4,500	2,320
400	13,600	8,100	4,500

917.3 Branch size.

Horizontal branches connecting to a single stack vent system shall be sized in accordance with Table 710.1(2). Not more than one water closet shall discharge into a 80 mm horizontal branch at a point within a developed length of 500 mm measured horizontally from the stack.

Where a water closet is within 500 mm measured horizontally from the stack and not more than one fixture with a drain size of not more than 40 mm connects to a 75 mm horizontal branch, the *branch* drain connection to the *stack* shall be made with a sanitary tee.

917.4 Length of horizontal branches.

The length of horizontal branches shall conform to the requirements of Sections 917.4.1 through 917.4.3.

917.4.1 Water closet connection.

Water closet connections shall be not greater than 12 m in developed length measured horizontally from the stack.

Exception: Where the connection is made with a sanitary tee, the maximum developed length shall be 24 m.

917.4.2 Fixture connections.

Fixtures other than water closets shall be located not greater than 36 m in developed length, measured horizontally from the stack.

917.4.3 Vertical piping in branch.

The length of vertical piping in a fixture drain connecting to a horizontal branch shall not be considered in computing the fixture's distance in developed length measured horizontally

from the stack.

917.5 Minimum vertical piping size from fixture.

The vertical portion of piping in a fixture drain to a horizontal branch shall be 50 mm. The minimum size of the vertical portion of piping for a water-supplied urinal or standpipe shall be 75 mm. The maximum vertical drop shall be 12 m. Fixture drains that are not increased in size, or have a vertical drop in excess of 12 m, shall be individually vented.

917.6 Additional venting required.

Additional venting shall be provided where more than one water closet discharges to a horizontal branch and where the distance from a fixture trap to the stack exceeds the limits in Section 917.4. Where additional venting is required, the fixture(s) shall be vented by individual vents, common vents, wet vents, circuit vents, or a combination waste and vent pipe. The dry vent extensions for the additional venting shall connect to a branch vent, vent stack, stack vent, air admittance valve, or shall terminate outdoors.

917.7 Stack offsets.

Where fixture drains are not connected below a horizontal offset in a stack, a horizontal offset shall not be required to be vented. Where horizontal branches or fixture drains are connected below a horizontal offset in a stack, the offset shall be vented in accordance with Section 907. Fixture connections shall not be made to a stack within 600 mm above or below a horizontal offset.

917.8 Prohibited lower connections.

Stacks greater than 2 branch intervals in height shall not receive the discharge of horizontal branches on the lower two floors. There shall be no connections to the stack between the lower two floors and a distance of not less than 10 pipe diameters downstream from the base of the single stack vented system.

917.9 Sizing building drains and sewers.

The building drain and building sewer receiving the discharge of a single stack vent system shall be sized in accordance with Table 710.1(1).

SECTION 918 AIR ADMITTANCE VALVES

918.1 General.

Vent systems utilizing air admittance valves shall comply with this section. Stack-type air admittance valves shall conform to ASSE 1050. Individual and branch type air admittance valves shall conform to ASSE 1051.

918.2 Installation.

The valves shall be installed in accordance with the requirements of this section and the manufacturer's instructions. Air admittance valves shall be installed after the DWV testing required by Section 312.2 or 312.3 has been performed.

918.3 Where permitted.

Individual, branch and circuit vents shall be permitted to terminate with a connection to an individual or branch-type air admittance valve in accordance with Section 918.3.1. Stack vents and vent stacks shall be permitted to terminate to stack-type air admittance valves in accordance with Section 918.3.2.

918.3.1 Horizontal branches.

Individual and branch-type air admittance valves shall vent only fixtures that are on the same floor level and connect to a horizontal branch drain. Where the horizontal branch is located more than four branch intervals from the top of the stack, the horizontal branch shall be provided with a relief vent that shall connect to a vent stack or stack vent, or extend outdoors to the open air. The relief vent shall connect to the horizontal branch drain between the stack and the most downstream fixture drain connected to the horizontal branch drain. The

relief vent shall be sized in accordance with Section 906.2 and installed in accordance with Section 905. The relief vent shall be permitted to serve as the vent for other fixtures.

918.3.2 Stack.

Stack-type air admittance valves shall be prohibited from serving as the vent terminal for vent stacks or stack vents that serve drainage stacks having more than six branch intervals.

918.4 Location.

Individual and branch-type air admittance valves shall be located a minimum of 100 mm above the horizontal branch drain or fixture drain being vented. Stack-type air admittance valves shall be located not less than 150 mm above the flood level rim of the highest fixture being vented. The air admittance valve shall be located within the maximum developed length permitted for the vent. The air admittance valve shall be installed not less than 150 mm above insulation materials.

918.5 Access and ventilation.

Access shall be provided to all air admittance valves. Such valves shall be installed in a location that allows air to enter the valve.

918.6 Size.

The air admittance valve shall be rated in accordance with the standard for the size of the vent to which the valve is connected.

918.7 Vent required.

Within each plumbing system, not less than one stack vent or vent stack shall extend outdoors to the open air.

918.8 Prohibited installations.

Air admittance valves shall not be installed in non-neutralized special waste systems as described in Chapter 8 except where such valves are in compliance with ASSE 1049, are constructed of materials approved in accordance with Section 702.5 and are tested for chemical resistance in accordance with ASTM F 1412. Air admittance valves shall not be located in spaces utilized as supply or return air plenums. Air admittance valves without an engineered design shall not be utilized to vent sumps or tanks of any type.

SECTION 919 ENGINEERED VENT SYSTEMS

919.1 General.

Engineered vent systems shall comply with this section and the design, submittal, approval, inspection and testing requirements of Section 105.4.

919.2 Individual branch fixture and individual fixture header vents.

The maximum developed length of individual fixture vents to vent branches and vent headers shall be determined in accordance with Table 919.2 for the minimum pipe diameters at the indicated vent airflow rates.

The individual vent airflow rate shall be determined in accordance with the following:

$$Q_{h,b} = N_{n,b} Q_v (0.4719)$$

(Equation 9-1)

where:

$N_{n,b}$ = Number of fixtures per header (or vent *branch*) ÷ total number of fixtures connected to vent *stack*.

$Q_{h,b}$ = Vent branch or vent header airflow rate (L/s).

Q_v = Total vent stack airflow rate (L/s).

$$Q_v \text{ (L/s)} = \frac{1}{3178.8} r_s^{2/3} (1 - r_s) D^{8/3}$$

where:

D = Drainage stack diameter (mm).

Q_w = Design discharge load (L/sec.).
Wastewater flow area to total area

r_s = (proportion of cross section occupied by falling water)

$$= 3178.80 \frac{Q_w}{D^{8/3}}$$

Individual vent airflow rates are obtained by equally distributing $Q_{h,b}$ into one-half the total number of fixtures on the *branch* or header for more than two fixtures; for an odd number of total fixtures, decrease by one; for one fixture, apply the full value of $Q_{h,b}$.

Individual vent *developed length* shall be increased by 20 percent of the distance from the vent *stack* to the fixture vent connection on the vent *branch* or header

SECTION 920 COMPUTERIZED VENT DESIGN

920.1 Design of vent system.

The sizing, design and layout of the vent system shall be permitted to be determined by approved computer program design methods.

920.2 System capacity.

The vent system shall be based on the air capacity requirements of the drainage system under a peak load condition.

TABLE 919.2 Minimum Diameter and Maximum Length of Individual Branch Fixture Vents and Individual Fixture Header Vents For Smooth Pipes

DIAMETER OF VENT PIPE (mm)	INDIVIDUAL VENT AIRFLOW RATE (Liters per minute)																			
	Maximum developed length of vent (m)																			
	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3	3.3	3.6	3.9	4.2	4.5	4.8	5.2	5.5	5.8	6
15	2690	708	368	227	142	113	85	57	28	28	28	28	28	28	28	28	28	28	28	28
20	2831	2492	1331	849	566	425	283	255	198	170	142	113	85	85	57	57	57	57	57	28
25	—	—	2813	2662	1840	1359	1048	821	680	566	481	396	340	311	255	227	198	198	170	170
32	—	—	—	—	—	—	—	2813	2463	2067	1755	1500	1302	1133	1019	906	821	736	651	595
40	—	—	—	—	—	—	—	—	—	—	—	2813	2718	2378	2124	1840	1699	1529	1387	1274
50	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	2813



CHAPTER 10 TRAPS, INTERCEPTORS AND SEPARATORS

SECTION 1001 GENERAL

1001.1 Scope.

This chapter shall govern the material and installation of traps, interceptors and separators.

SECTION 1002 TRAP REQUIREMENTS

1002.1 Fixture traps.

Each plumbing fixture shall be separately trapped by a liquid-seal trap, except as otherwise permitted by this code. The vertical distance from the fixture outlet to the trap weir shall not exceed 600 mm, and the horizontal distance shall not exceed 750 mm measured from the centerline of the fixture outlet to the centerline of the inlet of the trap. The height of a clothes washer standpipe above a trap shall conform to Section 802.3.3. A fixture shall not be double trapped.

Exceptions:

1. This section shall not apply to fixtures with integral traps.
2. A combination plumbing fixture is permitted to be installed on one trap, provided that one compartment is not more than 150 mm deeper than the other compartment and the waste outlets are not more than 750 mm apart.
3. A grease interceptor intended to serve as a fixture trap in accordance with the manufacturer's installation instructions shall be permitted to serve as the trap for a single fixture or a combination sink of not more than three compartments where the vertical distance from the fixture outlet to the inlet of the interceptor does not exceed 750 mm and the *developed length* of the waste pipe from the most upstream fixture outlet to the inlet of the interceptor does not exceed 1500 mm.
4. Floor drains in multilevel parking structures that discharge to a building storm sewer shall not be required to be individually trapped. Where floor drains in multilevel parking structures are required to discharge to a combined *building sewer* system, the floor drains shall not be required to be individually trapped

provided that they are connected to a main trap in accordance with Section 1103.1.

1002.2 Design of traps.

Fixture traps shall be self-scouring. Fixture traps shall not have interior partitions, except where such traps are integral with the fixture or where such traps are constructed of an approved material that is resistant to corrosion and degradation. Slip joints shall be made with an approved elastomeric gasket and shall be installed only on the trap inlet, trap outlet and within the trap seal.

1002.3 Prohibited traps.

The following types of traps are prohibited:

1. Traps that depend on moving parts to maintain the seal.
- 4 Bell traps.
- 5 Crown-vented traps.
- 6 Traps not integral with a fixture and that depend on interior partitions for the seal, except those traps constructed of an approved material that is resistant to corrosion and degradation.
- 7 "S" traps.
- 8 Drum traps.

Exception: Drum traps used as solids interceptors and drum traps serving chemical waste systems shall not be prohibited.

1002.4 Trap seals.

Each fixture trap shall have a liquid seal of not less than 50 mm and not more than 100 mm, or deeper for special designs relating to accessible fixtures.

1002.4.1 Trap seal protection.

Trap seals of emergency floor drain traps and trap seals subject to evaporation shall be protected by one of the methods in Sections 1002.4.1.1 through 1002.4.1.4.

1002.4.1.1 Potable water-supplied trap seal primer valve.

A potable water-supplied trap seal primer valve

shall supply water to the trap. Water-supplied trap seal primer valves shall conform to ASSE 1018. The discharge pipe from the trap seal primer valve shall connect to the trap above the trap seal on the inlet side of the trap.

1002.4.1.2 Reclaimed or gray water-supplied trap seal primer valve.

A reclaimed or gray water-supplied trap seal primer valve shall supply water to the trap. Water-supplied trap seal primer valves shall conform to ASSE 1018. The quality of reclaimed or gray water supplied to trap seal primer valves shall be in accordance with the requirements of the manufacturer of the trap seal primer valve. The discharge pipe from the trap seal primer valve shall connect to the trap above the trap seal, on the inlet side of the trap.

1002.4.1.3 Waste water-supplied trap primer device.

A waste water-supplied trap primer device shall supply water to the trap. Waste water-supplied trap primer devices shall conform to ASSE 1044. The discharge pipe from the trap seal primer device shall connect to the trap above the trap seal on the inlet side of the trap.

1002.4.1.4 Barrier-type trap seal protection device.

A barrier-type trap seal protection device shall protect the floor drain trap seal from evaporation. Barrier-type floor drain trap seal protection devices shall conform to ASSE 1072. The devices shall be installed in accordance with the manufacturer's instructions.

1002.5 Size of fixture traps.

Fixture trap size shall be sufficient to drain the fixture rapidly and not less than the size indicated in Table 709.1. A trap shall not be larger than the drainage pipe into which the trap discharges.

1002.6 Building traps.

Building (house) traps shall be prohibited.

1002.7 Trap setting and protection.

Traps shall be set level with respect to the trap seal and, where necessary, shall be protected from freezing.

1002.8 Recess for trap connection.

A recess provided for connection of the

underground trap, such as one serving a bathtub in slab-type construction, shall have sides and a bottom of corrosion-resistant, insect- and verminproof construction.

1002.9 Acid-resisting traps.

Where a vitrified clay or other brittle ware, acid-resisting trap is installed underground, such trap shall be embedded in concrete extending 150 mm beyond the bottom and sides of the trap.

1002.10 Plumbing in mental health centers.

In mental health centers, pipes and traps shall not be exposed.

SECTION 1003 INTERCEPTORS AND SEPARATORS

1003.1 Where required.

Interceptors and separators shall be provided to prevent the discharge of oil, grease, sand and other substances harmful or hazardous to the public sewer, the private sewage system or the sewage treatment plant or processes.

1003.2 Approval.

The size, type and location of each interceptor and of each separator shall be designed and installed in accordance with the manufacturer's instructions and the requirements of this section based on the anticipated conditions of use. Wastes that do not require treatment or separation shall not be discharged into any interceptor or separator.

1003.3 Grease interceptors.

Grease interceptors shall comply with the requirements of Sections 1003.3.1 through 1003.3.5.

1003.3.1 Grease interceptors and automatic grease removal devices required.

A grease interceptor or automatic grease removal device shall be required to receive the drainage from fixtures and equipment with grease laden waste located in food preparation areas, such as in restaurants, hotel kitchens, hospitals, school kitchens and factory cafeterias. Fixtures and equipment shall include pot sinks, prerinse sinks; soup kettles or similar devices; work stations; floor drains or sinks into which kettles are drained; automatic hood wash

units and dishwashers without prerinse sinks. Grease interceptors and automatic grease removal devices shall receive waste only from fixtures and equipment that allow fats, oils or grease to be discharged. Where lack of space or other constraints prevent the installation or replacement of a grease interceptor, one or more grease interceptors shall be permitted to be installed on or above the floor and upstream of an existing grease interceptor.

1003.3.2 Food waste disposers.

Where food waste disposers connect to grease interceptors, a solids interceptor shall separate the discharge before connecting to the grease interceptor. Solids interceptors and grease interceptors shall be sized and rated for the discharge of the food waste disposers. Emulsifiers, chemicals, enzymes and bacteria shall not discharge into the food waste disposer.

1003.3.3 Grease interceptors and automatic grease removal devices not required.

A grease interceptor or an automatic grease removal device shall not be required for individual dwelling units or any private living quarters.

1003.3.4 Hydro-mechanical grease interceptors, fats, oils and

greases disposal systems and automatic grease removal devices.

Hydro-mechanical grease interceptors; fats, oils, and greases disposal systems and automatic grease removal devices shall be sized in accordance with ASME A112.14.3, ASME A112.14.4, ASME A112.14.6, CSA B481.3 or PDI G101. Hydro-mechanical grease interceptors; fats, oils, and greases disposal systems and automatic grease removal devices shall be designed and tested in accordance with ASME A112.14.3, ASME A112.14.4, CSA B481.1, PDI G101 or PDI G102. Hydro-mechanical grease interceptors; fats, oils, and greases disposal systems and automatic grease removal devices shall be installed in accordance with the manufacturer's instructions. Where manufacturer's instructions are not provided, hydro-mechanical grease interceptors; fats, oils, and greases disposal systems and automatic grease removal devices shall be installed in compliance with ASME A112.14.3, ASME A112.14.4, ASME A112.14.6, CSA B481.3 or PDI G101.

1003.3.4.1 Grease interceptor capacity.

Grease interceptors shall have the grease retention capacity indicated in Table 1003.3.4.1 for the flow-through rates indicated.

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Table 1003.3.4.1 Capacity of Grease Interceptorsa

TOTAL FLOW-THROUGH RATING (Lpm)	GREASE RETENTION CAPACITY (kg)
15	3.6
23	5.5
26	6.5
34	8
38	9
45	11
53	13
57	14
68	16
76	18
95	23
132	32
189	45
284	68
380	91

- a. For total flow-through ratings greater than 380 (Lpm), double the flow-through rating to determine the grease retention capacity.

1003.3.4.2 Rate of flow controls.

Grease interceptors shall be equipped with devices to control the rate of water flow so that the water flow does not exceed the rated flow. The flow-control device shall be vented and terminate not less than 150 mm above the flood rim level or be installed in accordance with the manufacturer's instructions.

1003.3.5 Automatic grease removal devices.

Where automatic grease removal devices are installed, such devices shall be located downstream of each fixture or multiple fixtures in accordance with the manufacturer's instructions. The automatic grease removal device shall be sized to pre-treat the measured or calculated flows for all connected fixtures or equipment. Ready access shall be provided for inspection and maintenance.

1003.3.6 Gravity grease interceptors and gravity grease interceptors with fats, oils, and greases disposal systems.

The required capacity of gravity grease interceptors and gravity grease interceptors with fats, oils, and greases disposal systems shall be determined by multiplying the peak drain flow into the interceptor in liter per minute by a retention time of 30 minutes. Gravity grease interceptors shall be designed

and tested in accordance with IAPMO/ANSI Z1001. Gravity grease interceptors with fats, oils, and greases disposal systems shall be designed and tested in accordance with ASME A112.14.6 and IAPMO/ANSI Z1001. Gravity grease interceptors and gravity grease interceptors with fats, oils, and greases disposal systems shall be installed in accordance with manufacturer's instructions. Where manufacturer's instructions are not provided, gravity grease interceptors and gravity grease interceptors with fats, oils, and greases disposal systems shall be installed in compliance with ASME A112.14.6 and IAPMO/ANSI Z1001.

1003.3.7 Direct connection.

The discharge piping from a grease interceptor shall be directly connected to the sanitary drainage system.

1003.4 Oil separators required.

At repair garages where floor or trench drains are provided, car washing facilities, factories where oily and flammable liquid wastes are produced and hydraulic elevator pits, oil separators shall be installed into which oil-bearing, grease-bearing or flammable wastes shall be discharged before emptying into the building drainage system or other point of disposal.

Exception: An oil separator is not required in hydraulic elevator pits where an approved alarm system is installed. Such alarm systems

shall not terminate the operation of pumps utilized to maintain emergency operation of the elevator by fire fighters.

1003.4.1 Separation of liquids.

A mixture of treated or untreated light and heavy liquids with various specific gravities shall be separated in an approved receptacle.

1003.4.2 Oil separator design.

Oil separators shall be listed and labeled, or designed in accordance with Sections 1003.4.2.1 and 1003.4.2.2.

1003.4.2.1 General design requirements.

Oil separators shall have a depth of not less than 600 mm below the invert of the discharge drain. The outlet opening of the separator shall have not less than an 500mm water seal.

1003.4.2.2 Garages and service stations.

Where automobiles are serviced, greased, repaired or washed or where gasoline is dispensed, oil separators shall have a capacity of not less than 0.18 m³ for the first 10 m² of area to be drained, plus 0.03 m³ for each additional 10 m² of area to be drained into the separator. Parking garages in which servicing, repairing or washing is not conducted, and in which gasoline is not dispensed, shall not require a separator. Areas of commercial garages utilized only for storage of automobiles are not required to be drained through a separator.

1003.5 Sand interceptors in commercial establishments.

Sand and similar interceptors for heavy solids shall be designed and located so as to be provided with ready access for cleaning, and shall have a water seal of not less than 150 mm.

1003.6 Clothes washer discharge interceptor.

Clothes washers shall discharge through an interceptor that is provided with a wire basket or similar device, removable for cleaning, that prevents passage into the drainage system of solids 13 mm or larger in size, string, rags, buttons or other materials detrimental to the public sewage system.

Exceptions:

1. Clothes washers in individual dwelling units shall not be required to discharge through an interceptor.
2. A single clothes washer designed for use in individual dwelling units and installed in a location other than an individual dwelling unit shall not be required to discharge through an interceptor.

1003.7 Bottling establishments.

Bottling plants shall discharge process wastes into an interceptor that will provide for the separation of broken glass or other solids before discharging waste into the drainage system.

1003.8 Slaughterhouses.

Slaughtering room and dressing room drains shall be equipped with approved separators. The separator shall prevent the discharge into the drainage system of feathers, entrails and other materials that cause clogging.

1003.9 Venting of interceptors and separators.

Interceptors and separators shall be designed so as not to become air bound. Interceptors and separators shall be vented in accordance with one of the methods in Chapter 9.

1003.10 Access and maintenance of interceptors and separators.

Access shall be provided to each interceptor and separator for service and maintenance. Interceptors and separators shall be maintained by periodic removal of accumulated grease, scum, oil, or other floating substances and solids deposited in the interceptor or separator.

SECTION 1004 MATERIALS, JOINTS AND CONNECTIONS

1004.1 General.

The materials and methods utilized for the construction and installation of traps, interceptors and separators shall comply with this chapter and the applicable provisions of Chapters 4 and 7. The fittings shall not have ledges, shoulders or reductions capable of retarding or obstructing flow of the piping.

CHAPTER 11 STORM DRAINAGE

SECTION 1101 GENERAL

1101.1 Scope.

The provisions of this chapter shall govern the materials, design, construction and installation of storm drainage.

1101.2 Disposal.

Rainwater from roofs and storm water from paved areas, yards, courts and courtyards shall drain to an approved place of disposal. For one- and two-family dwellings, and where approved, storm water is permitted to discharge onto flat areas, such as streets or lawns, provided that the storm water flows away from the building.

1101.3 Prohibited drainage.

Storm water shall not be drained into sewers intended for sewage only.

1101.4 Tests.

The conductors and the building storm drain shall be tested in accordance with Section 312.

1101.5 Change in size.

The size of a drainage pipe shall not be reduced in the direction of flow.

1101.6 Fittings and connections.

All connections and changes in direction of the storm drainage system shall be made with approved drainage-type fittings in accordance with Table 706.3. The fittings shall not obstruct or retard flow in the system.

1101.7 Roof design.

Roofs shall be designed for the maximum possible depth of water that will pond thereon as determined by the relative levels of roof deck and overflow weirs, scuppers, edges or serviceable drains in combination with the deflected structural elements. In determining the maximum possible depth of water, all primary roof drainage means shall be assumed to be blocked. The maximum possible depth of water on the roof shall include the height of the water required above the inlet of the secondary roof drainage means to achieve the required flow rate of the secondary drainage means to accommodate the design rainfall rate as required by Section 1106.

1101.8 Cleanouts required.

Cleanouts shall be installed in the storm drainage system and shall comply with the provisions of this code for sanitary drainage pipe cleanouts.

Exception: Subsurface drainage system.

1101.9 Backwater valves.

Storm drainage systems shall be provided with backwater valves as required for sanitary drainage systems in accordance with Section 715.

SECTION 1102 MATERIALS

1102.1 General.

The materials and methods utilized for the construction and installation of storm drainage systems shall comply with this section and the applicable provisions of Chapter 7.

1102.2 Inside storm drainage conductors.

Inside storm drainage conductors installed above ground shall conform to one of the standards listed in Table 702.1.

1102.3 Underground building storm drain pipe.

Underground building storm drain pipe shall conform to one of the standards listed in Table 702.2.

1102.4 Building storm sewer pipe.

Building storm sewer pipe shall conform to one of the standards listed in Table 1102.4.

1102.5 Subsoil drain pipe.

Subsoil drains shall be open jointed, horizontally split or perforated pipe conforming to one of the standards listed in Table 1102.5.

1102.6 Roof Drains.

Roof drains shall conform to ASME A112.6.4 or ASME A112.3.1.

1102.7 Fittings.

Pipe fittings shall be approved for installation with the piping material installed, and shall conform to the respective pipe standards or one

of the standards listed in Table 1102.7. The fittings shall not have ledges, shoulders or reductions capable of retarding or obstructing

flow in the piping. Threaded drainage pipe fittings shall be of the recessed drainage type.

TABLE 1102.4 BUILDING STORM SEWER PIPE

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic pipe	ASTM D 2661; ASTM D 2751; ASTM F 628; CSA B181.1; CSA B182.1
Cast-iron pipe	ASTM A 74; ASTM A 888; CISPI 301
Concrete pipe	ASTM C 14; ASTM C 76; CSA A257.1M; CSA A257.2M
Copper or copper-alloy tubing (Type K, L, M or DWV)	ASTM B 75; ASTM B 88; ASTM B 251; ASTM B 306
Polyethylene (PE) plastic pipe	ASTM F 2306/F 2306M
Polyvinyl chloride (PVC) plastic pipe (Type DWV, SDR26, SDR35, SDR41, PS50 or PS100)	ASTM D 2665; ASTM D 3034; ASTM F 891; CSA B182.4; CSA B181.2; CSA B182.2
Vitrified clay pipe	ASTM C 4; ASTM C 700
Stainless steel drainage systems, Type 316L	ASME A112.3.1

TABLE 1102.5 SUBSOIL DRAIN PIPE

MATERIAL	STANDARD
Cast-iron pipe	ASTM A 74; ASTM A 888; CISPI 301
Polyethylene (PE) plastic pipe	ASTM F 405; CSA B182.1; CSA B182.6; CSA B182.8
Polyvinyl chloride (PVC) Plastic pipe (type sewer pipe, SDR35, PS25, PS50 or PS100)	ASTM D 2729; ASTM D 3034, ASTM F 891; CSA B182.2; CSA B182.4
Stainless steel drainage systems, Type 316L	ASME A 112.3.1
Vitrified clay pipe	ASTM C 4; ASTM C 700

TABLE 1102.7 PIPE FITTING

MATERIAL	STANDARD
Acrylonitrile butadiene styrene (ABS) plastic	ASTM D 2661; ASTM D 3311; CSA B181.1
Cast-iron	ASME B16.4; ASME B16.12; ASTM A 888; CISPI 301; ASTM A 74
Coextruded composite ABS and drain DR-PS in PS35, PS50, PS100, PS140, PS200	ASTM D 2751
Coextruded composite ABS DWV Schedule 40 IPS pipe (solid or cellular core)	ASTM D 2661; ASTM D 3311; ASTM F 628
Coextruded composite PVC DWV Schedule 40 IPS-DR, PS140, PS200 (solid or cellular core)	ASTM D 2665; ASTM D 3311; ASTM F 891
Coextruded composite PVC sewer and drain DR-PS in PS35, PS50, PS100, PS140, PS200	ASTM D 3034
Copper or copper alloy	ASME B16.15; ASME B16.18; ASME B16.22; ASME B16.23; ASME B16.26; ASME B16.29
Gray iron and ductile iron	AWWA C110/A21.10
Malleable iron	ASME B16.3
Plastic, general	ASTM F 409
Polyethylene (PE) plastic pipe	ASTM F 2306/F 2306M
Polyvinyl chloride (PVC) plastic	ASTM D 2665; ASTM D 3311; ASTM F 1866
Steel	ASME B16.9; ASME B16.11; ASME B16.28
Stainless steel drainage systems, Type 316L	ASME A112.3.1

SECTION 1103 TRAPS**1103.1 Main trap.**

Leaders and storm drains connected to a combined sewer shall be trapped. Individual storm water traps shall be installed on the storm water drain branch serving each conductor, or a single trap shall be installed in the main storm drain just before its connection with the combined building sewer or the public sewer. Leaders and storm drains connected to a building storm sewer shall not be required to be trapped.

1103.2 Material.

Storm water traps shall be of the same material as the piping system to which they are attached.

1103.3 Size.

Traps for individual conductors shall be the same size as the horizontal drain to which they are connected.

1103.4 Cleanout.

An accessible cleanout shall be installed on the building side of the trap.

SECTION 1104 CONDUCTORS AND CONNECTIONS**1104.1 Prohibited use.**

Conductor pipes shall not be used as soil, waste or vent pipes, and soil, waste or vent pipes shall not be used as conductors.

1104.2 Floor drains.

Floor drains shall not be connected to a storm drain.

SECTION 1105 ROOF DRAINS**1105.1 General.**

Roof drains shall be installed in accordance with the manufacturer's instructions. The inside opening for the roof drain shall not be obstructed by the roofing membrane material.

1105.2 Roof drain flow rate.

The published roof drain flow rate, based on the head of water above the roof drain, shall be used to size the storm drainage system in accordance with Section 1106. The flow rate used for sizing the storm drainage piping shall be based on the maximum anticipated ponding at the roof drain.

SECTION 1106 SIZE OF CONDUCTORS, LEADERS AND STORM DRAINS**1106.1 General.**

The size of the vertical conductors and leaders, building storm drains, building storm sewers, and any horizontal branches of such drains or sewers shall be based on the 100-year hourly rainfall rate indicated in Figure 1106.1 or on other rainfall rates determined from approved local weather data.

1106.2 Size of storm drain piping.

Vertical and horizontal storm drain piping shall be sized based on the flow rate through the roof drain. The flow rate in storm drain piping shall not exceed that specified in Table 1106.2.

1106.3 Vertical leader sizing.

Vertical leaders shall be sized based on the flow rate from horizontal gutters or the maximum flow rate through roof drains. The flow rate through vertical leaders shall not exceed that specified in Table 1106.3.

1106.4 Vertical walls.

In sizing roof drains and storm drainage piping, one-half of the area of any vertical wall that diverts rainwater to the roof shall be added to the projected roof area for inclusion in calculating the required size of vertical conductors, leaders and horizontal storm drainage piping.

1106.5 Parapet wall scupper location.

Parapet wall roof drainage scupper and overflow scupper location shall comply with the SBC 201.

1106.6 Size of roof gutters.

Horizontal gutters shall sized be based on the flow rate from the roof surface. The flow rate in horizontal gutters shall not exceed that specified in Table 1106.6 (1) and Table 1106.6 (2).

1106.7 Vertical conductors and leaders.

Vertical conductors and leaders shall be sized for the maximum projected roof area, in accordance with Table 1106.7.

1106.8 Building storm drains and sewers.

The size of the building storm drain, building storm sewer and their horizontal branches having a slope of 4-percent slope shall be based on the maximum projected roof area in accordance with Table 1106.8. The minimum

slope of horizontal branches shall be 1-percent slope unless otherwise approved.

TABLE 1106.2 STORM DRAIN PIPE SIZING

PIPE SIZE (mm)	VERTICAL DRAIN	CAPACITY (Liters per minute)			
		SLOPE OF HORIZONTAL DRAIN			
		5 mm per meter	10 mm per meter	20 mm per meter	40 mm per meter
50	130	60	80	120	170
80	330	150	210	300	420
100	680	310	440	620	870
125	1180	440	620	890	1250
150	2040	920	1300	1840	2610
200	4230	1910	2700	3820	5410
250	7760	3510	4960	7020	9930
300	12390	5600	7920	11200	15850
375	20980	9490	13420	18990	26850

TABLE 1106.3 VERTICAL LEADER SIZING

SIZE OF LEADER (mm)	CAPACITY Liters per minute
50	110
50 x 50	110
40 x 65	110
65	200
65 x 65	200
75	350
50 x 100	350
65 x 75	350
100	730
75 x 106	730
90 x 100	730
125	1360
100 x 125	1360
115 x 115	1360
150	2130
125 x 150	2130
140 x 140	2130
200	4570
150 x 200	4570

TABLE 1106.6 (1) Horizontal Gutter Sizing

GUTTER DIMENSIONS ^a (mm)	SLOPE (mm per meter)	CAPACITY (Liters per minute)
40 x 65	21	98
40 x 65	42	151
100	10	147
56 x 80	21	208
56 x 80	42	330
125	10	280
100 x 65	21	400
80 x 90	42	590
150	10	416
80 x 125	21	594
80 x 125	42	850
200	5	650
200	10	935
115 x 150	21	1317
115 x 150	42	1870
250	5	1253
250	10	1787
125 x 200	21	2465
100 x 250	42	4000

a. Dimensions are width by depth for rectangular shapes. Single dimensions are diameters of a semicircle.

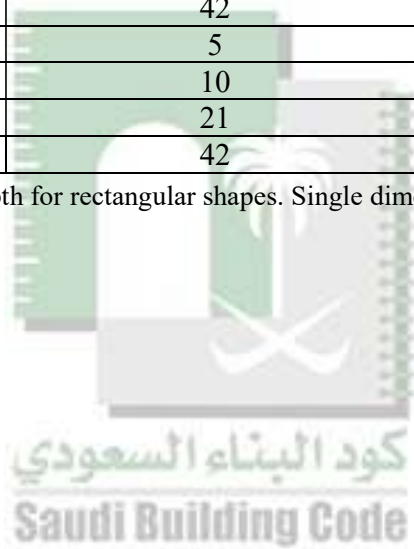


TABLE 1106.6 (2) SIZE SEMICIRCULAR ROOF GUTTERS

DIAMETER OF GUTTER ^a (mm)	HORIZONTALLY PROJECTED ROOF AREA (square meters)					
	Rainfall rate (mm per hour)					
	25	50	75	100	125	150
0-5 percent slope						
80	65	30	20	15	15	10
100	135	65	45	35	25	20
125	230	115	75	60	45	40
150	355	180	120	90	70	60
175	515	255	170	130	100	85
200	740	370	245	185	150	125
250	1340	670	445	335	270	225
1-percent slope						
80	90	45	30	20	20	15
100	190	95	65	45	40	30
125	325	165	110	80	65	55
150	505	255	170	125	100	85
175	725	360	240	180	145	120
200	1040	520	345	260	210	175
250	1895	950	630	475	380	315
2-percent slope						
80	125	65	40	30	25	20
100	270	135	90	65	55	45
125	465	230	155	115	95	75
150	715	355	240	180	145	120
175	1025	515	360	255	205	170
200	1480	740	495	370	295	245
250	2675	1340	890	670	535	445
4-percent slope						
80	180	90	60	45	35	30
100	380	190	125	95	75	65
125	660	330	220	165	130	110
150	1030	515	345	255	205	170
175	1450	725	485	360	290	240
200	2080	1040	695	520	415	345
250	3715	1860	1240	930	745	620

- a. Shown sizes are nominal sizes based on internal diameters. If the pipe nominal size is based on the external diameter, find the internal diameter (from SASO standards or pipe's manufacturer) and use a pipe size with an internal diameter equal to or greater
- b. than the size shown in the table.

TABLE 1106.7 SIZE OF SEMICIRCULAR ROOF GUTTERS

DIAMETER OF LEADER (mm) ^a	HORIZONTALLY PROJECTED ROOF AREA (Square meters)											
	Rainfall rate (mm per hour) ^b											
	25	50	75	100	125	150	175	200	225	250	275	300
50	270	135	90	67	53	45	38	33	30	27	24	22
80	818	409	272	204	164	137	117	102	91	82	74	68
100	1709	855	569	427	342	285	244	214	190	171	156	142
125	3214	1607	1071	804	643	536	459	402	357	321	292	268
150	5017	2508	1627	1254	1003	836	717	627	557	502	456	418
200	10776	5388	3592	2694	2155	1794	1539	1347	1197	1078	980	892

- a. Sizes indicated are the diameter of circular piping. This table is applicable to piping of other shapes provided the cross-sectional shape fully encloses a circle of the diameter indicated in this table.
- b. Shown sizes are nominal sizes based on internal diameters. If the pipe nominal size is based on the external diameter, find the internal diameter (from SASO standards or pipe's manufacturer) and use a pipe size with an internal diameter equal to or greater than the size shown in the table.



TABLE 1106.8
SIZE OF HORIZONTAL STORM DRAINAGE PIPING

SIZE OF HORIZONTAL PIPING ^a (mm)	HORIZONTALLY PROJECTED ROOF AREA (square meters)					
	Rainfall rate (mm per hour)					
	25	50	75	100	125	150
1-percent slope						
80	305	155	100	75	60	50
100	700	350	135	165	140	115
125	1240	620	415	310	250	205
150	1990	995	665	495	400	330
200	4275	2135	1425	1070	855	705
250	7690	3845	2565	1925	1540	1280
300	12375	6185	4125	3095	2475	2060
350	20250	10125	6765	5530	4420	3685
2-percent slope						
80	430	215	145	110	85	70
100	985	490	330	245	195	165
125	1755	875	585	440	350	290
150	2805	1405	935	700	560	470
200	6055	3030	2020	1515	1210	1010
250	10850	5425	3620	2715	2170	1805
300	17465	8735	5815	4365	3495	2910
350	31215	15605	10405	7805	6250	5200
4-percent slope						
80	610	305	215	155	120	100
100	1395	700	465	350	280	230
125	2480	1240	825	620	495	415
150	3975	1990	1275	995	795	665
200	8545	4275	2845	2135	1710	1425
250	15940	7970	5130	3845	3080	2565
300	24750	12375	8250	6185	4940	4125
350	44220	22110	14755	11055	8855	7360

- a. Shown sizes are nominal sizes based on internal diameters. If the pipe nominal size is based on the external diameter, find the internal diameter (from SASO standards or pipe's manufacturer) and use a pipe size with an internal diameter equal to or greater than the size shown in the table.

SECTION 1107 SIPHONIC ROOF DRAINAGE SYSTEMS

1107.1 General.

Siphonic roof drains and drainage systems shall be designed in accordance with ASME A112.6.9 and ASPE 45.

SECTION 1108 SECONDARY (EMERGENCY) ROOF DRAINS

1108.1 Secondary (emergency overflow) drains or scuppers.

Where roof drains are required, secondary (emergency overflow) roof drains or scuppers shall be provided where the roof perimeter construction extends above the roof in such a manner that water will be entrapped if the primary drains allow buildup for any reason. Where primary and secondary roof drains are manufactured as a single assembly, the inlet and outlet for each drain shall be independent.

1108.2 Separate systems required.

Secondary roof drain systems shall have the end point of discharge separate from the primary system. Discharge shall be above grade, in a location that would normally be observed by the building occupants or maintenance personnel.

1108.3 Sizing of secondary drains.

Secondary (emergency) roof drain systems shall be sized in accordance with Section 1106 based on the rainfall rate for which the primary system is sized. Scuppers shall be sized to prevent the depth of ponding water from exceeding that for which the roof was designed as determined by Section 1101.7. Scuppers shall have an opening dimension of not less than 100 mm. The flow through the primary system shall not be considered when sizing the secondary roof drain system.

SECTION 1109 COMBINED SANITARY AND STORM PUBLIC SEWER

1109.1 General.

Where the public sewer is a combined system for both sanitary and storm water, the storm sewer shall be connected independently to the public sewer.

SECTION 1110 CONTROLLED FLOW ROOF DRAIN SYSTEMS

1110.1 General.

The roof of a structure shall be designed for the storage of water where the storm drainage system is engineered for controlled flow. The controlled flow roof drain system shall be an engineered system in accordance with this section and the design, submittal, approval, inspection and testing requirements of Section 105.4. The controlled flow system shall be designed based on the required rainfall rate in accordance with Section 1106.1.

1110.2 Control devices.

The control devices shall be installed so that the rate of discharge of water per minute shall not exceed the values for continuous flow as indicated in Section 1110.1.

1110.3 Installation.

Runoff control shall be by control devices. Control devices shall be protected by strainers.

1110.4 Minimum number of roof drains.

Not less than two roof drains shall be installed in roof areas 900 m² or less and not less than four roof drains shall be installed in roofs over 900 m² in area.

SECTION 1111 SUBSOIL DRAINS

1111.1 Subsoil drains.

Subsoil drains shall be open-jointed, horizontally split or perforated pipe conforming to one of the standards listed in Table 1102.5. Such drains shall not be less than 100 mm in diameter. Where the building is subject to backwater, the subsoil drain shall be protected by an accessibly located backwater valve. Subsoil drains shall discharge to a trapped area drain, sump, dry well or approved location above ground. The subsoil sump shall not be required to have either a gas-tight cover or a vent. The sump and pumping system shall comply with Section 1113.1.

SECTION 1112 BUILDING SUBDRAINS

1112.1 Building subdrains.

Building subdrains located below the public sewer level shall discharge into a sump or receiving tank, the contents of which shall be automatically lifted and discharged into the

drainage system as required for building sumps. The sump and pumping equipment shall comply with Section 1113.1.

SECTION 1113 SUMPS AND PUMPING SYSTEMS

1113.1 Pumping system.

The sump pump, pit and discharge piping shall conform to Sections 1113.1.1 through 1113.1.4.

1113.1.1 Pump capacity and head.

The sump pump shall be of a capacity and head appropriate to anticipated use requirements.

1113.1.2 Sump pit.

The sump pit shall be not less than 500 mm in diameter and not less than 600 mm in depth, unless otherwise approved. The pit shall be accessible and located such that all drainage flows into the pit by gravity. The sump pit shall be constructed of tile, steel, plastic, cast iron,

concrete or other approved material, with a removable cover adequate to support anticipated loads in the area of use. The pit floor shall be solid and provide permanent support for the pump.

1113.1.3 Electrical.

Electrical service outlets, when required, shall meet the requirements of NFPA 70.

1113.1.4 Piping.

Discharge piping shall meet the requirements of Section 1102.2, 1102.3 or 1102.4 and shall include a gate valve and a full flow check valve. Pipe and fittings shall be the same size as, or larger than, the pump discharge tapping.

Exception: In one- and two-family dwellings, only a check valve shall be required, located on the discharge piping from the pump or ejector.



To be provided later

FIGURE 1106.1
100-YEAR, 1-HOUR RAINFALL (mm) of Saudi Arabia



CHAPTER 12 SPECIAL PIPING AND STORAGE SYSTEMS

SECTION 1201 GENERAL

1201.1 Scope.

The provisions of this chapter shall govern the design and installation of piping and storage systems for nonflammable medical gas systems and nonmedical oxygen systems. All maintenance and operations of such systems shall be in accordance with the SBC 801.

SECTION 1202 MEDICAL GASES

1202.1 Nonflammable medical gases.

Nonflammable medical gas systems, inhalation anesthetic systems and vacuum piping systems

shall be designed and installed in accordance with NFPA 99.

Exceptions:

1. This section shall not apply to portable systems or cylinder storage.
2. Vacuum system exhaust terminations shall comply with the *SBC 501*.

SECTION 1203 OXYGEN SYSTEMS

1203.1 Design and installation.

Nonmedical oxygen systems shall be designed and installed in accordance with NFPA 55 and NFPA 51.



CHAPTER 13 NON-POTABLE WATER SYSTEMS

SECTION 1301 GENERAL

1301.1 Scope.

The provisions of Chapter 13 shall govern the materials, design, construction and installation of systems for the collection, storage, treatment and distribution of non-potable water. The use and application of non-potable water shall comply with laws, rules and ordinances applicable in the jurisdiction.

1301.2 Water quality.

Non-potable water for each end use application shall meet the minimum water quality requirements as established for the intended application by the laws, rules and ordinances applicable in the jurisdiction. Where non-potable water from different sources is combined in a system, the system shall comply with the most stringent of the requirements of this code that are applicable to such sources.

1301.2.1 Residual disinfectants.

Where chlorine is used for disinfection, the non-potable water shall contain not more than 4 mg/L of chloramines or free chlorine when tested in accordance with ASTM D 1253. Where ozone is used for disinfection, the non-potable water shall not contain gas bubbles having elevated levels of ozone at the point of use.

Exception: Reclaimed water sources shall not be required to comply with these requirements.

1301.2.2 Filtration required.

Non-potable water utilized for water closet and urinal flushing applications shall be filtered by a 100-micron or finer filter.

Exception: Reclaimed water sources shall not be required to comply with these requirements.

1301.3 Signage required.

Non-potable water outlets such as hose connections, open ended pipes and faucets shall be identified at the point of use for each outlet with signage that reads as follows: "Nonpotable water is utilized for [application name]. CAUTION: NON-POTABLE WATER – DO NOT DRINK." The words shall be legibly and indelibly printed on a tag or sign

constructed of corrosion-resistant waterproof material or shall be indelibly printed on the fixture. The letters of the words shall be not less than 13 mm in height and in colors in contrast to the background on which they are applied. In addition to the required wordage, the pictograph shown in Figure 1301.3 shall appear on the signage required by this section.

1301.4 Permits.

Permits shall be required for the construction, installation, alteration and repair of non-potable water systems. Construction documents, engineering calculations, diagrams and other such data pertaining to the non-potable water system shall be submitted with each permit application.

1301.5 Potable water connections.

Where a potable system is connected to a non-potable water system, the potable water supply shall be protected against backflow in accordance with Section 608.

1301.6 Approved components and materials.

Piping, plumbing components and materials used in collection and conveyance systems shall be manufactured of material approved for the intended application and compatible with any disinfection and treatment systems used.

1301.7 Insect and vermin control.

The system shall be protected to prevent the entrance of insects and vermin into storage tanks and piping systems. Screen materials shall be compatible with contacting system components and shall not accelerate the corrosion of system components.

1301.8 Freeze protection.

Where sustained freezing temperatures occur, provisions shall be made to keep storage tanks and the related piping from freezing.

1301.9 Non-potable water storage tanks.

Non-potable water storage tanks shall comply with Sections 1301.9.1 through 1301.9.11.

1301.9.1 Sizing.

The holding capacity of the storage tank shall

be sized in accordance with the anticipated demand.

1301.9.2 Location.

Storage tanks shall be installed above or below grade. Above-grade storage tanks shall be protected from direct sunlight and shall be constructed using opaque, UV-resistant materials such as, but not limited to, heavily tinted plastic, fiberglass, lined metal, concrete, wood, or painted to prevent algae growth, or shall have specially constructed sun barriers including, but not limited to, installation in garages, crawl spaces or sheds. Storage tanks and their manholes shall not be located directly under soil piping, waste piping or any source of contamination.

1301.9.3 Materials.

Where collected on site, water shall be collected in an approved tank constructed of durable, nonabsorbent and corrosion-resistant materials. The storage tank shall be constructed of materials compatible with any disinfection systems used to treat water upstream of the tank and with any systems used to maintain water quality in the tank. Wooden storage tanks that are not equipped with a makeup water source shall be provided with a flexible liner.

1301.9.4 Foundation and supports.

Storage tanks shall be supported on a firm base capable of withstanding the weight of the storage tank when filled to capacity. Storage tanks shall be supported in accordance with the SBC 201.

1301.9.4.1 Ballast.

Where the soil can become saturated, an underground storage tank shall be ballasted, or otherwise secured, to prevent the tank from floating out of the ground when empty. The combined weight of the tank and hold down ballast shall meet or exceed the buoyancy force of the tank. Where the installation requires a foundation, the foundation shall be flat and shall be designed to support the weight of the storage tank when full, consistent with the bearing capability of adjacent soil.

1301.9.4.2 Structural support.

Where installed below grade, storage tank installations shall be designed to withstand earth and surface structural loads without damage and with minimal deformation when

empty or filled with water.

1301.9.5 Makeup water.

Where an uninterrupted supply is required for the intended application, potable or reclaimed water shall be provided as a source of makeup water for the storage tank. The makeup water supply shall be protected against backflow in accordance with Section 608. A full-open valve located on the makeup water supply line to the storage tank shall be provided. Inlets to the storage tank shall be controlled by fill valves or other automatic supply valves installed to prevent the tank from overflowing and to prevent the water level from dropping below a predetermined point. Where makeup water is provided, the water level shall not be permitted to drop below the source water inlet or the intake of any attached pump.

1301.9.6 Overflow.

The storage tank shall be equipped with an overflow pipe having a diameter not less than that shown in Table 606.5.4. The overflow pipe shall be protected from insects or vermin and shall discharge in a manner consistent with storm water runoff requirements of the jurisdiction. The overflow pipe shall discharge at a sufficient distance from the tank to avoid damaging the tank foundation or the adjacent property. Drainage from overflow pipes shall be directed to prevent freezing on roof walkways. The overflow drain shall not be equipped with a shutoff valve. A cleanout shall be provided on each overflow pipe in accordance with Section 708.

1301.9.7 Access.

Not less than one access opening shall be provided to allow inspection and cleaning of the tank interior. Access openings shall have an approved locking device or other approved method of securing access. Below-grade storage tanks, located outside of the building, shall be provided with a manhole either not less than 600 mm square or with an inside diameter not less than 600 mm. Manholes shall extend not less than 100 mm above ground or shall be designed to prevent water infiltration. Finished grade shall be sloped away from the manhole to divert surface water. Manhole covers shall be secured to prevent unauthorized access. Service ports in manhole covers shall be not less than 200 mm in diameter and shall be not less than 100 mm above the finished grade level. The

service port shall be secured to prevent unauthorized access.

Exception: Storage tanks less than 3028 L in volume and installed below grade shall not be required to be equipped with a manhole, but shall have a service port not less than 200 mm in diameter.

1301.9.8 Venting.

Storage tanks shall be provided with a vent sized in accordance with Chapter 9 and based on the aggregate diameter of all tank influent pipes. The reservoir vent shall not be connected to sanitary drainage system vents. Vents shall be protected from contamination by means of an approved cap or U-bend installed with the opening directed downward. Vent outlets shall extend not less than 100 mm above grade or as necessary to prevent surface water from entering the storage tank. Vent openings shall be protected against the entrance of vermin and insects in accordance with the requirements of Section 1301.7.

1301.9.9 Draining of tanks.

Where tanks require draining for service or cleaning, tanks shall be drained by using a pump or by a drain located at the lowest point in the tank. The tank drain pipe shall discharge as required for overflow pipes and shall not be smaller in size than specified in Table 606.5.7. Not less than one cleanout shall be provided on each drain pipe in accordance with Section 708.

1301.9.10 Marking and signage.

Each non-potable water storage tank shall be labeled with its rated capacity. The contents of storage tanks shall be identified with the words "CAUTION: NON-POTABLE WATER – DO NOT DRINK." Where an opening is provided that could allow the entry of personnel, the opening shall be marked with the words, "DANGER – CONFINED SPACE." Markings shall be indelibly printed on the tank or on a tag or sign constructed of corrosion-resistant waterproof material that is mounted on the tank. The letters of the words shall be not less than 13 mm in height and shall be of a color in contrast with the background on which they are applied.

1301.9.11 Storage tank tests.

Storage tanks shall be tested in accordance with the following:

Storage tanks shall be filled with water to the

overflow line prior to and during inspection. All seams and joints shall be left exposed and the tank shall remain water tight without leakage for a period of 24 hours.

1. After 24 hours, supplemental water shall be introduced for a period of 15 minutes to verify proper drainage of the overflow system and that there are no leaks.
- 9 The tank drain shall be observed for proper operation.
- 10 The makeup water system shall be observed for proper operation and successful automatic shutoff of the system at the refill threshold shall be verified.

1301.10 System abandonment.

If the owner of an on-site non-potable water reuse system or rainwater collection and conveyance system elects to cease use of, or fails to properly maintain such system, the system shall be abandoned and shall comply with the following:

- 11 All system piping connecting to a utility-provided water system shall be removed or disabled.
- 12 The distribution piping system shall be replaced with an *approved* potable water supply piping system. Where an existing potable pipe system is already in place, the fixtures shall be connected to the existing system.
- 13 The storage tank shall be secured from accidental access by sealing or locking tank inlets and access points, or filling with sand or equivalent.

1301.11 Trenching requirements for non-potable water piping.

Non-potable water collection and distribution piping and reclaimed water piping shall be separated from the building sewer and potable water piping underground by 1500 mm of undisturbed or compacted earth. Non-potable water collection and distribution piping shall not be located in, under or above cesspools, septic tanks, septic tank drainage fields or seepage pits. Buried non-potable water piping shall comply with the requirements of Section 306.

Exceptions:

1. The required separation distance shall not apply where the bottom of the non-potable water pipe within 1500 mm of the *sewer* is not less than 300 mm above the top of the highest point of the *sewer* and the pipe materials conform to Table 702.3.
2. The required separation distance shall not apply where the bottom of the potable water service pipe within 1500 mm of the non-potable water pipe is a minimum of 300 mm above the top of the highest point of the non-potable water pipe and the pipe materials comply with the requirements of Table 605.4.
3. Non-potable water pipe is permitted to be located in the same trench with a *building sewer*, provided that such *sewer* is constructed of materials that comply with the requirements of Table 702.2.
4. The required separation distance shall not apply where a non-potable water pipe crosses a sewer pipe, provided that the pipe is sleeved to at least 1500 mm horizontally from the sewer pipe centerline on both sides of such crossing, with pipe materials that comply with Table 702.2.
5. The required separation distance shall not apply where a potable water service pipe crosses a non-potable water pipe, provided that the potable water service pipe is sleeved for a distance of at least 1500 mm horizontally from the centerline of the non-potable pipe on both sides of such crossing, with pipe materials that comply with Table 702.2.
6. Irrigation piping located outside of a building and downstream of the backflow preventer is not required to meet the trenching requirements where non-potable water is used for outdoor applications.

1301.12 Outdoor outlet access.

Sillcocks, hose bibbs, wall hydrants, yard hydrants and other outdoor outlets supplied by non-potable water shall be located in a locked vault or shall be operable only by means of a removable key.

SECTION 1302 ON-SITE NON-POTABLE WATER REUSE SYSTEMS**1302.1 General.**

The provisions of Section 1302 shall govern the construction, installation, alteration and repair of on-site non-potable water reuse systems for the collection, storage, treatment and distribution of on-site sources of non-potable water as permitted by the jurisdiction.

1302.2 Sources.

On-site non-potable water reuse systems shall collect waste discharge from only the following sources: bathtubs, showers, lavatories, clothes washers and laundry trays. Water from other approved non-potable sources including swimming pool backwash operations, air conditioner condensate, rainwater, cooling tower blow-down water, foundation drain water, steam system condensate, fluid cooler discharge water, food steamer discharge water, combination oven discharge water, industrial process water and fire pump test water shall also be permitted to be collected for reuse by on-site non-potable water reuse systems, as approved by the code official and as appropriate for the intended application.

1302.2.1 Prohibited sources.

Waste water containing urine or fecal matter shall not be diverted to on-site non-potable water reuse systems and shall discharge to the sanitary drainage system of the building or premises in accordance with Chapter 7. Reverse osmosis system reject water, water softener discharge water, kitchen sink waste water, dishwasher waste water and waste water discharged from wet-hood scrubbers shall not be collected for reuse in an on-site non-potable water reuse system.

1302.3 Traps.

Traps serving fixtures and devices discharging waste water to on-site non-potable water reuse systems shall comply with Section 1002.4.

1302.4 Collection pipe.

On-site non-potable water reuse systems shall utilize drainage piping approved for use in plumbing drainage systems to collect and convey untreated water for reuse. Vent piping approved for use in plumbing venting systems shall be utilized for vents in the gray water

system. Collection and vent piping materials shall comply with Section 702.

1302.4.1 Installation.

Collection piping conveying untreated water for reuse shall be installed in accordance with Section 704.

1302.4.2 Joints.

Collection piping conveying untreated water for reuse shall utilize joints approved for use with the distribution piping and appropriate for the intended applications as specified in Section 705.

1302.4.3 Size.

Collection piping conveying untreated water for reuse shall be sized in accordance with drainage sizing requirements specified in Section 710.

1302.4.4 Labeling and marking.

Additional marking of collection piping conveying untreated water for reuse shall not be required beyond that required for sanitary drainage, waste and vent piping by Chapter 7.

1302.5 Filtration.

Untreated water collected for reuse shall be filtered as required for the intended end use. Filters shall be accessible for inspection and maintenance. Filters shall utilize a pressure gauge or other approved method to provide indication when a filter requires servicing or replacement. Filters shall be installed with shutoff valves immediately upstream and downstream to allow for isolation during maintenance.

1302.6 Disinfection and treatment.

Where the intended application for non-potable water collected on site for reuse requires disinfection or other treatment or both, it shall be disinfected as needed to ensure that the required water quality is delivered at the point of use. Non-potable water collected on site containing untreated gray water shall be retained in collection reservoirs for a maximum of 24 hours.

1302.6.1 Gray water used for fixture flushing.

Gray water used for flushing water closets and urinals shall be disinfected and treated by an on-site water reuse treatment system complying

with NSF 350.

1302.7 Storage tanks.

Storage tanks utilized in on-site non-potable water reuse systems shall comply with Sections 1301.9 and 1302.7.1 through 1302.7.3.

1302.7.1 Location.

Storage tanks shall be located with a minimum horizontal distance between various elements as indicated in Table 1302.7.1.

TABLE 1302.7.1 LOCATION OF NON-POTABLE WATER REUSE STORAGE TANKS

ELEMENT	MINIMUM HORIZONTAL DISTANCE FROM STORAGE TANK (m)
Critical root zone (CRZ) of protected trees	0.6
Lot line adjoining private lots	1.5
Seepage pits	1.5
Septic tanks	1.5
Water wells	15
Streams and lakes	15
Water service	1.5
Public water main	3.0

1302.7.2 Design and construction.

Storage tanks shall be designed and constructed in accordance with Chapters 16 through 22 of the SBC 201 and in accordance with the following standards, as appropriate for the material of the storage tank: AWWA D100, AWWA D115, AWWA D120, UL 58, UL 1746, UL 1316, UL 142, API 12F or API 12D.

1302.7.3 Outlets.

Outlets shall be located not less than 100 mm above the bottom of the storage tank and shall not skim water from the surface.

1302.8 Valves.

Valves shall be supplied on on-site non-potable water reuse systems in accordance with Sections 1302.8.1 and 1302.8.2.

1302.8.1 Bypass valve.

One three-way diverter valve listed and labeled

to NSF 50 or other approved device shall be installed on collection piping upstream of each storage tank, or drain field, as applicable, to divert untreated on-site reuse sources to the sanitary sewer to allow servicing and inspection of the system. Bypass valves shall be installed downstream of fixture traps and vent connections. Bypass valves shall be marked to indicate the direction of flow, connection and storage tank or drain field connection. Bypass valves shall be installed in accessible locations. Two shutoff valves shall not be installed to serve as a bypass valve.

1302.8.2 Backwater valve.

One or more backwater valves shall be installed on each overflow and tank drain pipe. Backwater valves shall be in accordance with Section 715.

1302.9 Pumping and control system.

Mechanical equipment including pumps, valves and filters shall be easily accessible and removable in order to perform repair, maintenance and cleaning. The minimum flow rate and flow pressure delivered by the pumping system shall be appropriate for the application and in accordance with Section 604.

1302.10 Water pressure-reducing valve or regulator.

Where the water pressure supplied by the pumping system exceeds 550 kPa static, a pressure-reducing valve shall be installed to reduce the pressure in the non-potable water distribution system piping to 550 kPa static or less. Pressure-reducing valves shall be specified and installed in accordance with Section 604.8.

1302.11 Distribution pipe.

Distribution piping utilized in on-site non-potable water reuse systems shall comply with Sections 1302.11.1 through 1302.11.3.

Exception: Irrigation piping located outside of the building and downstream of a backflow preventer.

1302.11.1 Materials, joints and connections.

Distribution piping shall conform to the standards and requirements specified in Section 605.

1302.11.2 Design.

On-site non-potable water reuse distribution

piping systems shall be designed and sized in accordance with Section 604 for the intended application.

1302.11.3 Marking.

On-site non-potable water distribution piping labeling and marking shall comply with Section 608.8.

1302.12 Tests and inspections.

Tests and inspections shall be performed in accordance with Sections 1302.12.1 through 1302.12.6.

1302.12.1 Collection pipe and vent test.

Drain, waste and vent piping used for on-site water reuse systems shall be tested in accordance with Section 312.

1302.12.2 Storage tank test.

Storage tanks shall be tested in accordance with Section 1301.9.11.

1302.12.3 Water supply system test.

The testing of makeup water supply piping and distribution piping shall be conducted in accordance with Section 312.5.

1302.12.4 Inspection and testing of backflow prevention assemblies.

The testing of backflow preventers and backwater valves shall be conducted in accordance with Section 312.10.

1302.12.5 Inspection of vermin and insect protection.

Inlets and vents to the system shall be inspected to verify that each is protected to prevent the entrance of insects and vermin into the storage tank and piping systems in accordance with Section 1301.7.

1302.12.6 Water quality test.

The quality of the water for the intended application shall be verified at the point of use in accordance with the requirements of the jurisdiction.

1302.13 Operation and maintenance manuals.

Operation and maintenance materials shall be supplied with non-potable on-site water reuse systems in accordance with Sections 1302.13.1 through 1302.13.4.

1302.13.1 Manual.

A detailed operations and maintenance manual shall be supplied in hardcopy form with all systems.

1302.13.2 Schematics.

The manual shall include a detailed system schematic, and the locations and a list of all system components, including manufacturer and model number.

1302.13.3 Maintenance procedures.

The manual shall provide a schedule and procedures for all system components requiring periodic maintenance. Consumable parts, including filters, shall be noted along with part numbers.

1302.13.4 Operations procedures.

The manual shall include system startup and shutdown procedures. The manual shall include detailed operating procedures for the system.

SECTION 1303 NON-POTABLE RAINWATER COLLECTION AND DISTRIBUTION SYSTEMS

1303.1 General.

The provisions of Section 1303 shall govern the construction, installation, alteration and repair of rainwater collection and conveyance systems for the collection, storage, treatment and distribution of rainwater for non-potable applications, as permitted by the jurisdiction.

1303.2 Collection surface.

Rainwater shall be collected only from above-ground impervious roofing surfaces constructed from approved materials. Collection of water from vehicular parking or pedestrian surfaces shall be prohibited except where the water is used exclusively for landscape irrigation. Overflow and bleed-off pipes from roof-mounted appliances including, but not limited to, evaporative coolers, water heaters, and solar water heaters shall not discharge onto rainwater collection surfaces.

1303.3 Debris excluders.

Downspouts and leaders shall be connected to a roof washer and shall be equipped with a debris excluder or equivalent device to prevent the contamination of collected rainwater with leaves, sticks, pine needles and similar material. Debris excluders and equivalent devices shall

be self-cleaning.

1303.4 Roof washer.

A sufficient amount of rainwater shall be diverted at the beginning of each rain event, and not allowed to enter the storage tank, to wash accumulated debris from the collection surface. The amount of rainfall to be diverted shall be field adjustable as necessary to minimize storage tank water contamination. The roof washer shall not rely on manually operated valves or devices, and shall operate automatically. Diverted rainwater shall not be drained to the roof surface, and shall be discharged in a manner consistent with the storm water runoff requirements of the jurisdiction. Roof washers shall be accessible for maintenance and service.

1303.5 Roof gutters and downspouts.

Gutters and downspouts shall be constructed of materials that are compatible with the collection surface and the rainwater quality for the desired end use. Joints shall be water tight.

1303.5.1 Slope.

Roof gutters, leaders and rainwater collection piping shall slope continuously toward collection inlets. Gutters and downspouts shall have a slope of not less than 10.4 mm/meter along their entire length, and shall not permit the collection or pooling of water at any point.

Exception: Siphonic drainage systems installed in accordance with the manufacturer's instructions shall not be required to have a slope.

1303.5.2 Size.

Gutters and downspouts shall be installed and sized in accordance with Section 1106.6 and local rainfall rates.

1303.5.3 Cleanouts.

Cleanouts shall be provided in the water conveyance system to allow access to all filters, flushes, pipes and downspouts.

1303.6 Drainage.

Water drained from the roof washer or debris excluder shall not be drained to the sanitary sewer. Such water shall be diverted from the storage tank and discharge in a location that will not cause erosion or damage to property in accordance with the SBC 201. Roof washers and debris excluders shall be provided with an

automatic means of self-draining between rain events, and shall not drain onto roof surfaces.

1303.7 Collection pipe.

Rainwater collection and conveyance systems shall utilize drainage piping approved for use within plumbing drainage systems to collect and convey captured rainwater. Vent piping approved for use within plumbing venting systems shall be utilized for vents within the rainwater system. Collection and vent piping materials shall comply with Section 702.

1303.7.1 Installation.

Collection piping conveying captured rainwater shall be installed in accordance with Section 704.

1303.7.2 Joints.

Collection piping conveying captured rainwater shall utilize joints approved for use with the distribution piping and appropriate for the intended applications as specified in Section 705.

1303.7.3 Size.

Collection piping conveying captured rainwater shall be sized in accordance with drainage sizing requirements specified in Section 710.

1303.7.4 Marking.

Additional marking of collection piping conveying captured rainwater for reuse

shall not be required beyond that required for sanitary drainage, waste and vent piping by

Chapter 7.

1303.8 Filtration.

Collected rainwater shall be filtered as required for the intended end use. Filters shall be accessible for inspection and maintenance. Filters shall utilize a pressure gauge or other approved method to provide indication when a filter requires servicing or replacement. Filters shall be installed with shutoff valves installed immediately upstream and downstream to allow for isolation during maintenance.

1303.9 Disinfection.

Where the intended application for rainwater requires disinfection or other treatment or both, it shall be disinfected as needed to ensure that the required water quality is delivered at the point of use. Where chlorine is used for disinfection or treatment, water shall be tested for residual chlorine in accordance with ASTM D 1253. The levels of residual chlorine shall not exceed that allowed for the intended use in accordance with the requirements of the jurisdiction.

1303.10 Storage tanks.

Storage tanks utilized in non-potable rainwater collection and conveyance systems shall comply with Sections 1301.9 and 1303.10.1 through 1303.10.3.

1303.10.1 Location.

Storage tanks shall be located with a minimum horizontal distance between various elements as indicated in Table 1303.10.1.

TABLE 1303.10.1 LOCATION OF RAINWATER STORAGE TANKS

ELEMENT	MINIMUM HORIZONTAL DISTANCE FROM STORAGE TANK (m)
Critical root zone (CRZ) of protected trees	0.6
Lot line adjoining private lots	1.5
Seepage pits	1.5
Septic tanks	1.5

1303.10.2 Inlets.

Storage tank inlets shall be designed to introduce collected rainwater into the tank with minimum turbulence, and shall be located and designed to avoid agitating the contents of the storage tank.

1303.10.3 Outlets.

Outlets shall be located at least 100 mm above the bottom of the storage tank and shall not skim water from the surface.

1303.11 Valves.

Valves shall be supplied on rainwater collection and conveyance systems in accordance with Section 1303.11.1.

1303.11.1 Backwater valve.

Backwater valves shall be installed on each overflow and tank drain pipe. Backwater valves shall be in accordance with Section 715.

1303.12 Pumping and control system.

Mechanical equipment including pumps, valves and filters shall be easily accessible and removable in order to perform repair, maintenance and cleaning. The minimum flow rate and flow pressure delivered by the pumping system shall be appropriate for the application and in accordance with Section 604.

1303.13 Water pressure-reducing valve or regulator.

Where the water pressure supplied by the pumping system exceeds 550 kPa static, a pressure-reducing valve shall be installed to reduce the pressure in the rainwater distribution system piping to 550 kPa static or less. Pressure-reducing valves shall be specified and installed in accordance with Section 604.8.

1303.14 Distribution pipe.

Distribution piping utilized in rainwater collection and conveyance systems shall comply with Sections 1303.14.1 through 1303.14.3.

Exception: Irrigation piping located outside of the building and downstream of a backflow preventer.

1303.14.1 Materials, joints and connections.

Distribution piping shall conform to the standards and requirements specified in Section 605 for non-potable water.

1303.14.2 Design.

Distribution piping systems shall be designed and sized in accordance with Section 604 for the intended application.

1303.14.3 Marking.

Non-potable rainwater distribution piping labeling and marking shall comply with Section 608.8.

1303.15 Tests and inspections.

Tests and inspections shall be performed in accordance with Sections 1303.15.1 through 1303.15.8.

1303.15.1 Roof gutter inspection and test.

Roof gutters shall be inspected to verify that the installation and slope is in accordance with Section 1303.5.1. Gutters shall be tested by pouring not less than 3.8 L of water into the end of the gutter opposite the collection point. The gutter being tested shall not leak and shall not retain standing water.

1303.15.2 Roof washer test.

Roof washers shall be tested by introducing water into the gutters. Proper diversion of the first quantity of water in accordance with the requirements of Section 1303.4 shall be verified.

1303.15.3 Collection pipe and vent test.

Drain, waste and vent piping used for rainwater collection and conveyance systems shall be tested in accordance with Section 312.

1303.15.4 Storage tank test.

Storage tanks shall be tested in accordance with Section 1301.9.11.

1303.15.5 Water supply system test.

The testing of makeup water supply piping and distribution piping shall be conducted in accordance with Section 312.5.

1303.15.6 Inspection and testing of backflow prevention assemblies.

The testing of backflow preventers and backwater valves shall be conducted in accordance with Section 312.10.

1303.15.7 Inspection of vermin and insect protection.

Inlets and vents to the system shall be inspected to verify that each is protected to prevent the entrance of insects and vermin into the storage tank and piping systems in accordance with Section 1301.7.

1303.15.8 Water quality test.

The quality of the water for the intended application shall be verified at the point of use in accordance with the requirements of the jurisdiction. Except where site conditions as specified in ASTM E 2727 affect the rainwater, collected rainwater shall be considered to have the parameters indicated in Table 1303.15.8.

TABLE 1303.15.8 RAINWATER QUALITY

PARAMETER	VALUE
pH	6.0-7.0
BOD	Not greater than 10 mg/L
NTU	Not greater than 2
Fecal coliform	No detectable fecal coli in 100 mL
Sodium	No detectable sodium in 100 mL
Chlorine	No detectable chlorine in 100 mL
Enteroviruses	No detectable enteroviruses in 100 mL

1303.16 Operation and maintenance manuals.

Operation and maintenance manuals shall be supplied with rainwater collection and conveyance systems in accordance with Sections 1303.16.1 through 1303.16.4.

1303.16.1 Manual.

A detailed operations and maintenance manual shall be supplied in hardcopy form with all systems.

1303.16.2 Schematics.

The manual shall include a detailed system schematic, and locations and a list of all system components, including manufacturer and model number.

1303.16.3 Maintenance procedures.

The manual shall provide a maintenance schedule and procedures for all system components requiring periodic maintenance. Consumable parts, including filters, shall be noted along with part numbers.

1303.16.4 Operations procedures.

The manual shall include system startup and shutdown procedures, as well as detailed operating procedures.

SECTION 1304 RECLAIMED WATER SYSTEMS

1304.1 General.

The provisions of this section shall govern the construction, installation, alteration and repair of systems supplying non-potable reclaimed water.

1304.2 Water pressure-reducing valve or regulator.

Where the reclaimed water pressure supplied to the building exceeds 550 kPa static, a pressure-reducing valve shall be installed to reduce the pressure in the reclaimed water distribution system piping to 550 kPa static or less. Pressure-reducing valves shall be specified and installed in accordance with Section 604.8.

1304.3 Reclaimed water systems.

The design of the reclaimed water systems shall conform to ASTM E 2635 and accepted engineering practice.

1304.3.1 Distribution pipe.

Distribution piping shall comply with Sections 1304.3.1.1 through 1304.3.1.3.

Exception: Irrigation piping located outside of the building and downstream of a backflow preventer.

1304.3.1.1 Materials, joints and connections.

Distribution piping conveying reclaimed water shall conform to standards and requirements specified in Section 605 for non-potable water.

1304.3.1.2 Design.

Distribution piping systems shall be designed and sized in accordance with Section 604 for the intended application.

1304.3.1.3 Labeling and marking.

Non-potable rainwater distribution piping labeling and marking shall comply with Section 608.8.

1304.4 Tests and inspections.

Tests and inspections shall be performed in accordance with Sections 1304.4.1 and 1304.4.2.

1304.4.1 Water supply system test.

The testing of makeup water supply piping and reclaimed water distribution piping shall be conducted in accordance with Section 312.5.

1304.4.2 Inspection and testing of backflow prevention assemblies.

The testing of backflow preventers shall be conducted in accordance with Section 312.10



Figures for chapter 13



FIGURE 1301.3
PICTOGRAPH—DO NOT DRINK



CHAPTER 14 SUBSURFACE LANDSCAPE IRRIGATION SYSTEMS

SECTION 1401 GENERAL

1401.1 Scope.

The provisions of Chapter 14 shall govern the materials, design, construction and installation of subsurface landscape irrigation systems connected to non-potable water from on-site water reuse systems.

1401.2 Materials.

Above-ground drain, waste and vent piping for subsurface landscape irrigation systems shall conform to one of the standards listed in Table 702.1. Subsurface landscape irrigation, underground building drainage and vent pipe shall conform to one of the standards listed in Table 702.2.

1401.3 Tests.

Drain, waste and vent piping for subsurface landscape irrigation systems shall be tested in accordance with Section 312.

1401.4 Inspections.

Subsurface landscape irrigation systems shall be inspected in accordance with Section 107.

1401.5 Disinfection.

Disinfection shall not be required for on-site non-potable water reuse for subsurface landscape irrigation systems.

1401.6 Coloring.

On-site non-potable water reuse for subsurface landscape irrigation systems shall not be required to be dyed.

SECTION 1402 SYSTEM DESIGN AND SIZING

1402.1 Sizing.

The system shall be sized in accordance with the sum of the output of all water sources connected to the subsurface irrigation system. Where gray water collection piping is connected to subsurface landscape irrigation systems, gray water output shall be calculated according to the liter-per-day-per-occupant number based on the type of fixtures connected.

The gray water discharge shall be calculated by the following equation:

$$C = A \times B \quad (\text{Equation 14-1})$$

where:

A = Number of occupants:

Residential—Number of occupants shall be determined by the actual number of occupants, but not less than two occupants for one bedroom and one occupant for each additional bedroom.

Commercial—Number of occupants shall be determined by the *SBC 201*.

B = Estimated flow demands for each occupant:

Residential—94.6 Liter per day per occupant for showers, bathtubs and lavatories and 56.7 Lpd per occupant for clothes washers or laundry trays.

Commercial—Based on type of fixture or water use records minus the discharge of fixtures other than those discharging gray water.

C = Estimated gray water discharge based on the total number of occupants.

1402.2 Percolation tests.

The permeability of the soil in the proposed absorption system shall be determined by percolation tests or permeability evaluation.

1402.2.1 Percolation tests and procedures.

At least three percolation tests in each system area shall be conducted. The holes shall be spaced uniformly in relation to the bottom depth of the proposed absorption system. More percolation tests shall be made where necessary, depending on system design.

1402.2.1.1 Percolation test hole.

The test hole shall be dug or bored. The test hole shall have vertical sides and a horizontal dimension of 100 mm to 200 mm. The bottom and sides of the hole shall be scratched with a

sharp-pointed instrument to expose the natural soil. All loose material shall be removed from the hole and the bottom shall be covered with 50 mm of gravel or coarse sand.

1402.2.1.2 Test procedure, sandy soils.

The hole shall be filled with clear water to a minimum of 300 mm above the bottom of the hole for tests in sandy soils. The time for this amount of water to seep away shall be determined, and this procedure shall be repeated if the water from the second filling of the hole seeps away in 10 minutes or less. The test shall proceed as follows: Water shall be added to a point not more than 150 mm above the gravel or coarse sand. Thereupon, from a fixed reference point, water levels shall be measured at 10-minute intervals for a period of 1 hour. Where 150 mm of water seeps away in less than 10 minutes, a shorter interval between measurements shall be used, but in no case shall the water depth exceed 150 mm. Where 150 mm of water seeps away in less than 2 minutes, the test shall be stopped and a rate of less than 7.2 sec/mm shall be reported. The final water level drop shall be used to calculate the percolation rate. Soils not meeting the above requirements shall be tested in accordance with Section 1303.7.1.3.

1402.2.1.3 Test procedure, other soils.

The hole shall be filled with clear water, and a minimum water depth of 300 mm shall be maintained above the bottom of the hole for a 4-hour period by refilling whenever necessary or by use of an automatic siphon. Water remaining in the hole after 4 hours shall not be removed. Thereafter, the soil shall be allowed to swell not less than 16 hours or more than 30 hours. Immediately after the soil swelling period, the measurements for determining the percolation rate shall be made as follows: any soil sloughed into the hole shall be removed and the water level shall be adjusted to 150 mm above the gravel or coarse sand. Thereupon, from a fixed reference point, the water level shall be measured at 30-minute intervals for a period of 4 hours, unless two successive water level drops do not vary by more than 1.59 mm. At least three water level drops shall be observed and recorded. The hole shall be filled with clear water to a point not more than 150 mm above the gravel or coarse sand whenever it becomes nearly empty. Adjustments of the water level shall not be made during the three

measurement periods except to the limits of the last measured water level drop. When the first 150 mm of water seeps away in less than 30 minutes, the time interval between measurements shall be 10 minutes and the test run for 1 hour. The water depth shall not exceed 125 mm at any time during the measurement period. The drop that occurs during the final measurement period shall be used in calculating the percolation rate.

1402.2.1.4 Mechanical test equipment.

Mechanical percolation test equipment shall be of an approved type.

1402.2.2 Permeability evaluation.

Soil shall be evaluated for estimated percolation based on structure and texture in accordance with accepted soil evaluation practices. Borings shall be made in accordance with Section 1402.2.1.1 for evaluating the soil.

1402.3 Subsurface landscape irrigation site location.

The surface grade of all soil absorption systems shall be located at a point lower than the surface grade of any water well or reservoir on the same or adjoining lot. Where this is not possible, the site shall be located so surface water drainage from the site is not directed toward a well or reservoir. The soil absorption system shall be located with a minimum horizontal distance between various elements as indicated in Table 1402.3. Private sewage disposal systems in compacted areas, such as parking lots and driveways, are prohibited. Surface water shall be diverted away from any soil absorption site on the same or neighboring lots.

SECTION 1403 INSTALLATION

1403.1 Installation.

Absorption systems shall be installed in accordance with Sections 1403.1.1 through 1403.1.5 to provide landscape irrigation without surfacing of water.

1403.1.1 Absorption area.

The total absorption area required shall be computed from the estimated daily gray water discharge and the design-loading rate based on the percolation rate for the site. The required absorption area equals the estimated gray water discharge divided by the design-loading rate

1403.1.2 Seepage trench excavations.

Seepage trench excavations shall be not less than 300 mm in width and not greater than 1500 mm in width. Trench excavations shall be spaced not less than 600 mm apart. The soil absorption area of a seepage trench shall be computed by using the bottom of the trench area (width) multiplied by the length of pipe. Individual seepage trenches shall be not greater than 30 m in developed length. from Table 1403.1.1.

1403.1.3 Seepage bed excavations.

Seepage bed excavations shall be not less than 1500 mm in width and have more than one distribution pipe. The absorption area of a seepage bed shall be computed by using the bottom of the trench area. Distribution piping in a seepage bed shall be uniformly spaced not greater than 1500 mm and not less than 3 m apart, and greater than 900 mm and not less than 300 mm from the sidewall or headwall.

1403.1.4 Excavation and construction.

The bottom of a trench or bed excavation shall be level. Seepage trenches or beds shall not be excavated where the soil is so wet that such material rolled between the hands forms a soil wire. All smeared or compacted soil surfaces in the sidewalls or bottom of seepage trench or bed excavations shall be scarified to the depth of smearing or compaction and the loose material removed. Where rain falls on an open

excavation, the soil shall be left until sufficiently dry so a soil wire will not form when soil from the excavation bottom is rolled between the hands. The bottom area shall then be scarified and loose material removed.

1403.1.5 Aggregate and backfill.

Not less than 150 mm in depth of aggregate, ranging in size from 13 mm to 64 mm, shall be laid into the trench below the distribution piping elevation. The aggregate shall be evenly distributed not less than 50 mm in depth over the top of the distribution pipe. The aggregate shall be covered with approved synthetic materials or 200 mm of un-compacted marsh hay or straw. Building paper shall not be used to cover the aggregate. Not less than 200 mm of soil backfill shall be provided above the covering.

1403.2 Distribution piping.

Distribution piping shall be not less than 75 mm in diameter. Materials shall comply with Table 1403.2. The top of the distribution pipe shall be not less than 200 mm below the original surface. The slope of the distribution pipes shall be not less than 50 mm and not greater than 100 mm per 30 m.

1403.2.1 Joints.

Joints in distribution pipe shall be made in accordance with Section 705 of this code.

TABLE 1403.3 LOCATION OF SUBSURFACE IRRIGATION SYSTEM

ELEMENT	MINIMUM HORIZONTAL DISTANCE	
	Storage tank (m)	Irrigation disposal field (m)
Buildings	1.5	0.6
Lot line adjoining private property	1.5	1.5
Water wells	15	30
Streams and lakes	15	15
Seepage pits	1.5	1.5
Septic tanks	0	1.5
Water service	1.5	1.5
Public water main	3.0	3.0

TABLE 1403.1.1 DESIGN LOADING RATE

PERCOLATION RATE (minutes per mm)	DESIGN LOADING FACTOR (Liters per square meter per day)
0 to less than 0.4	49
0.4 to less than 1.2	33
1.2 to less than 1.8	29
1.8 to 2.4	16

TABLE 1403.2 DISTRIBUTION PIPE

MATERIAL	STANDARD
Polyethylene (PE) plastic pipe	ASTM F 405
Polyvinyl chloride (PVC) plastic pipe	ASTM D 2729
Polyvinyl chloride (PVC) plastic pipe with a 88-mm O.D. and solid cellular core or composite wall	ASTM F 1488

CHAPTER 15 REFERENCED STANDARDS

These standards are listed here for guidance; however, for enforcement, they shall be adopted and/or approved by the appropriate governmental authority (SASO or SBCNC) as per their mandates, before considered part of this code.

ANSI	American National Standards Institute 25 West 43rd Street, Fourth Floor New York, NY 10036	
Standard reference number	Title	Referenced in code section number
A118.10—99	Specifications for Load Bearing, Bonded, Waterproof Membranes for Thin Set Ceramic Tile and Dimension Stone Installation	417.5.2.5, 417.5.2.6
Z4.3—95	Minimum Requirements for Nonsewered Waste-disposal Systems	311.1
Z21.22—99 (R2003)	Relief Valves for Hot Water Supply Systems with Addenda Z21.22a—2000 (R2003) and Z21.22b—2001 (R2003)	504.2, 504.4, 504.4.1
CSA B45.5—11/ IAPMO Z124—11	Plastic Plumbing Fixtures	407.1, 415.1, 416.1, 416.2, 417.1, 418.1, 419.1, 420.1

AHRI	Air-Conditioning, Heating, & Refrigeration Institute 4100 North Fairfax Drive, Suite 200 Arlington, VA 22203	
Standard reference number	Title	Referenced in code section number
1010—02	Self-contained, Mechanically Refrigerated Drinking-Water Coolers	410.1

API	American Petroleum Institute 1220 L Street NW Washington, DC 20005-4070	
Standard reference number	Title	Referenced in code section number
12D—2008	Specification for Field Welded Tanks for Storage of Production Liquids, effective April 1, 2009	1302.7.2
12F—2008	Specification for Shop Welded Tanks for Storage of Production Liquids, effective April 1 2009	1302.7.2

ASME	American Society of Mechanical Engineers Three Park Avenue New York, NY 10016-5990	
Standard reference number	Title	Referenced in code section number
A112.1.2—2004	Air Gaps in Plumbing Systems	406.1, 409.2, Table 608.1, 608.13.1
A112.1.3—2000 (R2011)	Air Gap Fittings for Use with Plumbing Fixtures, Appliances and Appurtenances	406.1, 409.2, Table 608.1, 608.13.1, 1102.6
A112.3.1—2007	Stainless Steel Drainage Systems for Sanitary, DWV, Storm and Vacuum Applications Above and Below Ground	412.1, Table 702.1, Table 702.2, Table 702.3, Table 702.4, Table 1102.4, Table 1102.5, 1102.6, Table 1102.7
ASME A112.3.4—2013/ CSA B45.9—2013	Macerating Toilet Systems and Related Components	712.4.1
A112.4.1—2009	Water Heater Relief Valve Drain Tubes	504.6
A112.4.2—2009	Water Closet Personal Hygiene Devices	424.9
A112.4.3—1999 (R2010)	Plastic Fittings for Connecting Water Closets to the Sanitary Drainage System	405.4
A112.4.14—2004 (R2010)	Manually Operated, Quarter-Turn Shutoff Valves for Use in Plumbing Systems	Table 605.7
A112.4-14.6—2010	FOG (Fats, Oils and Greases) Disposal Systems	1003.3.4, 1003.3.6
A112.6.1M—1997 (R2008)	Floor-affixed Supports for Off-the-floor Plumbing Fixtures for Public Use	405.4.3
A112.6.2—2000 (R2010)	Framing-affixed Supports for Off-the-floor Water Closets with Concealed Tanks	405.4.3
A112.6.3—2001 (R2007)	Floor and Trench Drains	412.1
A112.6.4—2003 (R2008)	Roof, Deck, and Balcony Drains	1102.6
A112.6.7—2010	Enameled and Epoxy-coated Cast-iron and PVC Plastic Sanitary Floor Sinks	427.1
A112.6.9—2005 (R2010)	Siphonic Roof Drains	1107.1
A112.14.1—2003	Backwater Valves	715.2
A112.14.3—2000	Grease Interceptors	1003.3.4
A112.14.4—2001 (Reaffirmed 2007)	Grease Removal Devices	1003.3.4

A112.18.1— 2012/ CSA B125.1— 2012	Plumbing Supply Fittings	424.1, 424.2, 424.3, 424.4, 424.6, 424.8, 605.7, 607.4, 608.2
A112.18.2— 2011/ CSA B125.2— 2011	Plumbing Waste Fittings	424.1.2
A112.18.3— 2002 (Reaffirmed 2008)	Performance Requirements for Backflow Protection Devices and Systems in Plumbing Fixture Fittings	424.2, 424.6
A112.18.6/ CSA B125.6— 2009	Flexible Water Connectors	605.6
A112.18.7— 1999 (Reaffirmed 2004)	Deck mounted Bath/Shower Transfer Valves with Integral Backflow Protection	424.8
A112.18.9— 2011	Protectors/Insulators for Exposed Waste and Supplies on Accessible Fixtures	404.3
A112.19.1/— 2013 CSA B45.2— 2013	Enameled Cast Iron and Enameled Steel Plumbing Fixtures	407.1, 410.1, 415.1, 416.1, 418.1
A112.19.2— 2013/ CSA B45.1—13	Ceramic Plumbing Fixtures	401.2, 405.9, 407.1, 408.1, 410.1, 415.1, 416.1, 417.1, 418.1, 419.1, 420.1
A112.19.3— 2008/ CSA B45.4— 08(R2013)	Stainless Steel Plumbing Fixtures	405.9, 407.1, 415.1, 416.1, 418.1, 420.1
A112.19.5/— 2011 CSA B45.15— 2011	Flush Valves and Spuds for Water-closets, Urinals, and Tanks	425.4
A112.19.7M— 2012/ CSA B45.10— 2012	Hydromassage Bathtub Systems	421.1, 421.4
A112.19.12— 2006	Wall Mounted and Pedestal Mounted, Adjustable, Elevating, Tilting and Pivoting Lavatory, Sink and Shampoo Bowl Carrier Systems and Drain Systems	416.4, 418.3
A112.19.14- 2006(R2011)	Six-Liter Water Closets Equipped with a Dual Flushing Device	420.1
A112.19.15— 2005	Bathtub/Whirlpool Bathtubs with Pressure Sealed Doors	407.4, 421.6
A112.19.19— 2006	Vitreous China Nonwater Urinals	419.1, 421.6

A112.21.3-1985(R2007)	Hydrants for Utility and Maintenance Use	Table 608.1, 608.13.6
A112.36.2M—1991 (R2008)	Cleanouts	708.1.10.2
ASSE 1016/ ASME A112.1016/ CSA B125.16-2011	Performance Requirements for Individual Thermostatic, Pressure Balancing and Combination Control Valves for Individual Fixture Fittings	424.3, 424.4, 607.4
B1.20.1—1983(R2006)	Pipe Threads, General Purpose	605.10.3, 605.11.3, 605.13.4, 605.15.3, 605.18.1, 605.22.4, 705.2.3, 705.3.3, 705.6.4, 705.19.1, 705.11.3
B 16.3—2011	Malleable Iron Threaded Fittings Classes 150 and 300	Table 702.4, Table 1102.7
B 16.4—2011	Gray Iron Threaded Fittings Classes 125 and 250	Table 605.5, Table 702.4, Table 1102.7
B 16.9—2007	Factory-made Wrought Steel Buttwelding Fittings	Table 605.5, Table 702.4, Table 1102.7
B 16.11—2011	Forged Fittings, Socket-welding and Threaded	Table 605.5, Table 702.4, Table 1102.7
B 16.12—2009	Cast-iron Threaded Drainage Fittings	Table 702.4, Table 1102.7
B 16.15—2011	Cast Bronze Threaded Fittings	Table 605.5, Table 702.4, Table 1102.7
B 16.18—2012	Cast Copper Alloy Solder Joint Pressure Fittings	Table 605.5, Table 702.4, Table 1102.7
B 16.22—2001(R2010)	Wrought Copper and Copper Alloy Solder Joint Pressure Fittings	Table 605.5, Table 702.4, Table 1102.7
B 16.23—2011	Cast Copper Alloy Solder Joint Drainage Fittings DWV	Table 702.4, Table 1102.7
B 16.26—2011	Cast Copper Alloy Fittings for Flared Copper Tubes	Table 605.5, Table 702.4, Table 1102.7
B 16.28—1994	Wrought Steel Buttwelding Short Radius Elbows and Returns	Table 605.5, Table 702.4, Table 1102.7
B 16.29—2012	Wrought Copper and Wrought Copper Alloy Solder Joint Drainage Fittings (DWV)	Table 702.4, Table 1102.7
B16.34—2009	Valves Flanged, Threaded and Welding End	Table 605.7
B16.51—2011	Copper and Copper Alloy Press-Connect Pressure Fittings	Table 605.5

ASPE	American Society of Plumbing Engineers 8614 Catalpa Avenue, Suite 1007 Chicago, IL 60656-1116	
Standard reference number	Title	Referenced in code section number
45—2013	Siphonic Roof Drainage Systems	1107.1

ASSE	American Society of Sanitary Engineering 901 Canterbury Road, Suite A Westlake, OH 44145	
Standard reference number	Title	Referenced in code section number
1001—08	Performance Requirements for Atmospheric Type Vacuum Breakers	425.2, Table 608.1, 608.13.6, 608.16.4.1
1002—08	Performance Requirements for Antisiphon Fill Valves (Ballcocks) for Gravity Water Closet Flush Tanks	425.3.1, Table 608.1
1003—09	Performance Requirements for Water Pressure Reducing Valves	604.8
1004—08	Performance Requirements for Backflow Prevention Requirements for Commercial Dishwashing Machines	409.1
1005—99	Performance Requirements for Water Heater Drain Valves	501.3
1008—06	Performance Requirements for Plumbing Aspects of Food Waste Disposer Units	413.1
1010—04	Performance Requirements for Water Hammer Arresters	604.9
1011—04	Performance Requirements for Hose Connection Vacuum Breakers	Table 608.1, 608.13.6
1012—09	Performance Requirements for Backflow Preventers with Intermediate Atmospheric Vent	Table 608.1, 608.13.3, 608.16.2
1013—09	Performance Requirements for Reduced Pressure Principle Backflow Preventers and Reduced Pressure Principle Fire Protection Backflow Preventers	Table 608.1, 608.13.2, 608.16.2
1015—09	Performance Requirements for Double Check Backflow Prevention Assemblies and Double Check Fire Protection Backflow Prevention Assemblies	Table 608.1, 608.13.7
ASSE 1016/ ASME A112.1016/ CSA B125.16— 2011	Performance Requirements for Individual Thermostatic, Pressure Balancing and Combination Control Valves for Individual Fixture Fittings	424.3, 424.4, 607.4
1017—2010	Performance Requirements for Temperature Actuated Mixing Valves for Hot Water Distribution Systems	51.2, 613.1
1018—2010	Performance Requirements for Trap Seal Primer Valves; Potable Water Supplied	1002.4.1.2

1019—2010	Performance Requirements for Vacuum Breaker Wall Hydrants, Freeze Resistant, Automatic Draining Type	Table 608.1, 608.13.6, 608.13.10
1020—04	Performance Requirements for Pressure Vacuum Breaker Assembly	Table 608.1, 608.13.5
1022—03	Performance Requirements for Backflow Preventer for Beverage Dispensing Equipment	Table 608.1, 608.16.1, 608.16.10
1024—04	Performance Requirements for Dual Check Valve Type Backflow Preventers (for Residential Supply Service or Individual Outlets)	605.3.1, Table 608.1, 608.13.10
1035—08	Performance Requirements for Laboratory Faucet Backflow Preventers	Table 608.1, 608.13.6
1037—90	Performance Requirements for Pressurized Flushing Devices for Plumbing Fixtures	425.2
1044—2010	Performance Requirements for Trap Seal Primer Devices Drainage Types and Electronic Design Types	1002.4.1.3
1047—2009	Performance Requirements for Reduced Pressure Detector Fire Protection Backflow Prevention Assemblies	Table 608.1, 608.13.2
1048—2009	Performance Requirements for Double Check Detector Fire Protection Backflow Prevention Assemblies	Table 608.1, 608.13.7
1049—2009	Performance Requirements for Individual and Branch Type Air Admittance Valves for Chemical Waste Systems	901.3, 918.8
1050—2009	Performance Requirements for Stack Air Admittance Valves for Sanitary Drainage Systems	918.1
1051—2009	Performance Requirements for Individual and Branch Type Air Admittance Valves for Sanitary Drainage Systems-fixture and Branch Devices	918.1
1052—04	Performance Requirements for Hose Connection Backflow Preventers	Table 608.1, 608.13.6
1055—2009	Performance Requirements for Chemical Dispensing Systems	608.13.9
1056—2010	Performance Requirements for Spill Resistant Vacuum Breaker	Table 608.1, 608.13.5, 608.13.8
1060—2006	Performance Requirements for Outdoor Enclosures for Fluid Conveying Components	608.14.1
1061—2010	Performance Requirements for Removable and Nonremovable Push Fit Fittings	Table 605.5
1062—2006	Performance Requirements for Temperature Actuated, Flow Reduction Valves to Individual Supply Fittings	424.7
1066—2009	Performance Requirements for Individual Pressure Balancing In-line Valves for Individual Fixture Fittings	604.11
1069—05	Performance Requirements for Automatic Temperature Control Mixing Valves	424.4
1070—04	Performance Requirements for Water-temperature Limiting Devices	408.3, 416.5, 623.3, 424.5, 607.1.2

1072—07	Performance Requirements for Barrier Type Floor Drain Tap Seal Protection Devices	1002.4.14
1079—2005	Performance Requirements for Dielectric Pipe Unions	605.24.1, 605.24.3
5013—2009	Performance Requirements for Testing Reduced Pressure Principle Backflow Prevention Assembly (RPA) and Reduced Pressure Fire Protection Backflow Preventers (RFP)	312.10.2
5015—2009	Performance Requirements for Testing Double Check Valve Backflow Prevention Assemblies (DC) and Double Check Fire Protection Backflow Prevention Assemblies (DCF)	312.10.2
5020—2009	Performance Requirements for Testing Pressure Vacuum Breaker Assemblies (PVBA)	312.10.2
5047—98	Performance Requirements for Testing Reduced Pressure Detector Fire Protection Backflow Prevention Assemblies (RPDA)	312.10.2
5048—2009	Performance Requirements for Testing Double Check Valve Detector Assembly (DCDA)	312.10.2
5052—98	Performance Requirements for Testing Hose Connection Backflow Preventers	312.10.2
5056—98	Performance Requirements for Testing Spill Resistant Vacuum Breaker (SRVB)	312.10.2

ASTM	ASTM International 100 Barr Harbor Drive West Conshohocken, PA 19428-2959	
Standard reference number	Title	Referenced in code section number
A 53/A 53M—12	Specification for Pipe, Steel, Black and Hot-dipped, Zinc-coated Welded and Seamless	Table 605.3, Table 605.4, Table 702.1
A 74—13A	Specification for Cast-iron Soil Pipe and Fittings	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 708.2, 708.7, Table 1102.4, Table 1102.5, Table 1102.7
A 312/A 312M—13a	Specification for Seamless, Welded, And Heavily Cold Worked Austenitic Stainless Steel Pipes	Table 605.3, Table 605.4, Table 605.5, 605.23.2
A 733—2003(2009)e1	Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples	Table 605.8
A 778—01(2009)e1	Specification for Welded Unannealed Austenitic Stainless Steel Tubular Products	Table 605.3, Table 605.4, Table 605.5
A 888—13A	Specification for Hubless Cast-iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Application	Table 702.1, Table 702.2, Table 702.3,

		Table 702.4, 708.7, Table 1102.4, Table 1102.5, Table 1102.7
B 32—08	Specification for Solder Metal	605.13.3, 605.14.6, 705.6.3, 705.17.3
B 42—10	Specification for Seamless Copper Pipe, Standard Sizes	Table 605.3, Table 605.4, Table 702.1
B 43—09	Specification for Seamless Red Brass Pipe, Standard Sizes	Table 605.3, Table 605.4, Table 702.1
B 75—11	Specification for Seamless Copper Tube	Table 605.3, Table 605.4, Table 702.1, Table 702.2, Table 702.3, Table 1102.4
B 88—09	Specification for Seamless Copper Water Tube	Table 605.3, Table 605.4, Table 702.1, Table 702.2, Table 702.3, Table 1102.4
B 152/B 152M— 13	Specification for Copper Sheet, Strip Plate and Rolled Bar	402.3, 417.5.2.4, 425.3.3, 902.2
B 251—10	Specification for General Requirements for Wrought Seamless Copper and Copper-alloy Tube	Table 605.3, Table 605.4, Table 702.1, Table 702.2, Table 702.3, Table 1102.4
B 302—12	Specification for Threadless Copper Pipe, Standard Sizes	Table 605.3, Table 605.4, Table 702.1
B 306—09	Specification for Copper Drainage Tube (DWV)	Table 702.1, Table 702.2, Table 1102.4
B 447—12a	Specification for Welded Copper Tube	Table 605.3, Table 605.4
B 687—99(2011)	Specification for Brass, Copper and Chromium-plated Pipe Nipples	Table 605.8
B 813—10	Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube	605.13.3, 605.14.6, 705.6.3, 705.7.3
B 828—02(2010)	Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings	605.13.3, 605.14.6, 705.6.3, 705.17.3
C 4—04(2009)	Specification for Clay Drain Tile and Perforated Clay Drain Tile	Table 702.3, Table 1102.4, Table 1102.5

C 14—11	Specification for Nonreinforced Concrete Sewer, Storm Drain and Culvert Pipe	Table 702.3, Table 1102.4
C 76—13a	Specification for Reinforced Concrete Culvert, Storm Drain and Sewer Pipe	Table 702.3, Table 1102.4
C 425—04(2009)	Specification for Compression Joints for Vitrified Clay Pipe and Fittings	705.12, 705.16
C 443—12	Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets	705.5, 705.16
C 564—12	Specification for Rubber Gaskets for Cast-iron Soil Pipe and Fittings	705.4.2, 705.4.3, 705.16
C 700—13	Specification for Vitrified Clay Pipe, Extra Strength, Standard Strength, and Perforated	Table 702.3, Table 702.4, Table 1102.4, Table 1102.5
C 1053—00(2010)	Specification for Borosilicate Glass Pipe and Fittings for Drain, Waste, and Vent (DWV) Applications	Table 702.1, Table 702.4
C 1173—10e1	Specification for Flexible Transition Couplings for Underground Piping System	705.2.1, 705.5, 705.11.1, 705.12, 705.13.2, 705.16
C 1277—12	Specification for Shielded Coupling Joining Hubless Cast-iron Soil Pipe and Fittings	705.4.3
C 1440—08	Specification for Thermoplastic Elastomeric (TPE) Gasket Materials for Drain, Waste, and Vent (DWV), Sewer, Sanitary and Storm Plumbing Systems	705.16
C 1460—08	Specification for Shielded Transition Couplings for Use with Dissimilar DWV Pipe and Fittings Above Ground	705.16
C 1461—08	Specification for Mechanical Couplings Using Thermoplastic Elastomeric (TPE) Gaskets for Joining Drain, Waste and Vent (DWV) Sewer, Sanitary and Storm Plumbing Systems for Above and Below Ground Use	705.19
C 1540—11	Specification for Heavy Duty Shielded Couplings Joining Hubless Cast-iron Soil Pipe and Fittings	705.4.3
C 1563—13	Standard Test Method for Gaskets for Use in Connection with Hub and Spigot Cast Iron Soil Pipe and Fittings for Sanitary Drain, Waste, Vent and Storm Piping Applications	705.4.2
D 1253—08	Standard Test Method For Residual Chlorine in Water	1301.2.1, 1303.9
D 1527—99(2005)	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe, Schedules 40 and 80	Table 605.3
D 1785—12	Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80 and 120	Table 605.3
D 2235—04(2011)	Specification for Solvent Cement for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe and Fittings	605.10.2, 705.2.2, 705.7.2
D 2239—12a	Specification for Polyethylene (PE) Plastic Pipe (SIDR-PR) Based on Controlled Inside Diameter	Table 605.3
D 2241—09	Specification for Poly (Vinyl Chloride) (PVC) Pressure-rated Pipe (SDR-Series)	Table 605.3

D 2282—99(2005)	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe (SDR-PR)	Table 605.3
D 2464—06	Specification for Threaded Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80	Table 605.5
D 2466—06	Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40	Table 605.5
D 2467—06	Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80	Table 605.5
D 2468—96a	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Plastic Pipe Fittings, Schedule 40	Table 605.5
D 2564—12	Specification for Solvent Cements for Poly (Vinyl Chloride) (PVC) Plastic Piping Systems	605.22.3, 705.11.2, 705.14.2
D 2609—02(2008)	Specification for Plastic Insert Fittings for Polyethylene (PE) Plastic Pipe	Table 605.5
D 2657—07	Practice for Heat Fusion-joining of Polyolefin Pipe and Fitting Waste, and Vent Pipe and Fittings	605.19.2, 705.13.1
D 2661—11	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Schedule 40 Plastic Drain, Waste, and Vent Pipe and Fittings	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 705.2.2, 705.7.2, Table 1102.4, Table 1102.7
D 2665—12	Specification for Poly (Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings	Table 702.1, Table 702.2, Table 702.3, Table 702.4, Table 1102.4, Table 1102.7
D 2672—96a(2009)	Specification for Joints for IPS PVC Pipe Using Solvent Cement	Table 605.3
D 2683—10e1	Standard Specification for Socket-type Polyethylene fittings for Outside Diameter-controlled Polyethylene Pipe and Tubing	Table 605.5, 717.5
D 2729—11	Specification for Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings	Table 1102.5, Table 1403.2
D 2737—2012a	Specification for Polyethylene (PE) Plastic Tubing	Table 605.3
D 2751—05	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Sewer Pipe and Fittings	Table 702.3, Table 702.4, Table 1102.4, Table 1102.7
D 2846/D 2846M—09be1	Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Hot and Cold Water Distribution Systems	Table 605.3, Table 605.4, Table 605.5, 605.15.2, 605.16.2
D 2855—96(2010)	Standard Practice for Making Solvent-cemented Joints with Poly (Vinyl Chloride) (PVC) Pipe and Fittings	605.22.3, 705.11.2

D 2949—10	Specification for 3.25-in Outside Diameter Poly (Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings	Table 702.1, Table 702.2, Table 702.3, Table 702.4, Table 1102.7
D 3034—08	Specification for Type PSM Poly (Vinyl Chloride) (PVC) Sewer Pipe and Fittings	Table 702.3, Table 702.4, Table 1102.4, Table 1102.5, Table 1102.7
D 3035—2012e1	Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter	Table 605.3
D 3139— 98(2011)	Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals	605.10.1, 605.22.1
D 3212—07D	Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals	705.2.1, 705.11.1, 705.13.2
D 3261—12	Specification for Butt Heat Fusion Polyethylene (PE) Plastic fittings for Polyethylene (PE) Plastic Pipe and Tubing	Table 605.5
D 3311—11	Specification for Drain, Waste and Vent (DWV) Plastic Fittings Patterns	Table 1102.7
D 4068—09	Specification for Chlorinated Polyethylene (CPE) Sheeting for Concealed Water-containment Membrane	417.5.2.2
D 4551—12	Specification for Poly (Vinyl Chloride) (PVC) Plastic Flexible Concealed Water-containment Membrane	417.5.2.1
E 2727—10	Standard Practice for the Assessment of Rainwater Quality	1302.8.1
F 405—05	Specification for Corrugated Polyethylene (PE) Pipe and Fittings	Table 1102.5, Table 1403.2
F 409—12	Specification for Thermoplastic Accessible and Replaceable Plastic Tube and Tubular Fittings	424.1.2, Table 1102.7
F 437—09	Specification for Threaded Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80	Table 605.5
F 438—09	Specification for Socket-type Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 40	Table 605.5
F 439—12	Standard Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80	Table 605.5
F 441/F 441M— 13	Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80	Table 605.3, Table 605.4
F 442/F 442M— 13	Specification for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe (SDR-PR)	Table 605.3, Table 605.4
F 477—10	Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe	605.24, 705.16

F 493—10	Specification for Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings	605.15.2, 605.16.2
F 628—08	Specification for Acrylonitrile-Butadiene-Styrene (ABS) Schedule 40 Plastic Drain, Waste, and Vent Pipe with a Cellular Core	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 705.2.2, Table 1102.4, Table 1102.7
F 656—10	Specification for Primers for Use in Solvent Cement Joints of Poly (Vinyl Chloride) (PVC) Plastic Pipe and Fittings	605.22.3, 705.11.2, 705.14.2
F 714—2013	Specification for Polyethylene (PE) Plastic Pipe (SDR-PR) Based on Outside Diameter	Table 702.3, 7017.4
F 876—2013	Specification for Cross-linked Polyethylene (PEX) Tubing	Table 605.3, Table 605.4
F 877—2011a	Specification for Cross-linked Polyethylene (PEX) Plastic Hot and Cold Water Distribution Systems	Table 605.3, Table 605.4, Table 605.5
F 891—10	Specification for Coextruded Poly (Vinyl Chloride) (PVC) Plastic Pipe with a Cellular Core	Table 702.1 Table 702.2, Table 702.3, Table 1102.4, Table 1102.5, Table 1102.7
F 1055—13	Standard Specification for Electrofusion Type Polyethylene Fittings for Outside Diameter Controlled Polyethylene and Crosslinked Polyethylene Pipe and Tubing	Table 605.5
F 1281—11	Specification for Cross-linked Polyethylene/Aluminum/ Cross-linked Polyethylene (PEX-AL-PEX) Pressure Pipe	Table 605.3, Table 605.4, Table 605.5, 605.21.1
F 1282—10	Specification for Polyethylene/Aluminum/Polyethylene (PE-AL-PE) Composite Pressure Pipe	Table 605.3, Table 605.4, Table 605.5, 605.21.1
F 1412—09	Specification for Polyolefin Pipe and Fittings for Corrosive Waste Drainage	Table 702.1, Table 702.2, Table 702.4, 705.14.1, 901.3
F 1476—07	Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications	Table 605.5, 605.14.3, 605.18.3, 605.22.2, 605.23.3
F 1488—09e1	Specification for Coextruded Composite Pipe	Table 702.1, Table 702.2, Table 702.3, Table 1403.2
F 1548—01(2006)	Standard Specification for the Performance of Fittings for Use with	Table 605.5

	Gasketed Mechanical Couplings Used in Piping Applications	
F 1673—10	Polyvinylidene Fluoride (PVDF) Corrosive Waste Drainage Systems	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 705.15.1
F 1807—13	Specification for Metal Insert Fittings Utilizing a Copper Crimp Ring for SDR9 Cross-linked Polyethylene (PEX) Tubing and SDR9 Polyethylene of Raised Temperature (PE-RT) Tubing	Table 605.5
F 1866—07	Specification for Poly (Vinyl Chloride) (PVC) Plastic Schedule 40 Drainage and DWV Fabricated Fittings	Table 702.4, Table 1102.7
F 1960—12	Specification for Cold Expansion Fittings with PEX Reinforcing Rings for use with Cross-linked Polyethylene (PEX) Tubing	Table 605.5
F 1970—12	Special Engineered Fittings, Appurtenances or Valves for use in Poly (Vinyl Chloride) (PVC) OR Chlorinated Poly (Vinyl Chloride) (CPVC) Systems	Table 605.7
F 1974—09	Specification for Metal Insert Fittings for Polyethylene/Aluminum/Polyethylene and Cross-linked Polyethylene/Aluminum/Cross-linked Polyethylene Composite Pressure Pipe	Table 605.5, 605.21.1
F 1986—01(2011)	Specification for Multilayer Pipe, Type 2, Compression Fittings and Compression Joints for Hot and Cold Drinking Water Systems	Table 605.3, Table 605.4, Table 605.5
F 2080—12	Specifications for Cold-expansion Fittings with Metal Compression-sleeves for Cross-linked Polyethylene (PEX) Pipe	Table 605.5
F 2098—08	Standard specification for Stainless Steel Clamps for Securing SDR9 Cross-linked Polyethylene (PEX) Tubing to Metal and Plastic Insert Fittings	Table 605.5
F 2159—11	Specification for Plastic Insert Fittings Utilizing a Copper Crimp Ring for SDR9 Cross-linked Polyethylene (PEX) Tubing and SDR9 Polyethylene of Raised Temperature (PE-RT) Tubing	Table 605.5
F 2262—09	Specification for Cross-linked Polyethylene/Aluminum/Cross-linked Polyethylene Tubing OD Controlled SDR9	Table 605.3, Table 605.4
F 2306/F 2306M—13	12" to 60" Annular Corrugated Profile-wall Polyethylene (PE) Pipe and Fittings for Gravity Flow Storm Sewer and Subsurface Drainage Applications	Table 1102.4, Table 1102.7
F 2389—10	Specification for Pressure-rated Polypropylene (PP) Piping Systems	Table 605.3, Table 605.4, Table 605.5, Table 605.7, 605.20.1
F 2434—09	Standard Specification for Plastic Insert Fittings Utilizing a	Table 605.5

	Copper Crimp Ring for SDR9 Cross-linked Polyethylene (PEX) Tubing and SDR9 Cross-linked Polyethylene/Aluminum/ Cross-linked Polyethylene (PEX AL-PEX) Tubing	
F 2735—09	Standard Specification for Plastic Insert Fittings for SDR9 Cross-linked Polyethylene (PEX) and Polyethylene of Raised Temperature (PE-RT) Tubing	Table 605.5
F 2769—10	Polyethylene or Raised Temperature (PE-RT) Plastic Hot and Cold Water Tubing and Distribution Systems	Table 605.3, Table 605.4, Table 605.5
F 2831—11	Standard Practice for Internal Non Structural Epoxy Barrier Coating Material Used In Rehabilitation of Metallic Pressurized Piping Systems	601.5
F 2855—11	Specification Poly (Vinyl Chloride)/Aluminum/Poly (Vinyl Chloride) (CPVC/AL/CPVC) Composite Pressure Tubing	Table 605.3, Table 605.4

AWS	American Welding Society 8669 NW 36 Street, #130 Doral, FL 33166	
Standard reference number	Title	Referenced in code section number
A5.8M/A5.8—2011	Specifications for Filler Metals for Brazing and Braze Welding	605.11.1, 605.13.1, 605.14.1, 705.3.1, 705.6.1, 705.7.1

AWWA	American Water Works Association 6666 West Quincy Avenue Denver, CO 80235	
Standard reference number	Title	Referenced in code section number
C104/A21.4-08	Cement-mortar Lining for Ductile-iron Pipe and Fittings for Water	605.3, 605.5
C110/A21.10—12	Ductile-iron and Gray-iron Fittings	Table 605.5, Table 702.4, Table 1102.7
C111/A21.11-12	Rubber-gasket Joints for Ductile-iron Pressure Pipe and Fittings	605.12
C115/A21.15—11	Flanged Ductile-iron Pipe with Ductile-iron or Gray-iron Threaded Flanges	Table 605.3, Table 605.4
C151/A21.51—09	Ductile-iron Pipe, Centrifugally Cast for Water	Table 605.3, Table 605.4
C153—00/A21.53—11	Ductile-iron Compact Fittings for Water Service	Table 605.5
C500—09	Standard for Metal-Seated Gate Valves for Water Supply Service	Table 605.7
C504—10	Standard for Rubber-Seated Butterfly Valves	Table 605.7
C507—11	Standard for Ball Valves, 6 In. Through 60 In	Table 608.1, 608.13.7

C510—07	Double Check Valve Backflow Prevention Assembly	Table 608.1, 608.13.7
C511—07	Reduced-pressure Principle Backflow Prevention Assembly	Table 608.1, 608.13.2, 608.16.2
C651—05	Disinfecting Water Mains	610.1
C652—11	Disinfection of Water-storage Facilities	610.1
C901—08	Polyethylene (PE) Pressure Pipe and Tubing 1/2 inch (13 mm) Through (75 mm) for Water Service	Table 605.3
C904—08	Cross-linked Polyethylene (PEX) Pressure Pipe 1/2 inch (13 mm) Through 75 mm) for Water Service	Table 605.3
D100—05	Standard for Welded Carbon Steel Tanks for Water Storage	1302.7.2
D115—06	Standard for Tendon Prestressed-Concrete Water Tanks	1302.7.2
D120—09	Standard for Thermosetting Fiberglass-Reinforced Plastic Tanks	1302.7.2

CISPI	Cast Iron Soil Pipe Institute 5959 Shallowford Road, Suite 419 Chattanooga, TN 37421	
Standard reference number	Title	Referenced in code section number
301—04a	Specification for Hubless Cast-iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste and Vent Piping Applications	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 708.7, Table 1102.4, Table 1102.5, Table 1102.7
310—04	Specification for Coupling for Use in Connection with Hubless Casts-iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste and Vent Piping Applications	705.4.3

CSA	CSA Group 8501 East Pleasant Valley Cleveland, OH 44131-5516	
Standard reference number	Title	Referenced in code section number
A257.1M—2009	Circular Concrete Culvert, Storm Drain, Sewer Pipe and Fittings	Table 702.3, Table 1102.4
A257.2M—2009	Reinforced Circular Concrete Culvert, Storm Drain, Sewer Pipe and Fittings	Table 702.3, Table 1102.4
A257.3M—2009	Joints for Circular Concrete Sewer and Culvert Pipe, Manhole Sections and Fittings Using Rubber Gaskets	705.5, 705.16
ASME A112.18.1— 2012/	Plumbing Supply Fittings	424.1, 424.2, 424.3, 424.4, 424.6, 424.8,

CSA B125.1— 2012		605.7, 607.4, 608.2
ASME A112.18.2— 2011/ CSA B125.2— 2011	Plumbing Waste Fittings	424.1.2
ASME A112.19.2— 2013/ B45.1—2013	Ceramic Plumbing Fixtures	401.2, 405.9, 407.1, 408.1, 410.1, 415.1, 416.1, 417.1, 418.1, 419.1, 420.1
ASME A112.19.1— 2013/ CSA B45.2— 2013	Enameled Cast-iron and Enameled Steel Plumbing Fixtures	407.1, 410.1, 415.1, 416.1, 418.1
ASME A112.19.3— 2008/ CSA B45.4— 08(R2013)	Stainless-steel Plumbing Fixtures	405.9, 407.1, 415.1, 416.1, 418.1, 420.1
ASME A112.19.5— 2011/ CSA B45.15— 2011	Flush Valves and Spuds for Water Closets, Urinals and Tanks	425.4
ASME A112.19.7— 2012/ CSA B45.10— 2012	Hydromassage Bathtub Systems	421.1, 421.4
CSA B45.5—11/ IAPMO Z124- 2011	Plastic Plumbing Fixtures	407.1, 415.1, 416.1, 416.2, 417.1, 419.1, 420.1
ASME A112.3.4—2013/ CSA B45.9— 2013	Macerating Systems and Related Components	712.4.1
ASSE 1016/ ASME A112.1016/ CSA B125.16— 2011	Performance Requirements for Individual Thermostatic, Pressure Balancing and Combination Control Valves for Individual Fixture Fittings	424.3, 424.4, 607.4
B64.1.1—11	Vacuum Breakers, Atmospheric Type (AVB)	425.2, Table 608.1, 608.13.6, 608.16.4.1
B64.1.2—11	Pressure Vacuum Breakers, (PVB)	Table 608.1, 608.13.5
B64.1.3—11	Spill Resistant Pressure Vacuum Breaks (SRPVB)	608.13.8
B64.2—11	Vacuum Breakers, Hose Connection Type (HCVB)	Table 608.1, 608.13.6

B64.2.1—11	Vacuum Breakers, Hose Connection (HCVB) with Manual Draining Feature	Table 608.1, 608.13.6
B64.2.1.1—11	Hose Connection Dual Check Vacuum Breakers, (HCDVB)	Table 608.1, 608.13.6
B64.2.2—11	Vacuum Breakers, Hose Connection Type (HCVB) with Automatic Draining Feature	Table 608.1, 608.13.6
B64.3—11	Backflow Preventers, Dual Check Valve Type with Atmospheric Port (DCAP)	Table 608.1, 608.13.3, 608.16.2
B64.4—11	Backflow Preventers, Reduced Pressure Principle Type (RP)	Table 608.1, 608.13.2, 608.16.2
B64.4.1—11	Reduced Pressure Principle for Fire Sprinklers (RPF)	Table 608.1, 608.13.2
B64.5—11	Double Check Backflow Preventers (DCVA)	Table 608.1, 608.13.7
B64.5.1—11	Double Check Valve Backflow Preventer for Fire Systems (DCVAF)	Table 608.1, 608.13.7
B64.6—11	Dual Check Backflow Preventers Valve (DuC)	605.3.1, Table 608.1, 608.13.10
B64.7—11	Laboratory Faucet Vacuum Breakers (LFVB)	Table 608.1, 608.13.6
B64.10—11	Manual for the Selection and Installation of Backflow Prevention Devices	312.10.2
B64.10.1—11	Maintenance and Field Testing of Backflow Preventers	312.10.2
B79—08(R2013)	Commercial and Residential Drains, and Cleanouts	412.1
B125.3—2012	Plumbing Fittings	408.3, 416.5, 623.3, 424.4, 424.5, 425.2, 425.3.1, Table 605.7, Table 608.1
B137.1—13	Polyethylene (PE) Pipe, Tubing and Fittings for Cold Water Pressure Services	Table 605.3, Table 605.5
B137.2—13	Polyvinylchloride PVC Injection-moulded Gasketed Fittings for Pressure Applications	Table 605.5
B137.3—13	Rigid Poly (Vinyl Chloride) (PVC) Pipe for Pressure Applications	Table 605.3, Table 605.5, 605.22.3, 705.11.2, 705.14.2
B137.5—13	Cross-linked Polyethylene (PEX) Tubing Systems for Pressure Applications	Table 605.3, Table 605.4, Table 605.5
B137.6—13	CPVC Pipe, Tubing and Fittings for Hot and Cold Water Distribution Systems	Table 605.3, Table 605.4, Table 605.5
B137.9—13	Polyethylene/Aluminum/ Polyethylene Composite Pressure Pipe Systems	Table 605.3, Table 605.5, 605.21.1
B137.10—13	Cross-linked Polyethylene/Aluminum/ Polyethylene Composite Pressure Pipe Systems	Table 605.3, Table 605.4, Table 605.5, 605.21.1

B137.11—13	Polypropylene (PP-R) Pipe and Fittings for Pressure Applications	Table 605.3, Table 605.4, Table 605.5
B181.1—11	Acrylonitrile-butadiene-styrene ABS Drain, Waste and Vent Pipe and Pipe Fittings	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 705.2.2, 7015.2, Table 1102.4, Table 1102.7
B 181.2—11	Polyvinylchloride PVC and chlorinated polyvinylchloride (CPVC) Drain, Waste, and Vent Pipe and Pipe Fittings	Table 702.1 Table 702.2, 705.11.2, 705.14.2, 715.2
B 181.3—11	Polyolefin and Polyvinylidene Fluoride (PVDF) Laboratory Drainage Systems	Table 702.1, Table 702.2, Table 702.3, Table 702.4, 705.14.1
B182.1—11	Plastic Drain and Sewer Pipe and Pipe Fittings	705.11.2, 705.14.2, Table 1102.4, Table 1102.5
B182.2—11	PSM Type Polyvinylchloride PVC Sewer Pipe and Fittings	Table 702.3, Table 1102.4, Table 1102.5
B182.4—11	Profile Polyvinylchloride PVC Sewer Pipe and Fittings	Table 702.3, Table 1102.4, Table 1102.5
B182.6—11	Profile Polyethylene (PE) Sewer Pipe and Fittings for Leak-proof Sewer Applications	Table 1102.5
B182.8—11	Profile Polyethylene (PE) Storm Sewer and Drainage Pipe and Fittings	Table 1102.5
B356—10	Water Pressure Reducing Valves for Domestic Water Systems	604.8
B483.1—14	Drinking Water Treatment Units	611.1, 611.2
B481.1—12	Testing and Rating of Grease interceptors Using Lard	1003.3.4
B481.3—12	Sizing, Selection, Location and Installation of Grease Interceptors	1003.3.4
B602—10	Mechanical Couplings for Drain, Waste and Vent Pipe and Sewer Pipe	705.2.1, 705.4.3, 705.5, 705.11.1, 705.12, 705.13.2, 705.16

IAPMO	IAPMO Group 4755 E. Philadelphia Ontario, CA 91761	
Standard reference number	Title	Referenced in code section number
Z1001-2007	Prefabricated Gravity Grease Interceptors	1003.3.6
CSA B45.5—11/ IAPMO Z124- 2011	Plastic Plumbing Fixtures	407.1, 415.1, 416.1, 416.2, 417.1, 419.1,

		420.1
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ICC	International Code Council, Inc. 500 New Jersey Ave, NW 6th Floor Washington, DC 20001	
Standard reference number	Title	Referenced in code section number
A117.1—2009	Accessible and Usable Buildings and Facilities	404.2
SBC 201	Saudi Building Code	201.3, 307.1, 307.2, 307.3, 308.2, 309.1, 309.2, 310.1, 310.3, 315.1, 403.1, Table 403.1, 403.4, 404.1, 407.3, 417.6, 502.4, 606.5.2, 1106.5, 1301.9.4, 1307.7.2, 1303.6, 1402.1
SBC 601	Saudi Energy Conservation Code	313.1, 607.5
SBC 801	Saudi Fire Code	201.3, 1201.1
SBC 1201	Saudi Fuel Gas Code	101.2, 201.3, 502.1, 502.1.1
SBC 501	Saudi Mechanical Cod	201.3, 307.7, 310.1, 422.9, 502.1, 502.1.1, 612.1, 1202.1
SBC 702	Saudi Private Sewage Disposal Code	701.2
SBC 1101	Saudi Residential Code®	101.2

ISEA	International Safety Equipment Association 1901 N. Moore Street, Suite 808 Arlington, VA 22209	
Standard reference number	Title	Referenced in code section number
ANSI/ISEA Z358.1—2009	Emergency Eyewash and Shower Equipment	411.1

MSS	Manufacturers Standardization Society Of the Valve and Fittings Industry, Inc. 127 Park St. NE Vienna, VA 22180-4602	
Standard reference number	Title	Referenced in code section number
SP-67—2011	Butterfly Valves	Table 605.7
SP-70—2011	Gray Iron Gate Valves, Flanged and Threaded Ends	Table 605.7

SP-71—2011	Gray Iron Swing Check Valves, Flanged and Threaded Ends	Table 605.7
SP-72—2010	Ball Valves with Flanged or Butt-Welding Ends for General Service	Table 605.7
SP-78—2011	Cast Iron Plug Valves, Flanged and Threaded Ends	Table 605.7
SP-80—2008	Bronze Gate, Globe, Angle and Check Valves	Table 605.7
SP-110—2010	Ball Valves, Threaded, Socket Welded, Solder Joint, Grooved and Flared Ends	Table 605.7

NFPA	National Fire Protection Association 1 Batterymarch Park Quincy, MA 02169-7471	
Standard reference number	Title	Referenced in code section number
51—13	Design and Installation of Oxygen-fuel Gas Systems for Welding, Cutting and Allied Processes	1203.1
55—13	Compressed Gases and Cryogenic Fluids Code	1203.1
70—14	National Electric Code	502.1, 504.3, 11013.1.3
99—15	Health Care Facilities Code	1202.1

NSF	NSF International 789 Dixboro Road Ann Arbor, MI 48105	
Standard reference number	Title	Referenced in code section number
3—2010	Commercial Warewashing Equipment	409.1
14—2011	Plastic Piping System Components and Related Materials	303.3, 611.3
18—2012	Manual Food and Beverage Dispensing Equipment	426.1
42—2011	Drinking Water Treatment Units—Aesthetic Effects	611.1, 611.3
44—2012	Residential Cation Exchange Water Softeners	611.1, 611.3
50—2012	Equipment for Swimming Pools, Spas, Hot Tubs and other Recreational Facilities	1302.8.1
53—2011a	Drinking Water Treatment Units—Health Effects	611.1, 611.3
58—2012	Reverse Osmosis Drinking Water Treatment Systems	611.2, 611.3
61—2012	Drinking Water System Components—Health Effects	410.1, 424.1, 605.3, 605.4, 605.5, 605.7, 611.3
62—2012	Drinking Water Distillation Systems	611.1
350—2011	Onsite Residential and Commercial Water Reuse Treatment Systems	1302.6.1
359—2011	Valves for Crosslinked Polyethylene (PEX) Water Distribution Tubing Systems	Table 605.7
372—2010	Drinking Water Systems Components—Lead Content	605.2.1

PDI	Plumbing and Drainage Institute 800 Turnpike Street, Suite 300 North Andover, MA 01845	
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Standard reference number	Title	Referenced in code section number
PDI G101 (2012)	Testing and Rating Procedure for Grease Interceptors with Appendix of Sizing and Installation Data	1003.3.4
PDI G102 (2009)	Testing and Certification for Grease Interceptors with Fog Sensing and Alarm Devices	1003.3.4

UL	UL LLC 333 Pfingsten Road Northbrook, IL 60062-2096	
Standard reference number	Title	Referenced in code section number
58—1996	Steel Underground Tanks for Flammable and Combustible Liquids— with revisions through July 27, 1998	1302.7.2
142—2006	Steel Aboveground Tanks for Flammable and Combustible Liquids— with revisions through February 12, 2010	1302.7.2
399—2008	Drinking-Water Coolers—with revisions through January 14, 2011	410.1
430—2009	Waste Disposers—with revisions through March 23, 2011	413.1
508—99	Industrial Control Equipment—with revisions through March 2013	314.2.3
515—2011	Electrical Resistance Heat Tracing for Commercial and Industrial Applications including—with revisions through November 30, 2011	607.2.1.3.2
1316—1994	Glass0Fiber Reinforced Plastic Underground Storage Tanks for Petroleum Products, Alcohols and Alcohol Gasoline Mixtures— with revisions through May 12, 2006	1302.7.2
1746—2007	External Corrosion Protection Systems for Steel Underground Storage Tanks	1302.7.2
1795—2009	Hydromassage Bathtubs including revisions through August 23, 2011	4021.1

APPENDIX A. PLUMBING PERMIT FEE SCHEDULE

The provisions contained in this appendix are not mandatory unless specifically referenced in the adopting ordinance.

Permit Issuance

1.	For issuing each permit	
2.	For issuing each supplemental permit	

Unit Fee Schedule

1.	For each plumbing fixture or trap or set of fixtures on one trap (including water, drainage piping and backflow protection thereof)	
2.	For each building sewer and each trailer park sewer	
3.	Rainwater systems—per drain (inside building)	
4.	For each cesspool (where permitted)	
5.	For each private sewage disposal system	
6.	For each water heater and/or vent	
7.	For each industrial waste pretreatment interceptor including its trap and vent, excepting kitchen-type grease interceptors functioning as fixture traps	
8.	For installation, alteration or repair of water-piping and/or water-treating equipment, each	
9.	For repair or alteration of drainage or vent piping, each fixture	
10.	For each lawn sprinkler system on any one meter including backflow protection devices therefor	
11.	For atmospheric-type vacuum breakers not included in Item 2	
	1 to 5	
	over 5, each	
12.	For each backflow protective device other than atmospheric-type vacuum breakers:	
	(50 mm) and smaller	
	Over (50 mm)	

Other Inspections and Fees

1.	Inspections outside of normal business hours (minimum charge 2 hours)	per hour
2.	Reinspection fee assessed under provisions of Section 107.4.3	each
3.	Inspections for which no fee is specifically indicated (minimum charge one-half hour)	per hour
4.	Additional plan review required by changes, additions or revisions to approved plans (minimum charge one-half hour)	per hour

APPENDIX B. Rates of Rainfall for Various Cities

Rainfall rates, in mm per hour, are based on a storm of 1-hour duration and a 100-year return period. The rainfall rates shown

in the appendix are derived from Figure 1106.1



APPENDIX C. STRUCTURAL SAFETY

SECTION C101 CUTTING, NOTCHING AND BORING IN WOOD MEMBERS

C101.1 Joist notching.

Notches on the ends of joists shall not exceed one-fourth the joist depth. Holes bored in joists shall not be within 50 mm of the top or bottom of the joist, and the diameter of any such hole shall not exceed one-third the depth of the joist. Notches in the top or bottom of joists shall not exceed one-sixth the depth and shall not be located in the middle third of the span.

C101.2 Stud cutting and notching.

In exterior walls and bearing partitions, any wood stud is permitted to be cut or notched to a depth not exceeding 25 percent of its width. Cutting or notching of studs to a depth not greater than 40 percent of the width of the stud is permitted in nonbearing partitions supporting no loads other than the weight of the partition.

C101.3 Bored holes.

The diameter of bored holes in wood studs shall not exceed 40 percent of the stud depth. The diameter of bored holes in wood studs shall not exceed 60 percent of the stud depth in nonbearing partitions. The diameter of bored holes in wood studs shall not exceed 60 percent of the stud depth in any wall where each stud is doubled, provided that not more than two such successive doubled studs are so bored. The edge of the bored hole shall not be closer than 16 mm to the edge of the stud. Bored holes shall be not located at the same section of stud as a cut or notch.

C101.4 Cutting, notching and boring holes in structural steel framing.

The cutting, notching and boring of holes in structural steel framing members shall be as prescribed by the registered design professional.

C101.5 Cutting, notching and boring holes in cold-formed steel framing.

Flanges and lips of load-bearing cold-formed steel framing members shall not be cut or notched. Holes in webs of load-bearing cold-formed steel framing members shall be permitted along the centerline of the web of the framing member and shall not exceed the dimensional limitations, penetration spacing or minimum hole edge distance as prescribed by the registered design professional. Cutting, notching and boring holes of steel floor/roof decking shall be as prescribed by the registered design professional.

C101.6 Cutting, notching and boring holes in nonstructural cold-formed steel wall framing.

Flanges and lips of nonstructural cold-formed steel wall studs shall not be cut or notched. Holes in webs of nonstructural cold-formed steel wall studs shall be permitted along the centerline of the web of the framing member, shall not exceed 40 mm in width or 100 mm in length, and the holes shall not be spaced less than 600 mm center to center from another hole or less than 250 mm from the bearing end

APPENDIX D. Degree Day and Design Temperatures

This appendix is informative and is not part of the code.



APPENDIX E. Sizing of Water Piping System

The provisions contained in this appendix are not mandatory unless specifically referenced in the adopting ordinance.

SECTION E101 GENERAL

E101.1 Scope.

E101.1.1 This appendix outlines two procedures for sizing a water piping system (see Sections E103.3 and E201.1). The design procedures are based on the minimum static pressure available from the supply source, the head changes in the system caused by friction and elevation, and the rates of flow necessary for operation of various fixtures.

E101.1.2 Because of the variable conditions encountered in hydraulic design, it is impractical to specify definite and detailed rules for sizing of the water piping system. Accordingly, other sizing or design methods conforming to good engineering practice standards are acceptable alternatives to those presented herein.

SECTION E102 INFORMATION REQUIRED

E102.1 Preliminary. Obtain the necessary information regarding the minimum daily static service pressure in the area where the building is to be located. If the building supply is to be metered, obtain information regarding friction loss relative to the rate of flow for meters in the range of sizes likely to be used. Friction loss data can be obtained from most manufacturers of water meters.

E102.1.1 Demand load. E102.2.1 Estimate the supply demand of the building main and the principal branches and risers of the system by totaling the corresponding demand from the applicable part of Table E103.3(3).

E102.1.2 Estimate continuous supply demands in liters per minute (L/min) for lawn sprinklers, air conditioners, etc., and add the sum to the total demand for fixtures. The result is the estimated supply demand for the building supply.

SECTION E103 SELECTION OF PIPE SIZE

E103.1 General. Decide from Table 604.3 what is the desirable minimum residual pressure that should be maintained at the highest fixture in the supply system. If the highest group of fixtures contains flushometer valves, the pressure for the group should be not less than 103 kPa flowing. For flush tank supplies, the available pressure should be not less than 55 kPa flowing, except blowout action fixtures must be not less than 172 kPa flowing.

E103.2 Pipe sizing.

E103.2.1 Pipe sizes can be selected according to the following procedure or by other design methods conforming to acceptable engineering practice and approved by the administrative authority. The sizes selected must not be less than the minimum required by this code.

E103.2.2 Water pipe sizing procedures are based on a system of pressure requirements and losses, the sum of which must not exceed the minimum pressure available at the supply source. These pressures are as follows:

1. Pressure required at fixture to produce required flow. See Sections 604.3 and 604.5.
2. Static pressure loss or gain (due to head) is computed at 9.8 kPa/meter of elevation change.

Example: Assume that the highest fixture supply outlet is 6 m above or below the supply source. This produces a static pressure differential of 6 m by 9.8 kPa/meter and an 58.8 kPa loss.

1. Loss through water meter. The friction or pressure loss can be obtained from meter manufacturers.

2. Loss through taps in water main.
3. Losses through special devices such as filters, softeners, backflow prevention devices and pressure regulators. These values must be obtained from the manufacturers.
4. Loss through valves and fittings. Losses for these items are calculated by converting to equivalent length of piping and adding to the total pipe length.
5. Loss due to pipe friction can be calculated when the pipe size, the pipe length and the flow through the pipe are known. With these three items, the friction loss can be determined. For piping flow charts not included, use manufacturers' tables and velocity recommendations.

E103.3 Segmented loss method. The size of water service mains, branch mains and risers by the segmented loss method, must be determined according to water supply demand [L/min], available water pressure [kPa] and friction loss caused by the water meter and developed length of pipe [m], including equivalent length of fittings. This design procedure is based on the following parameters:

- Calculates the friction loss through each length of the pipe.
- Based on a system of pressure losses, the sum of which must not exceed the minimum pressure available at the street main or other source of supply.
- Pipe sizing based on estimated peak demand, total pressure losses caused by difference in elevation, equipment, developed length and pressure required at most remote fixture, loss through taps in water main, losses through fittings, filters, backflow prevention devices, valves and pipe friction.

Because of the variable conditions encountered in hydraulic design, it is impractical to specify definite and detailed rules for sizing of the water piping system. Current sizing methods do not address the differences in the probability of use and flow characteristics of fixtures between types of occupancies. Creating an exact model of predicting the demand for a building is impossible and final studies assessing the impact of water conservation on demand are not

yet complete. The following steps are necessary for the segmented loss method.

1. **Preliminary.** Obtain the necessary information regarding the minimum daily static service pressure in the area where the building is to be located. If the building supply is to be metered, obtain information regarding friction loss relative to the rate of flow for meters in the range of sizes to be used. Friction loss data can be obtained from manufacturers of water meters. It is essential that enough pressure be available to overcome all system losses caused by friction and elevation so that plumbing fixtures operate properly. Section 604.6 requires the water distribution system to be designed for the minimum pressure available taking into consideration pressure fluctuations. The lowest pressure must be selected to guarantee a continuous, adequate supply of water. The lowest pressure in the public main usually occurs in the summer because of lawn sprinkling and supplying water for air-conditioning cooling towers. Future demands placed on the public main as a result of large growth or expansion should also be considered. The available pressure will decrease as additional loads are placed on the public system.

2. **Demand load.** Estimate the supply demand of the building main and the principal branches and risers of the system by totaling the corresponding demand from the applicable part of Table E103.3(3). When estimating peak demand sizing methods typically use water supply fixture units (w.s.f.u.) [see Table E103.3(2)]. This numerical factor measures the load-producing effect of a single plumbing fixture of a given kind. The use of such fixture units can be applied to a single basic probability curve (or table), found in the various sizing methods [Table E103.3(3)]. The fixture units are then converted into L/min flow rate for estimating demand.

- 2.1. Estimate continuous supply demand in L/min for lawn sprinklers, air conditioners, etc., and add the sum to the total demand for fixtures. The result is the estimated supply demand for the building supply. Fixture

units cannot be applied to constant use fixtures such as hose bibbs, lawn sprinklers and air conditioners. These types of fixtures must be assigned the L/min value.

- 14 Selection of pipe size.** This water pipe sizing procedure is based on a system of pressure requirements and losses, the sum of which must not exceed the minimum pressure available at the supply source. These pressures are as follows:

2.2. Pressure required at the fixture to produce required flow. See Section 604.3 and Section 604.5.

2.3. Static pressure loss or gain (because of head) is computed at 9.8 kPa/meter of elevation change.

2.4. Loss through a water meter. The friction or pressure loss can be obtained from the manufacturer.

2.5. Loss through taps in water main [see Table E103.3(4)].

2.6. Losses through special devices such as filters, softeners, backflow prevention devices and pressure regulators. These values must be obtained from the manufacturers.

2.7. Loss through valves and fittings [see Tables E103.3(5) and E103.3(6)]. Losses for these items are calculated by converting to equivalent length of piping and adding to the total pipe length.

2.8. Loss caused by pipe friction can be calculated when the pipe size, the pipe length and the flow through the pipe are known. With these three items, the friction loss can be determined as demonstrated in the solution of the next problem. For piping flow charts not included, use manufacturers' tables and velocity recommendations. Before attempting to size any water supply system, it is necessary to gather preliminary information which includes available pressure, piping material, select design velocity, elevation differences and developed length to most remote fixture. The water supply system is divided into sections at major changes

in elevation or where branches lead to fixture groups. The peak demand must be determined in each part of the hot and cold water supply system which includes the corresponding water supply fixture unit and conversion to L/min flow rate to be expected through each section. Sizing methods require the determination of the "most hydraulically remote" fixture to compute the pressure loss caused by pipe and fittings. The hydraulically remote fixture represents the most downstream fixture along the circuit of piping requiring the most available pressure to operate properly. Consideration must be given to all pressure demands and losses, such as friction caused by pipe, fittings and equipment, elevation and the residual pressure required by Table 604.3. The two most common and frequent complaints about the water supply system operation are lack of adequate pressure and noise.

Problem: What size Type L copper water pipe, service and distribution will be required to serve a two-story factory building having on each floor, back-to-back, two toilet rooms each equipped with hot and cold water? The highest fixture is 6400 mm above the street main, which is tapped with a 50 mm corporation cock at which point the minimum pressure is 380 kPa. In the building basement, 50 mm meter with a maximum pressure drop of 75 kPa and 80 mm reduced pressure principle backflow preventer with a maximum pressure drop of 62 kPa are to be installed. The system is shown by Figure E103.3(1). To be determined are the pipe sizes for the service main and the cold and hot water distribution pipes.

Solution: A tabular arrangement such as shown in Table E103.3(1) should first be constructed. The steps to be followed are indicated by the tabular arrangement itself as they are in sequence, columns 1 through 10 and lines A through L.

Step 1

Columns 1 and 2: Divide the system into sections breaking at major changes in elevation or where branches lead to fixture groups. After point B [see Figure E103.3(1)], separate

consideration will be given to the hot and cold water piping. Enter the sections to be considered in the service and cold water piping in Column 1 of the tabular arrangement. Column 1 of Table E103.3(1) provides a line-by-line recommended tabular arrangement for use in solving pipe sizing.

The objective in designing the water supply system is to ensure an adequate water supply and pressure to all fixtures and equipment. Column 2 provides the kPa to be considered separately from the minimum pressure available at the main. Losses to take into consideration are the following: the differences in elevations between the water supply source and the highest water supply outlet, meter pressure losses, the tap in main loss, special fixture devices such as water softeners and backflow prevention devices and the pressure required at the most remote fixture outlet. The difference in elevation can result in an increase or decrease in available pressure at the main. Where the water supply outlet is located above the source, this results in a loss in the available pressure and is subtracted from the pressure at the water source. Where the highest water supply outlet is located below the water supply source, there will be an increase in pressure that is added to the available pressure of the water source.

Column 3: According to Table E103.3(3), determine the L/min of flow to be expected in each section of the system. These flows range from 95 – 309 L/min. Load values for fixtures must be determined as water supply fixture units and then converted to a liter-per-minute (L/min) rating to determine peak demand. When calculating peak demands, the water supply fixture units are added and then converted to the liter-per-minute rating. For continuous flow fixtures such as hose bibbs and lawn sprinkler systems, add the liter-per-minute demand to the intermittent demand of fixtures. For example, a total of 120 water supply fixture units is converted to a demand of 182 liter per minute. Two hose bibbs $\times$ 20 L/min demand = 40 L/min. Total L/min rating = 182 L/min + 40 L/min = 222 L/min demand.

Step 2

Line A: Enter the minimum pressure available at the main source of supply in Column 2. This is 380 kPa. The local water authorities generally keep records of pressures at different

times of day and year. The available pressure can also be checked from nearby buildings or from fire department hydrant checks.

Line B: Determine from Table 604.3 the highest pressure required for the fixtures on the system, which is 103.4 kPa, to operate a flushometer valve. The most remote fixture outlet is necessary to compute the pressure loss caused by pipe and fittings, and represents the most downstream fixture along the circuit of piping requiring the available pressure to operate properly as indicated by Table 604.3.

Line C: Determine the pressure loss for the meter size given or assumed. The total water flow from the main through the service as determined in Step 1 will serve to aid in the meter selected. There are three common types of water meters; the pressure losses are determined by the American Water Works Association Standards for displacement type, compound type and turbine type. The maximum pressure loss of such devices takes into consideration the meter size, safe operating capacity (L/min) and maximum rates for continuous operations (L/min). Typically, equipment imparts greater pressure losses than piping.

Line D: Select from Table E103.3(4) and enter the pressure loss for the tap size given or assumed. The loss of pressure through taps and tees are based on the total liter-per-minute flow rate and size of the tap.

Line E: Determine the difference in elevation between the main and source of supply and the highest fixture on the system. Multiply this figure, expressed in meter, by 9.8 kPa. Enter the resulting kPa loss on Line E. The difference in elevation between the water supply source and the highest water supply outlet has a significant impact on the sizing of the water supply system. The difference in elevation usually results in a loss in the available pressure because the water supply outlet is generally located above the water supply source. The loss is caused by the pressure required to lift the water to the outlet. The pressure loss is subtracted from the pressure at the water source. Where the highest water supply outlet is located below the water source, there will be an increase in pressure which is added to the available pressure of the water source.

Lines F, G and H: The pressure losses through

filters, backflow prevention devices or other special fixtures must be obtained from the manufacturer or estimated and entered on these lines. Equipment such as backflow prevention devices, check valves, water softeners, instantaneous or tankless water heaters, filters and strainers can impart a much greater pressure loss than the piping. The pressure losses can range from 50 to 200 kPa.

Step 3

Line I: The sum of the pressure requirements and losses that affect the overall system (Lines B through H) is entered on this line. Summarizing the steps, all of the system losses are subtracted from the minimum water pressure. The remainder is the pressure available for friction, defined as the energy available to push the water through the pipes to each fixture. This force can be used as an average pressure loss, as long as the pressure available for friction is not exceeded. Saving a certain amount for available water supply pressures as an area incurs growth, or because of aging of the pipe or equipment added to the system is recommended.

Step 4

Line J: Subtract Line I from Line A. This gives the pressure that remains available from overcoming friction losses in the system. This figure is a guide to the pipe size that is chosen for each section, incorporating the total friction losses to the most remote outlet (measured length is called developed length).

Exception: When the main is above the highest fixture, the resulting kPa must be considered a pressure gain (static head gain) and omitted from the sums of Lines B through H and added to Line J.

The maximum friction head loss that can be tolerated in the system during peak demand is the difference between the static pressure at the highest and most remote outlet at no-flow conditions and the minimum flow pressure required at that outlet. If the losses are within the required limits, then every run of pipe will also be within the required friction head loss. Static pressure loss in the most remote outlet is 9.8 kPa per meter = loss in kPa caused by elevation differences.

Step 5

Column 4: Enter the length of each section from

the main to the most remote outlet (at Point E). Divide the water supply system into sections breaking at major changes in elevation or where branches lead to fixture groups.

Step 6

Column 5: When selecting a trial pipe size, the length from the water service or meter to the most remote fixture outlet must be measured to determine the developed length. However, in systems having a flushometer valve or temperature controlled shower at the top most floors the developed length would be from the water meter to the most remote flushometer valve on the system. A rule of thumb is that size will become progressively smaller as the system extends farther from the main source of supply. Trial pipe size may be arrived at by the following formula:

To determine the equivalent length of run to the most remote outlet, the developed length is determined and added to the friction losses for fittings and valves. The developed lengths of the designated pipe sections are as follows:

A – B	16 m
B – C	2.4 m
C – D	4 m
D – E	45 m

Total developed length = 67.4 m

The equivalent length of the friction loss in fittings and valves must be added to the developed length (most remote outlet). Where the size of fittings and valves is not known, the added friction loss should be approximated. A general rule that has been used is to add 50 percent of the developed length to allow for fittings and valves. For example, the equivalent length of run equals the developed length of run ($67.4 \times 1.5 = 101.1$ m). The total equivalent length of run for determining a trial pipe size is 101.1 m.

For trial pipe size, passed on 66.3 kPa (6.77 m) pressure available to overcome pipe friction, total flow 409 L/min (0.007 m³/s), 101.1 m total equivalent length of run, and C value of 150 for copper pipe, and using Hazen-Williams equation, the estimated diameter 65 mm pipe.

Step 7

Column 6: Select from Table E103.3(6) the equivalent lengths for the trial pipe size of

fittings and valves on each pipe section. Enter the sum for each section in Column 6. (The number of fittings to be used in this example must be an estimate.) The equivalent length of piping is the developed length plus the equivalent lengths of pipe corresponding to friction head losses for fittings and valves. Where the size of fittings and valves is not known, the added friction head losses must be approximated. An estimate for this example is found in Table E.1.

Step 8

Column 7: Add the figures from Column 4 and Column 6, and enter in Column 7. Express the sum in meter.

Step 9

Column 8: Use Hazen-Williams equation to calculate the friction loss, for copper pipe $C = 150$, for the L/min flow in a section (Column 3) and trial pipe size (Column 5) (inside diameter should be used in calculation). In addition, if head loss charts is available from pipe manufacture it is recommended be used in calculating the head loss. Maximum friction head loss is determined on the basis of total pressure available for friction head loss and the longest equivalent length of run. The selection is based on the liter-per-minute demand, the uniform friction head loss, and the maximum design velocity. Where the size indicated by hydraulic table indicates a velocity in excess of the selected velocity, a size must be selected which produces the required velocity.

Step 10

Line K: Enter the sum of the values in Column 8. The value is the total friction loss in equivalent length for each designated pipe section.

Note: Section C-F should be considered in the total pipe friction losses only if greater loss occurs in Section C-F than in pipe section D-E. Section C-F is not considered in the total developed length.

Step 11

Line L: Subtract Line J from Line K and enter in Column 9.

Note: 1 m head of water = 9.8 kPa

The result should always be a positive or plus figure. If it is not, repeat the operation using Columns 5, 6, and 8 until a balance or near balance is obtained. If the difference between Lines J and K is a high positive number, it is an indication that the pipe sizes are too large and should be reduced, thus saving materials. In such a case, the operations using Columns 5, 6, and 8 should again be repeated.

The total friction losses are determined and subtracted from the pressure available to overcome pipe friction for trial pipe size. This number is critical as it provides a guide to whether the pipe size selected is too large and the process should be repeated to obtain an economically designed system.

Answer: The final figures entered in Column 5 become the design pipe size for the respective sections. Repeating this operation a second time using the same sketch but considering the demand for hot water, it is possible to size the hot water distribution piping. This has been worked up as a part of the overall problem in the tabular arrangement used for sizing the service and water distribution piping. Note that consideration must be given to the pressure losses from the street main to the water heater (Section A-B) in determining the hot water pipe sizes.

TABLE E.1

Cold Water Pipe Section	Fittings/Valves	Pressure Loss Expressed as Equivalent Length of Tube (m)	Hot Water Pipe Section	Fittings/Valves	Pressure Loss Expressed as Equivalent of Tube (M)
A-B	80 – 65 mm Gate valves	0.91	A-B	80 – 65 mm Gate valves	0.91
	25 – 65 mm Side branch tee	3.65		25 – 65 mm Side branch tee	3.65
B-C	25 – 65 mm Straight run tee	0.15	B-C	25 – 50 mm Straight run tee	2.13
				25 – 50 mm 90-degree ell	0.15
C-F	25 – 65 mm Side branch tee	3.65	C-F	25 – 40 mm Side branch tee	2.13
C-D	25 – 65 mm 90-degree ell	2.13	C-D	25 – 40 mm 90-degree ell	1.22
D-E	25 – 65 mm Side branch tee	3.65	D-E	25 – 40 mm Side branch tee	2.13



TABLE E103.3(1)
RECOMMENDED TABULAR ARRANGEMENT FOR USE IN SOLVING PIPE SIZING PROBLEMS

Column	1	2	3	3'	4	5	5'	6	7	8	9
Line	Description	Pressure (kPa)	L/min through section	cubic meter per second through section	Length of section (m)	Trial pipe size (mm)	Trial pipe size inside diameter (mm)	Equivalent length of fittings and valves (m)	Total equivalent length col. 4 and col. 6 (m)	Friction loss in equivalent length (m)	Excess pressure over friction losses (kPa)
A	Minimum pressure available at main	380.0									
B	Highest pressure required at a fixture (Table 604.3)	103.0									
C	Meter loss 50 mm meter	75.0									
D	Tap in main loss 50 mm tap (Table E103A)	11.0									
E	Static head loss 6.4 m $\times$ 9.8 kPa/m	62.7									
F	Special fixture loss backflow preventer	62.0									
G	Special fixture loss—Filter	0.0									
H	Special fixture loss—Other	0.0									
I	Total overall losses and requirements (Sum of Lines B through H)	313.7									
J	Pressure available to overcome pipe friction (Line A minus Lines B to H)	66.3									
K	Pipe section (from diagram)	Pipe wsfu									
	AB	288	409.0	0.007	16	65	62.7	4.57	20.57	1.47	
	BC	264	396.0	0.007	2.4	65	62.7	0.15	2.55	0.17	
	CD	132	291.0	0.005	4	65	62.7	2.13	6.13	0.23	
	CF	132	291.0	0.005	45	65	62.7	3.65	48.65	1.85	
	DE	132	291.0	0.005	45	65	62.7	3.65	48.65	1.85	
K	Total pipe friction losses (cold)									3.72	
L	Difference (Line J minus Line K)										29.9
K	Pipe section (from diagram) hot water	Pipe section (from diagram) hot water									
	AB'	288	409	0.007	16	65	62.7	3.65	19.65	1.40	
	BC'	24	144	0.002	2.4	50	52.5	2.29	4.69	0.12	
	CD'	12	106	0.002	4	40	40.9	1.22	5.22	0.25	
	CF'	12	95	0.002	45	40	40.9	2.13	47.13	1.81	
	DE'	12	106	0.002	45	40	40.9	2.13	47.13	2.21	
K	Total pipe friction losses (hot)									3.97	
L	Difference (Line J minus Line K)										27.4

SECTION E201 SELECTION OF PIPE SIZE

E201.1 Size of water-service mains, branch mains and risers. The minimum size water service pipe shall be 20 mm. The size of water service mains, branch mains and risers shall be determined according to water supply demand [L/min], available water pressure [(kPa)] and friction loss due to the water meter and developed length of pipe [(m)], including equivalent length of fittings. The size of each water distribution system shall be determined according to the procedure outlined in this section or by other design methods conforming to acceptable engineering practice and approved by the code official:

1. Supply load in the building water-distribution system shall be determined by total load on the pipe being sized, in terms of water-supply fixture units (w.s.f.u.), as shown in Table E103.3(2). For fixtures not listed, choose a w.s.f.u. value of a fixture with similar flow characteristics.

2. Obtain the minimum daily static service pressure [(kPa)] available (as determined by the local water authority) at the water meter or other source of supply at the installation location. Adjust this minimum daily static pressure [(kPa)] for the following conditions:

2.1. Determine the difference in elevation between the source of supply and the highest water supply outlet. Where the highest water supply outlet is located above the source of supply, deduct 3.4 kPa for each 0.3 m of difference in elevation. Where the highest water supply outlet is located below the source of supply, add 3.4 kPa for each 0.3 m of difference in elevation.

2.2. Where a water pressure reducing valve is installed in the water distribution system, the minimum daily static water pressure available is 80 percent of the minimum daily static water pressure at the source of supply or the set pressure downstream of the pressure reducing valve, whichever is smaller.

2.3. Deduct all pressure losses due to special equipment such as a backflow preventer, water filter and water softener. Pressure loss data for each piece of equipment shall be obtained through the manufacturer of such devices.

2.4. Deduct the pressure in excess of (55 kPa) due to installation of the special plumbing fixture, such as temperature controlled shower and flushometer tank water closet. Using the resulting minimum available pressure, find the corresponding pressure range in Table E201.1.

15 The maximum developed length for water piping is the actual length of pipe between the source of supply and the most remote fixture, including either hot (through the water heater) or cold water branches multiplied by a factor of 1.2 to compensate for pressure loss through fittings. Select the appropriate column in Table E201.1 equal to or greater than the calculated maximum developed length.

16 To determine the size of water service pipe, meter and main distribution pipe to the building using the appropriate table, follow down the selected "maximum developed length" column to a fixture unit equal to, or greater than the total installation demand calculated by using the "combined" water supply fixture unit column of Table E103.3(2). Read the water service pipe and meter sizes in the first left-hand column and the main distribution pipe to the building in the second left-hand column on the same row.

17 To determine the size of each water distribution pipe, start at the most remote outlet on each branch (either hot or cold branch) and, working back toward the main distribution pipe to the building, add up the water supply fixture unit demand passing through each segment of the distribution system using the related hot or cold column of Table E103.3(2). Knowing demand, the size of each segment shall be read from the second left-hand column of the same table and maximum

developed length column selected in Steps 1 and 2, under the same or next smaller size meter row. In no case does the size of any branch or main need to be larger than the size of the main distribution pipe to the building established in Step 4.

SECTION E202 DETERMINATION OF PIPE VOLUMES

E202.1 Determining volume of piping systems. Where required for engineering design purposes, Table E202.1 shall be used to determine the approximate internal volume of water distribution piping.



TABLE E103.3(2)
LOAD VALUES ASSIGNED TO FIXTURES^(a)

Fixture	Occupancy	Type of Supply Control	Load Values, in Water Supply Fixture Units (wsfu)		
			Cold	Hot	Total
Bathroom group	Private	Flush tank	2.7	1.5	3.6
Bathroom group	Private	Flushometer valve	6.0	3.0	8.0
Bathtub	Private	Faucet	1.0	1.0	1.4
Bathtub	Public	Faucet	3.0	3.0	4.0
Bidet	Private	Faucet	1.5	1.5	2.0
Combination fixture	Private	Faucet	2.25	2.25	3.0
Dishwashing machine	Private	Automatic	—	1.4	1.4
Drinking fountain	Offices, etc.	10 mm valve	0.25	—	0.25
Kitchen sink	Private	Faucet	1.0	1.0	1.4
Kitchen sink	Hotel, restaurant	Faucet	3.0	3.0	4.0
Laundry trays (1 to 3)	Private	Faucet	1.0	1.0	1.4
Lavatory	Private	Faucet	0.5	0.5	0.7
Lavatory	Public	Faucet	1.5	1.5	2.0
Service sink	Offices, etc.	Faucet	2.25	2.25	3.0
Shower head	Public	Mixing valve	3.0	3.0	4.0
Shower head	Private	Mixing valve	1.0	1.0	1.4
Urinal	Public	25 mm flushometer valve	10.0	—	10.0
Urinal	Public	20 mm flushometer valve	5.0	—	5.0
Urinal	Public	Flush tank	3.0	—	3.0
Washing machine (4 kg)	Private	Automatic	1.0	1.0	1.4
Washing machine (4 kg)	Public	Automatic	2.25	2.25	3.0
Washing machine (7 kg)	Public	Automatic	3.0	3.0	4.0
Water closet	Private	Flushometer valve	6.0	—	6.0
Water closet	Private	Flush tank	2.2	—	2.2
Water closet	Public	Flushometer valve	10.0	—	10.0
Water closet	Public	Flush tank	5.0	—	5.0
Water closet	Public or private	Flushometer tank	2.0	—	2.0

- a. For fixtures not listed loads should be assumed by comparing the fixture to one listed using water in similar quantities and at similar rates. The assigned loads for fixtures with both hot and cold water supplies are given for separate hot and cold water loads and for total load. The separate hot and cold water loads being three-fourths of the total load for the fixture in each case.

TABLE E103.3(3)
TABLE FOR ESTIMATING DEMAND

Supply Systems Predominantly for Flush Tanks			Supply Systems Predominantly for Flushometer Valves		
Load	Demand		Load	Demand	
(Water supply fixture units)	(Liter per minute)	(Cubic meter per second)	(Water supply fixture units)	(Liter per minute)	(Cubic meter per second)
1	11.4	0.000189	—	—	—
2	18.9	0.000315	—	—	—
3	24.6	0.000410	—	—	—
4	30.3	0.000505	—	—	—
5	35.6	0.000593	5	56.8	0.000946
6	40.5	0.000675	6	65.9	0.001098
7	44.7	0.000744	7	74.9	0.001249
8	48.4	0.000807	8	84.0	0.001400
9	51.9	0.000864	9	93.1	0.001552
10	55.3	0.000921	10	102.2	0.001703
11	58.3	0.000971	11	105.2	0.001754
12	60.6	0.001009	12	108.3	0.001804
13	62.5	0.001041	13	111.3	0.001855
14	64.3	0.001072	14	114.3	0.001905
15	66.2	0.001104	15	117.3	0.001956
16	68.1	0.001136	16	120.4	0.002006
17	69.6	0.001161	17	123.4	0.002057
18	71.2	0.001186	18	126.4	0.002107
19	72.7	0.001211	19	129.4	0.002157
20	74.2	0.001236	20	132.5	0.002208
25	81.4	0.001356	25	143.8	0.002397
30	88.2	0.001470	30	159.0	0.002650
35	94.2	0.001571	35	166.5	0.002776
40	99.5	0.001659	40	174.1	0.002902
45	104.8	0.001747	45	181.7	0.003028
50	110.1	0.001836	50	189.3	0.003154
60	121.1	0.002019	60	204.4	0.003407
70	132.5	0.002208	70	219.5	0.003659
80	143.8	0.002397	80	231.6	0.003861
90	155.2	0.002586	90	243.4	0.004056
100	164.6	0.002744	100	255.5	0.004258
120	181.7	0.003028	120	276.3	0.004605
140	198.7	0.003312	140	291.4	0.004857
160	215.7	0.003596	160	306.6	0.005110
180	230.9	0.003848	180	323.6	0.005394
200	246.0	0.004100	200	340.7	0.005678
225	265.0	0.004416	225	361.5	0.006024
250	283.9	0.004731	250	382.3	0.006371
275	302.8	0.005047	275	395.5	0.006592
300	321.7	0.005362	300	408.8	0.006813
400	397.4	0.006624	400	480.7	0.008012
500	469.3	0.007822	500	541.3	0.009021
750	643.5	0.010724	750	669.9	0.011166
1,000	787.3	0.013121	1,000	787.3	0.013121
1,250	904.6	0.015077	1,250	904.6	0.015077
1,500	1018.2	0.016969	1,500	1018.2	0.016969
1,750	1124.1	0.018736	1,750	1124.1	0.018736
2,000	1230.1	0.020502	2,000	1230.1	0.020502
2,500	1438.3	0.023972	2,500	1438.3	0.023972
3,000	1638.9	0.027315	3,000	1638.9	0.027315
4,000	1987.1	0.033119	4,000	1987.1	0.033119
5,000	2244.5	0.037408	5,000	2244.5	0.037408

TABLE E103.3(4)
LOSS OF PRESSURE THROUGH TAPS AND TEES IN kPa

Liter per Minute	Size of Tap or Tee (mm)						
	16	20	25	32	40	50	80
37.9	9.3	4.4	1.2	0.6	—	—	—
75.7	37.1	17.5	5.3	2.1	1.0	—	—
113.6	83.4	39.4	11.2	4.8	2.3	0.7	—
151.4	—	70.3	21.2	8.5	4.0	1.2	—
189.3	—	109.6	31.0	13.2	6.3	1.9	—
227.1	—	—	44.5	19.0	9.0	2.8	—
265.0	—	—	60.6	25.9	12.3	3.8	0.7
302.8	—	—	79.3	33.8	16.0	5.0	0.9
340.7	—	—	100.0	42.8	20.3	6.3	1.1
378.5	—	—	123.7	52.9	25.0	7.7	1.4
454.2	—	—	177.9	75.8	36.1	11.1	2.1
529.9	—	—	242.7	103.4	49.1	15.2	2.8
567.8	—	—	—	118.6	56.3	17.4	3.2
605.6	—	—	—	135.1	64.1	20.1	3.7
681.3	—	—	—	171.0	81.4	25.0	4.7
757.0	—	—	—	211.7	100.0	30.9	5.8
851.6	—	—	—	267.5	126.9	38.6	7.3
946.3	—	—	—	330.3	156.5	48.3	9.0
1040.9	—	—	—	—	188.9	53.1	11.0
1135.5	—	—	—	—	224.8	69.6	13.0

TABLE E103.3(5)
ALLOWANCE IN EQUIVALENT LENGTHS OF PIPE FOR FRICTION LOSS IN VALVES AND
THREADED FITTINGS (meter)

Fitting or Valve	Pipe Size (mm)							
	15	20	25	32	40	50	65	80
45-degree elbow	0.4	0.5	0.6	0.7	0.9	1.2	1.5	1.8
90-degree elbow	0.6	0.8	0.9	1.2	1.5	2.1	2.4	3.1
Tee, run	0.2	0.2	0.3	0.4	0.5	0.6	0.8	0.9
Tee, branch	0.9	1.2	1.5	1.8	2.1	3.1	3.7	4.6
Gate valve	0.1	0.2	0.2	0.2	0.3	0.4	0.5	0.6
Balancing valve	0.2	0.3	0.5	0.6	0.7	0.9	1.1	1.4
Plug-type cock	0.2	0.3	0.5	0.6	0.7	0.9	1.1	1.4
Check valve, swing	1.7	2.6	3.4	4.3	5.1	6.8	8.5	10.3
Globe valve	4.6	6.1	7.6	10.7	13.7	16.8	19.8	24.4
Angle valve	2.4	3.7	4.6	5.5	6.7	8.5	10.4	12.2

TABLE E103.3(6)
PRESSURE LOSS IN FITTINGS AND VALVES EXPRESSED AS EQUIVALENT LENGTH OF
TUBE^(a) (meter)

Nominal or Standard Size (mm)	Fittings				Coupling	Valves			
	Standard EII		90-Degree Tee			Ball	Gate	Butterfly	Check
	90 Degree	45 Degree	Side Branch	Straight Run					
10	0.2	—	0.5	—	—	—	—	—	0.5
15	0.3	0.2	0.6	—	—	—	—	—	0.6
16	0.5	0.2	0.6	—	—	—	—	—	0.8
20	0.6	0.2	0.9	—	—	—	—	—	0.9
25	0.8	0.3	1.4	—	—	0.2	—	—	1.4
32	0.9	0.3	1.7	0.2	0.2	0.2	—	—	1.7
40	1.2	0.5	2.1	0.2	0.2	0.2	—	—	2.0
50	1.7	0.6	2.7	0.2	0.2	0.2	0.2	2.3	2.7
65	2.1	0.8	3.7	0.2	0.2	—	0.3	3.1	3.5
80	2.7	1.1	4.6	0.3	0.3	—	0.5	4.7	4.4
90	2.7	1.1	4.3	0.3	0.3	—	0.6	—	3.8
100	3.8	1.5	6.4	0.3	0.3	—	0.6	4.9	5.6
125	4.9	1.8	8.2	0.5	0.5	—	0.9	3.5	7.2
150	5.8	2.1	10.4	0.6	0.6	—	1.1	4.1	8.1
200	8.8	3.4	15.2	0.9	0.9	—	1.5	3.8	11.9

a. Allowances are for streamlined soldered fittings and recessed threaded fittings. For threaded fittings, double the allowances shown in the table. The equivalent lengths presented above are based on a C factor of 150 in the Hazen-Williams friction loss formula. The lengths shown are rounded to the nearest half-foot.

TABLE E201.1
MINIMUM SIZE OF WATER METERS, MAINS AND DISTRIBUTION PIPING
BASED ON WATER SUPPLY FIXTURE UNIT VALUES (w.s.f.u.)

Meter and Service Pipe (mm)	Distribution Pipe (mm)	Maximum Development Length (meter)									
Pressure Range 200 to 269 kPa		40	60	80	100	150	200	250	300	400	500
20	15 ^(a)	0.8	0.6	0.5	0.5	0.3	0.3	0.2	0.2	0.0	0.0
20	20	2.9	2.3	1.8	1.7	1.2	1.1	0.9	0.8	0.6	0.5
20	25	9.8	7.6	6.1	5.0	3.4	2.7	2.4	2.0	1.7	1.4
25	25	9.8	9.8	8.2	6.4	4.1	3.0	2.4	2.1	1.7	1.5
20	32	9.8	9.8	9.8	9.8	9.1	7.3	6.1	5.2	4.0	3.2
25	32	24.4	24.4	21.3	18.6	13.7	10.4	8.2	6.7	4.9	3.7
40	32	24.4	24.4	24.4	22.9	16.5	12.2	9.4	7.6	5.3	4.0
25	40	26.5	26.5	26.5	26.5	25.6	22.3	19.5	17.1	13.7	11.0
40	40	46.0	46.0	46.0	46.0	35.7	28.0	24.1	21.0	16.5	13.1
50	40	46.0	46.0	46.0	46.0	39.0	30.2	25.3	21.9	17.1	13.7
25	50	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.2
40	50	83.8	83.8	83.8	83.8	78.6	68.0	59.7	53.0	43.9	37.2
50	50	111.3	111.3	111.3	111.3	96.9	81.1	69.8	61.3	48.8	40.8
50	65	162.5	162.5	162.5	162.5	162.5	150.9	136.6	124.7	107.6	94.8

Meter and Service Pipe (mm)	Distribution Pipe (mm)	Maximum Development Length (meter)									
Pressure Range 270 to 339 kPa		40	60	80	100	150	200	250	300	400	500
20	15 ^(a)	0.9	0.8	0.6	0.5	0.5	0.3	0.3	0.2	0.2	0.2
20	20	2.9	2.9	2.6	2.1	1.7	1.4	1.1	0.9	0.8	0.6
20	25	9.8	9.8	9.8	7.9	5.5	4.1	3.2	2.7	2.3	1.8
25	25	9.8	9.8	9.8	9.8	6.4	4.6	3.5	2.9	2.3	2.0
20	32	9.8	9.8	9.8	9.8	9.8	9.8	9.8	8.2	6.4	5.0
25	32	24.4	24.4	24.4	24.4	19.8	15.8	12.8	10.7	7.9	6.1
40	32	24.4	24.4	24.4	24.4	22.9	18.0	14.6	11.9	8.5	6.4
25	40	26.5	26.5	26.5	26.5	26.5	26.5	26.5	23.8	19.8	16.8
40	40	46.0	46.0	46.0	46.0	46.0	39.6	33.2	28.3	22.9	19.2
50	40	46.0	46.0	46.0	46.0	46.0	42.4	35.1	29.9	23.5	19.5
25	50	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5
40	50	83.8	83.8	83.8	83.8	83.8	83.8	80.5	72.5	60.4	51.5
50	50	111.3	111.3	111.3	111.3	111.3	106.4	92.7	82.3	67.1	56.4
50	65	162.5	162.5	162.5	162.5	162.5	162.5	162.5	160.9	139.0	122.8

Meter and Service Pipe (mm)	Distribution Pipe (mm)	Maximum Development Length (meter)									
Pressure Range 340 to 415 kPa		40	60	80	100	150	200	250	300	400	500
20	15 ^(a)	0.9	0.9	0.8	0.6	0.5	0.3	0.3	0.3	0.2	0.2
20	20	2.9	2.9	2.9	2.6	2.0	1.5	1.4	1.2	0.9	0.8
20	25	9.8	9.8	9.8	9.8	7.6	5.6	4.4	3.7	2.9	2.4
25	25	9.8	9.8	9.8	9.8	9.1	6.7	5.0	4.0	3.0	2.4
20	32	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	8.8	7.3
25	32	24.4	24.4	24.4	24.4	24.4	20.7	17.4	14.6	10.7	8.5
40	32	24.4	24.4	24.4	24.4	24.4	22.9	19.2	16.2	11.9	8.8
25	40	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	25.0	21.3
40	40	46.0	46.0	46.0	46.0	46.0	46.0	42.4	36.6	28.7	24.1
50	40	46.0	46.0	46.0	46.0	46.0	46.0	44.5	38.4	29.6	24.7
25	50	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5
40	50	83.8	83.8	83.8	83.8	83.8	83.8	83.8	83.8	75.3	64.9
50	50	111.3	111.3	111.3	111.3	111.3	111.3	111.3	100.3	82.9	70.7
50	65	162.5	162.5	162.5	162.5	162.5	162.5	162.5	162.5	162.5	148.1

Meter and Service Pipe (mm)	Distribution Pipe (mm)	Maximum Development Length (meter)									
Pressure Range Over 415 kPa		40	60	80	100	150	200	250	300	400	500
20	15 ^(a)	0.9	0.9	0.9	0.8	0.6	0.5	0.5	0.3	0.3	0.2
20	20	2.9	2.9	2.9	2.9	2.3	1.8	1.5	1.4	1.1	0.9
20	25	9.8	9.8	9.8	9.8	9.8	7.3	5.9	4.7	3.5	2.9
25	25	9.8	9.8	9.8	9.8	9.8	8.5	8.5	5.2	3.7	2.9
20	32	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.8	9.1
25	32	24.4	24.4	24.4	24.4	24.4	24.4	21.0	18.3	14.0	11.0
40	32	24.4	24.4	24.4	24.4	24.4	24.4	23.2	19.8	15.2	11.6
25	40	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	25.6
40	40	46.0	46.0	46.0	46.0	46.0	46.0	46.0	43.9	34.7	28.7
50	40	46.0	46.0	46.0	46.0	46.0	46.0	46.0	46.0	36.0	29.6
25	50	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5	26.5
40	50	83.8	83.8	83.8	83.8	83.8	83.8	83.8	83.8	83.8	76.8
50	50	111.3	112.2	112.2	112.2	112.2	112.2	112.2	112.2	96.9	83.2
50	65	162.5	162.5	162.5	162.5	162.5	162.5	162.5	162.5	162.5	162.5

a. Minimum size for building supply is 20 mm pipe.



TABLE E202.1
INTERNAL VOLUME OF VARIOUS WATER DISTRIBUTION TUBING

Size Nominal (mm)	Liter of Water per Meter of Tube						
	Copper Type M	Copper Type L	Copper Type K	CPVC CTS SDR 11	CPVC SCH 40	Composite ASTM F1281	PEX CTS SDR 9
10	0.104	0.095	0.083	N/A	0.115	0.062	0.063
15	0.166	0.153	0.143	0.123	0.186	0.129	0.116
20	0.338	0.317	0.285	0.263	0.333	0.334	0.231
25	0.572	0.540	0.509	0.436	0.544	0.547	0.385
32	0.856	0.823	0.796	0.651	0.951	0.836	0.572
40	1.199	1.164	1.127	0.907	1.299	1.366	0.796
50	2.075	2.026	1.972	1.554	2.154	2.114	1.364

HOT WATER
COLD WATER
M = METER
BFP = BACKFLOW PREVENTER

↖ = 90 DEGREE ELBOW
T = "T"
V = VALVE

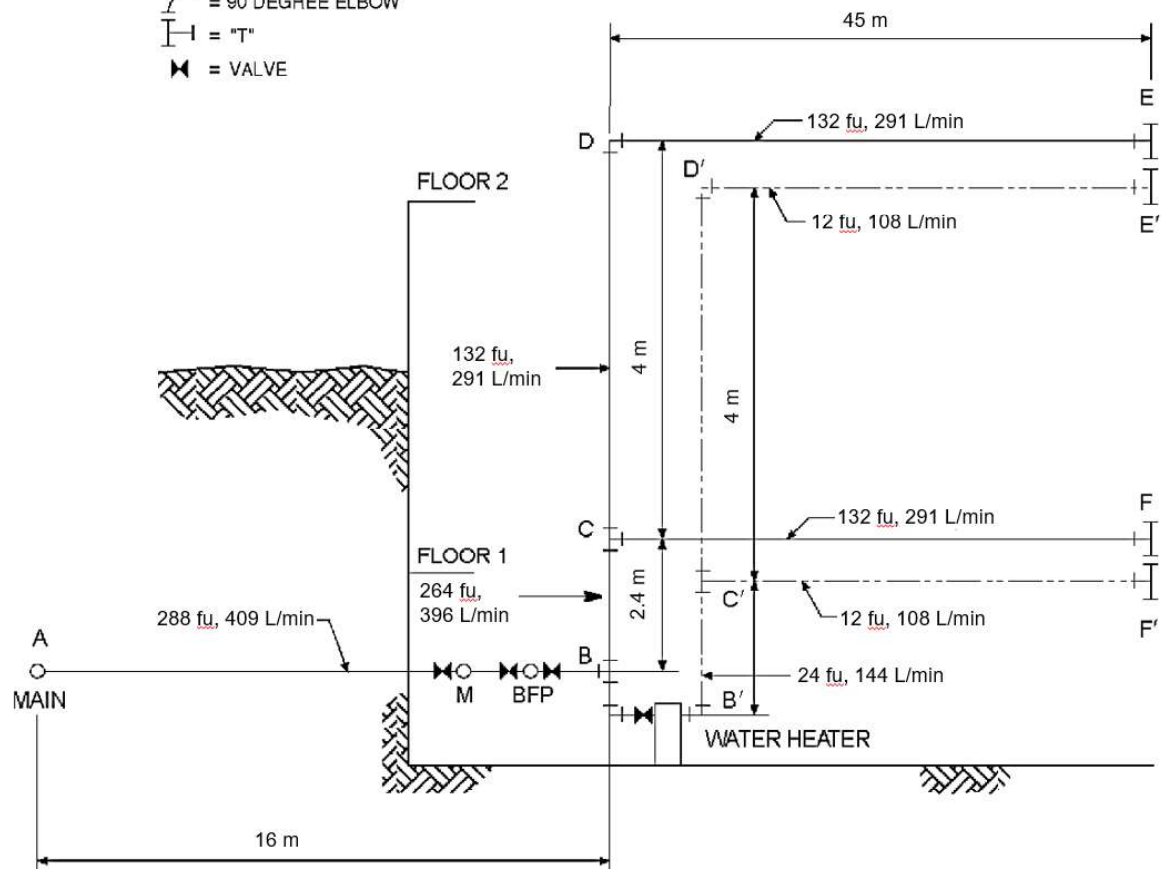


FIGURE E103.3(1)
EXAMPLE—SIZING